

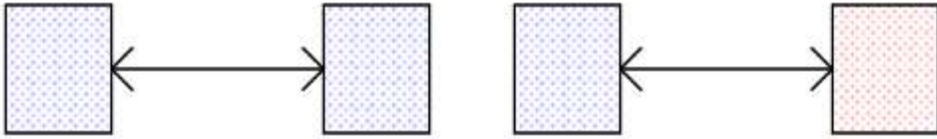
3조

조장	유경환
곽대현	전제현
김규민	박재호
전민석	이준민

Definition

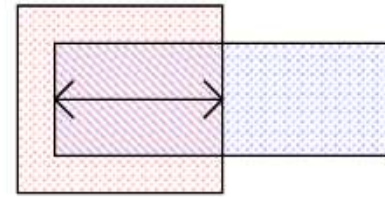
● Spacing

- 한 도형의 외곽경계에서 다른 도형의 외곽까지의 거리



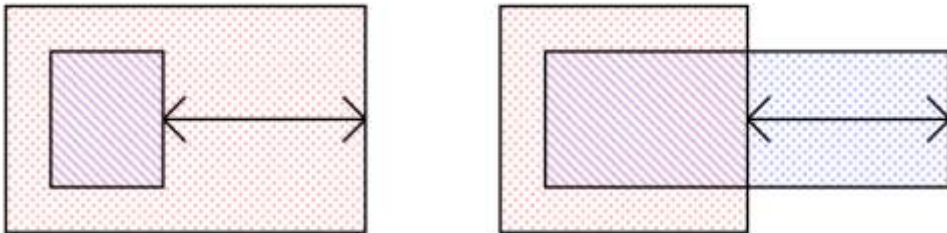
● Overlap

- 두 도형이 서로 겹쳐 있는 영역의 거리



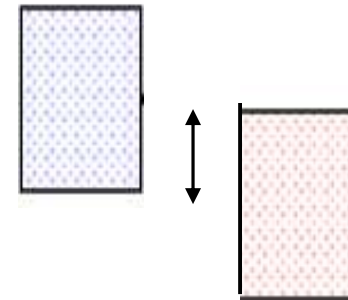
● Enclosure

- 한 도형이 다른 도형을 얼마나 감싸고 있는지를 나타내는 거리



● Parallel length

- 두 도형의 서로 마주보는 가장자리가 겹쳐 있는 길이



Layer colors

STI

Substrate

Nwell

POLY / Gate

METAL

VIA

IMD

ILD

Sillicide

P+

N+

CONTACT

OXIDE

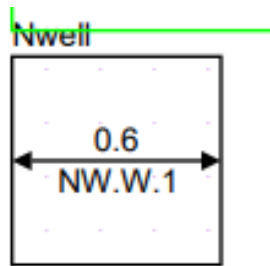
Thick OXIDE

SIPROT

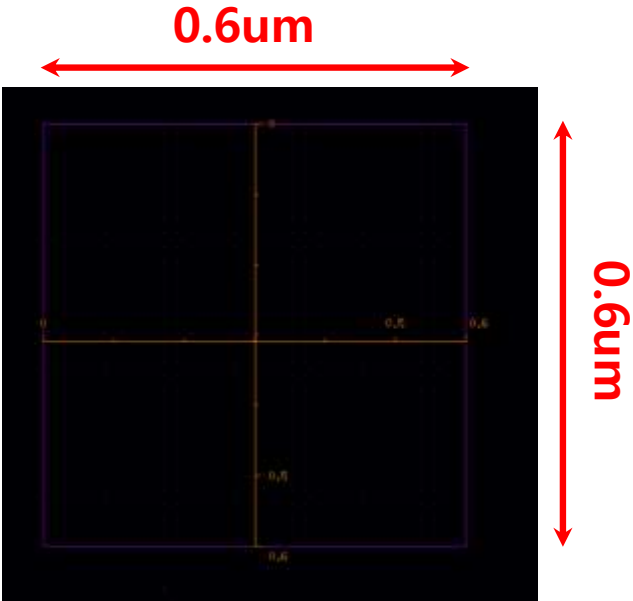
Nwell

Rule Name	Value	Description
NW.W.1	0.6um	Minimum Nwell width

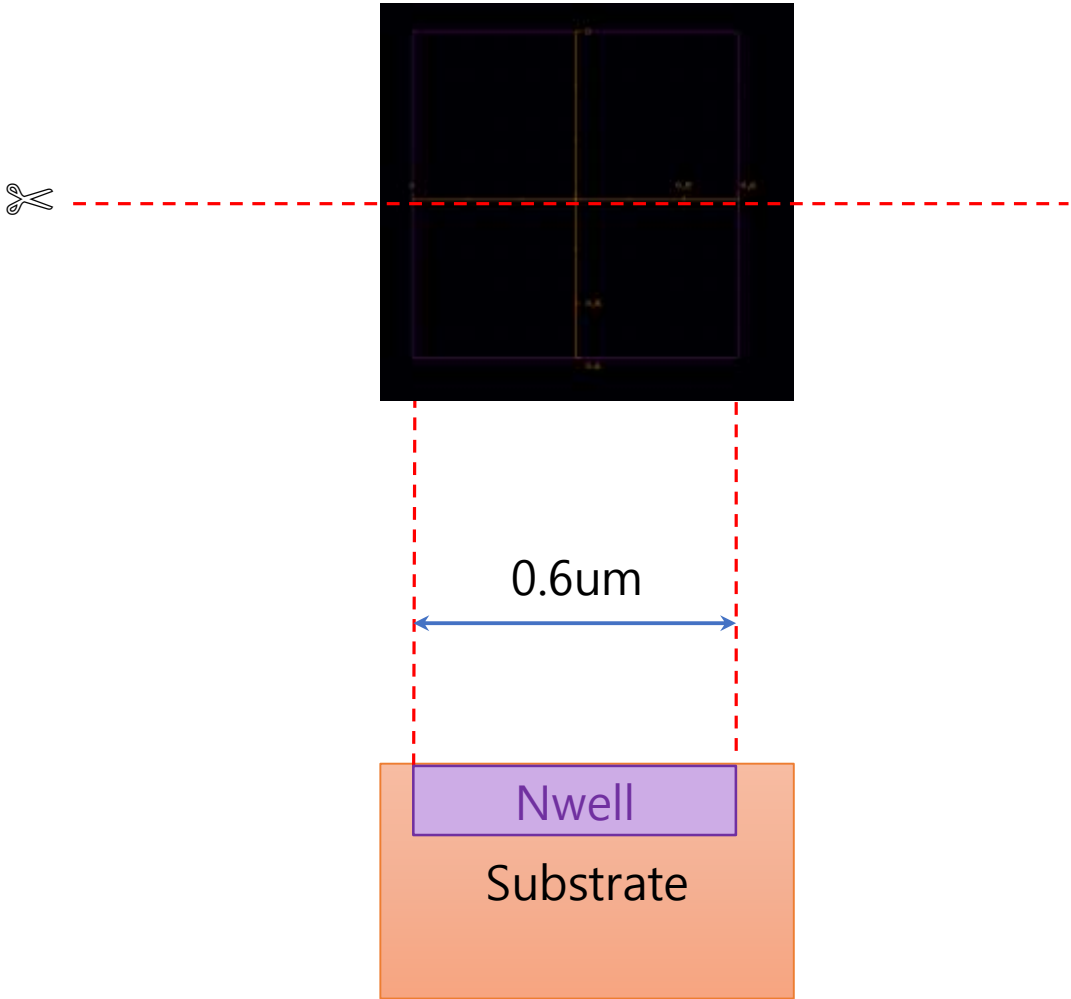
✓ Schematic in Design Rule Manual



Nwell의 폭이 너무 좁으면 공정 오차로 인해 well이 정상적으로 형성되지 않을 수 있다.

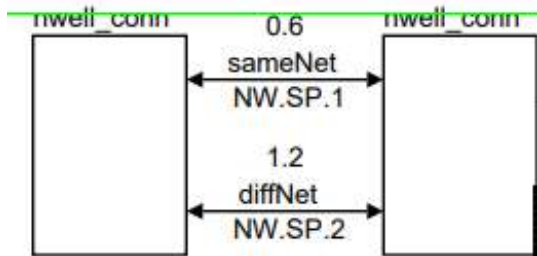


✓ Cadence Reproducing and vertical structure

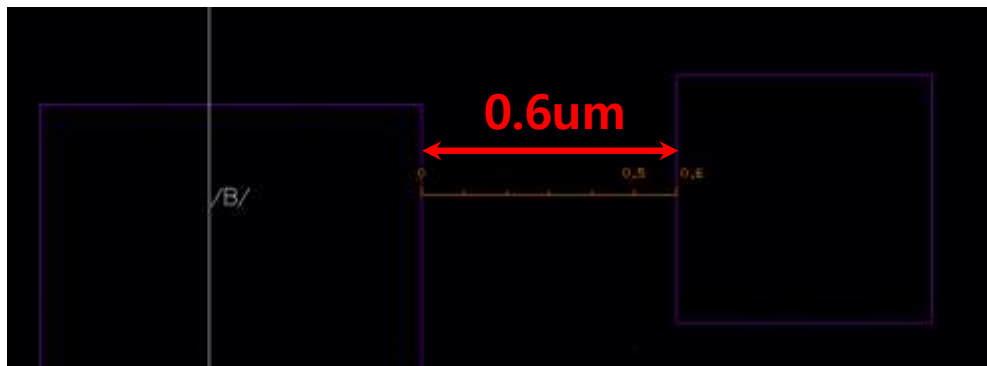


Rule Name	Value	Description
NW.SP.1	0.6um	Minimum Nwell spacing to Nwell (same potential)

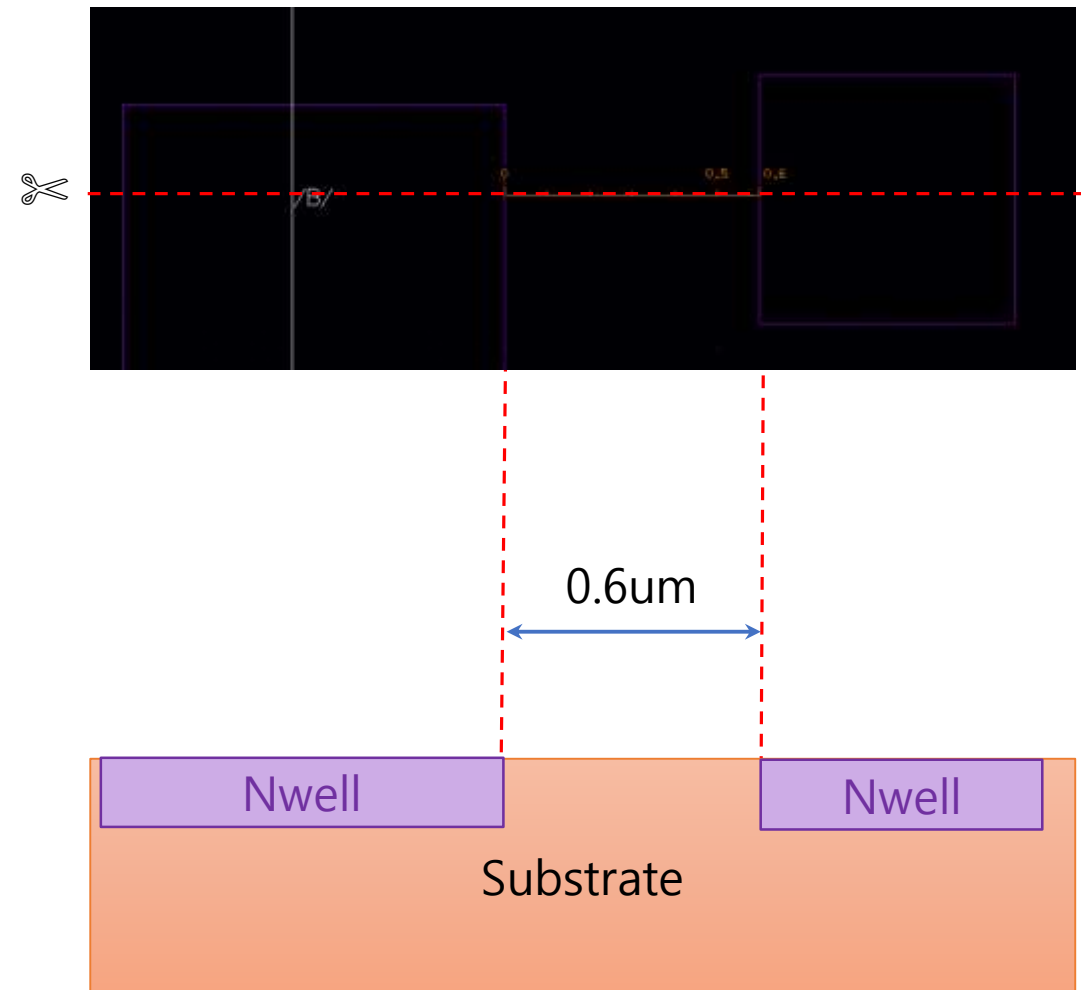
✓ Schematic in Design Rule Manual



Nwell의 간격이 너무 좁으면 공정 오차로 인해 well이 서로 붙거나 형상이 변형돼 punch-through 현상이나 latch-up 현상이 일어날 수 있다.

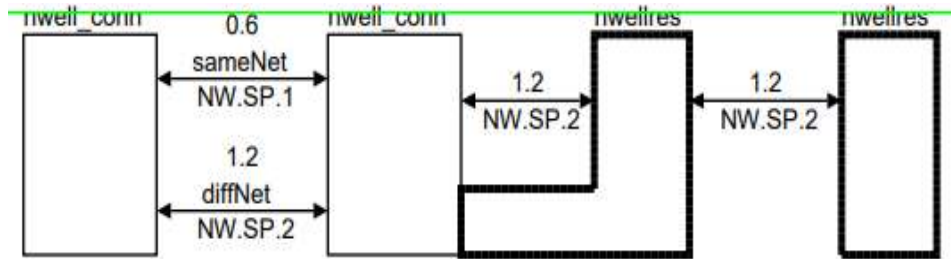


✓ Cadence Reproducing and vertical structure



Rule Name	Value	Description
NW.SP.2	1.2um	Minimum Nwell spacing to Nwell (different potential)

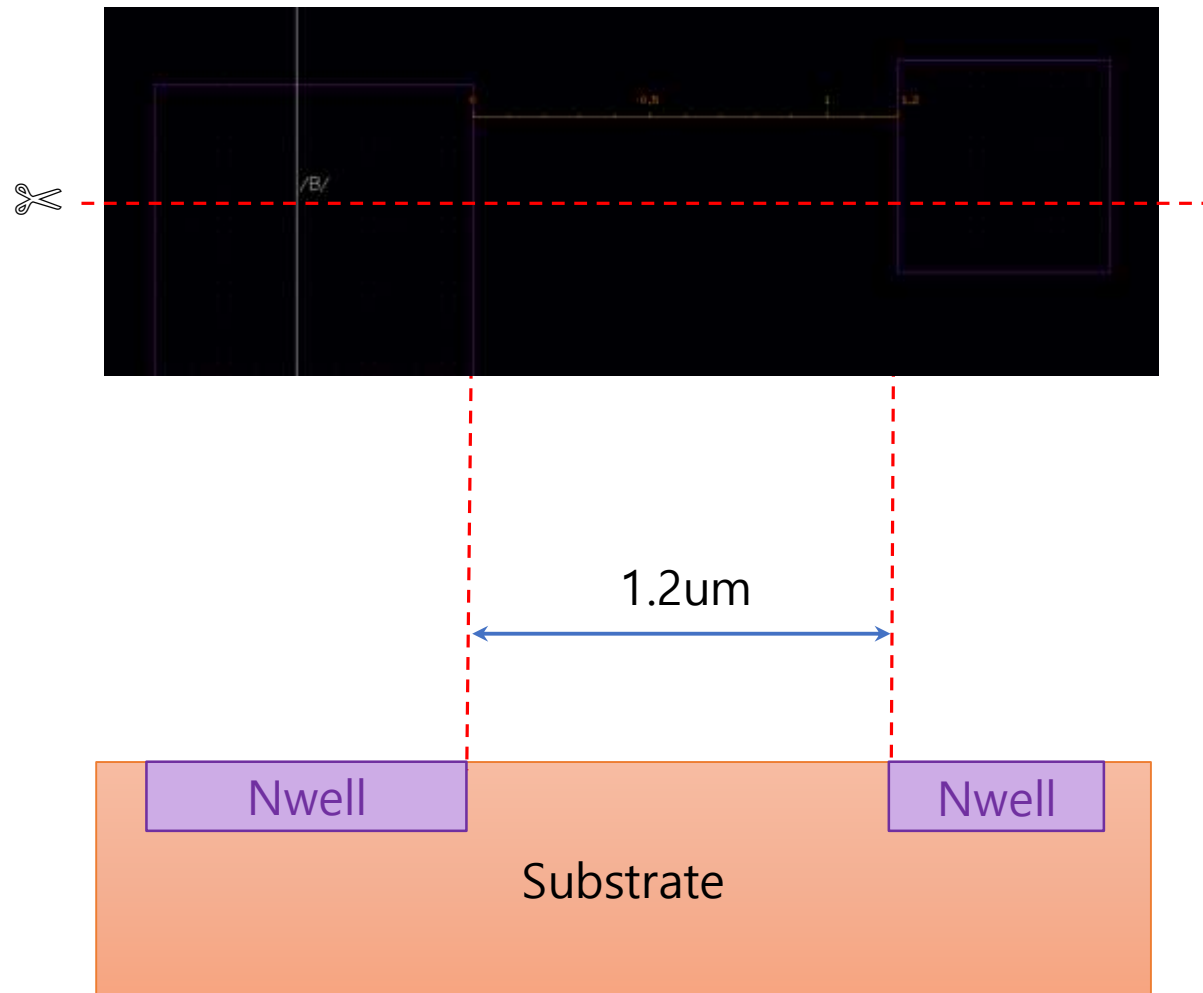
✓ Schematic in Design Rule Manual



서로 다른 전압이 인가되면 넓어지는 depletion 영역으로 인해 leakage path가 발생할 가능성이 더 커지므로 더 큰 최소 간격이 필요하다.

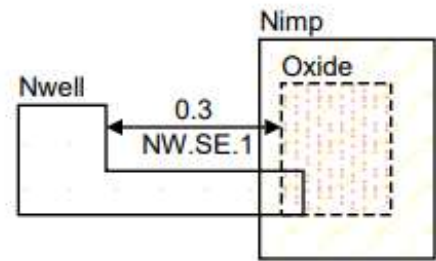


✓ Cadence Reproducing and vertical structure

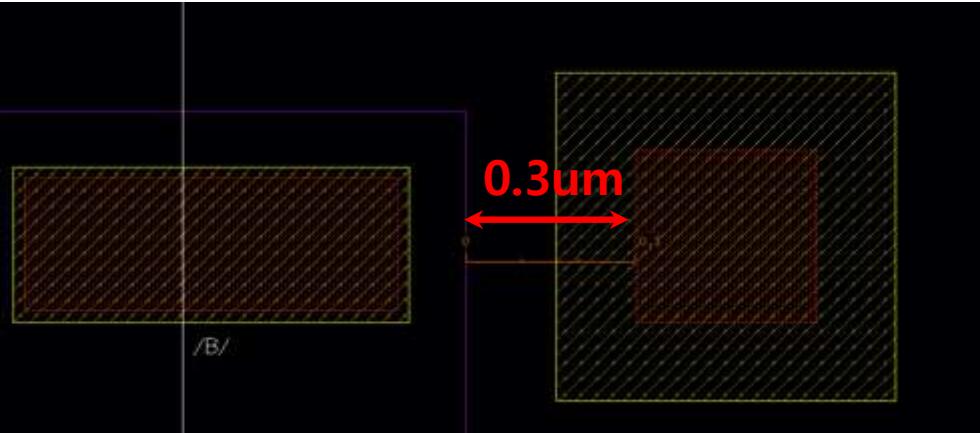


Rule Name	Value	Description
NW.SE.1	0.3um	Minimum Nwell spacing to N+ Active Area

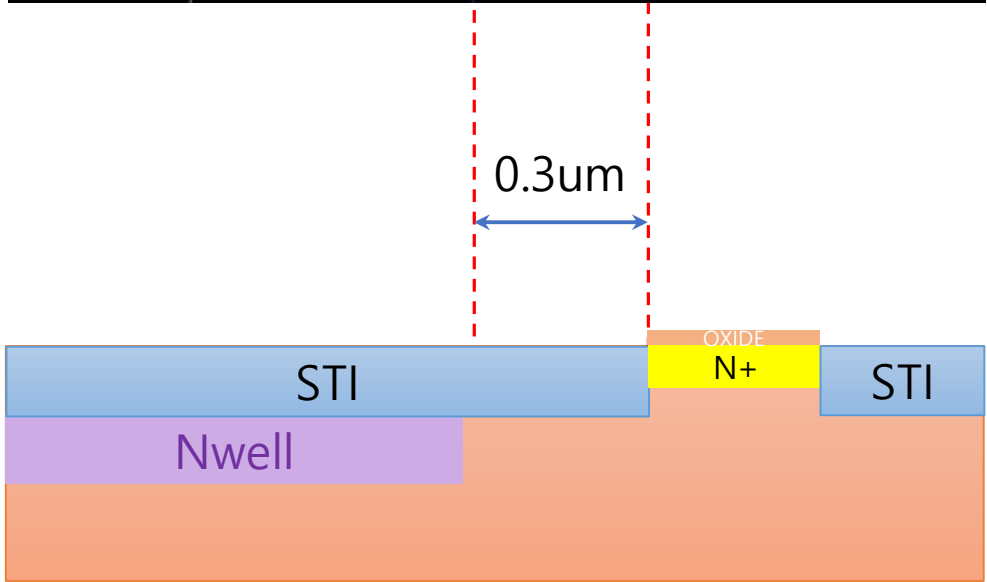
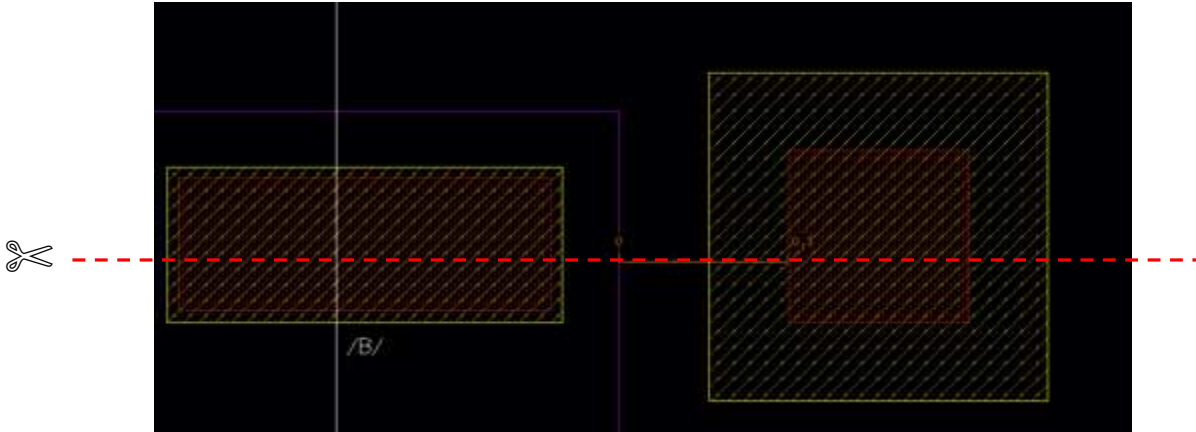
✓ Schematic in Design Rule Manual



두 영역이 너무 가까우면
latch up으로 인한 leakage path 현상이 발생할 수 있다.

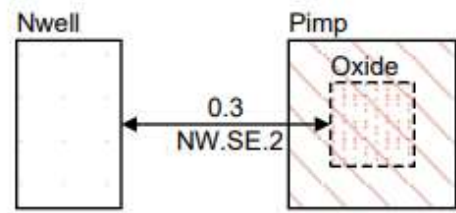


✓ Cadence Reproducing and vertical structure

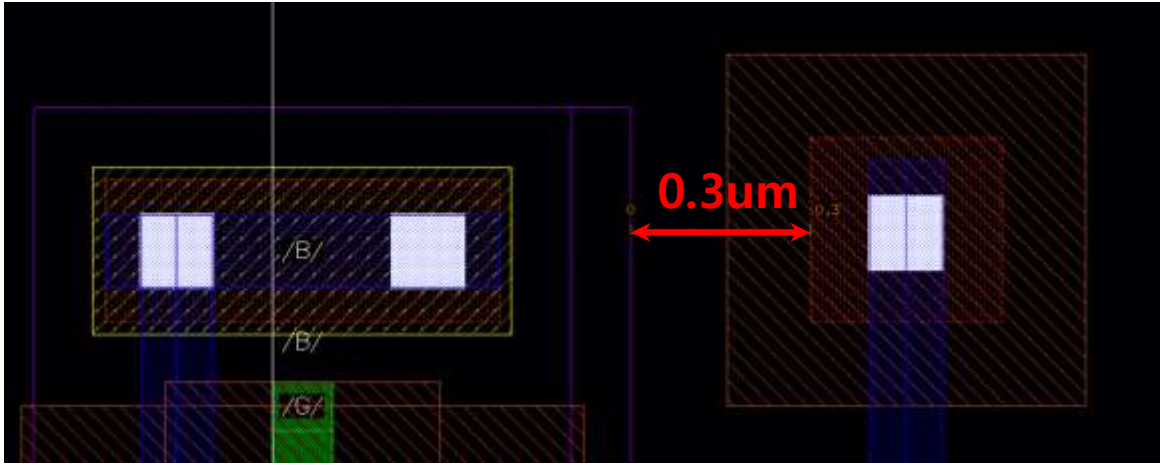


Rule Name	Value	Description
NW.SE.2	0.3um	Minimum Nwell spacing to P+ Active Area

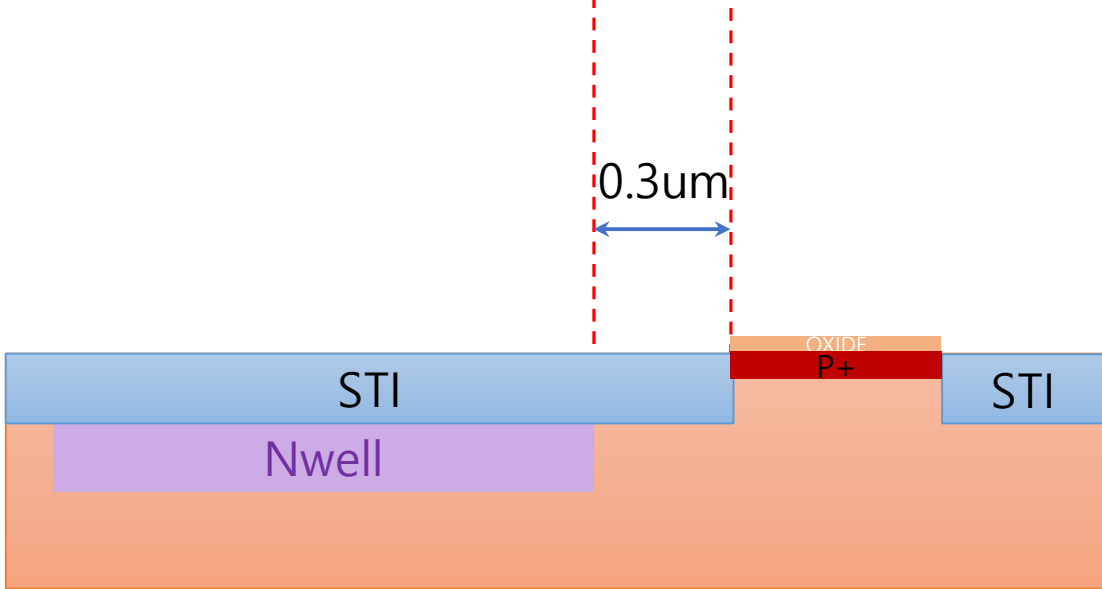
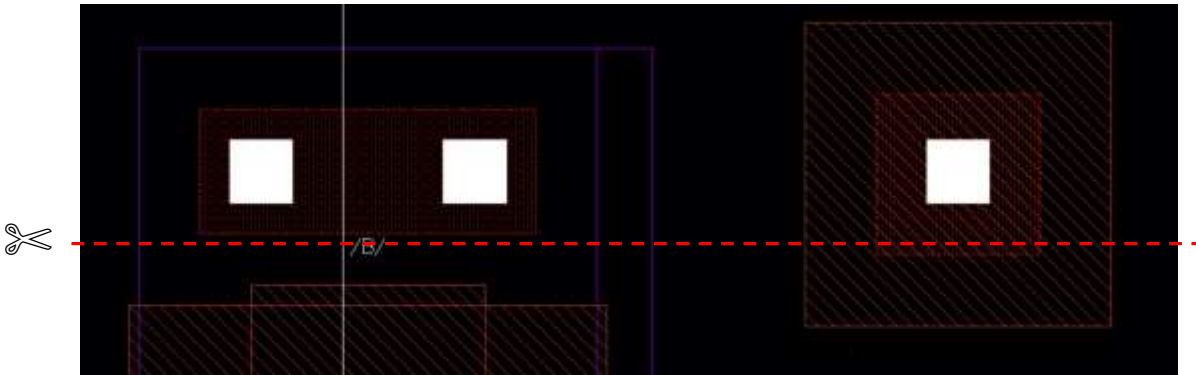
✓ Schematic in Design Rule Manual



두 영역이 너무 가까우면
latch up으로 인한 leakage path 현상이 발생할 수 있다.

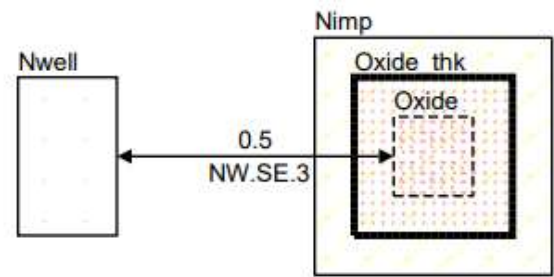


✓ Cadence Reproducing and vertical structure

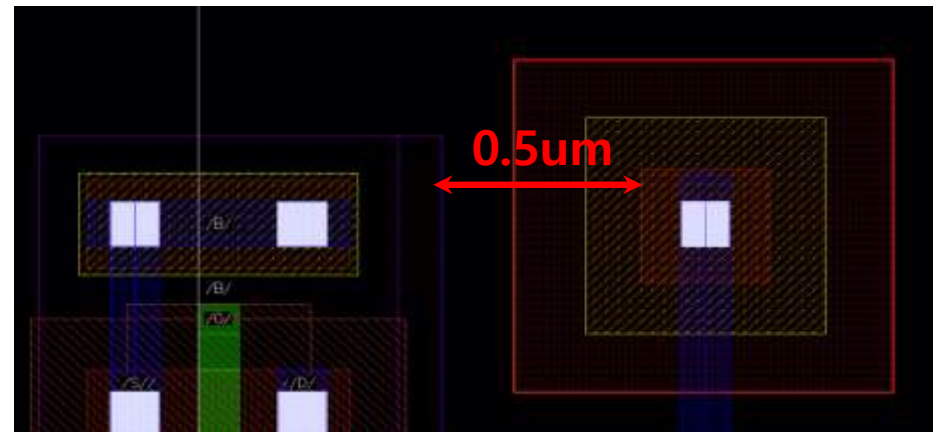


Rule Name	Value	Description
NW.SE.3	0.5um	Minimum Nwell spacing to N+ 2.5V Active Area

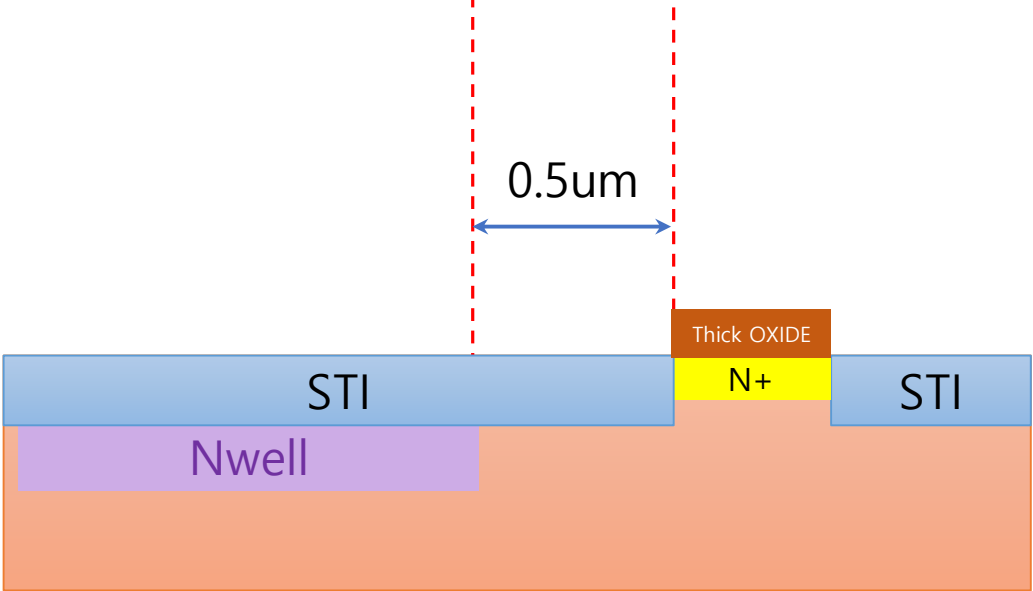
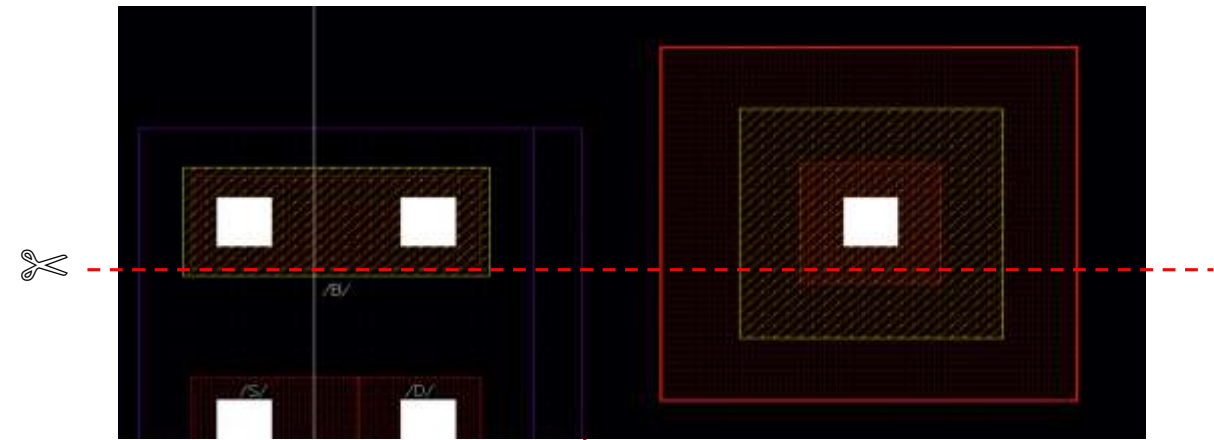
✓ Schematic in Design Rule Manual



더 큰 전압이 인가되어 더 큰 간격을 확보해야 leakage path 현상이 발생하지 않는다.

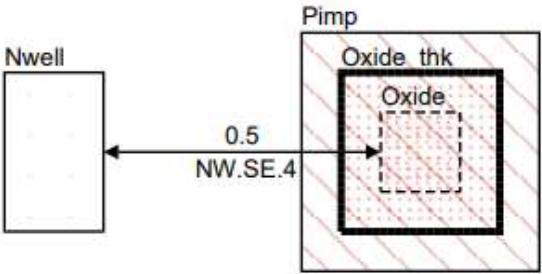


✓ Cadence Reproducing and vertical structure



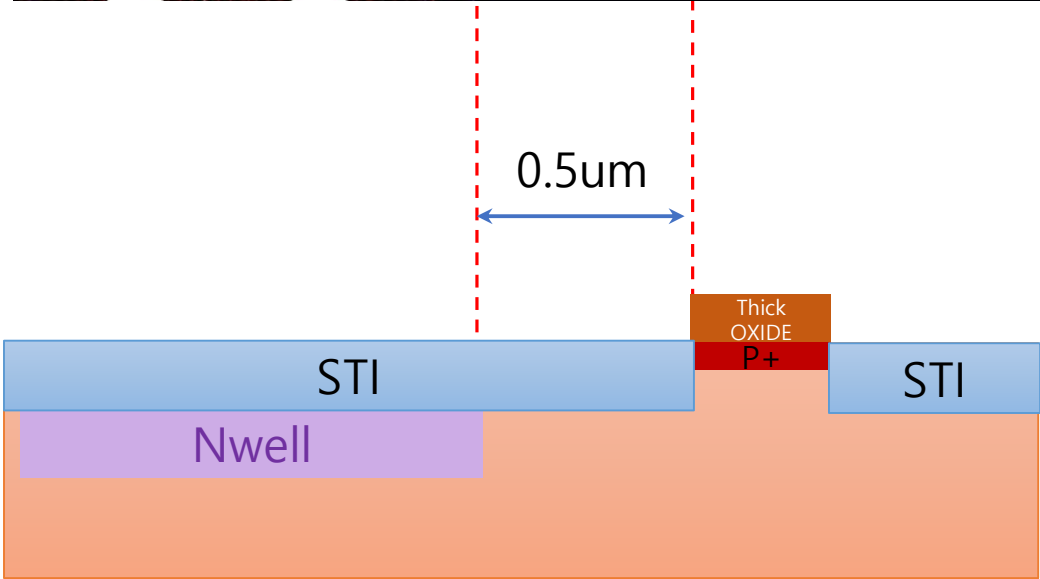
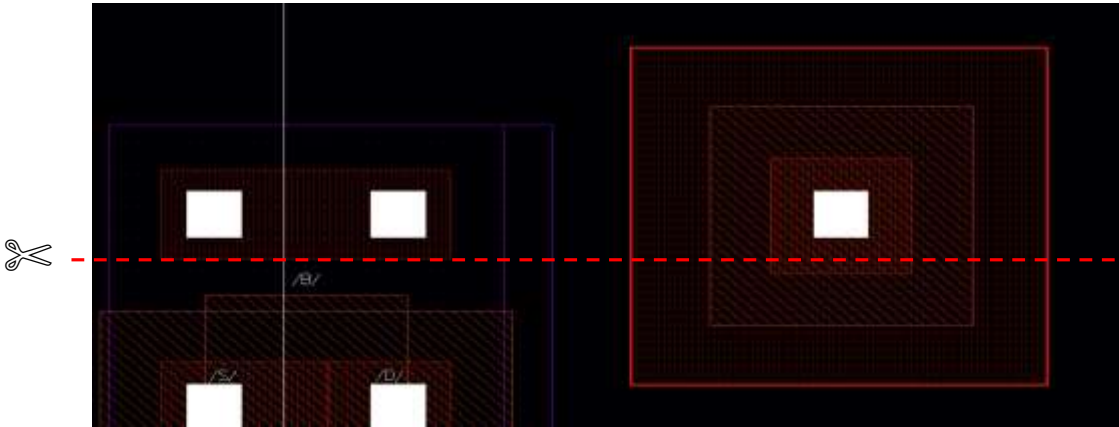
Rule Name	Value	Description
NW.SE.4	0.5um	Minimum Nwell spacing to P+ 2.5V Active Area

✓ Schematic in Design Rule Manual



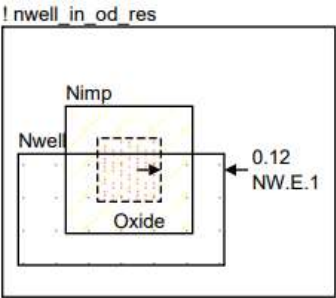
더 큰 전압이 인가되어 더 큰 간격을 확보해야 leakage path 현상이 발생하지 않는다.

✓ Cadence Reproducing and vertical structure

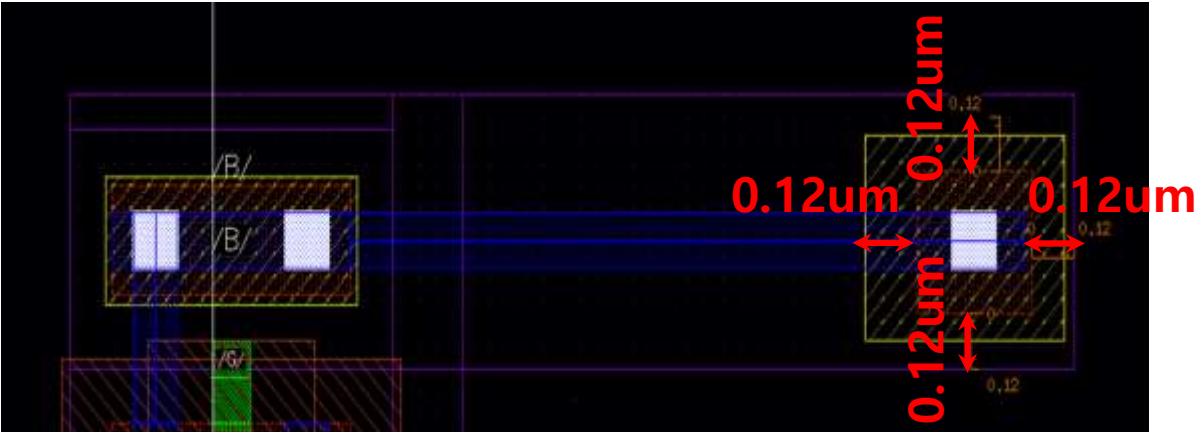


Rule Name	Value	Description
NW.E.1	0.12um	Minimum Nwell enclosure of N+ Active Area

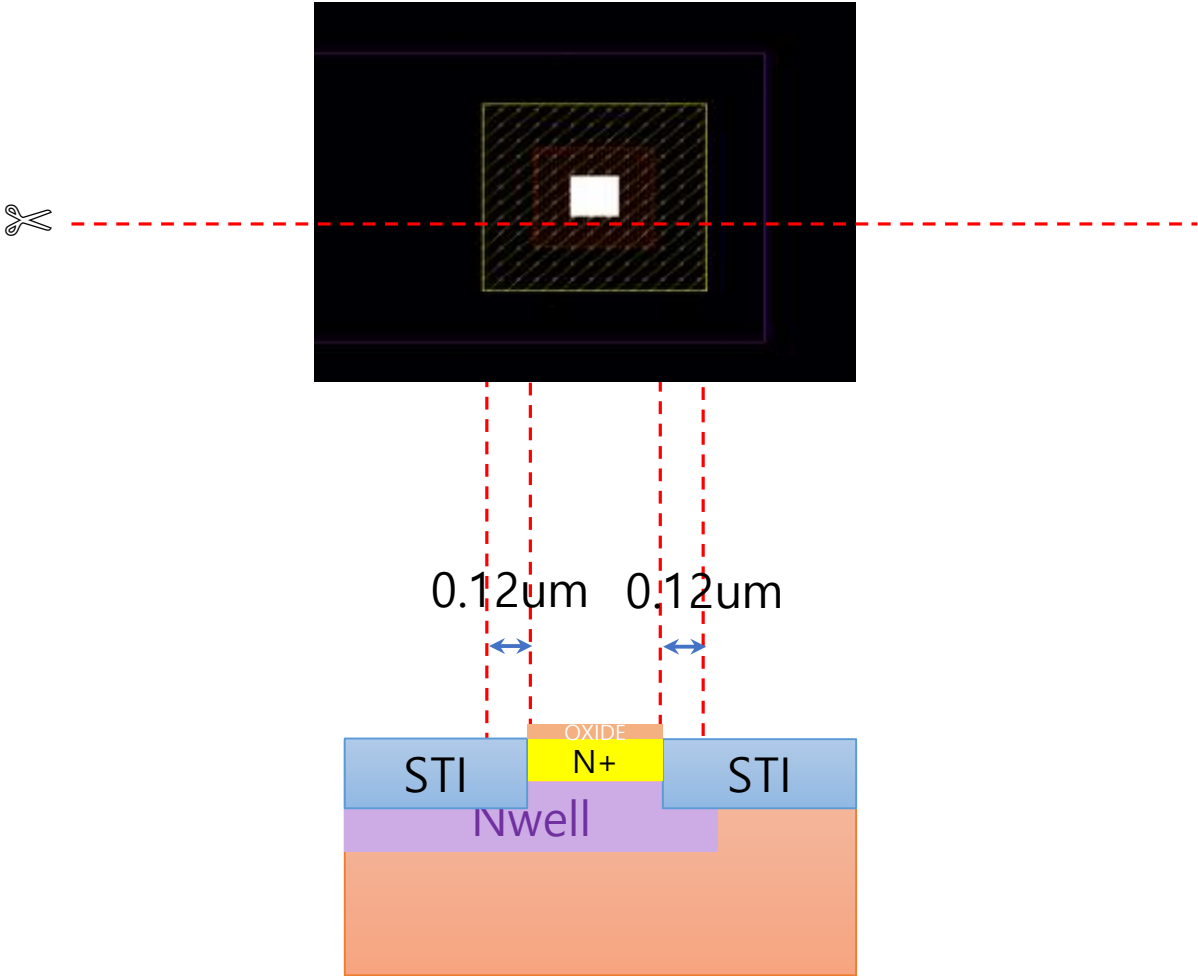
✓ Schematic in Design Rule Manual



enclosure가 부족하면 공정 오차로 인해 active area가 Nwell 밖으로 벗어날 수 있어 안정된 active를 형성하기 위해 필요하다.

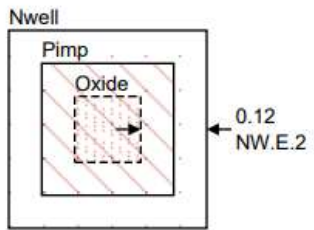


✓ Cadence Reproducing and vertical structure

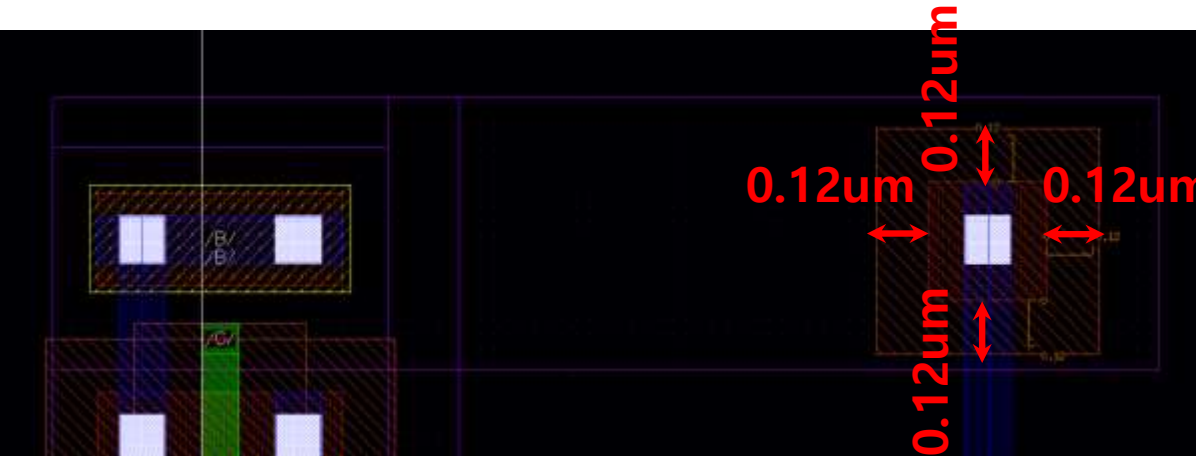


Rule Name	Value	Description
NW.E.2	0.12um	Minimum Nwell spacing to P+ 2.5V Active Area

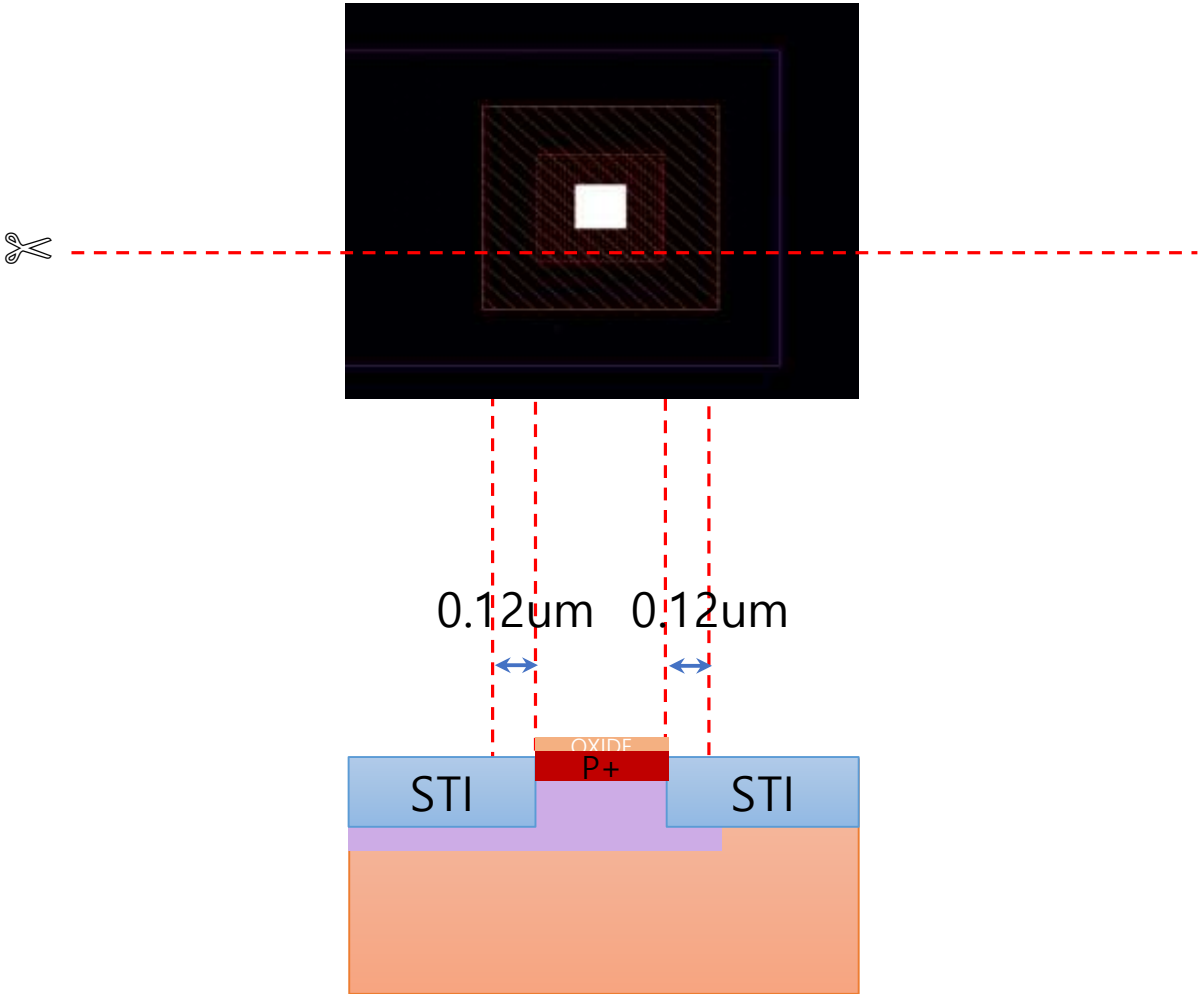
✓ Schematic in Design Rule Manual



enclosure가 부족하면 공정 오차로 인해 active area가 Nwell 밖으로 벗어날 수 있어 안정된 active를 형성하기 위해 필요하다.

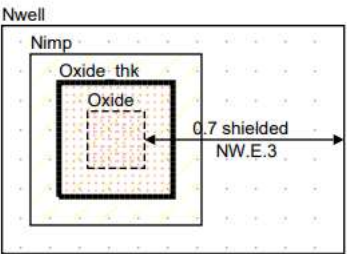


✓ Cadence Reproducing and vertical structure

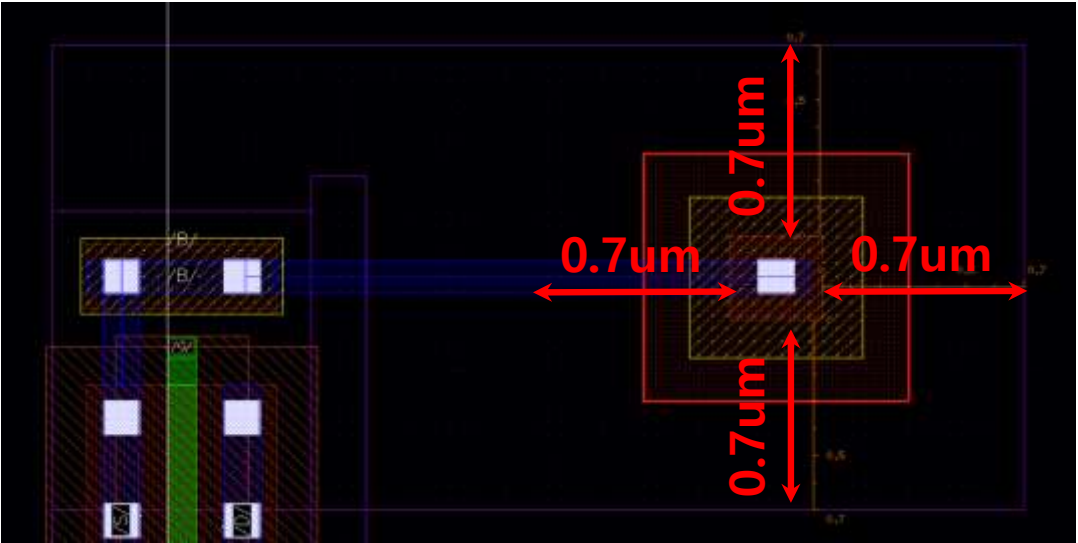


Rule Name	Value	Description
NW.E.3	0.7um	Minimum Nwell enclosure of N+ 2.5V Active Area

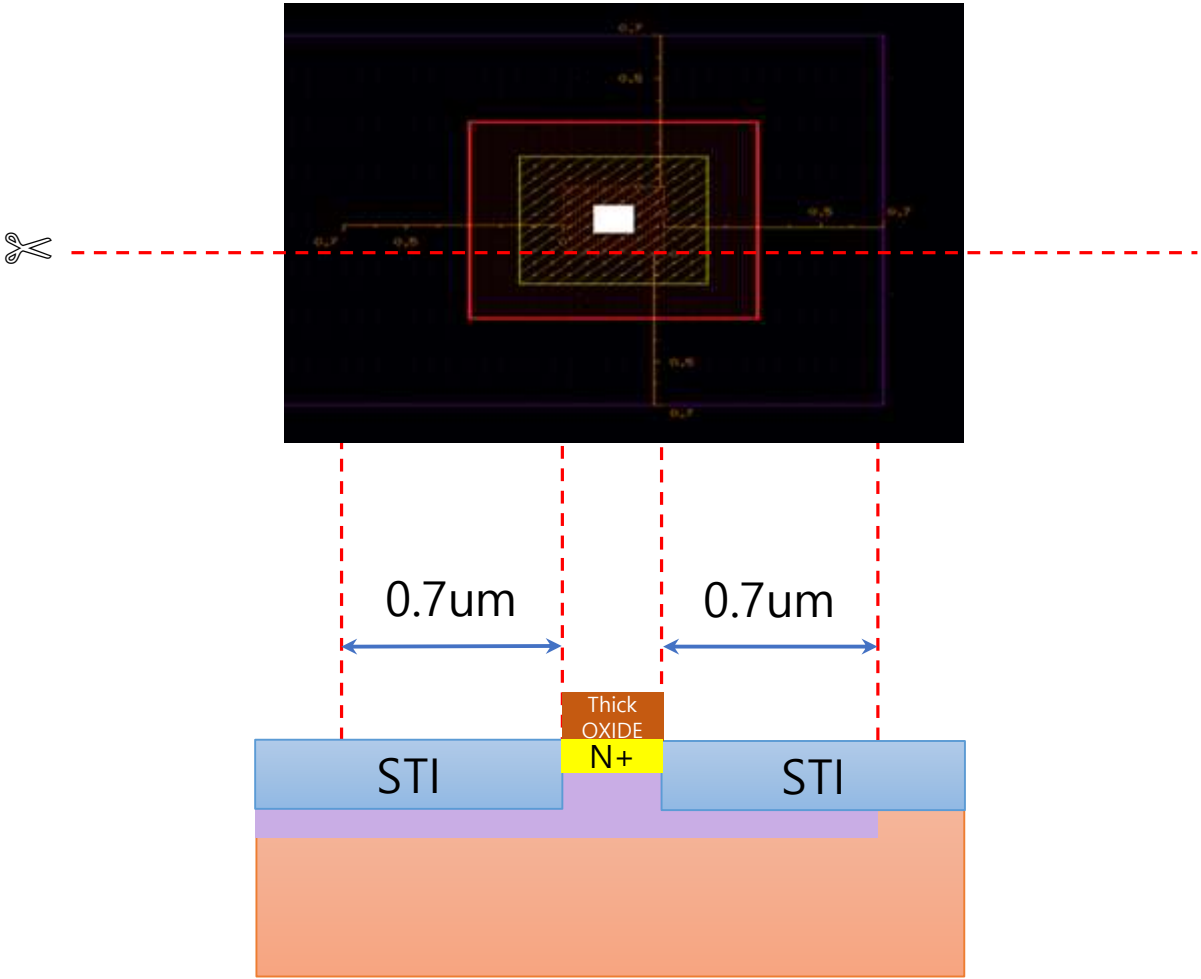
✓ Schematic in Design Rule Manual



더 큰 전압이 인가되므로 보다 넓은 영역의 Nwell을 형성해야 안정된 active가 형성될 수 있다.

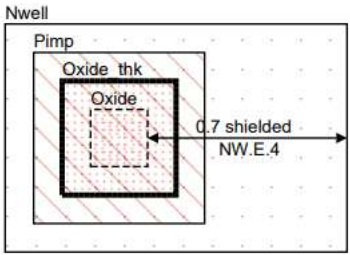


✓ Cadence Reproducing and vertical structure

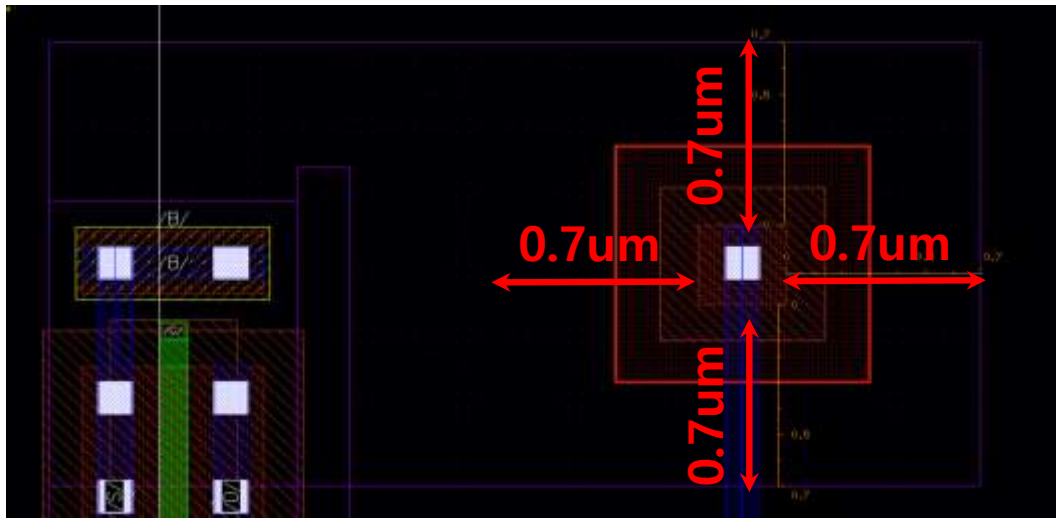


Rule Name	Value	Description
NW.E.4	0.7um	Minimum Nwell enclosure of P+ 2.5V Active Area

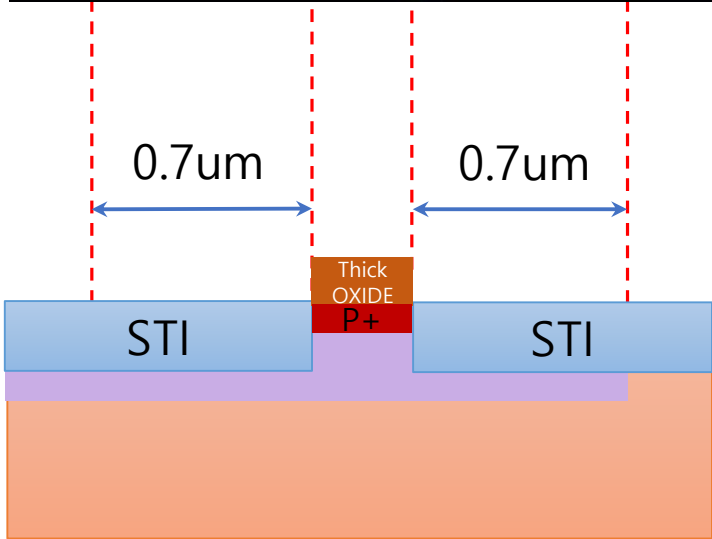
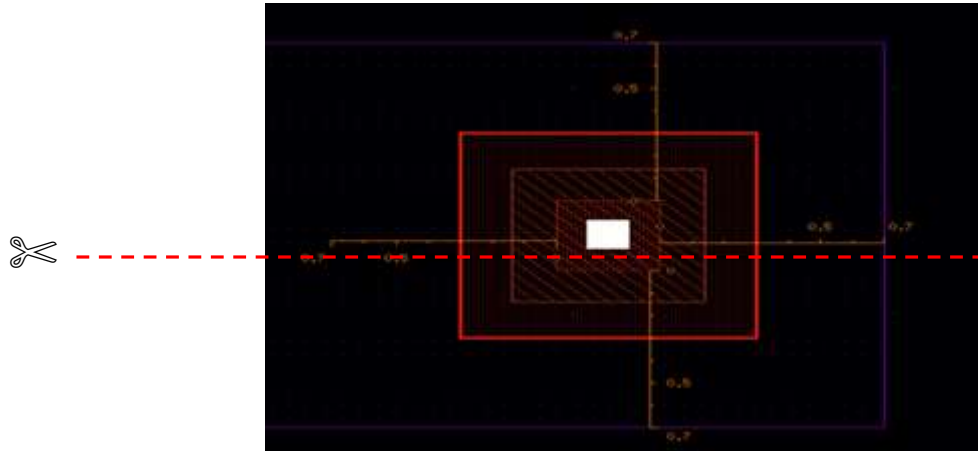
✓ Schematic in Design Rule Manual



더 큰 전압이 인가되므로 보다 넓은 영역의 Nwell을 형성해야 안정된 active가 형성될 수 있다.



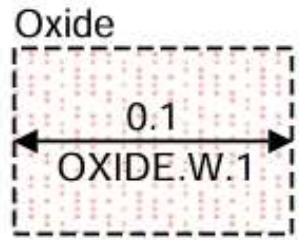
✓ Cadence Reproducing and vertical structure



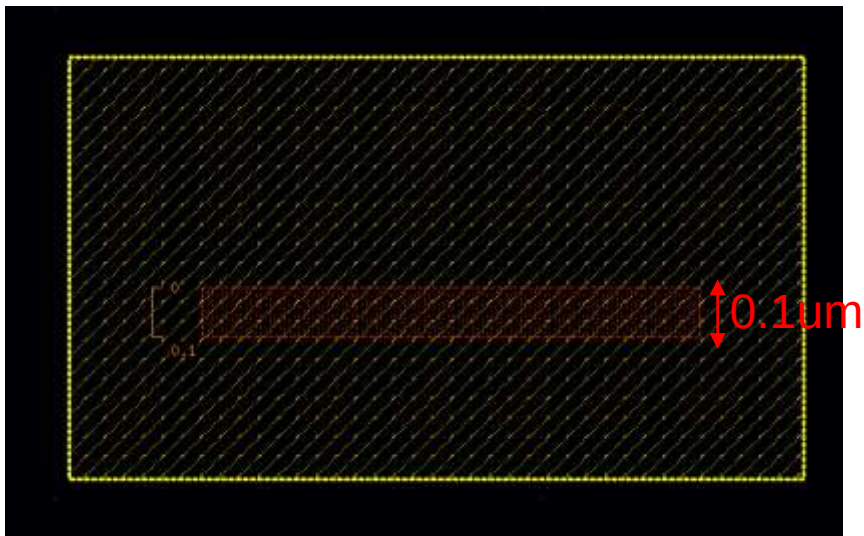
Oxide

Rule Name	Value	Description
OXIDE.W.1	0.1um	Minimum Active Area width

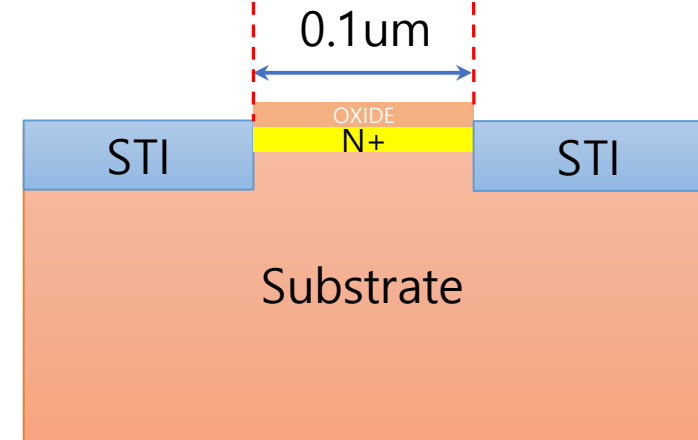
✓ Schematic in Design Rule Manual



빛의 회절 한계로 인해 패터닝 가능한 최소 실리콘 영역, oxide를 균일하게 채워 넣을 수 있는 최소 물리적 공간을 의미

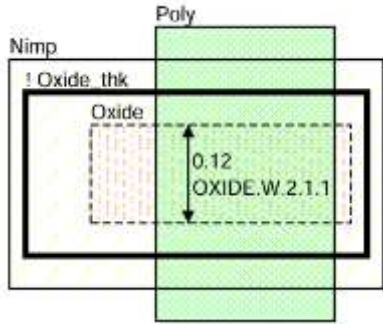


✓ Cadence Reproducing and vertical structure

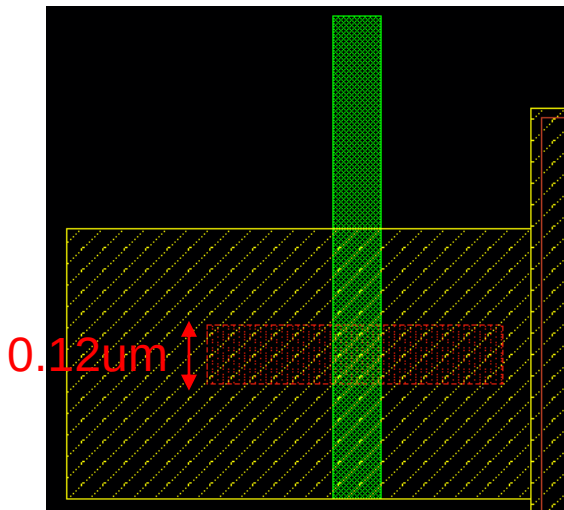


Rule Name	Value	Description
OXIDE.W.2.1.1	0.12um	Minimum 1.2V N-channel gate width

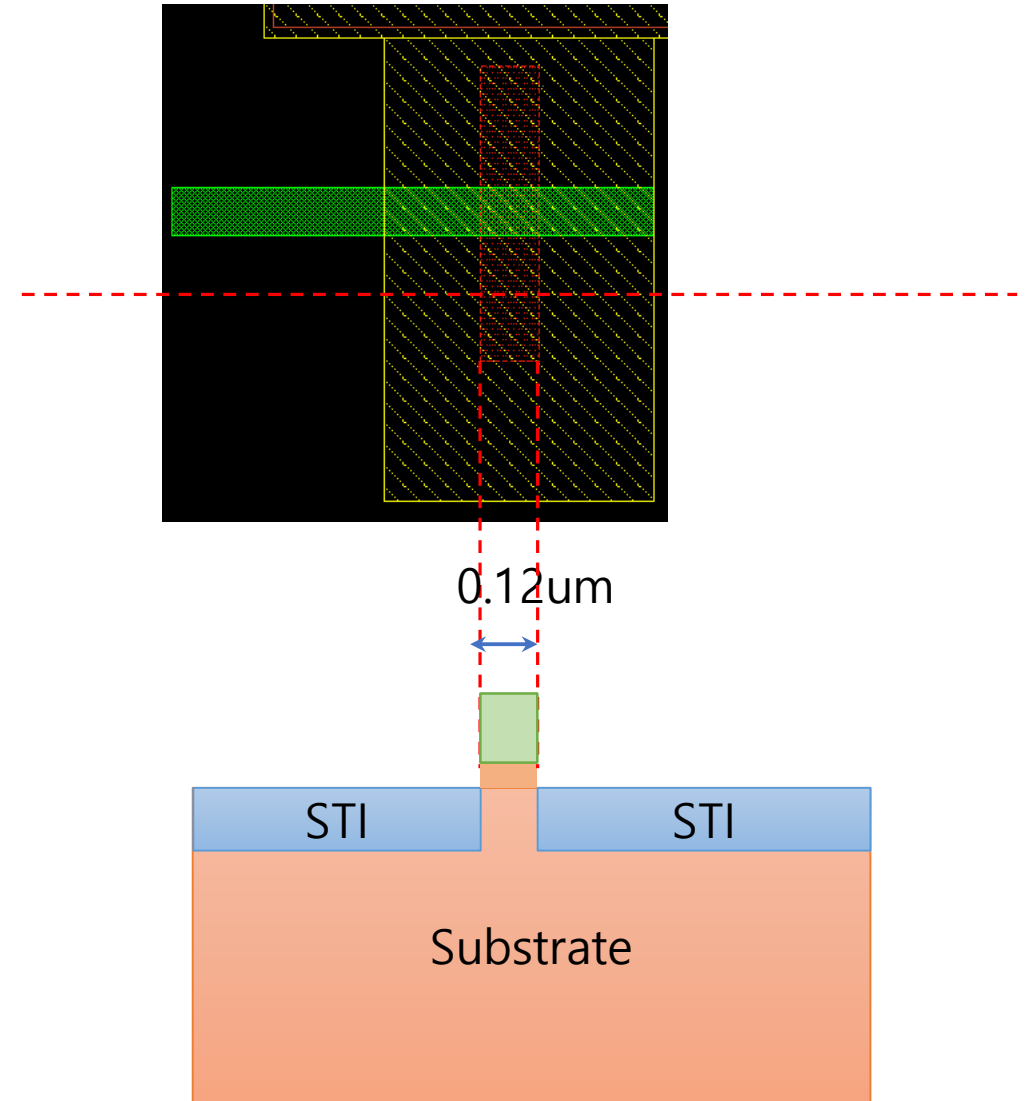
✓ Schematic in Design Rule Manual



폭이 너무 좁아지면 문턱 전압 변동성이 커지고 누설 전류가 증가하므로 이를 방지하고 안정적인 전류를 확보하기 위한 최소 채널 폭

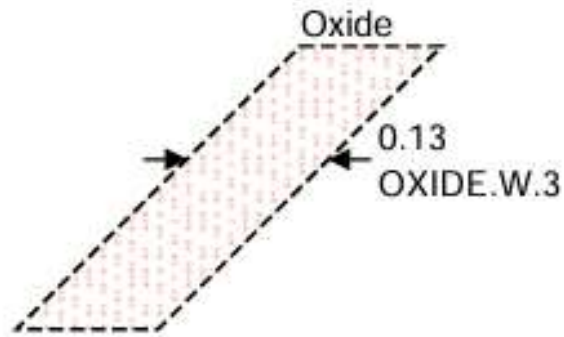


✓ Cadence Reproducing and vertical structure

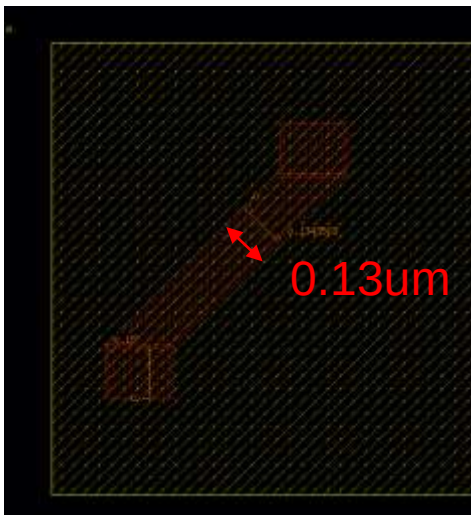


Rule Name	Value	Description
OXIDE.W.3	0.13um	Minimum Active Area bent 45 degrees width

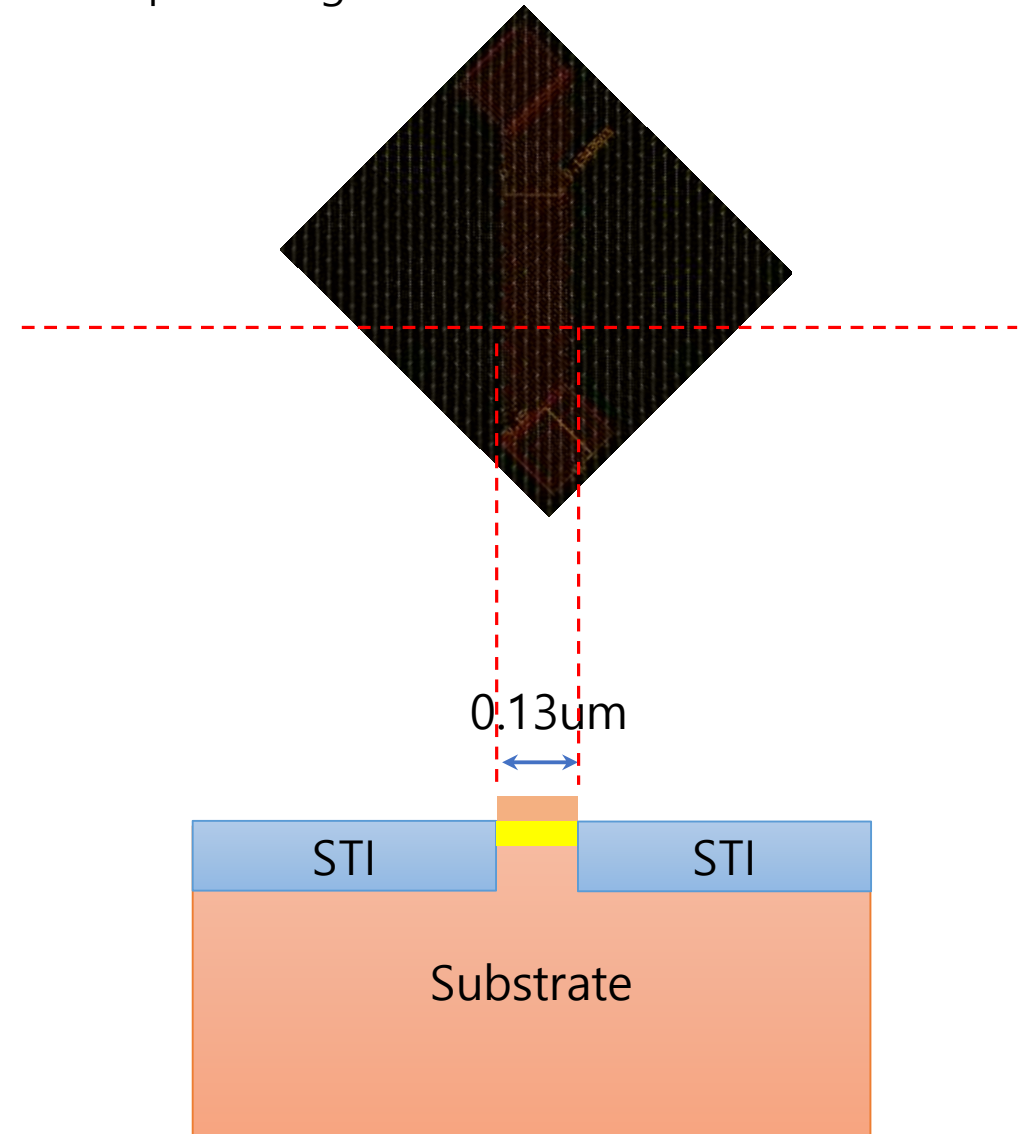
✓ Schematic in Design Rule Manual



포토리소그래피 공정에서 45도 사선 마스크 패턴은 격자배치 및 노광 시 감광액이 잘려 나가는 현상이 발생해서 0.13um보다 두꺼운 설계 필요

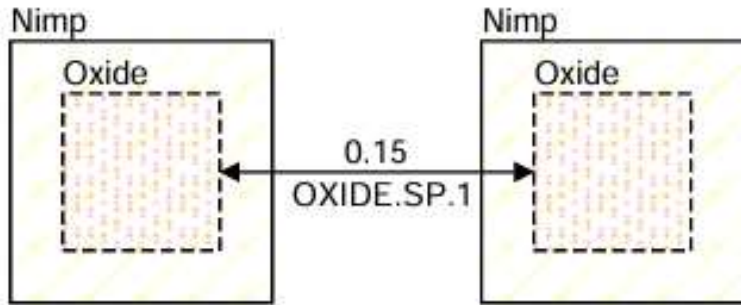


✓ Cadence Reproducing and vertical structure

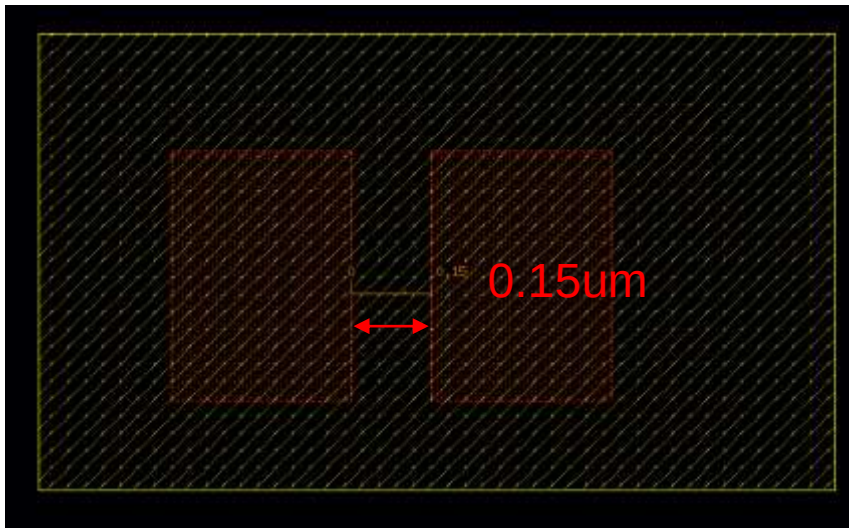


Rule Name	Value	Description
OXIDE.SP.1	0.15um	Minimum N+ Active Area to N+ Active Area spacing

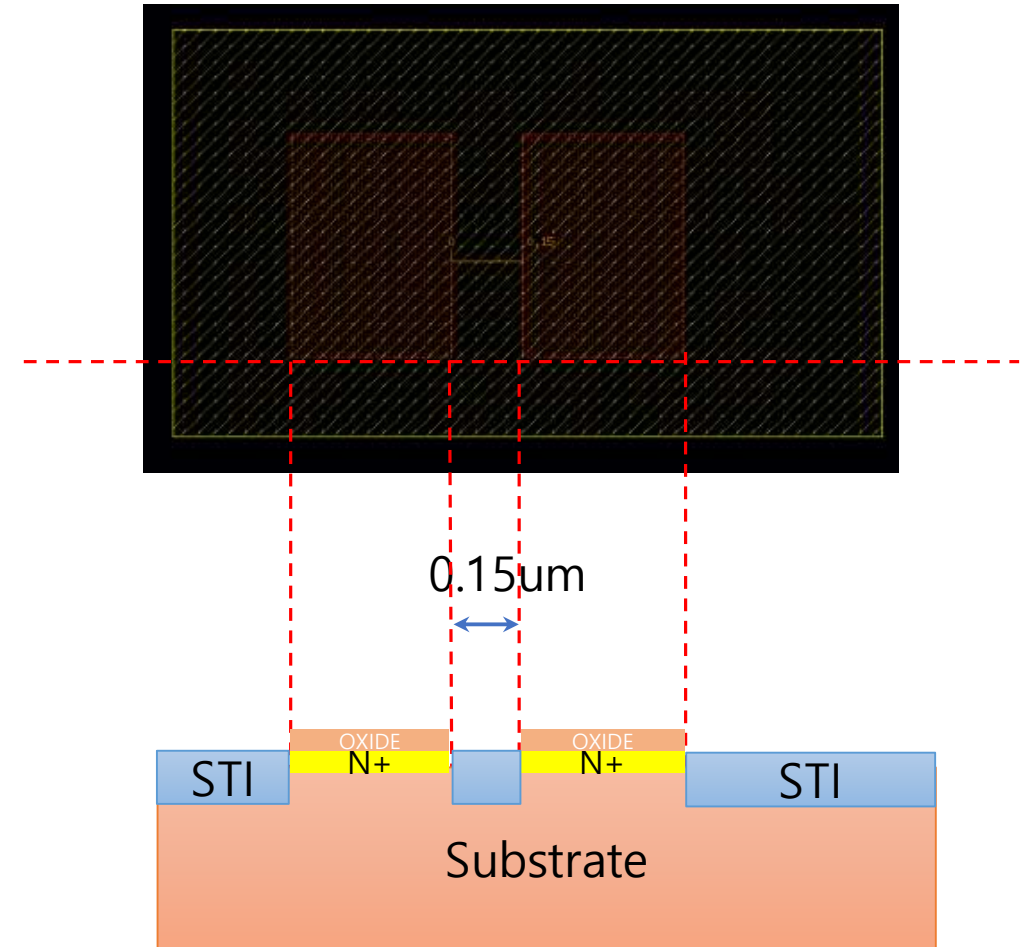
✓ Schematic in Design Rule Manual



두 Active 패턴 사이의 거리가 너무 가까우면
STI 식각 후 Oxide를 채워 넣을 때
내부에 빈 공간이 생겨 short가 발생할 수 있음

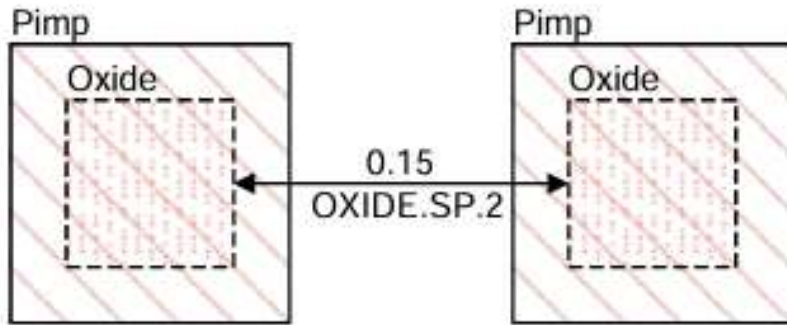


✓ Cadence Reproducing and vertical structure

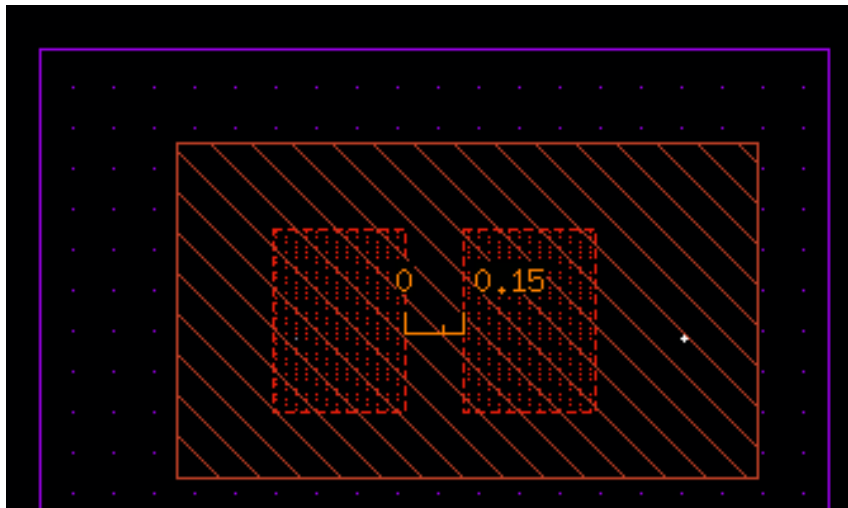


Rule Name	Value	Description
OXIDE.SP.2	0.15um	Minimum P+ Active Area to P+ Active Area spacing

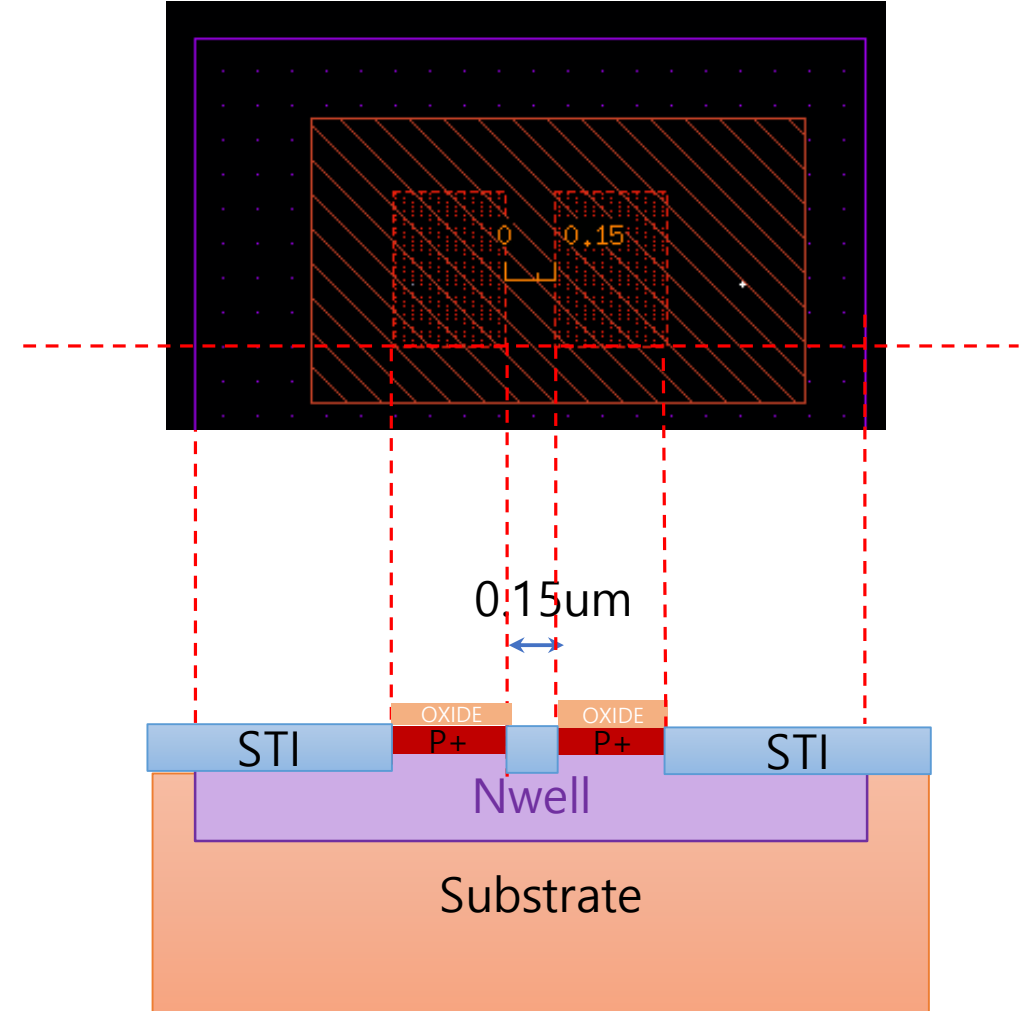
✓ Schematic in Design Rule Manual



두 Active 패턴 사이의 거리가 너무 가까우면
STI 식각 후 Oxide를 채워 넣을 때
내부에 빈 공간이 생겨 short가 발생할 수 있음

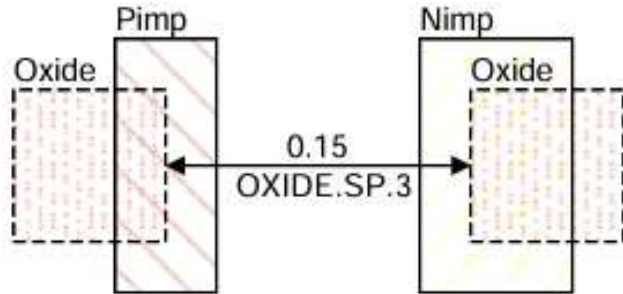


✓ Cadence Reproducing and vertical structure

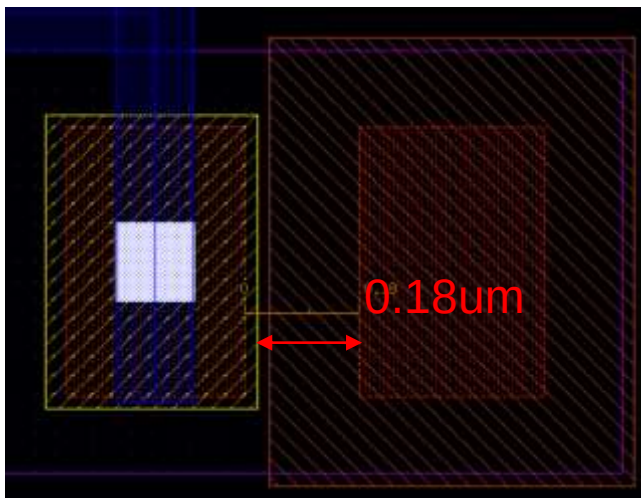


Rule Name	Value	Description
OXIDE.SP.3	0.15um	Minimum N+ Active Area to P+ Active Area spacing

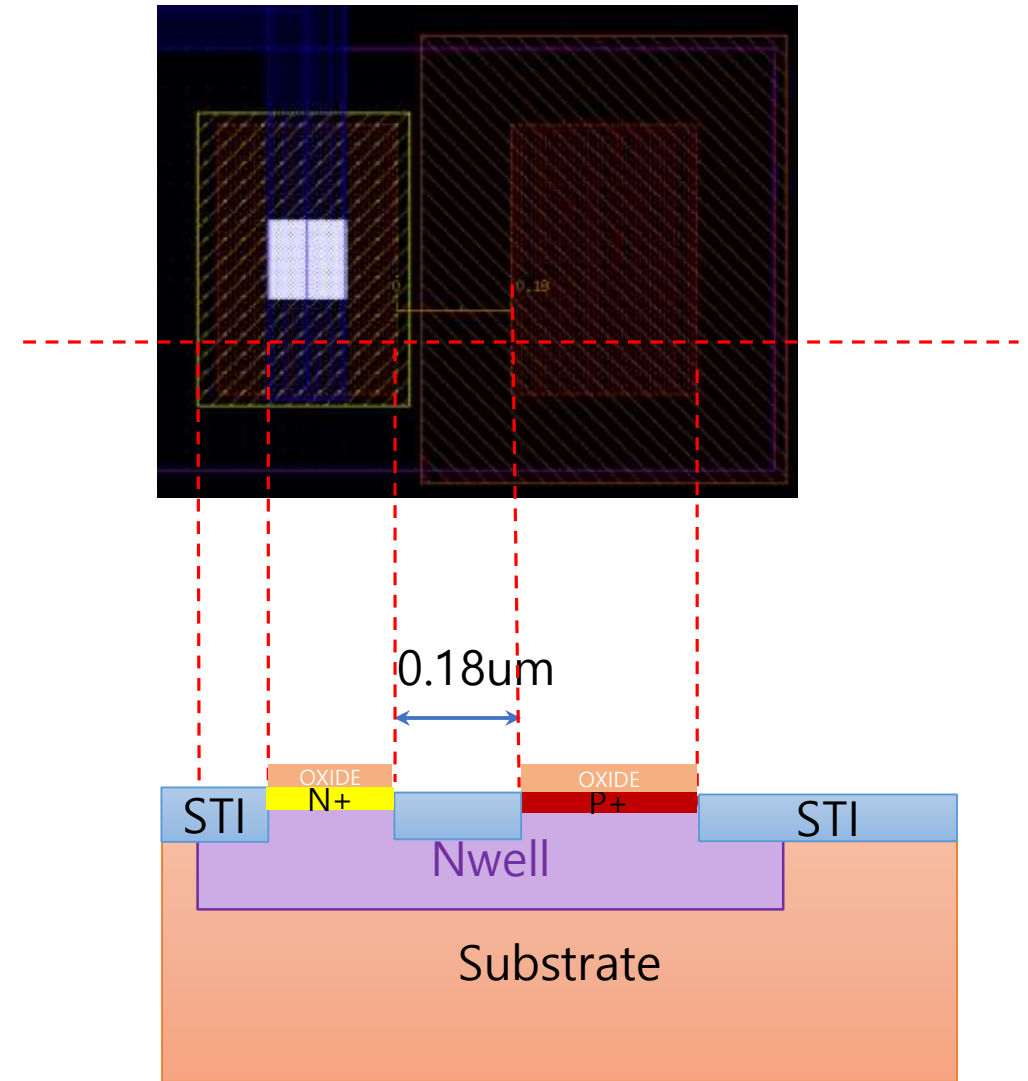
✓ Schematic in Design Rule Manual



두 Active 패턴 사이의 거리가 너무 가까우면
STI 식각 후 Oxide를 채워 넣을 때
내부에 빈 공간이 생겨 short가 발생할 수 있음

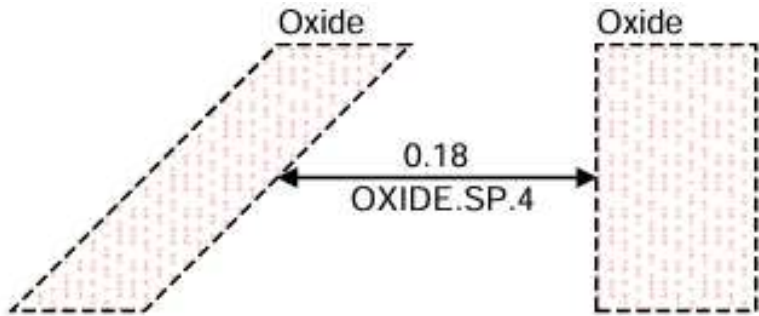


✓ Cadence Reproducing and vertical structure

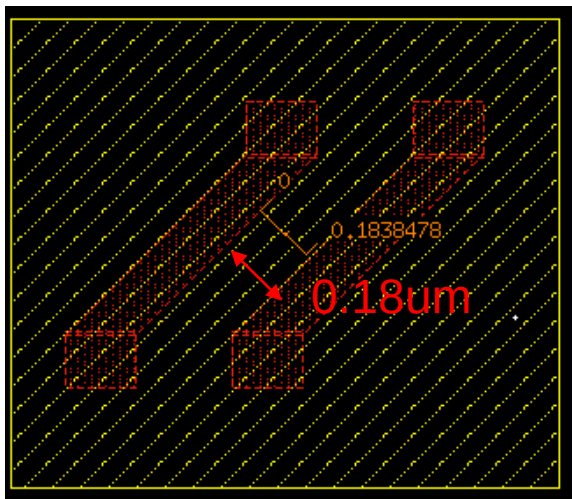


Rule Name	Value	Description
OXIDE.SP.4	0.18um	Minimum Active Area bent 45 degrees to Active Area spacing

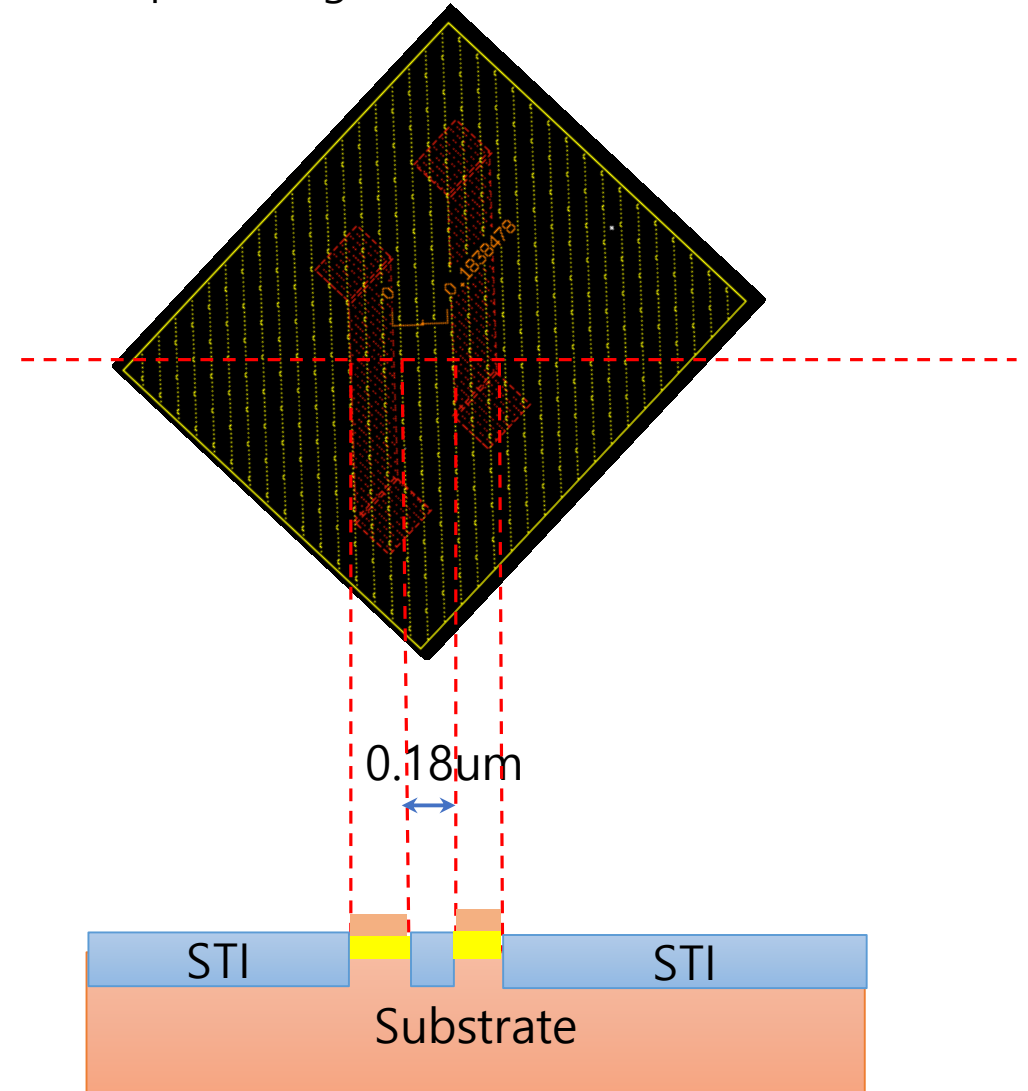
✓ Schematic in Design Rule Manual



45도 기울어진 패턴이 맞닿는 부분은
빛의 간섭 현상으로 인해
마스크 패턴이 쉽게 뭉개지거나 연결될 위험이 큼

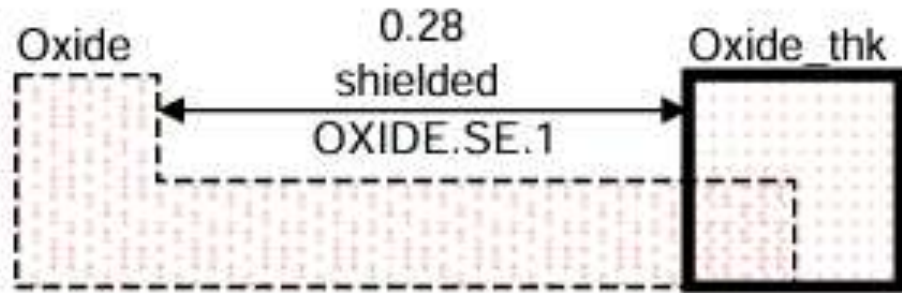


✓ Cadence Reproducing and vertical structure

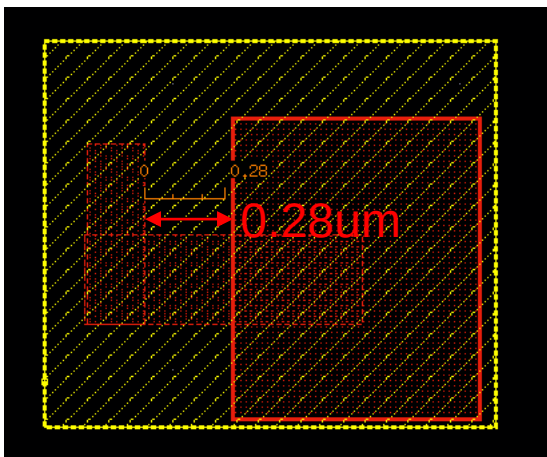


Rule Name	Value	Description
OXIDE.SE.1	0.28um	Minimum Active Area to Thick Active Area spacing

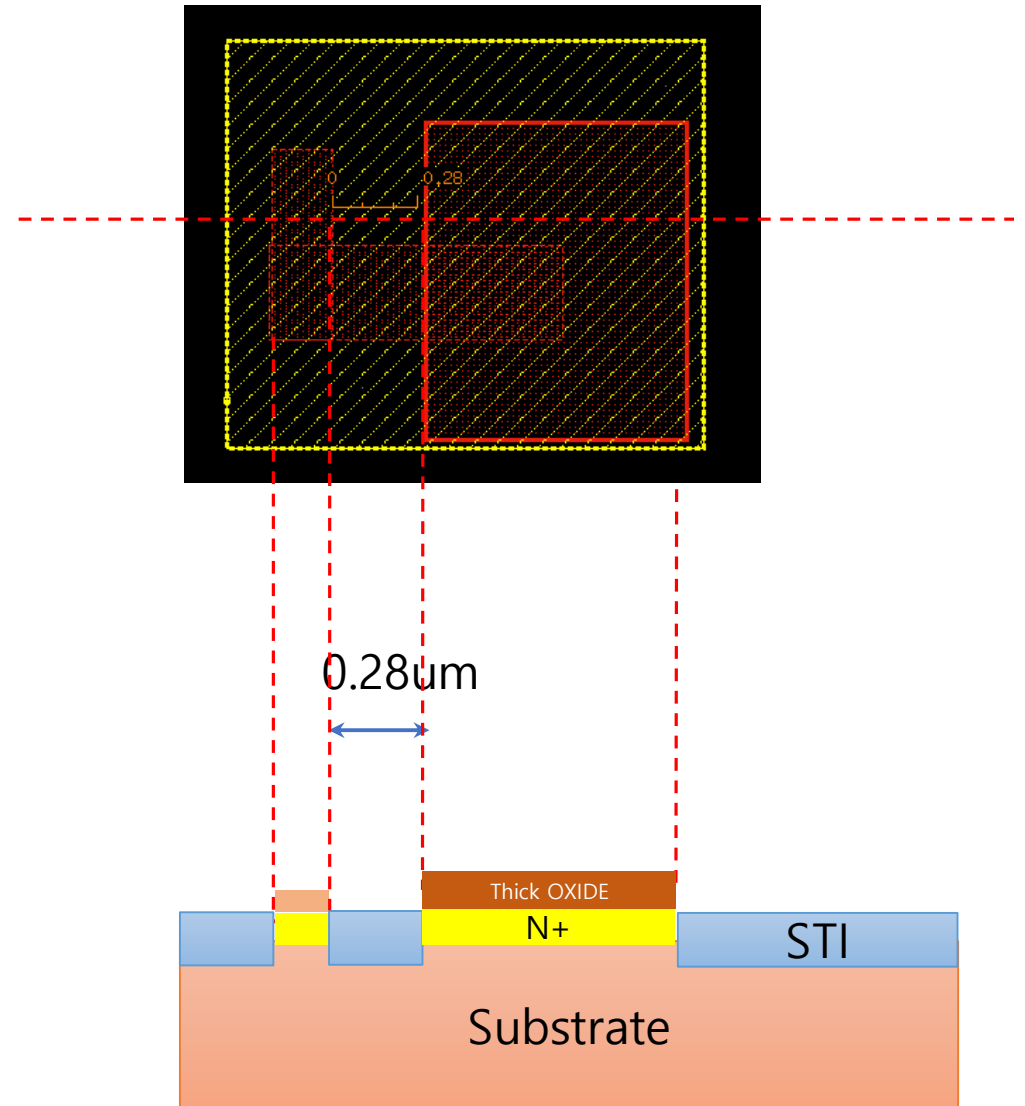
✓ Schematic in Design Rule Manual



얇은 oxide와 두꺼운 oxide 영역을 구분하기 위해 'Dual Oxide Mask'를 사용하는 추가 Etch 공정이 들어가며 두 영역 간의 마스크 정렬 오차 안정성을 보장하기 위해 일반 간격보다 훨씬 넓은 마진(0.28um)이 필요

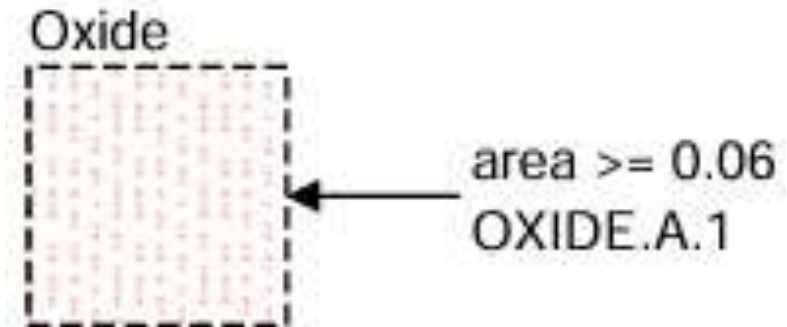


✓ Cadence Reproducing and vertical structure

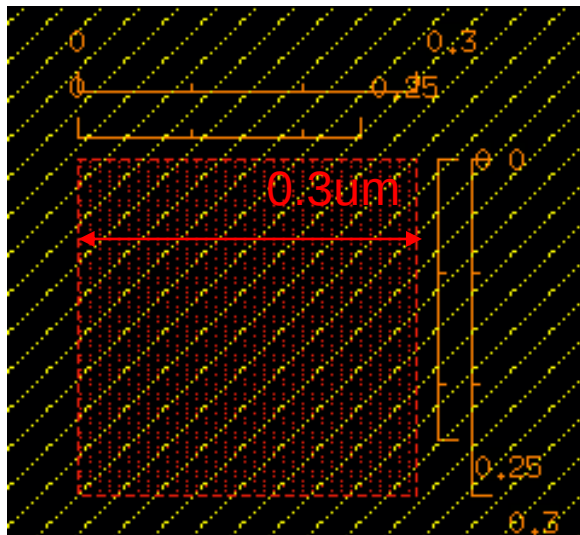


Rule Name	Value	Description
OXIDE.A.1	0.06um	Minimum area for Active Area

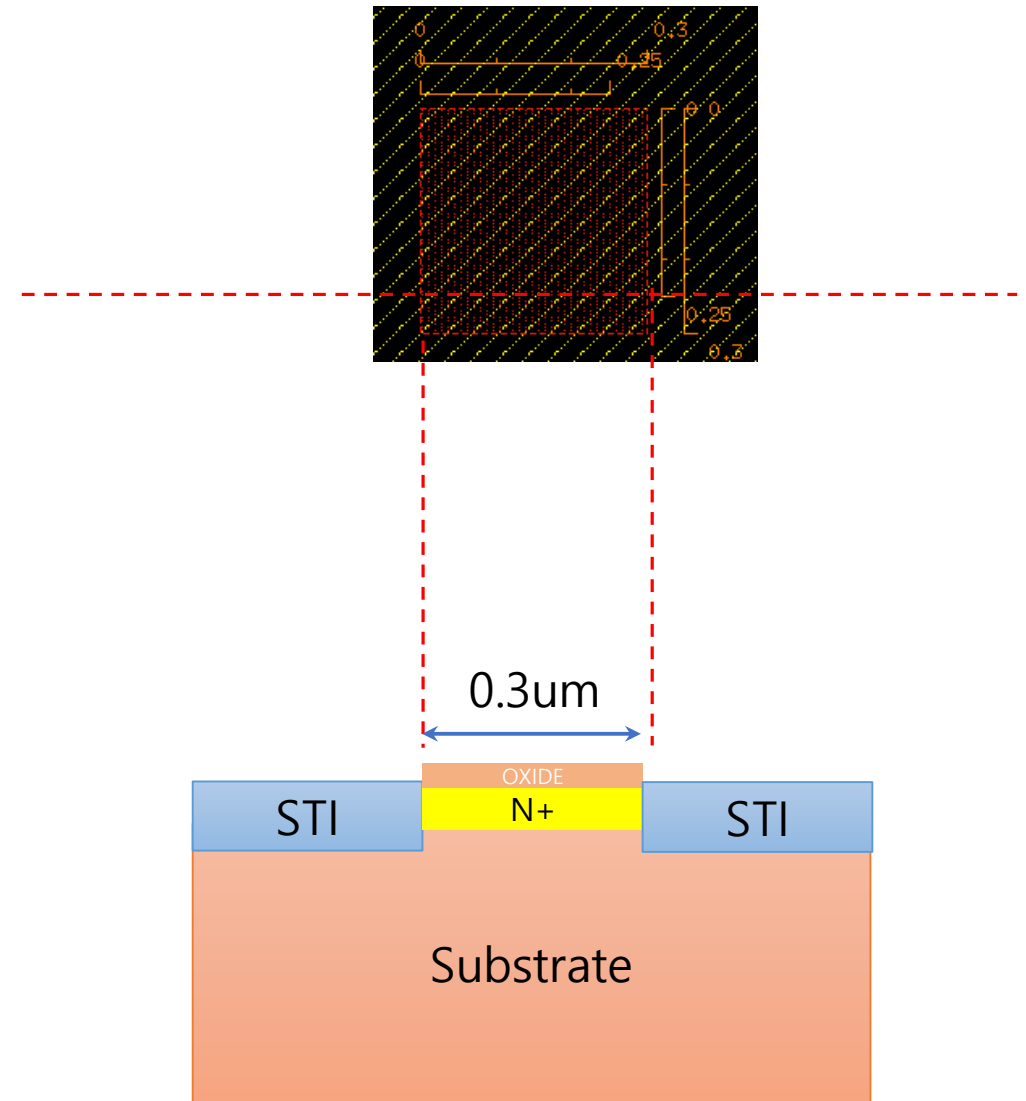
✓ Schematic in Design Rule Manual



Active 면적이 너무 작은 형태로 존재하면
CMP 공정 중에 연마 패드의 강한 마찰력을
견디지 못하고 패턴이 탈락되어 웨이퍼 오염을 유발

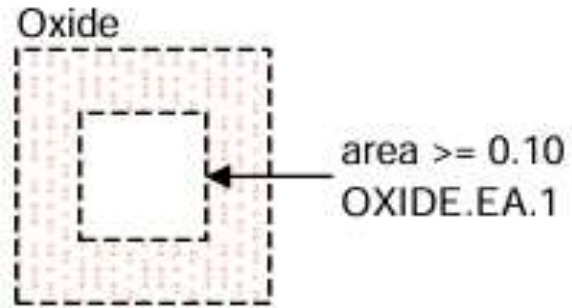


✓ Cadence Reproducing and vertical structure

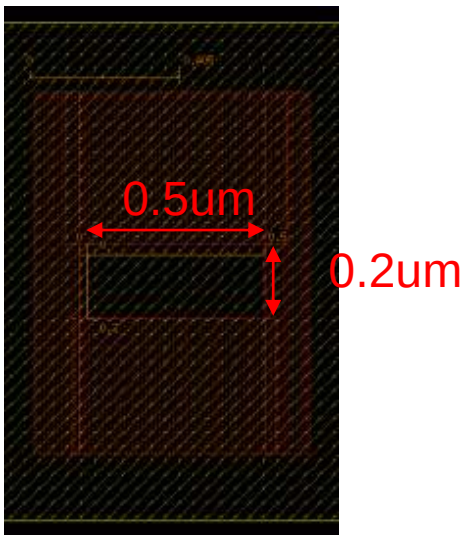


Rule Name	Value	Description
OXIDE.EA.1	0.1um	Minimum Active Area enclosed area ("donut" hole surrounded by Active Area)

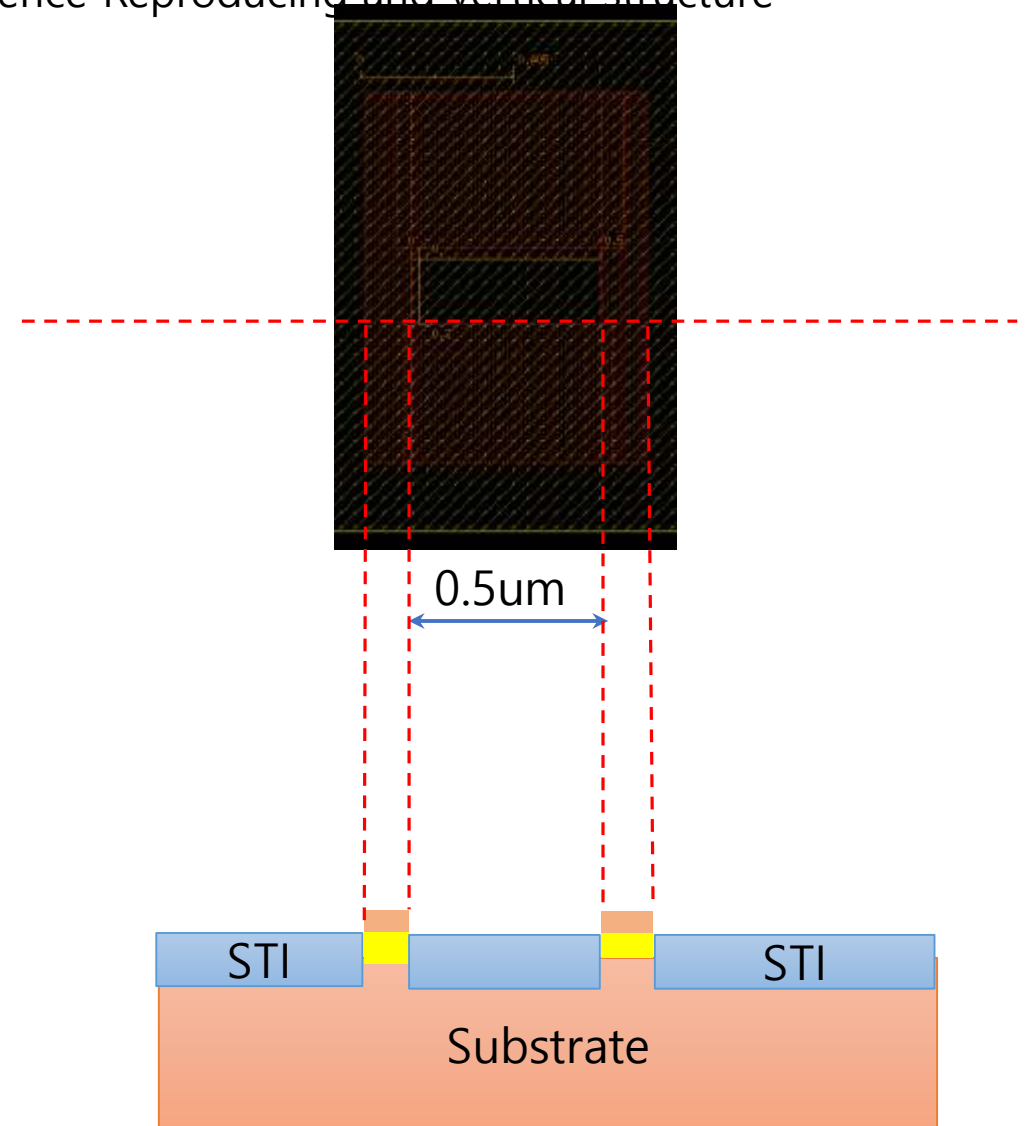
✓ Schematic in Design Rule Manual



Active 영역이 도넛 형태로 둘러싸고 있는 내부 구멍이 너무 작으면 식각 액체나 세정 케미컬의 표면장력 때문에 액체가 갇히므로 이로 인해 미식각 영역이 남는 결함이 발생

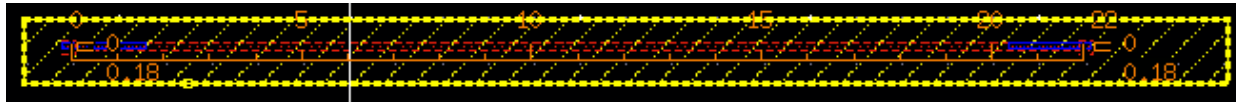


✓ Cadence Reproducing and vertical structure

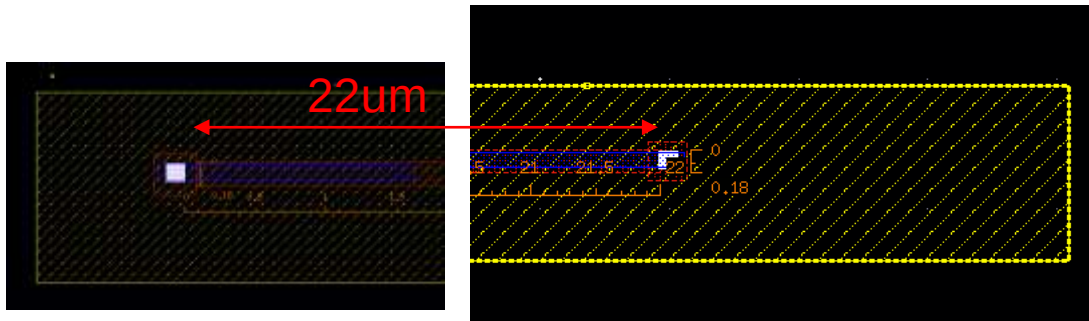


Rule Name	Value	Description
OXIDE.L.1	22um	Maximum Oxide length between two contacts when the Oxide width is $\leq 0.18\mu\text{m}$.

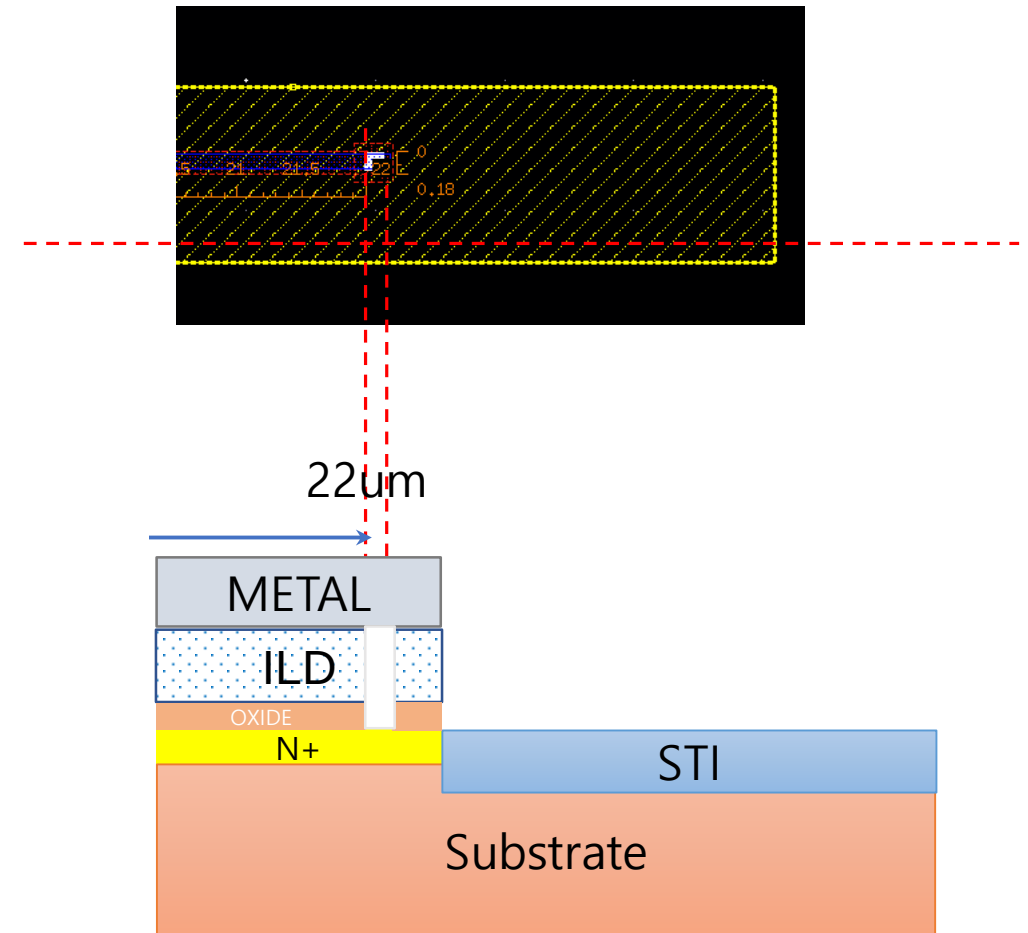
✓ Schematic in Design Rule Manual



0.18 μm 이하의 얇은 Active 선이
컨택(Contact) 없이 너무 길게 늘어서면
심각한 전압 강화(IR Drop) 및 RC delay가 발생
Contact ~ Contact 사이 거리 22um

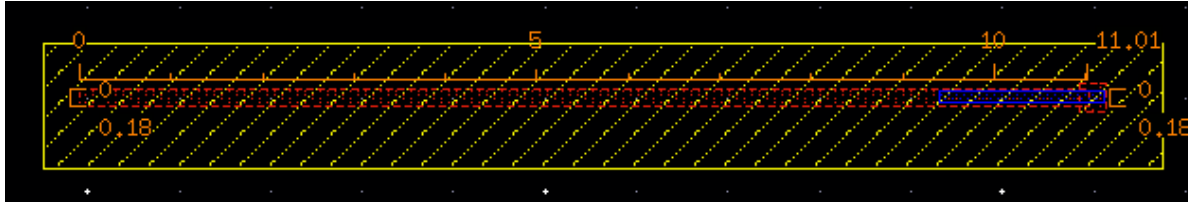


✓ Cadence Reproducing and vertical structure

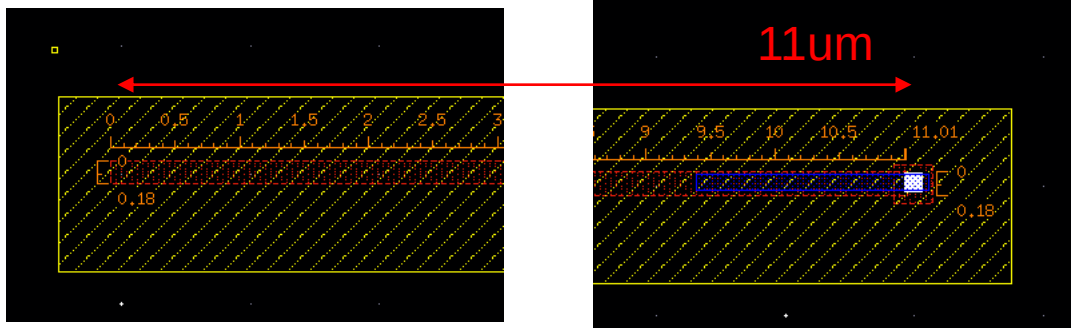


Rule Name	Value	Description
OXIDE.L.2	11um	Maximum Oxide length between one contact and the end of the Oxide line when the Oxide width is $\leq 0.18\mu\text{m}$

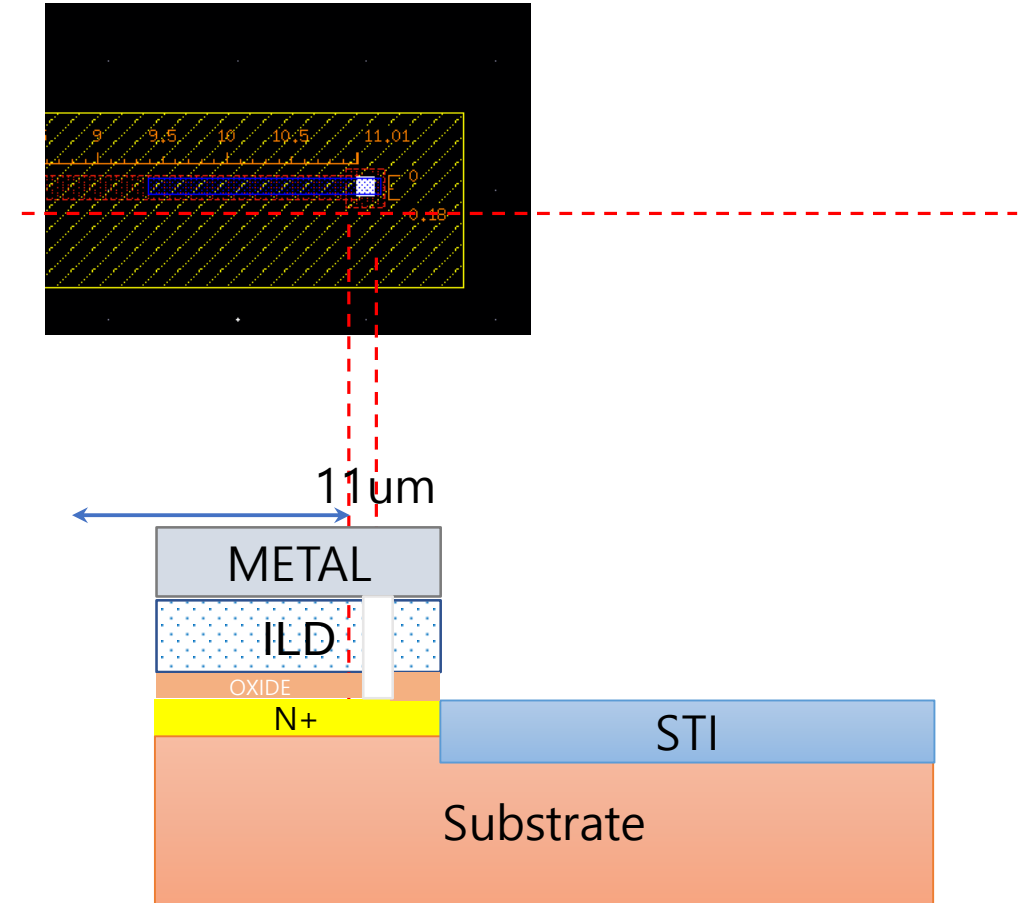
✓ Schematic in Design Rule Manual



0.18 μm 이하의 얇은 Active 선이
컨택(Contact) 없이 너무 길게 늘어서면
심각한 전압 강하(IR Drop) 및 RC delay가 발생
Contact ~ Oxide 끝 사이 거리 11um



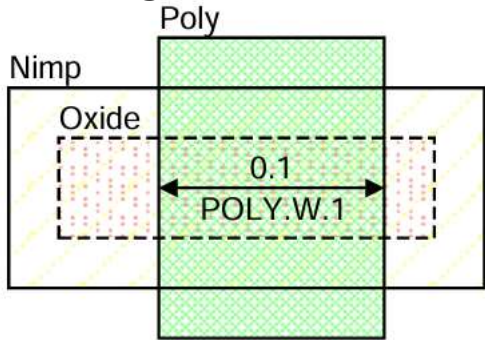
✓ Cadence Reproducing and vertical structure



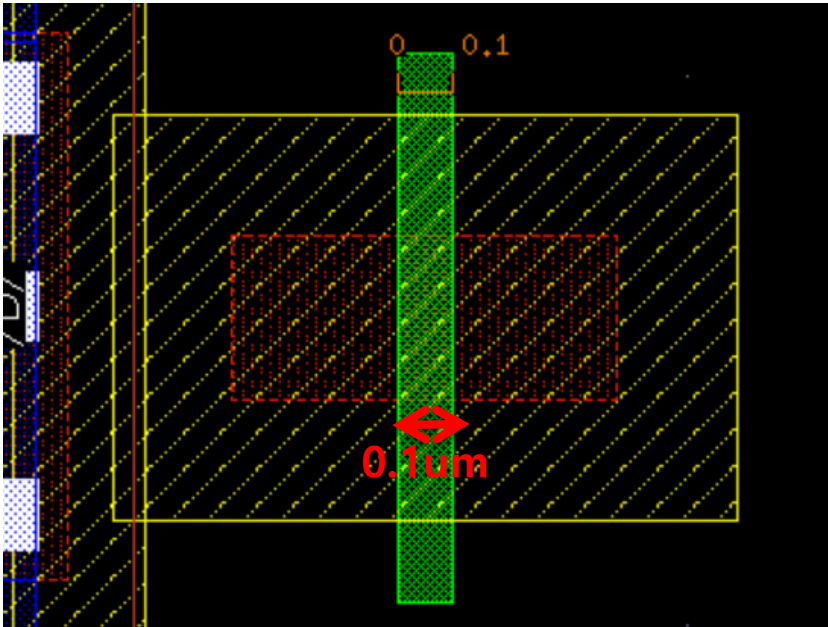
POLY

Rule Name	Value	Description
POLY.W.1	0.1um	Minimum 1.2V N-channel gate length

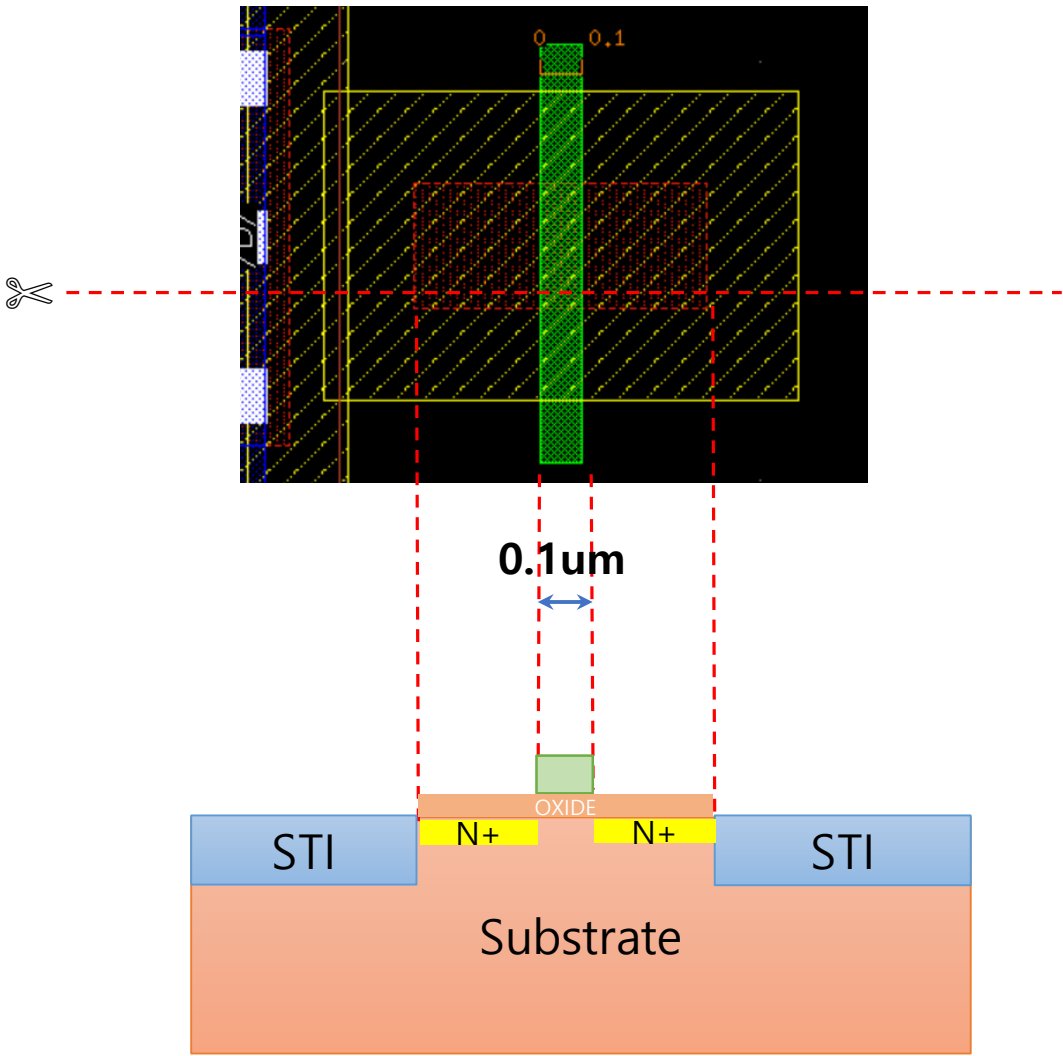
✓ Schematic in Design Rule Manual



“Short Channel Effect와 누설전류 증가를 방지하기 위해 NMOS의 최소 Gate Length를 제한”

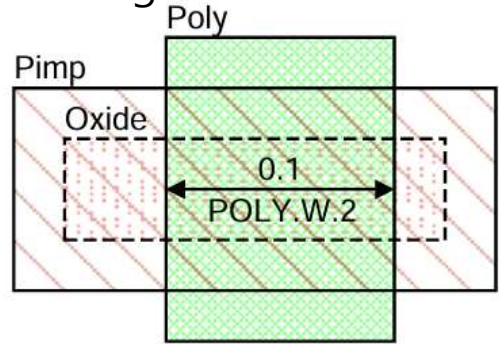


✓ Cadence Reproducing and vertical structure

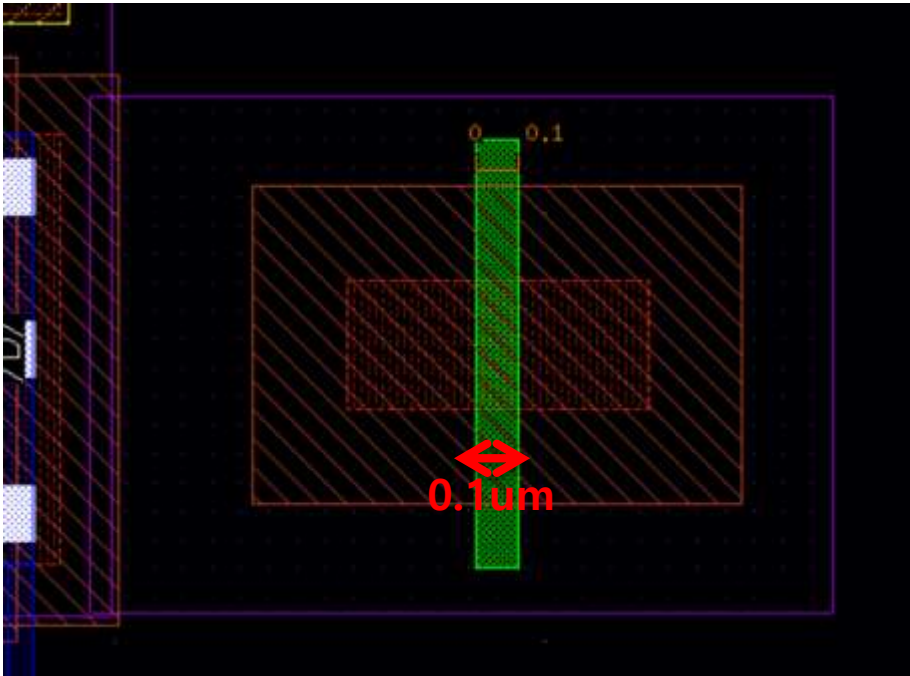


Rule Name	Value	Description
POLY.W.2	0.1um	Minimum 1.2V P-channel gate length

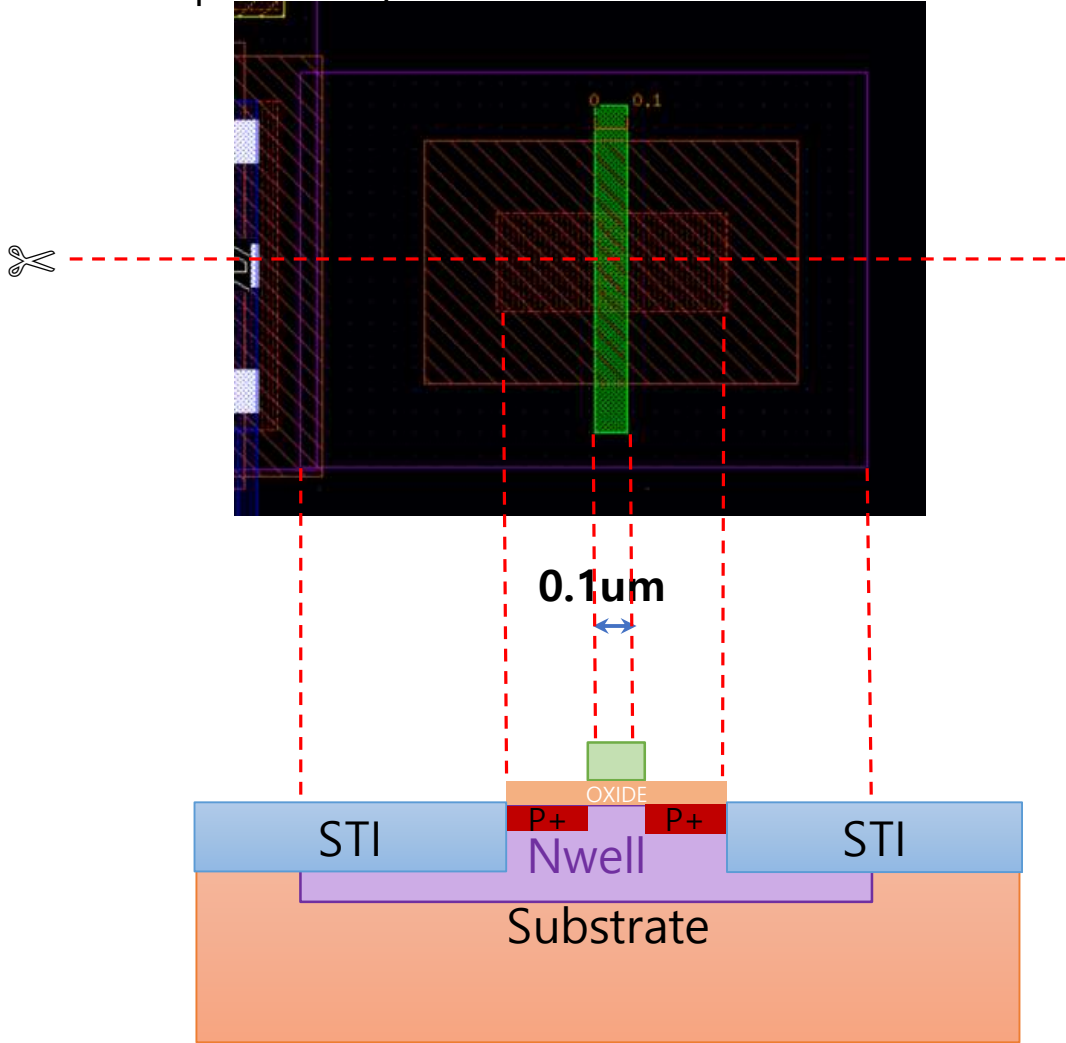
✓ Schematic in Design Rule Manual



“PMOS의 전기적 특성과 신뢰성을 유지하기 위해 최소 Gate Length를 제한”

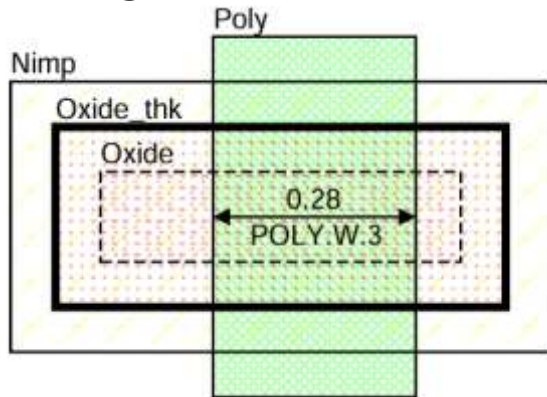


✓ Cadence Reproducing and vertical structure

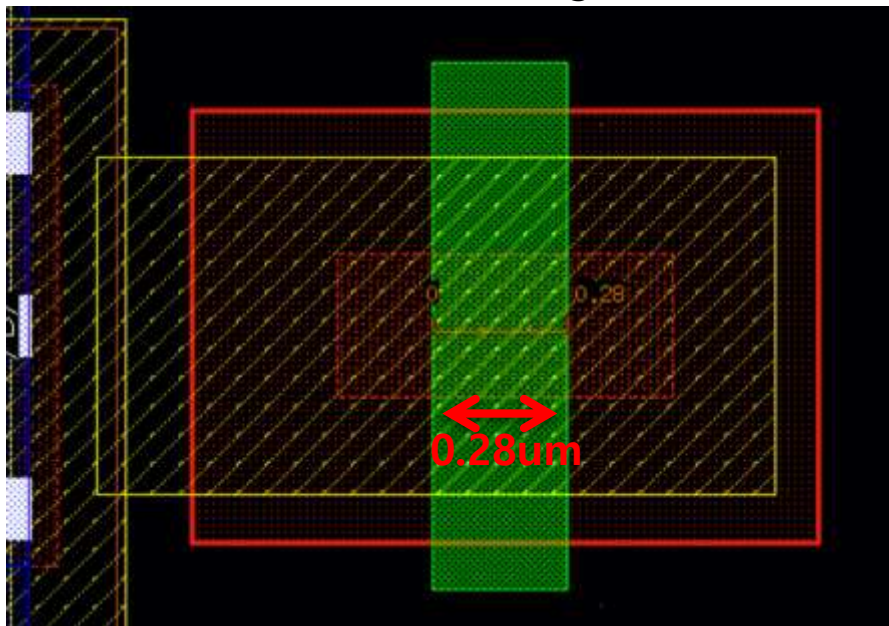


Rule Name	Value	Description
POLY.W.3	0.28um	Minimum 2.5V N-channel gate length

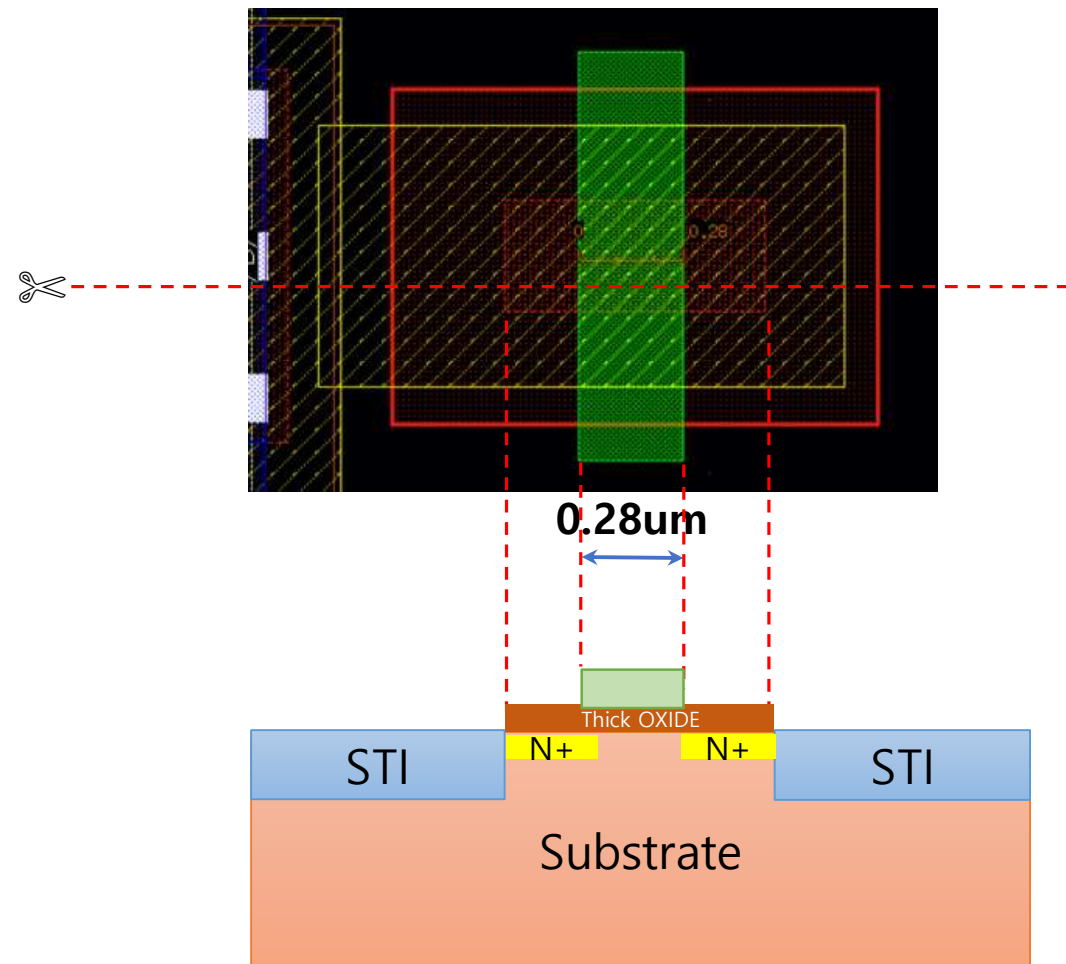
✓ Schematic in Design Rule Manual



“고전압 동작 시 Hot Carrier Injection과 Gate Oxide 열화를 방지하기 위해 최소 Gate Length를 제한”

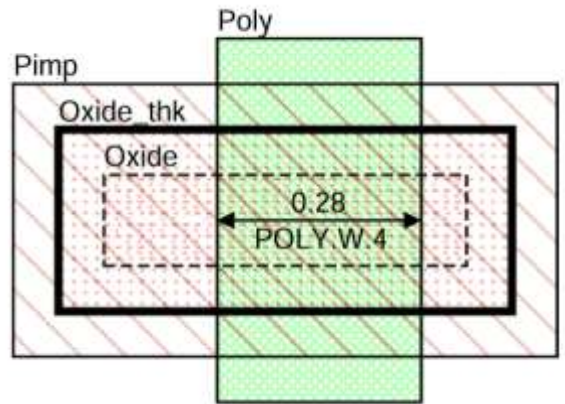


✓ Cadence Reproducing and vertical structure

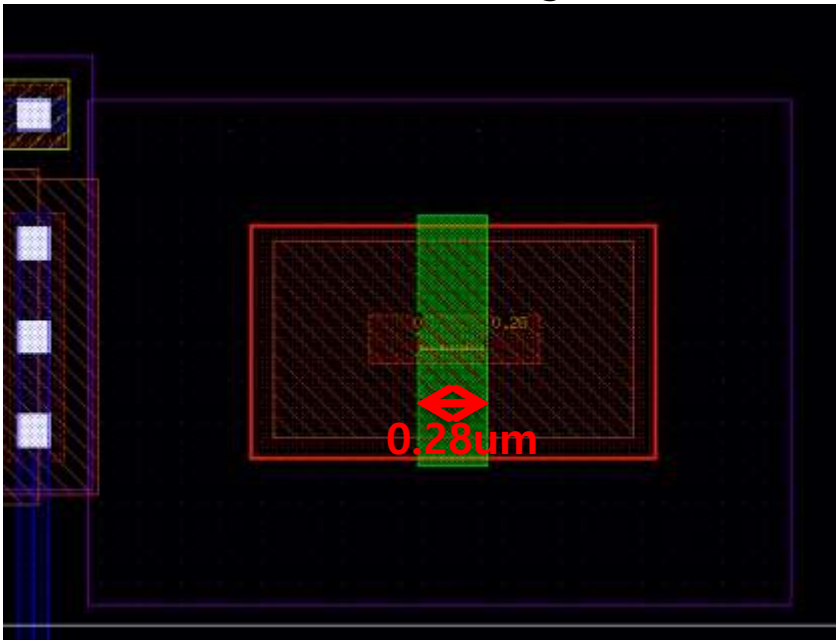


Rule Name	Value	Description
POLY.W.4	0.28um	Minimum 2.5V P-channel gate length

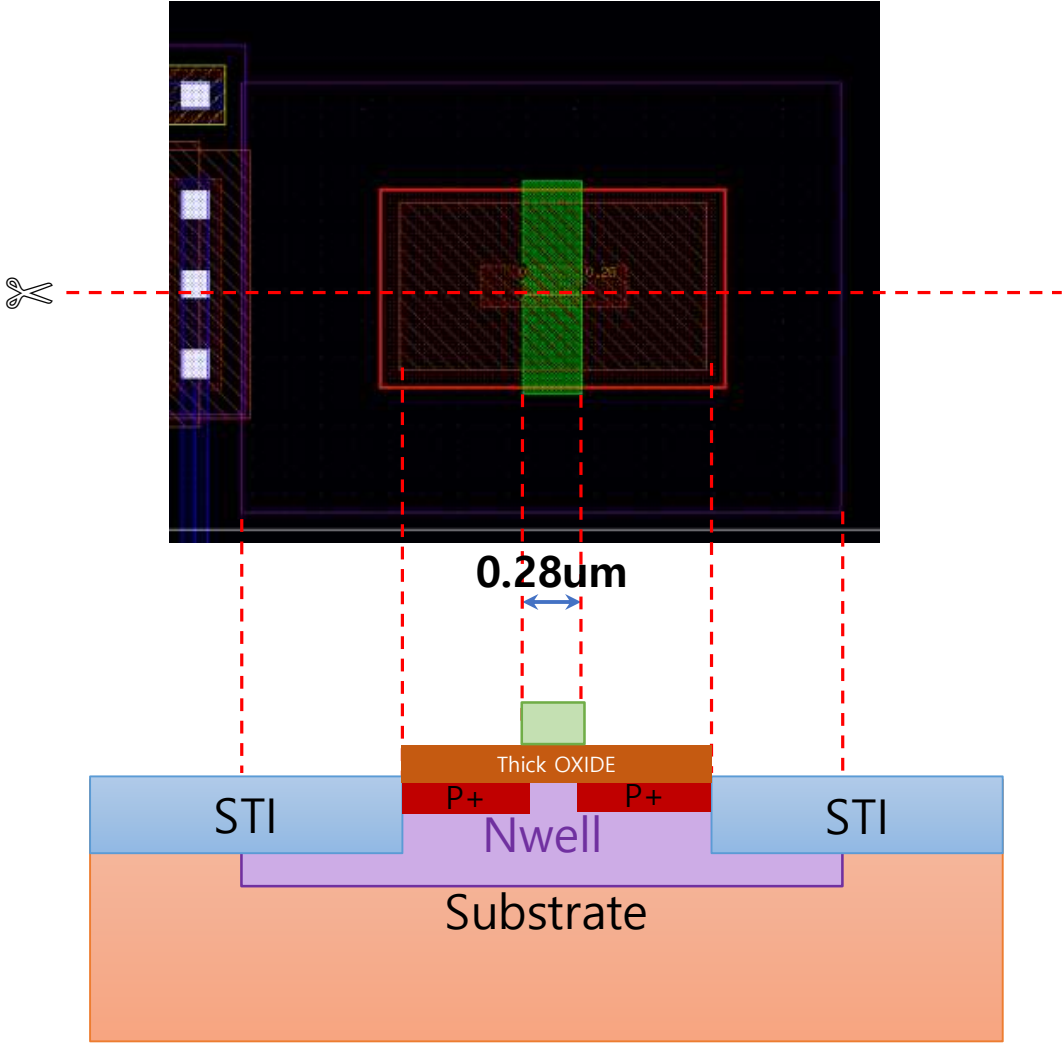
✓ Schematic in Design Rule Manual



“고전압 동작 시 Hot Carrier Injection과 Gate Oxide 열화를 방지하기 위해 최소 Gate Length를 제한”

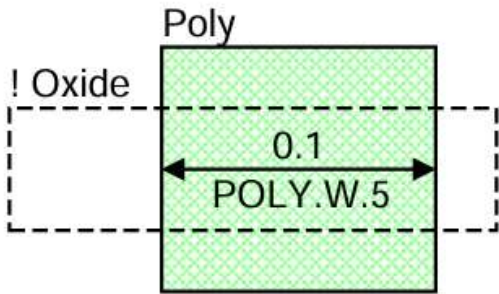


✓ Cadence Reproducing and vertical structure

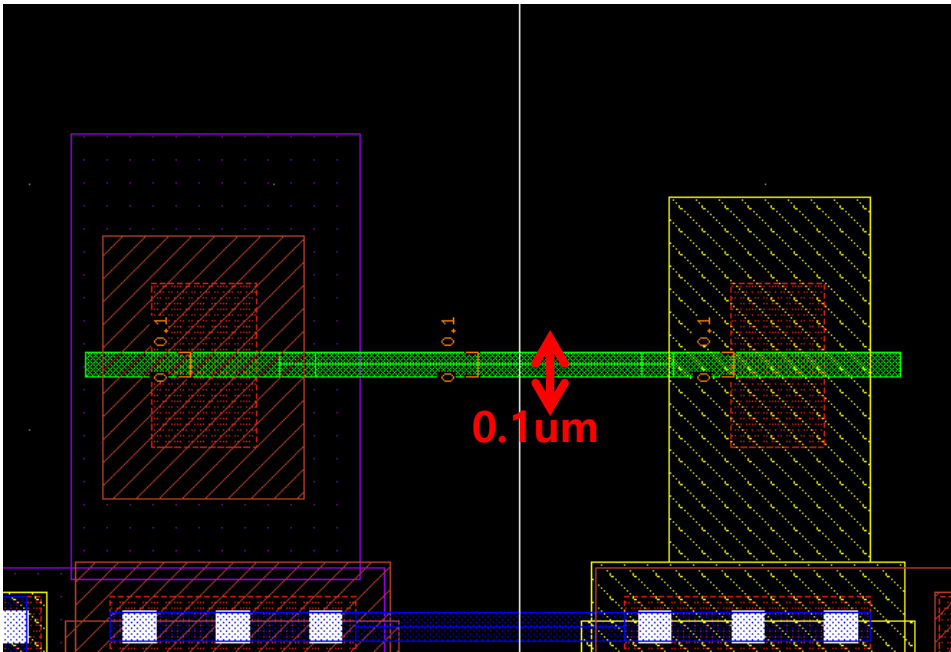


Rule Name	Value	Description
POLY.W.5	0.1um	Minimum Poly interconnect width

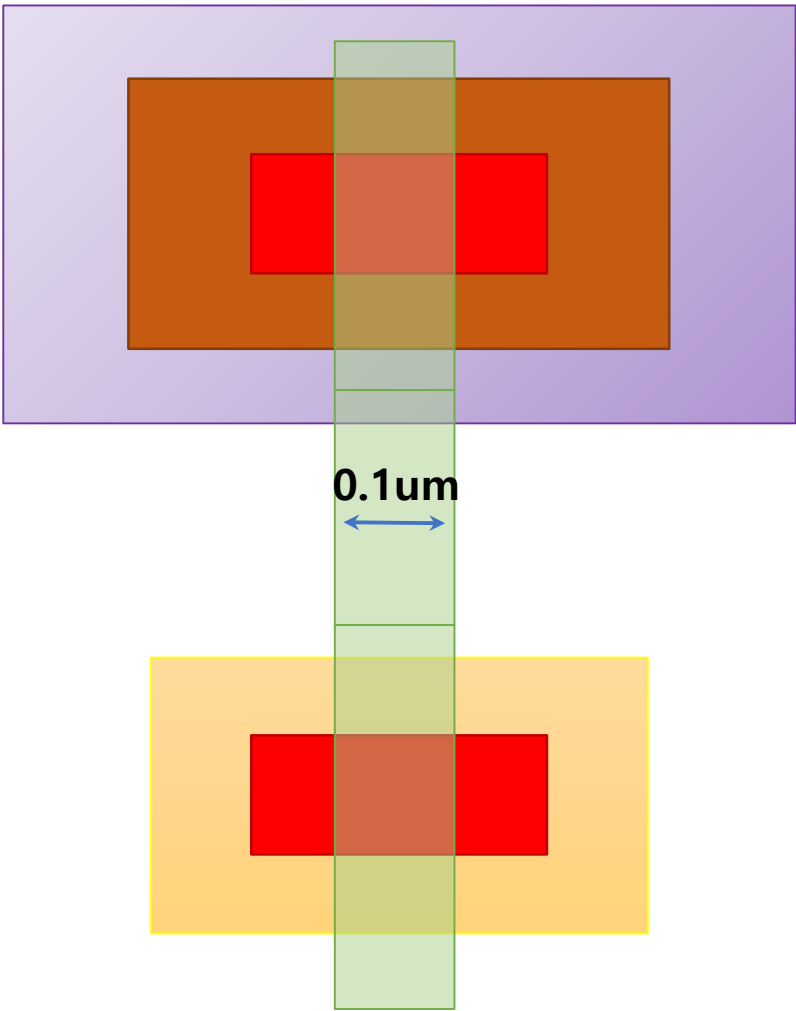
✓ Schematic in Design Rule Manual



“Poly 배선의 단선과 저항 증가를 방지하기 위해 최소 Poly 폭을 규정”

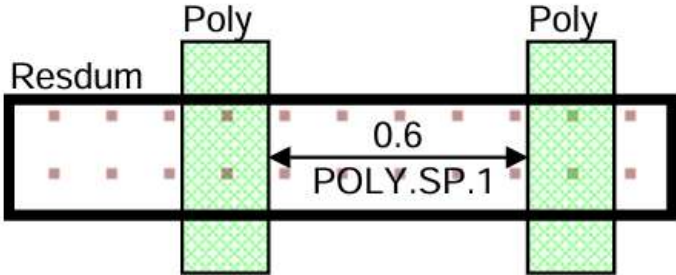


✓ Cadence Reproducing and vertical structure

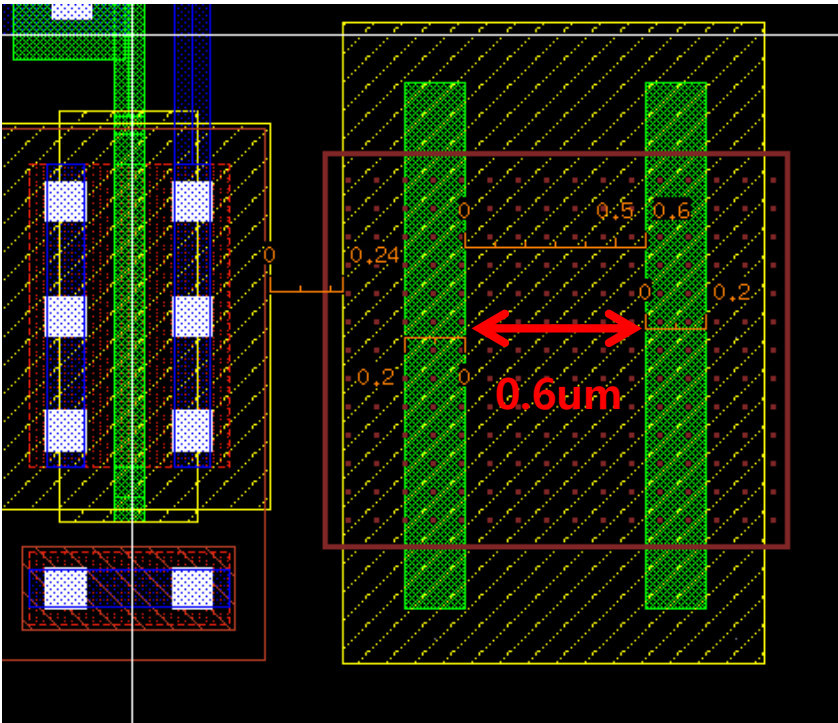


Rule Name	Value	Description
POLY.SP.1	0.6um	Minimum Poly resistor space

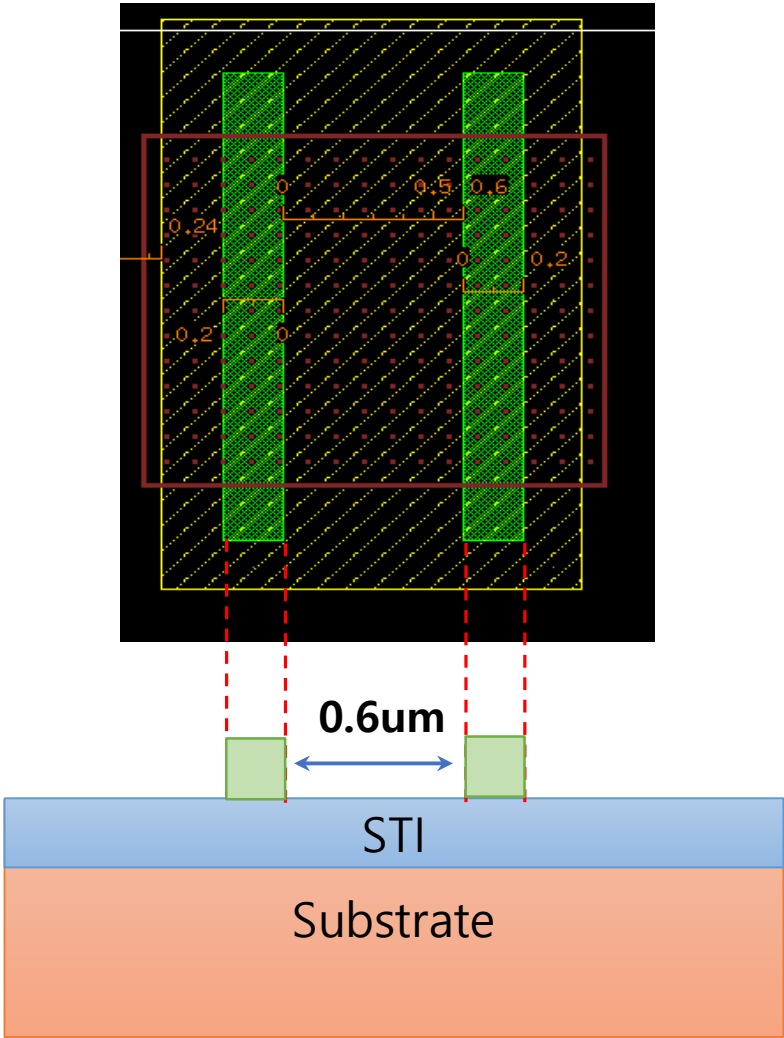
✓ Schematic in Design Rule Manual



“Poly 저항의 저항값 오차와 공정 간섭을 줄이기 위해 최소 간격을 확보”

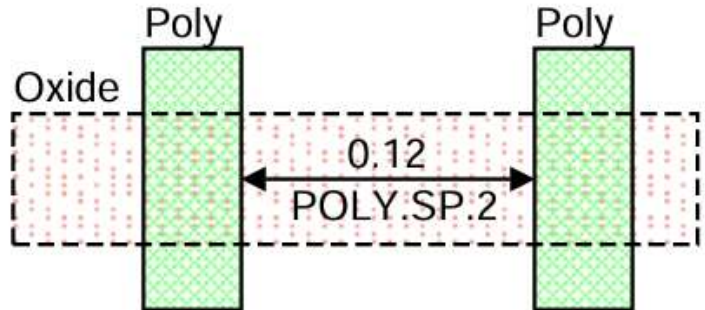


✓ Cadence Reproducing and vertical structure

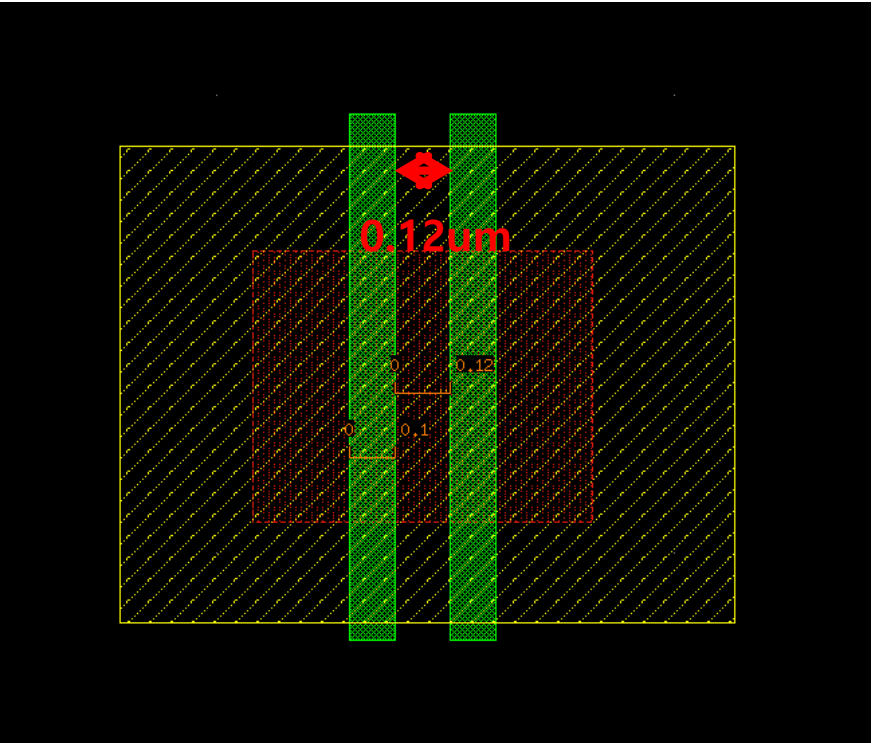


Rule Name	Value	Description
POLY.SP.2	0.12um	Minimum gate space

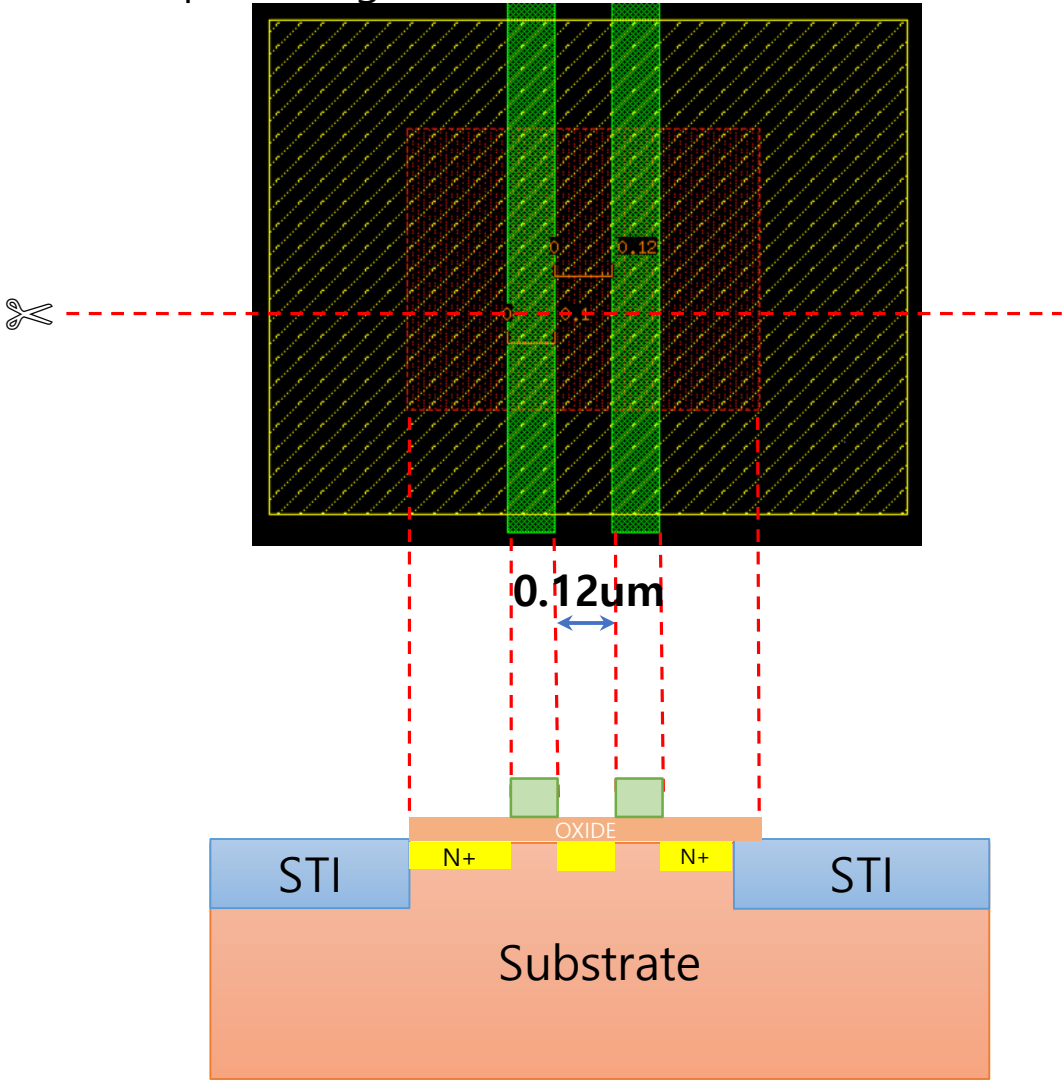
✓ Schematic in Design Rule Manual



“인접한 Gate의 브리징과 기생 capacitor 증가를 방지하기 위해 최소 Gate 간격을 유지.”

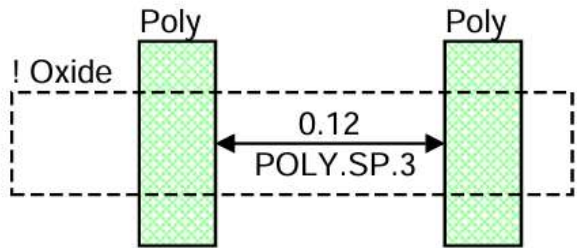


✓ Cadence Reproducing and vertical structure

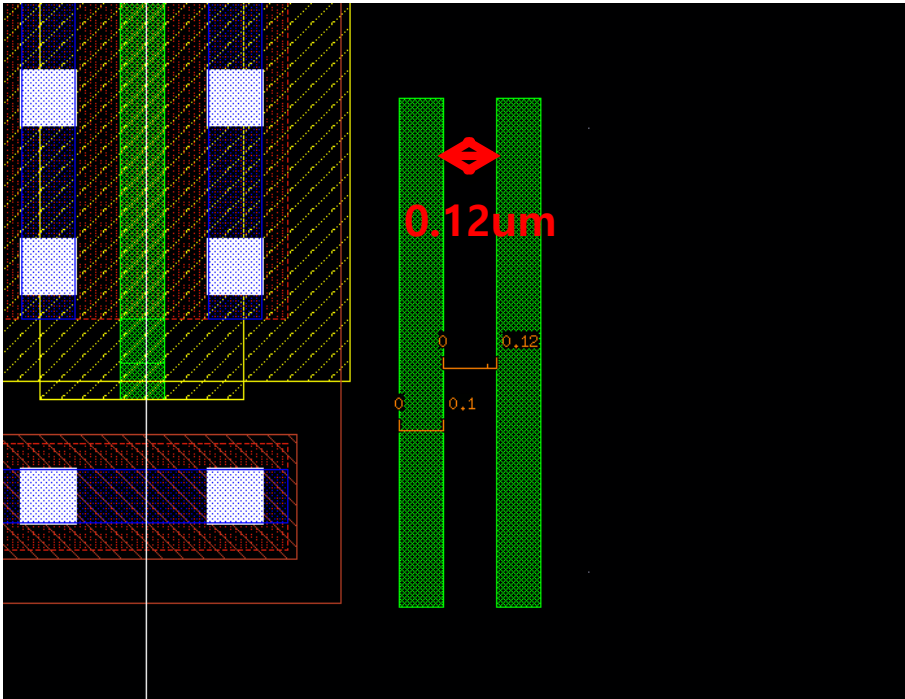


Rule Name	Value	Description
POLY.SP.3	0.12um	Minimum Poly interconnect space

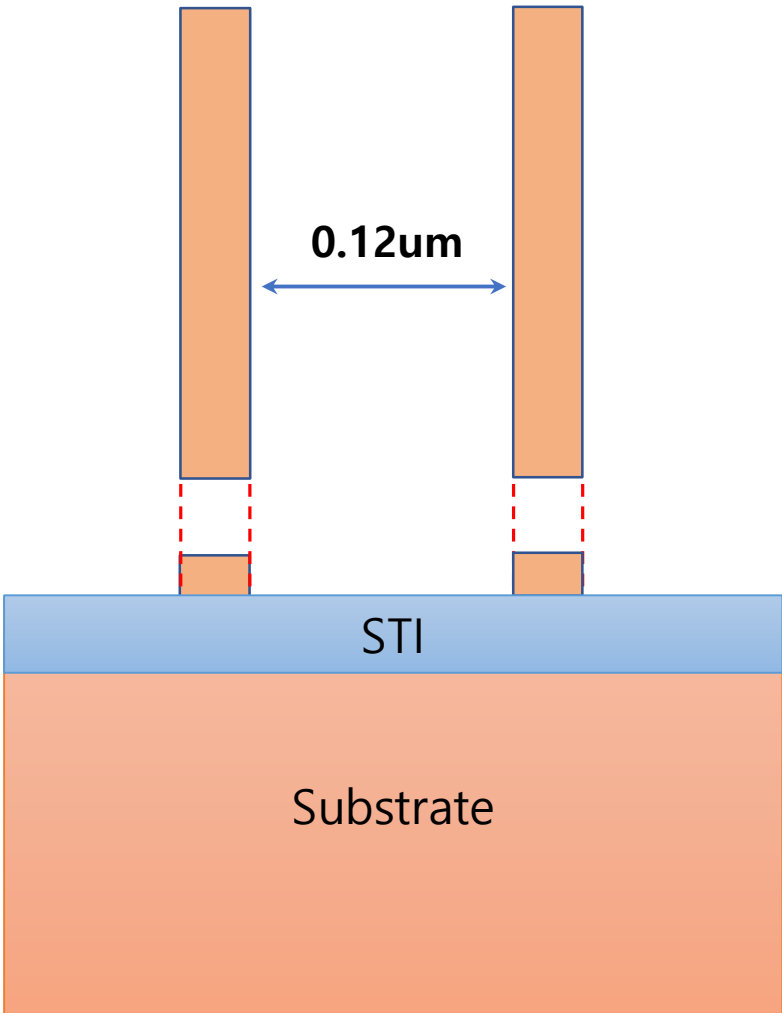
✓ Schematic in Design Rule Manual



“Poly 배선 간 브리징과 signal 간섭을 방지하기 위해 최소 배선 간격을 유지”

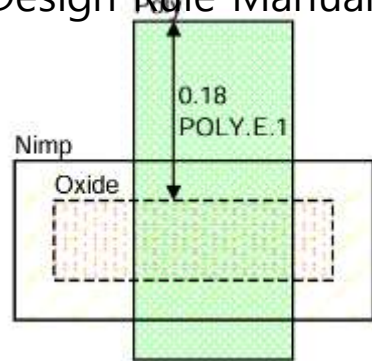


✓ Cadence Reproducing and vertical structure

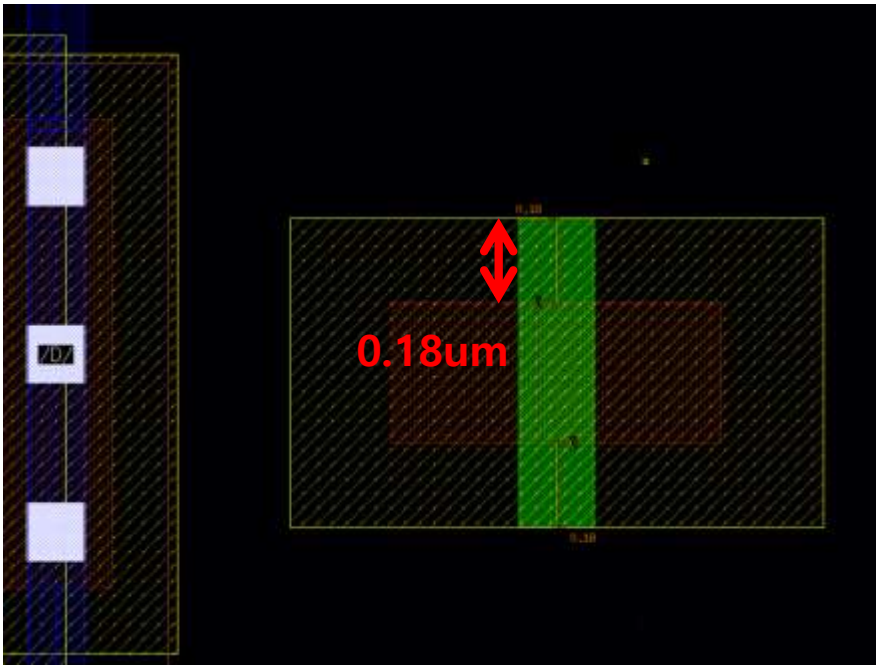


Rule Name	Value	Description
POLY.E.1	0.18um	Minimum N-channel gate extension beyond Active Area

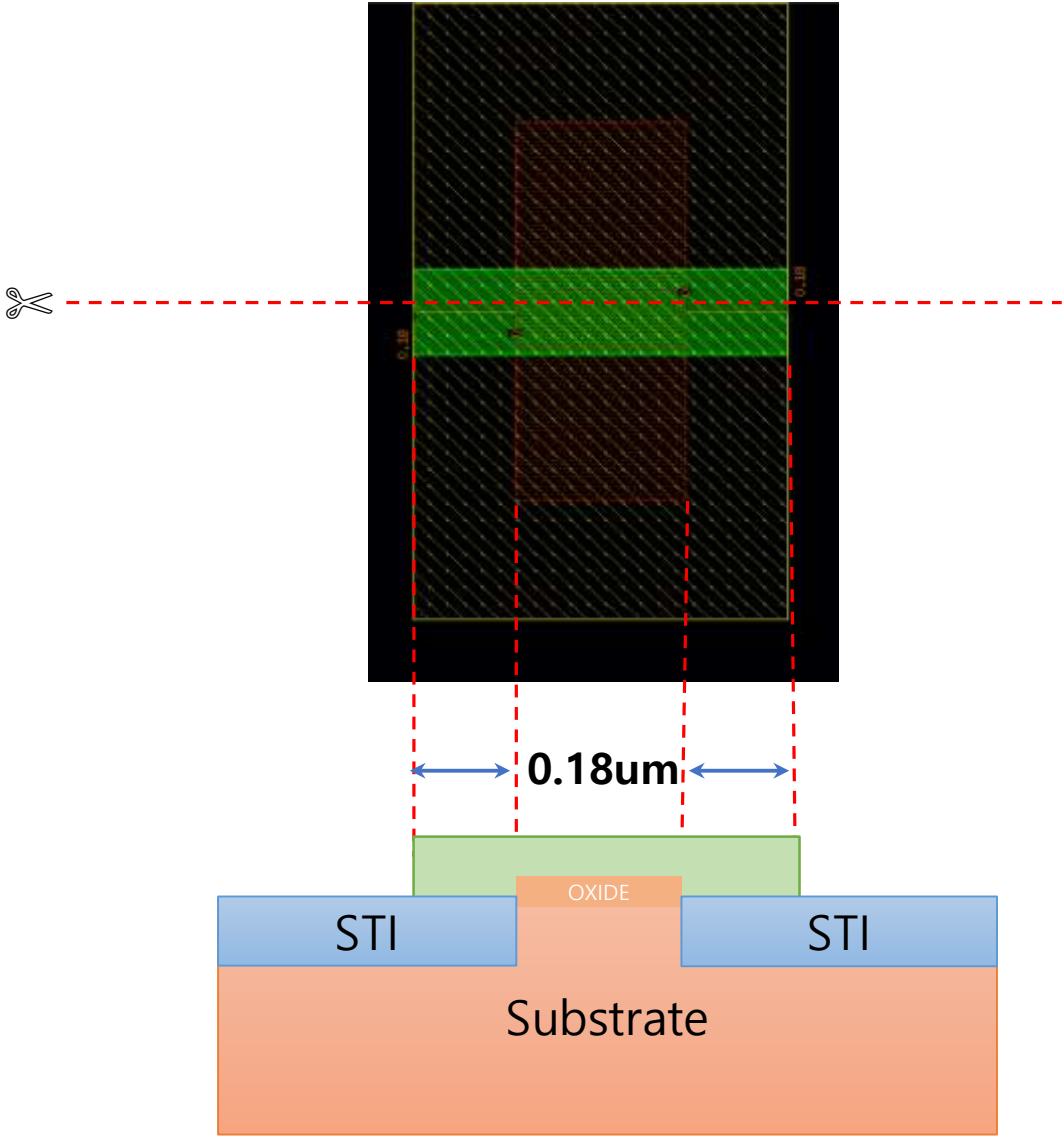
✓ Schematic in Design Rule Manual



“Gate Contact 배치 공간을 확보하고 Overlay 오차를 보상하기 위해 Gate Poly를 Active 밖으로 충분히 연장”

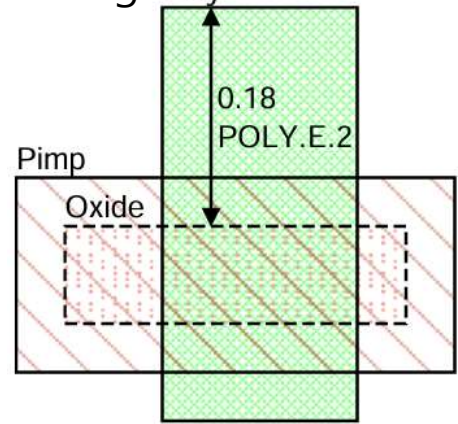


✓ Cadence Reproducing and vertical structure

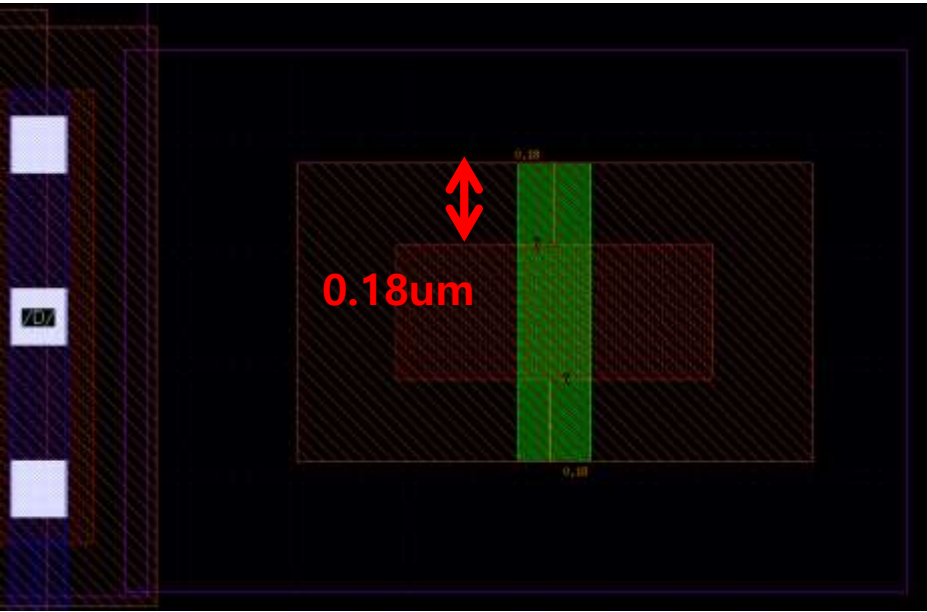


Rule Name	Value	Description
POLY.E.2	0.18um	Minimum P-channel gate extension beyond Active Area

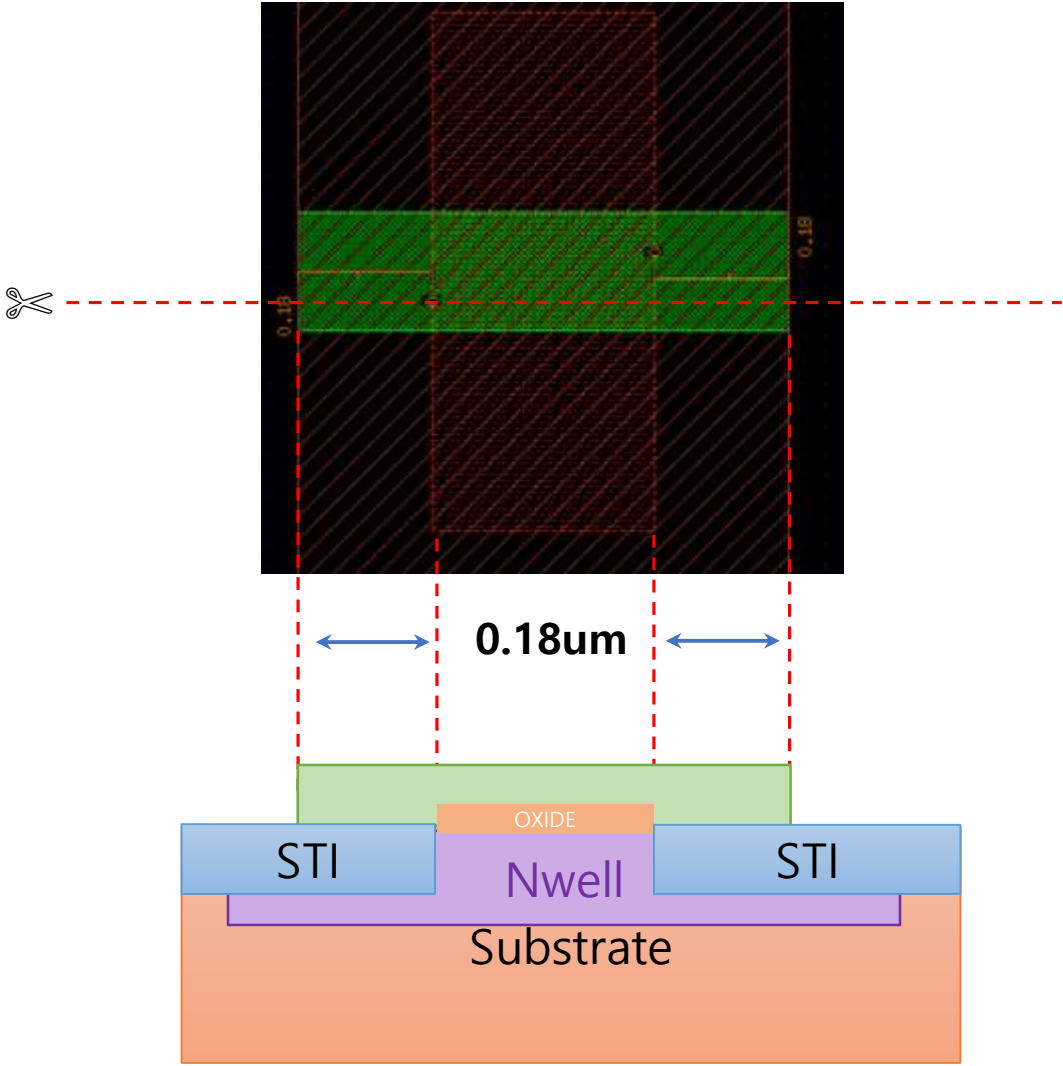
✓ Schematic in Design Rule Manual



“Gate Contact 배치 공간을 확보하고 Overlay 오차를 보상하기 위해 Gate Poly를 Active 밖으로 충분히 연장”

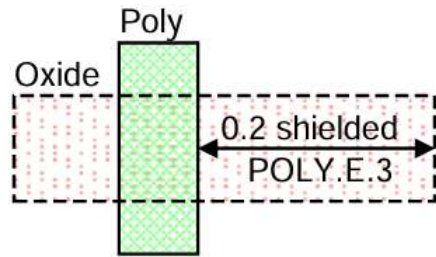


✓ Cadence Reproducing and vertical structure

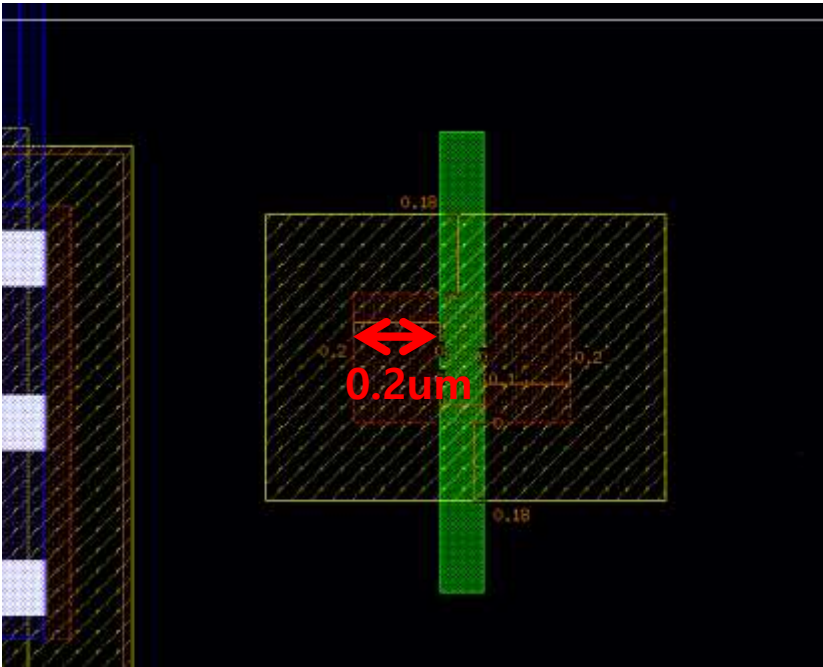


Rule Name	Value	Description
POLY.E.3	0.2um	Minimum Active Area (source/drain) to gate enclosure

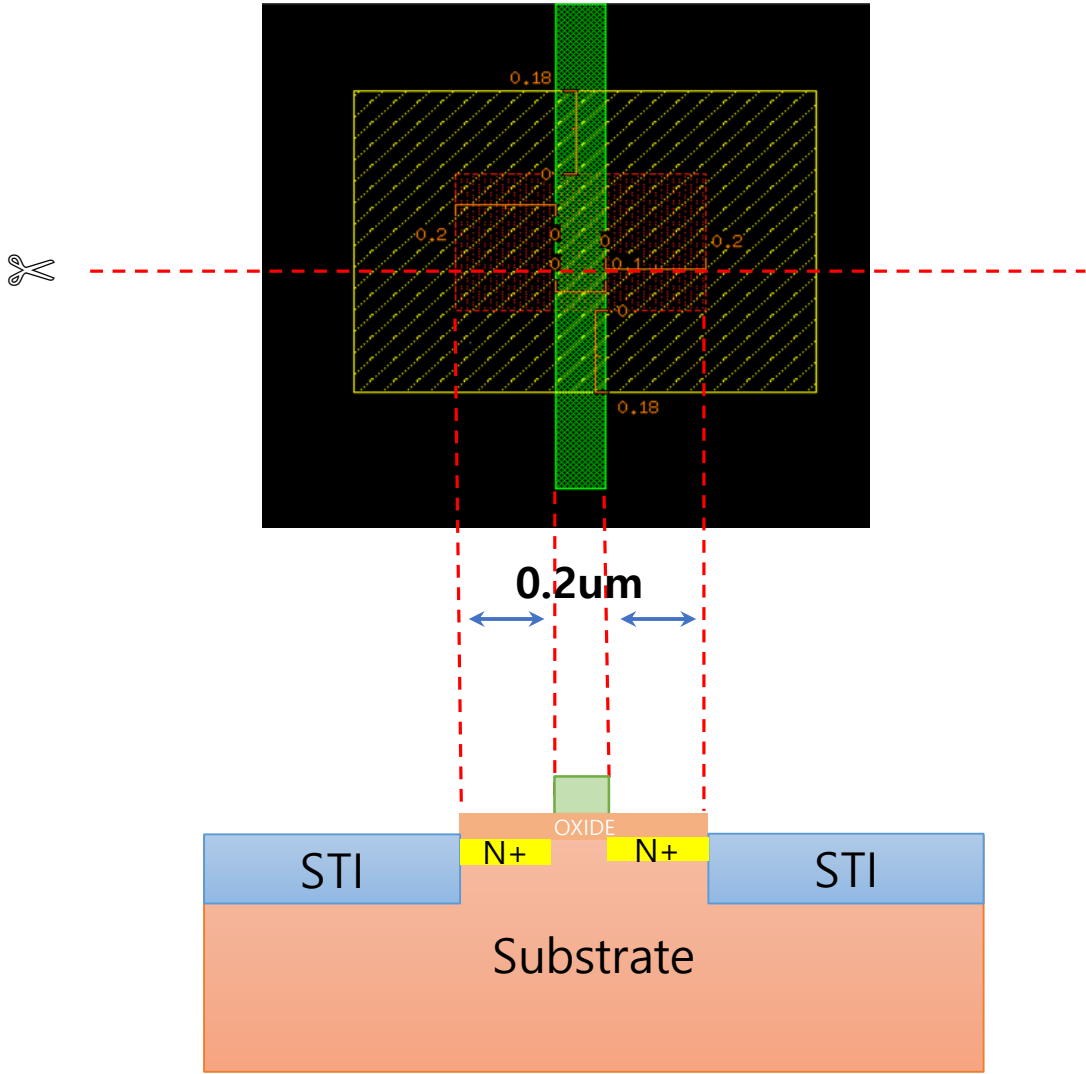
✓ Schematic in Design Rule Manual



“Source/Drain과 Gate의 정렬 오차를 보상하고 안정적인 채널 형성을 위해 충분한 Active Enclosure를 확보”

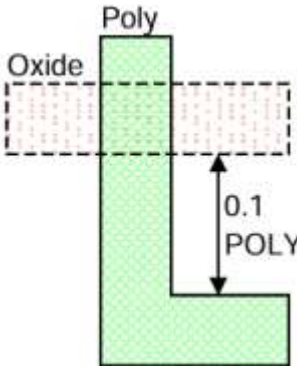


✓ Cadence Reproducing and vertical structure

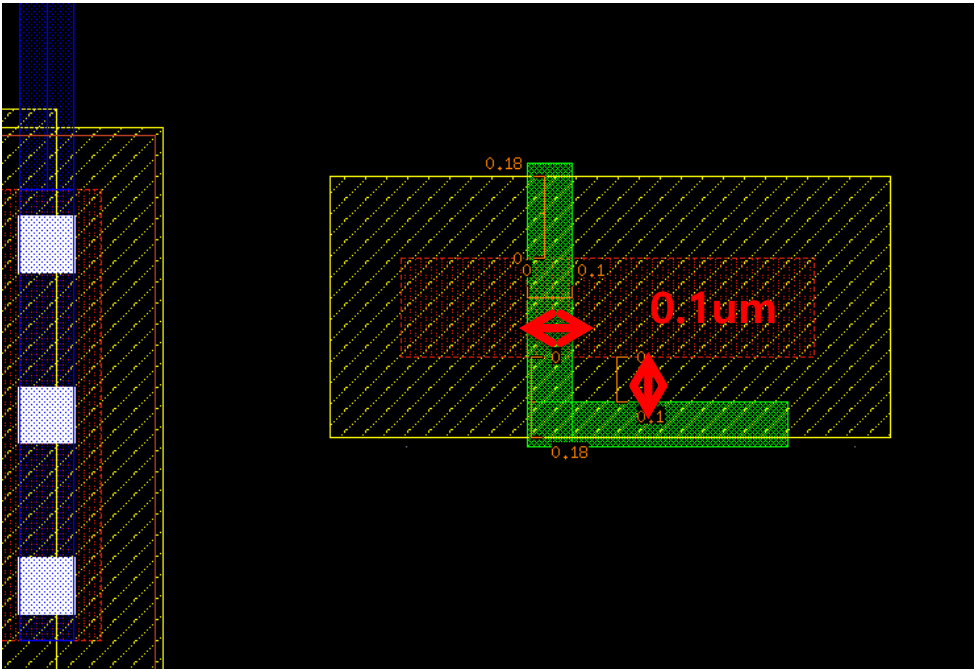


Rule Name	Value	Description
POLY.SE.1_2	0.1um	Minimum Poly interconnect to unrelated/related Active Area space

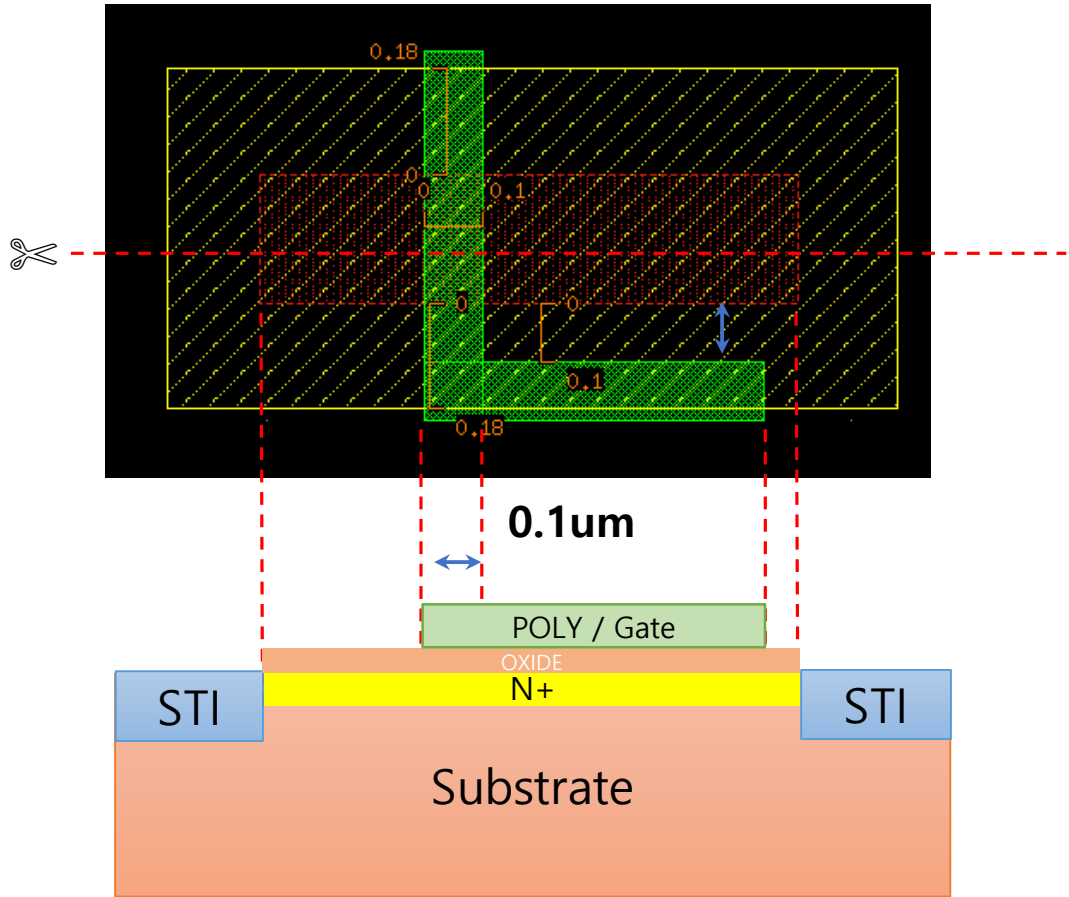
✓ Schematic in Design Rule Manual



“의도하지 않은 MOS 채널 형성과 단락을 방지하기 위해 Poly와 Active 사이의 최소 간격을 유지”

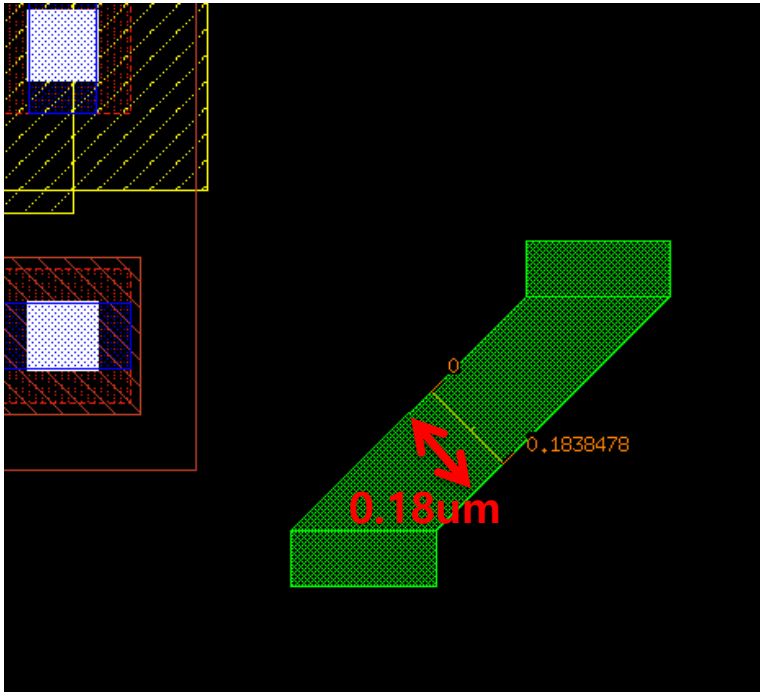
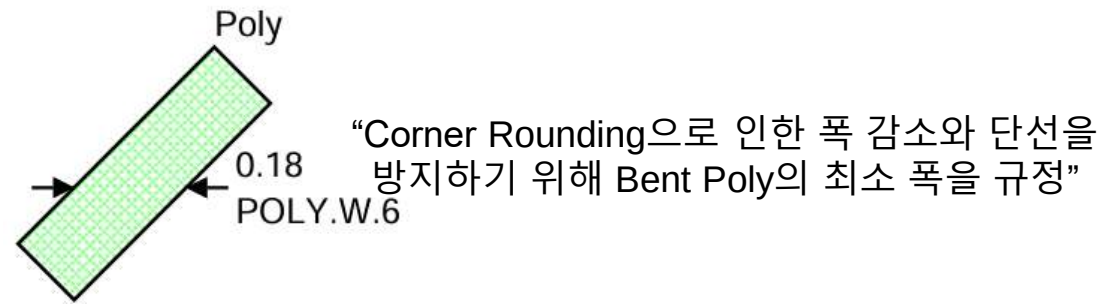


✓ Cadence Reproducing and vertical structure

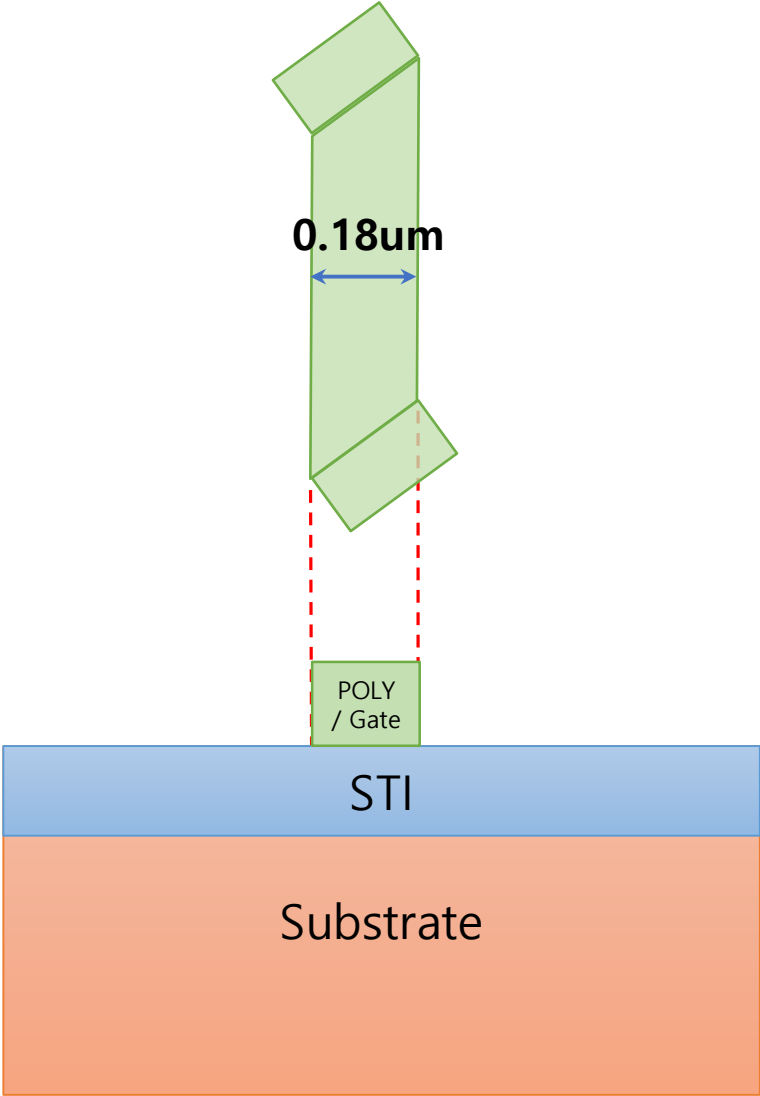


Rule Name	Value	Description
POLY.W.6	0.18um	Minimum bent Poly width

✓ Schematic in Design Rule Manual

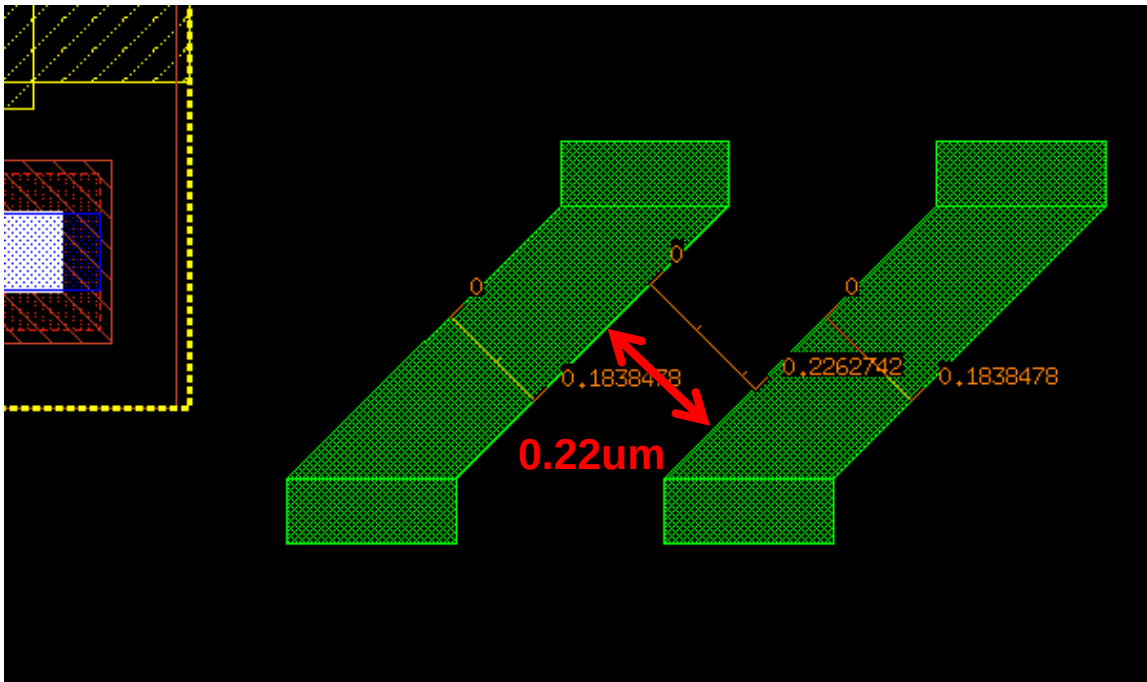
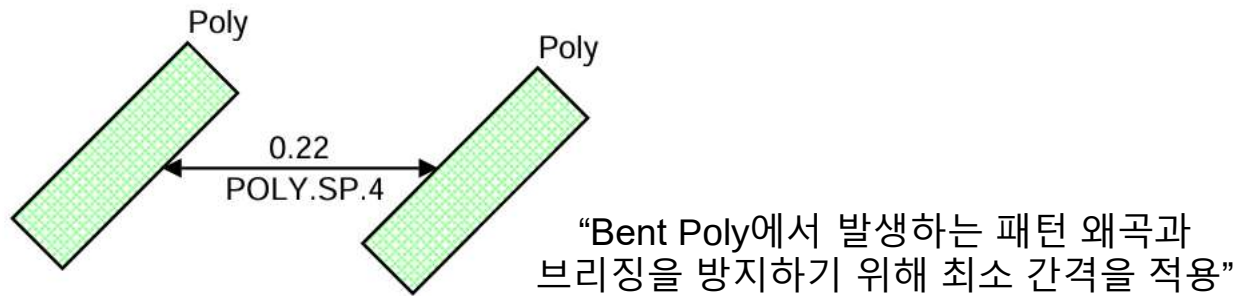


✓ Cadence Reproducing and vertical structure

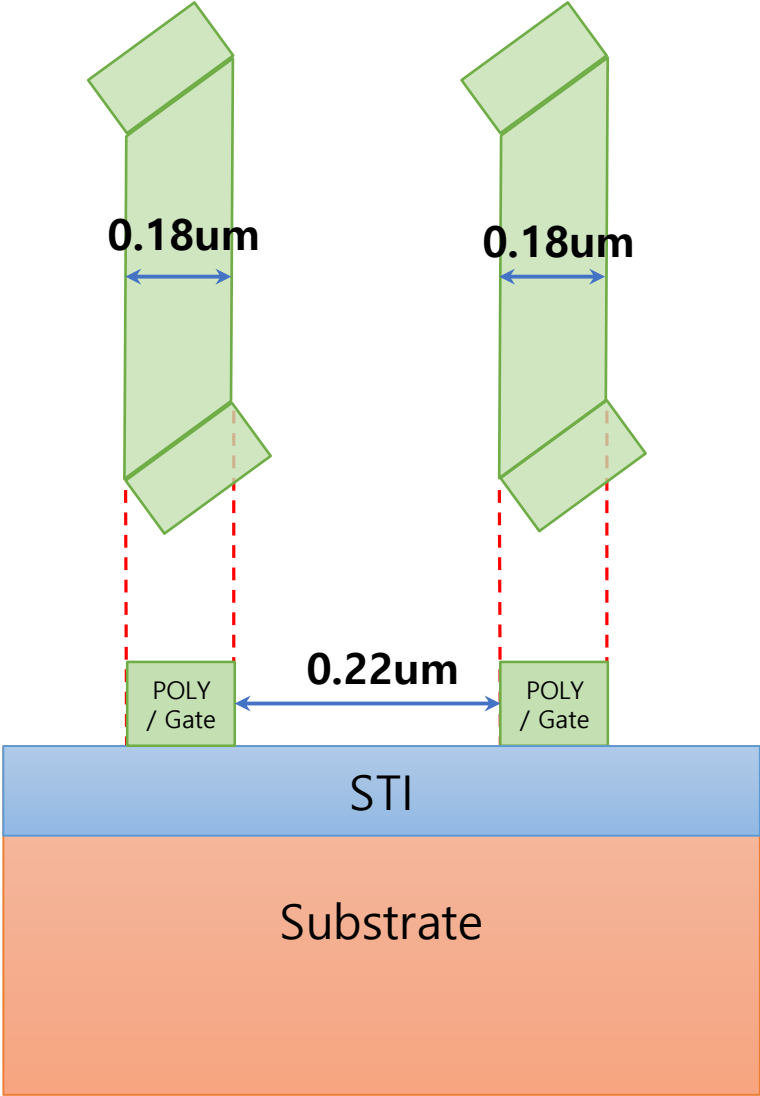


Rule Name	Value	Description
POLY.SP.4	0.22um	Minimum bent Poly space

✓ Schematic in Design Rule Manual

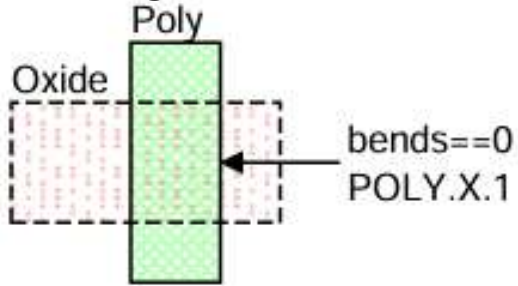


✓ Cadence Reproducing and vertical structure

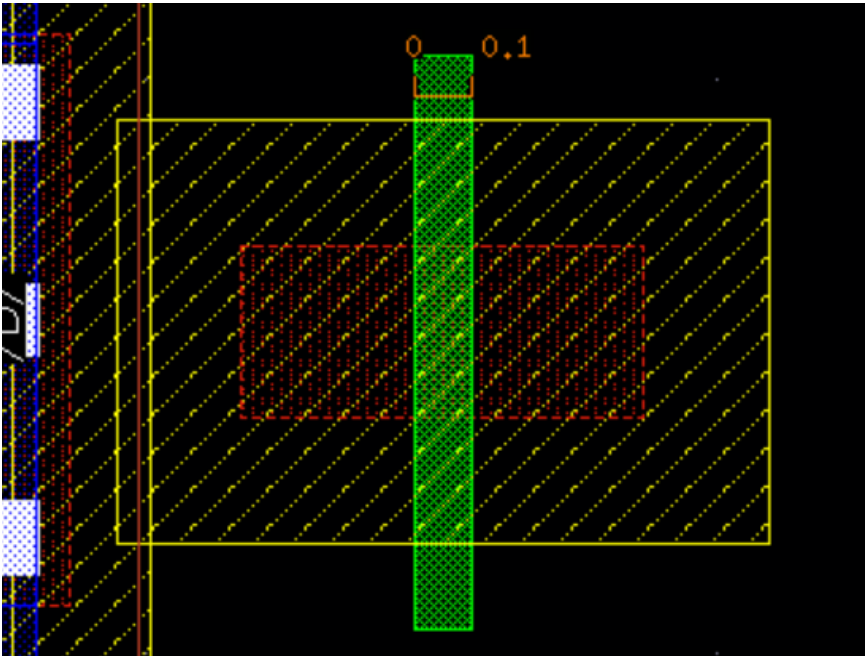


Rule Name	Value	Description
POLY.X.1	---	Bent gate is not allowed

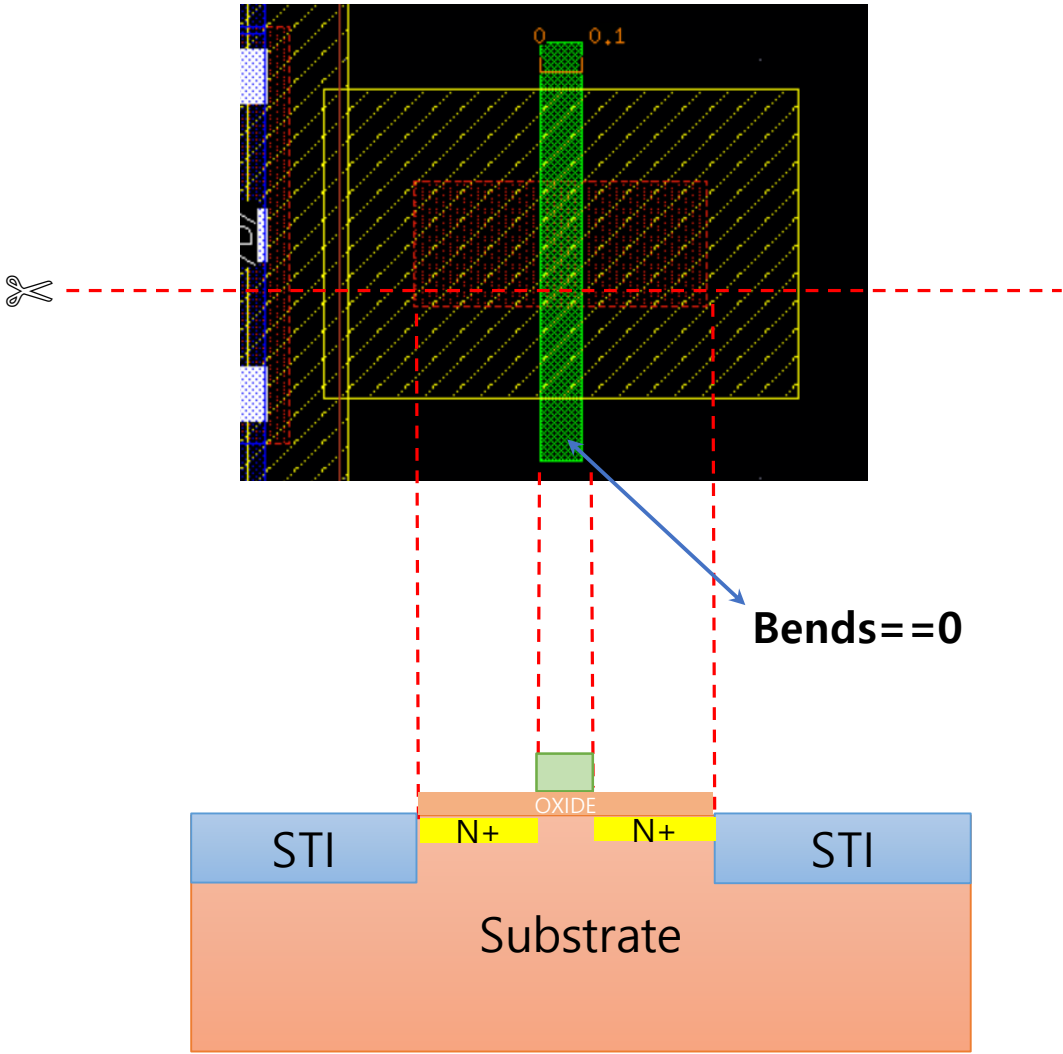
✓ Schematic in Design Rule Manual



“균일한 채널 형성과 안정적인 소자 특성을 위해 Bent Gate를 금지. 직선 구조만 사용”

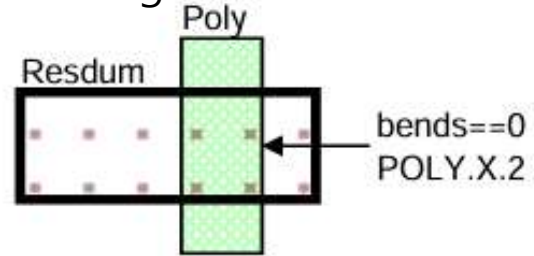


✓ Cadence Reproducing and vertical structure

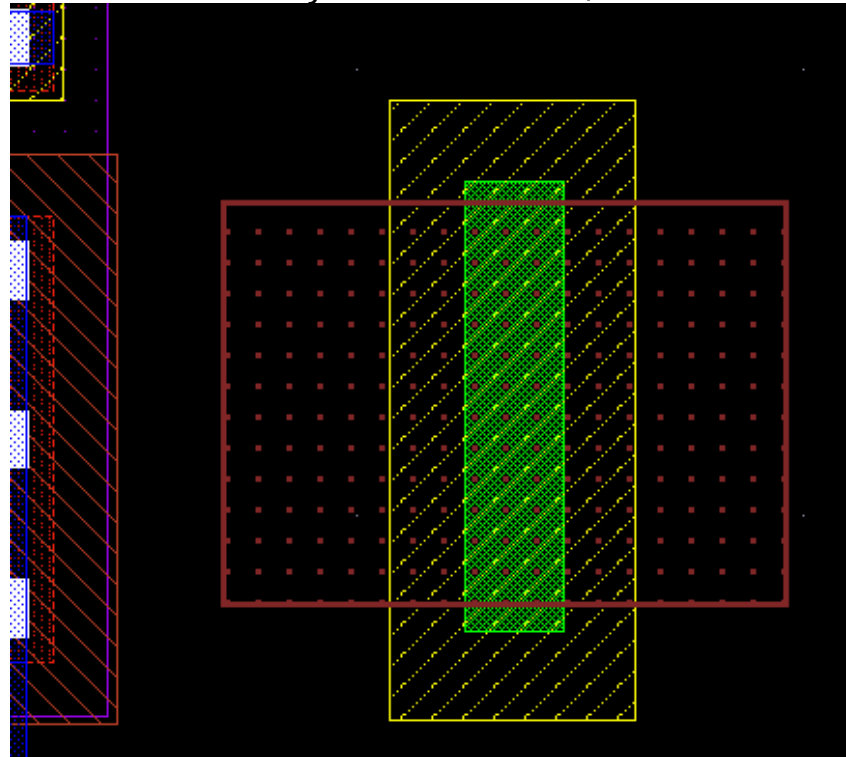


Rule Name	Value	Description
POLY.X.2	---	Bent Poly resistor is not allowed

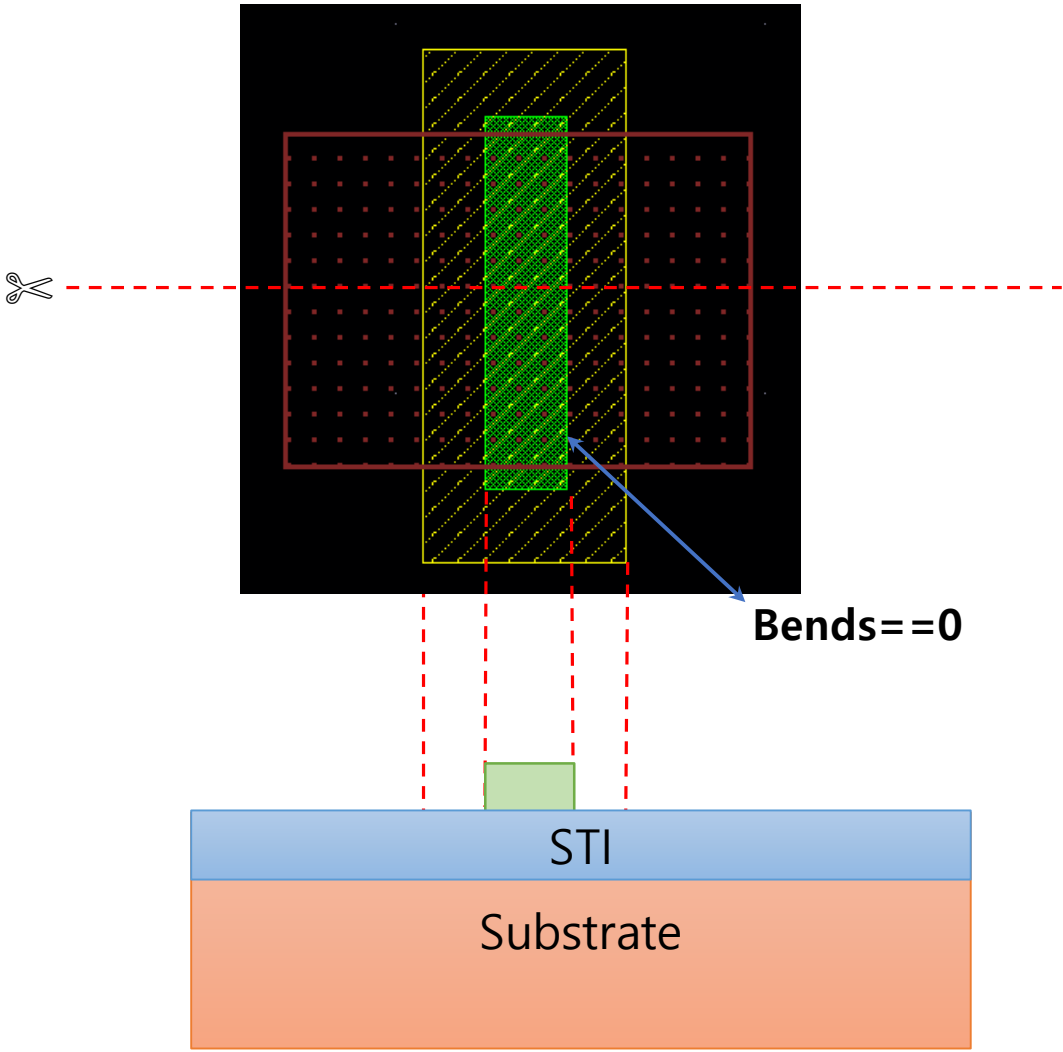
✓ Schematic in Design Rule Manual



“Current Crowding과 저항값 오차를 방지하기 위해
Bent Poly Resistor를 금지”



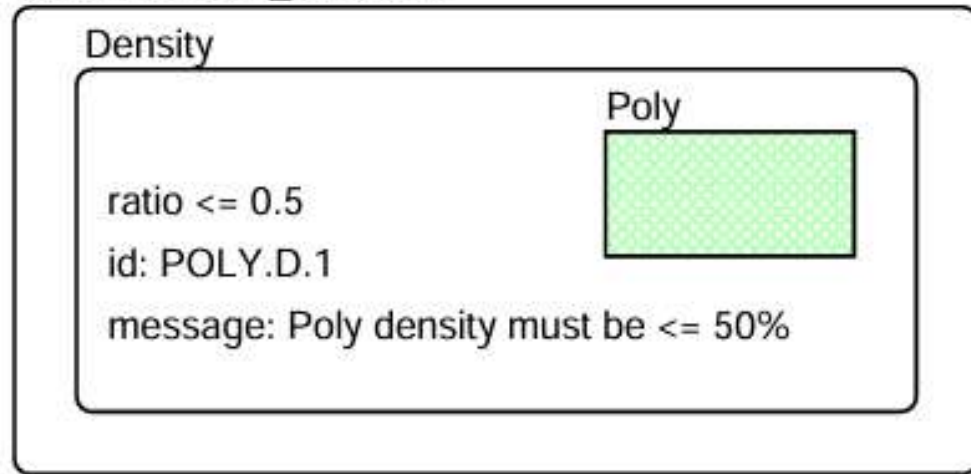
✓ Cadence Reproducing and vertical structure



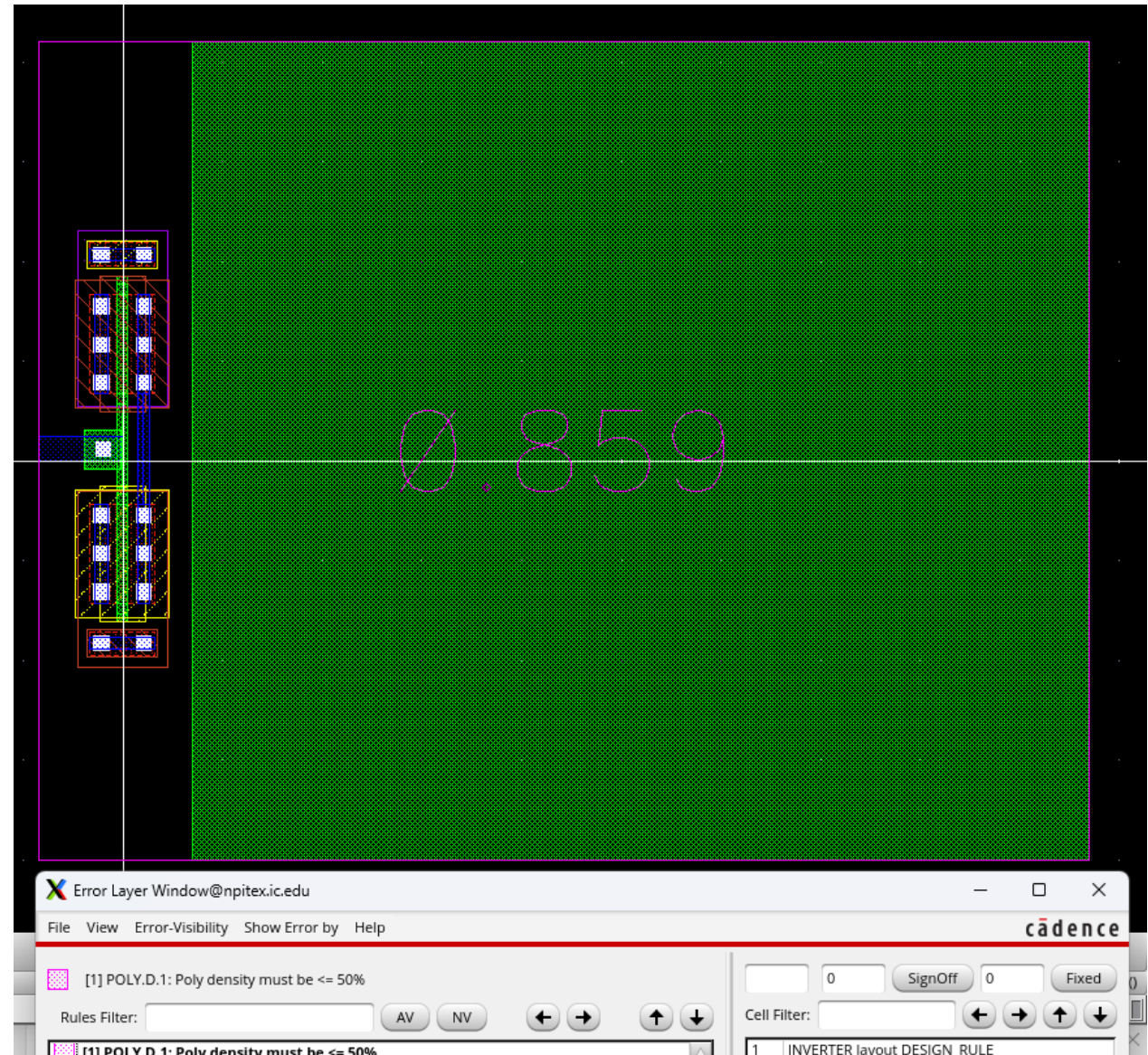
Rule Name	Value	Description
POLY.D.1	50%	Maximum Poly density across full chip

✓ Schematic in Design Rule Manual

switch CHECK_DENSITY

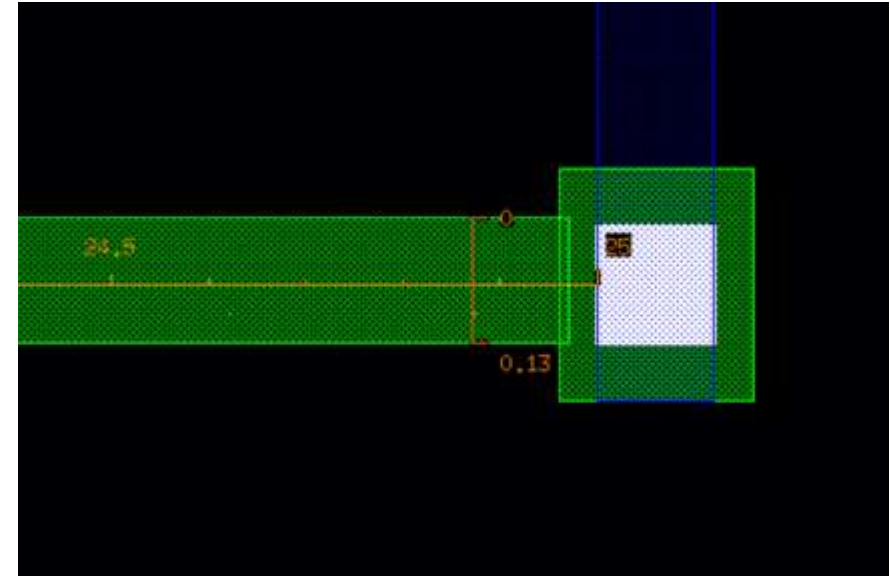
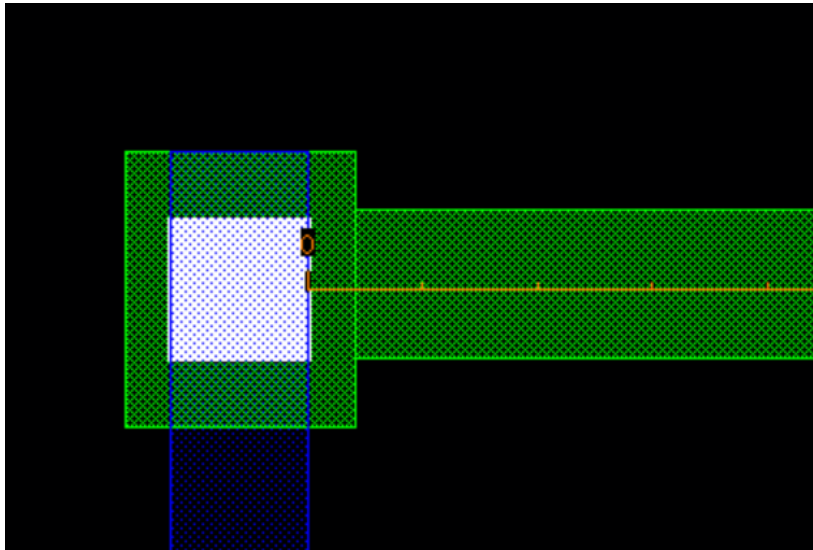


“Poly Density Ratio를 제한하여 CMP와 식각 공정의 균 일성을 확보”

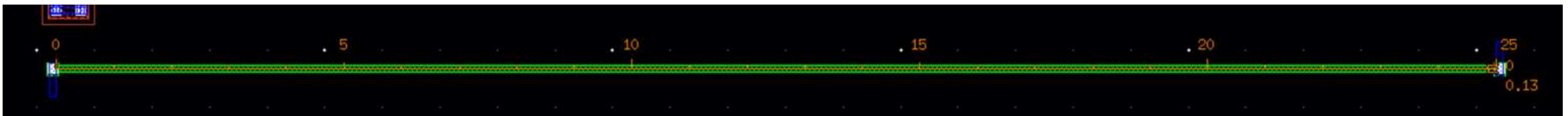


Rule Name	Value	Description
POLY.SE.3	25.0	Minimum Poly segment length (width < 0.14) between two contacts

✓ Schematic in Design Rule Manual

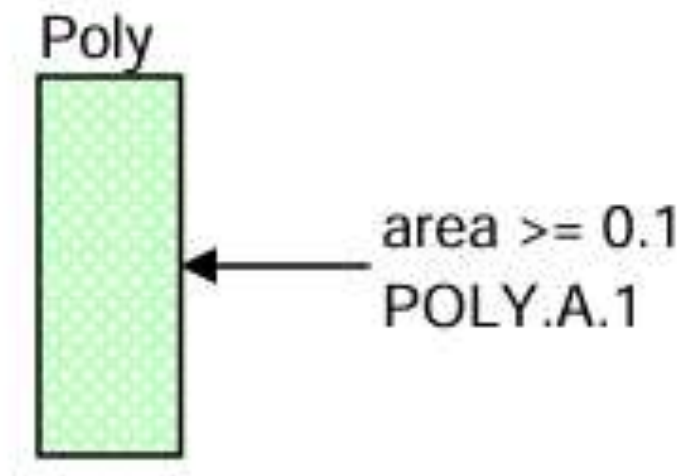


“좁고 긴 Poly의 높은 저항과 IR Drop을 방지하기 위해
최대 Poly Segment Length를 제한”

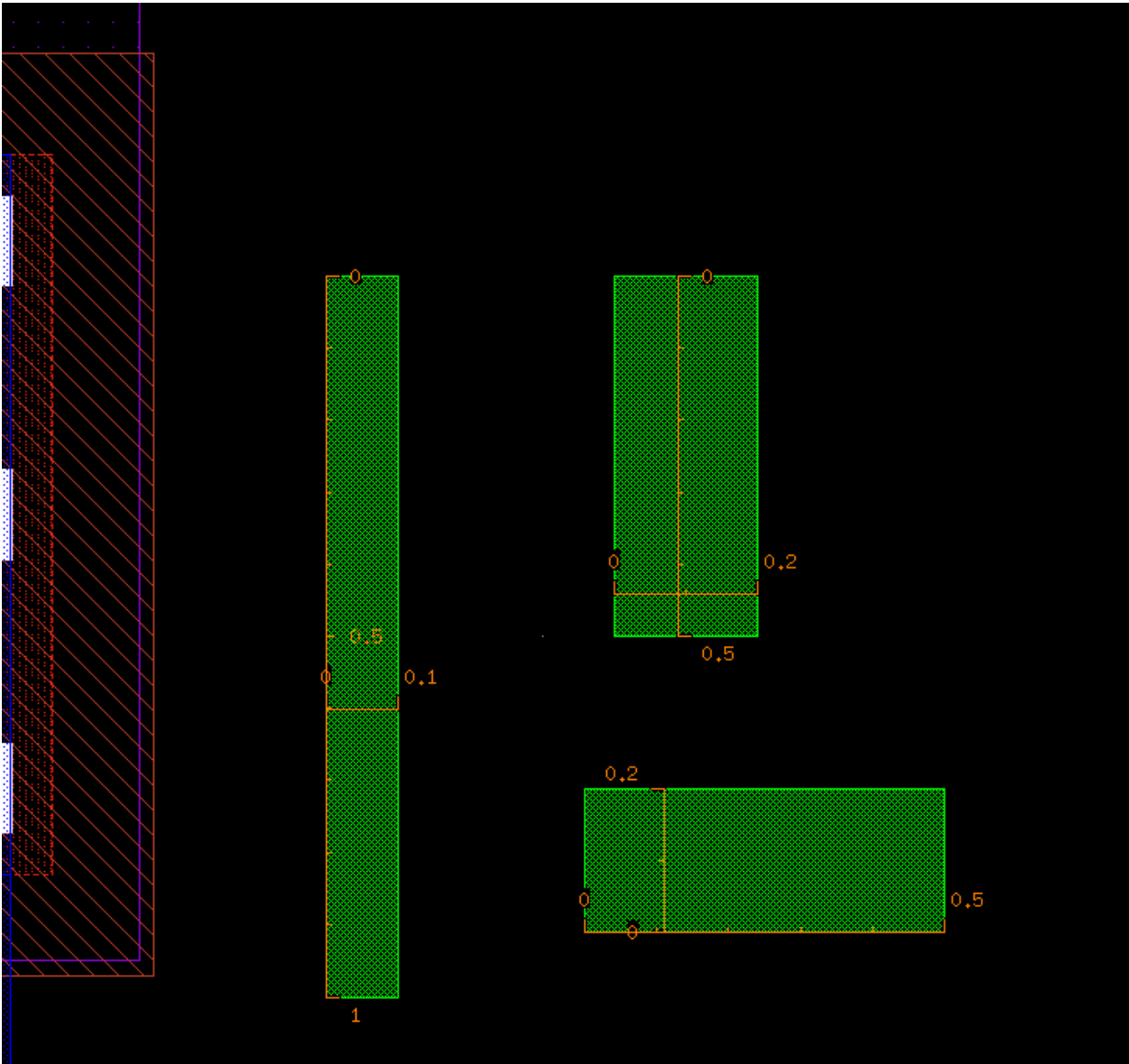


Rule Name	Value	Description
POLY.A.1	0.1	Minimum area for Poly interconnect

✓ Schematic in Design Rule Manual



“공정에서 안정적으로 패턴을 형성하기 위해 Poly의 최소 면적을 확보”

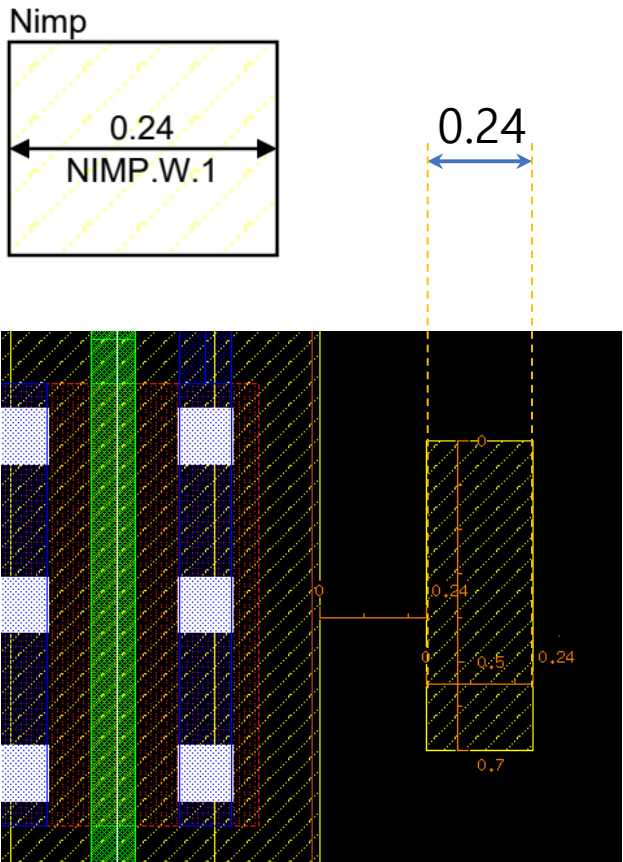


Implant

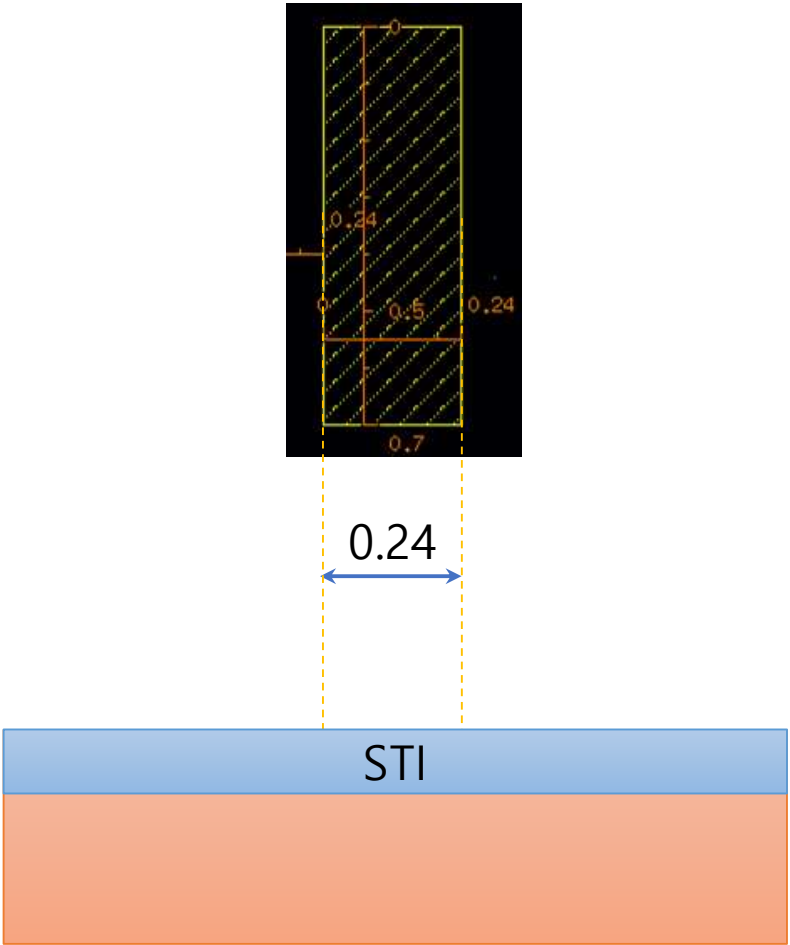
Rule Name	Value	Description
NIMP.W.1	0.24um	Minimum N+ Implant width

✓ Schematic in Design Rule Manual

도핑 균일성과 Source/Drain 특성 확보를 위한
최소 N+ Implant width



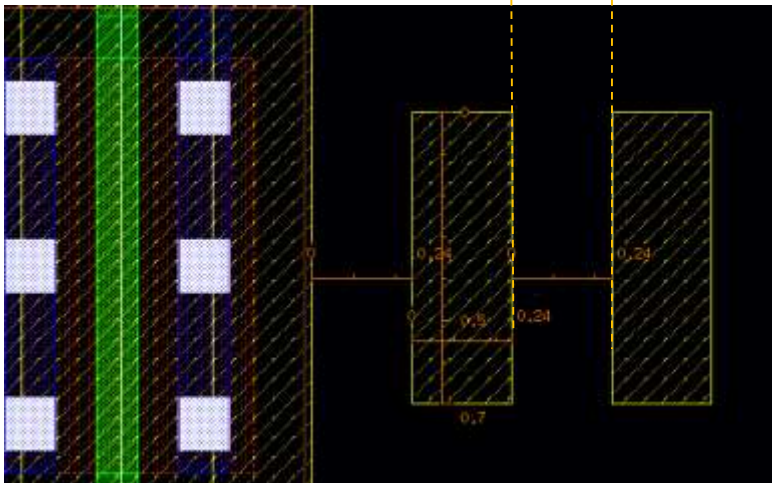
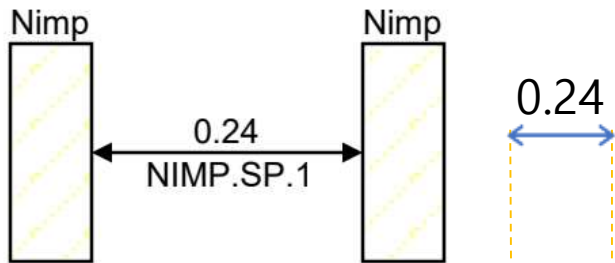
✓ Cadence Reproducing and vertical structure



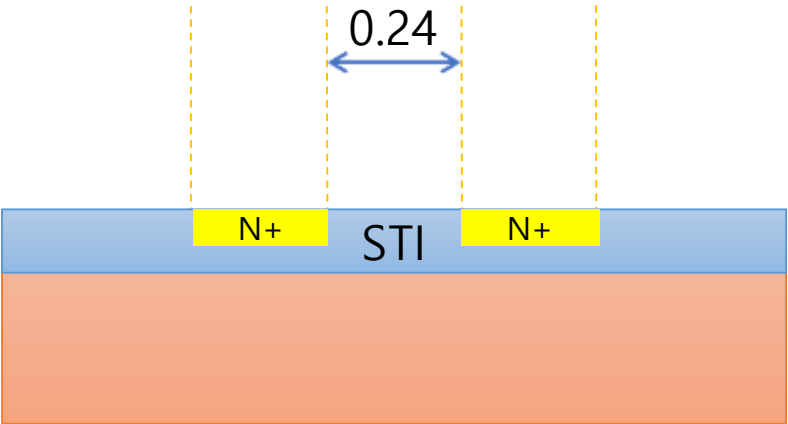
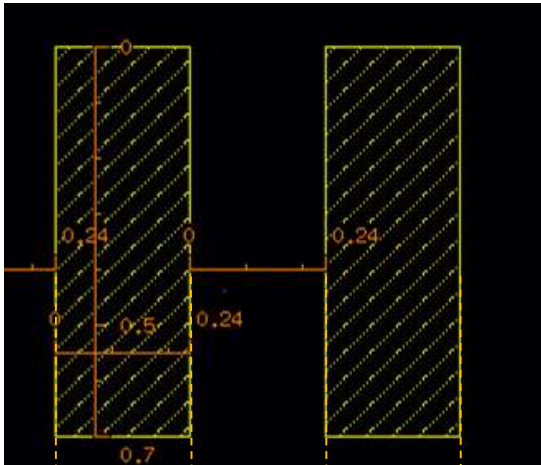
Rule Name	Value	Description
NIMP.SP.1	0.24um	Minimum N+ Implant space

✓ Schematic in Design Rule Manual

도핑 영역 간 간섭 및 Short 방지를 위한
최소 Implant space



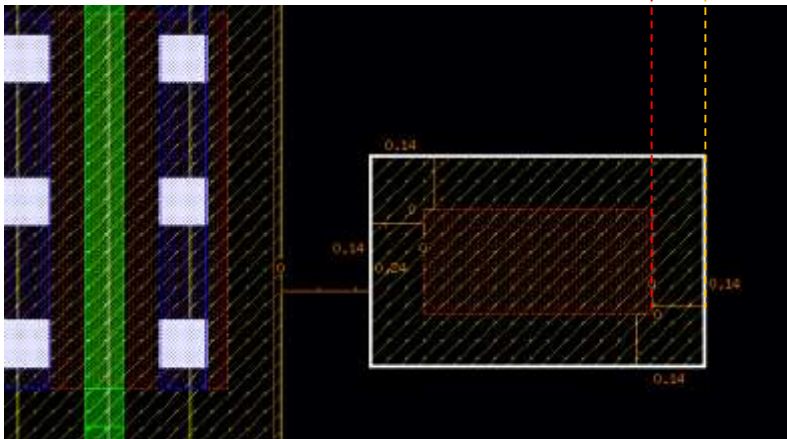
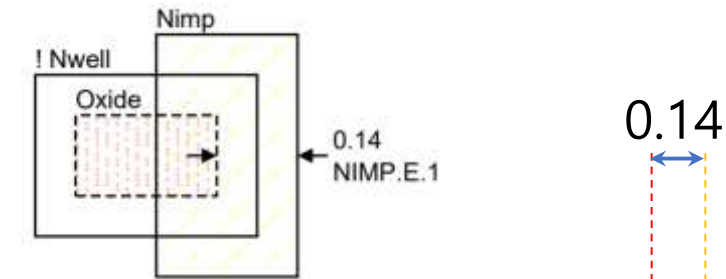
✓ Cadence Reproducing and vertical structure



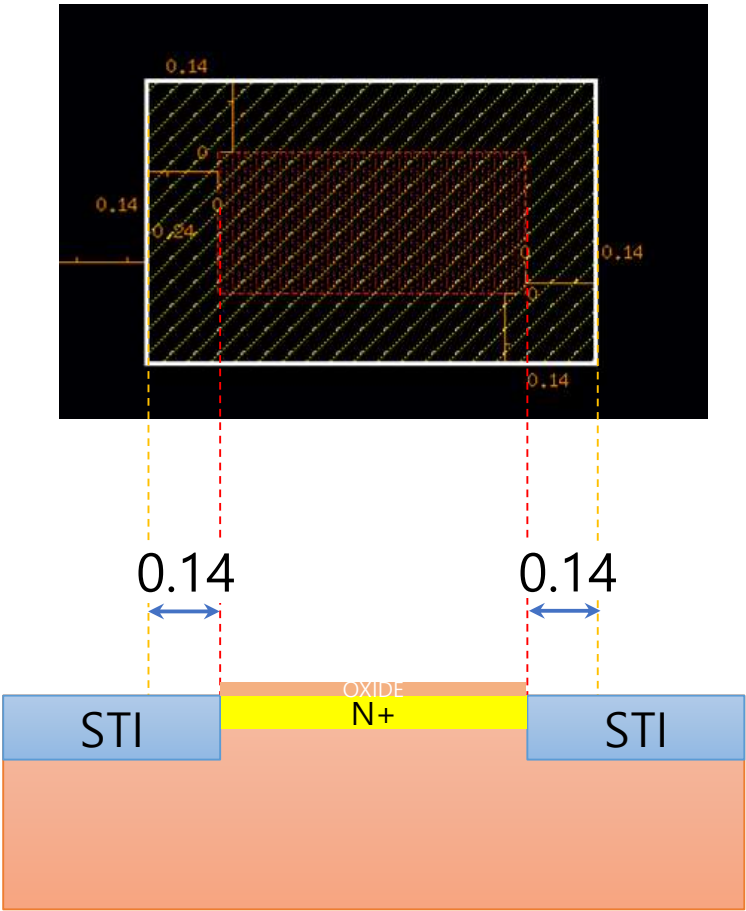
Rule Name	Value	Description
NIMP.E.1	0.14um	Minimum N+ Implant to Active Area enclosure

✓ Schematic in Design Rule Manual

Active Area의 안정적인 도핑을 위한
최소 Enclosure



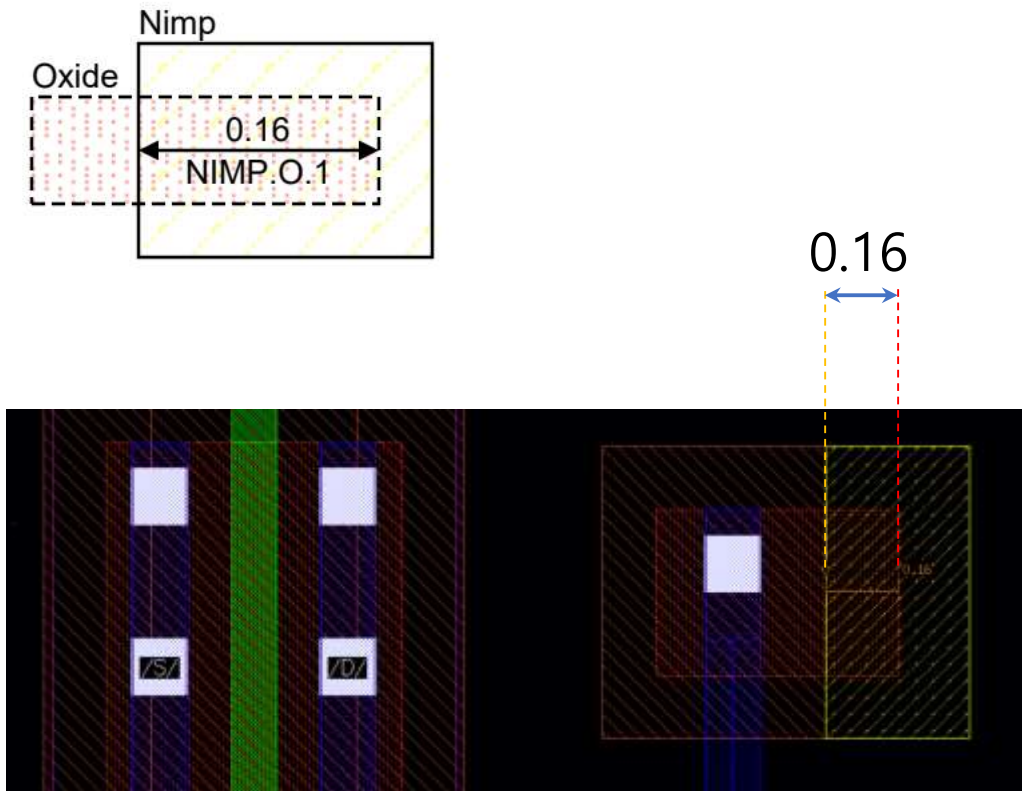
✓ Cadence Reproducing and vertical structure



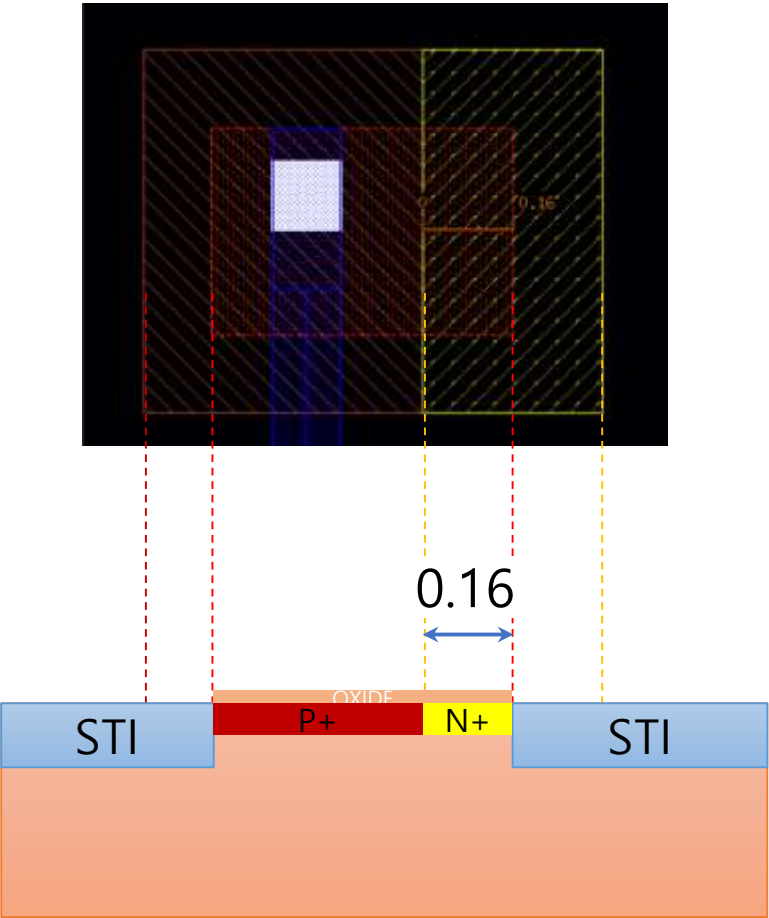
Rule Name	Value	Description
NIMP.O.1	0.16um	Minimum N+ Implant to Active Area overlap

✓ Schematic in Design Rule Manual

공정 오차를 고려한 Active Area와 Implant의
최소 Overlap



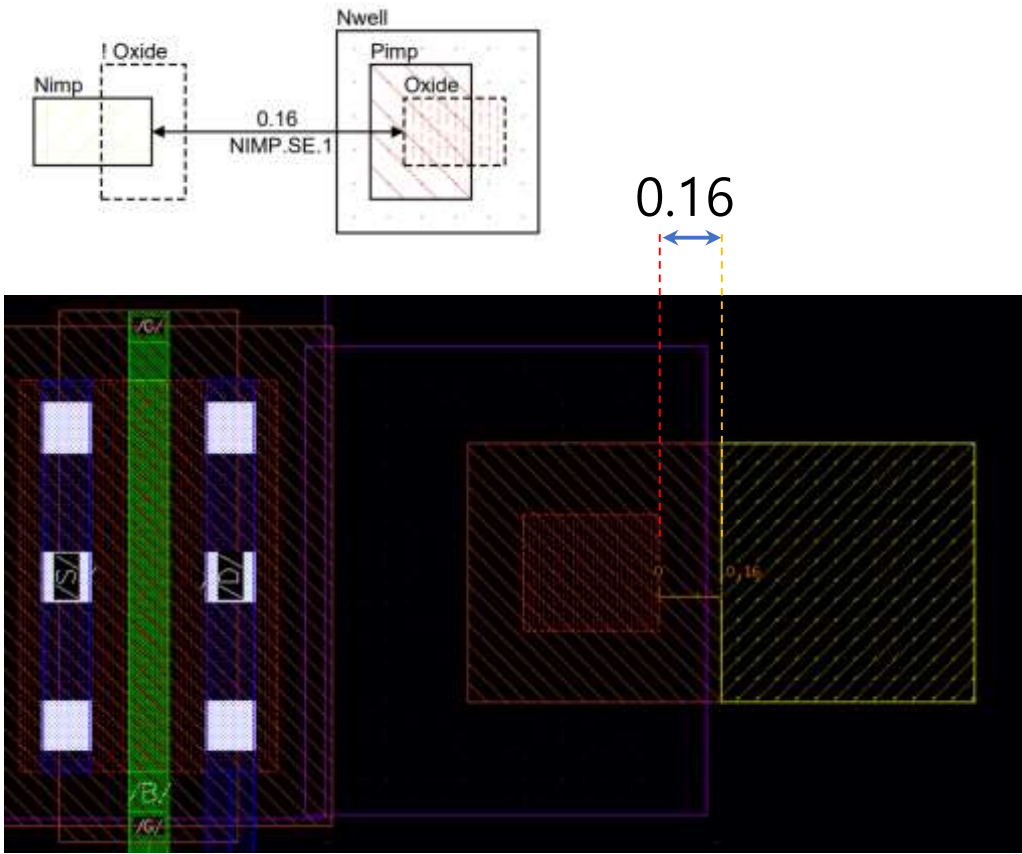
✓ Cadence Reproducing and vertical structure



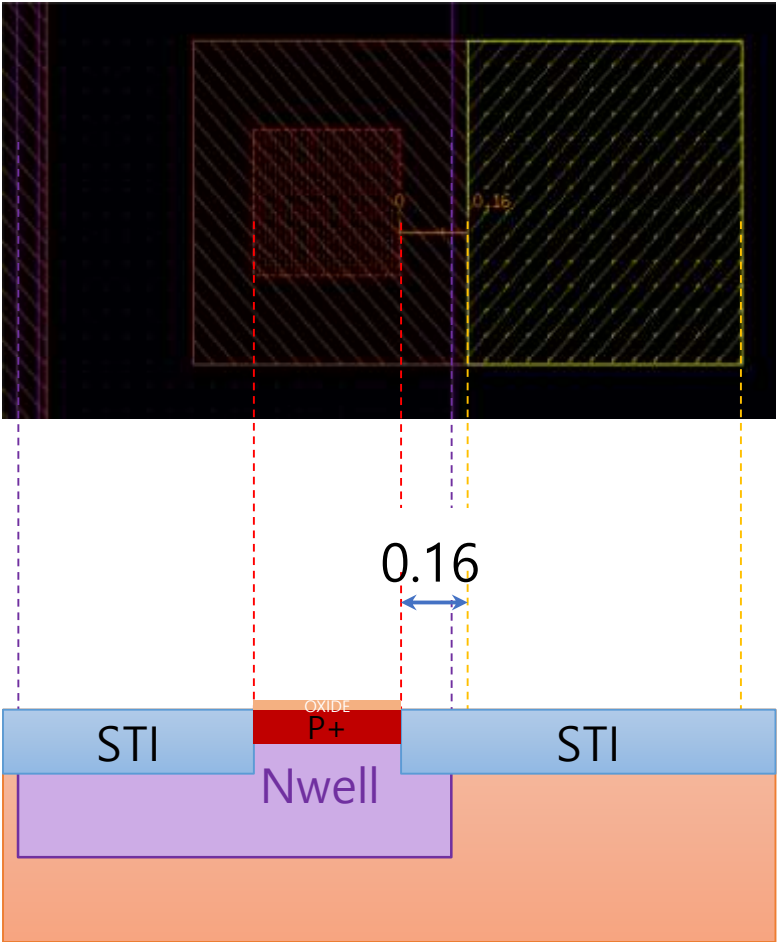
Rule Name	Value	Description
NIMP.SE.1	0.16um	Minimum N+ Implant to P+ Active (inside Nwell) Area spacing

✓ Schematic in Design Rule Manual

Nwell 내부 N+/P+ 간 누설전류 방지를 위한
최소 space



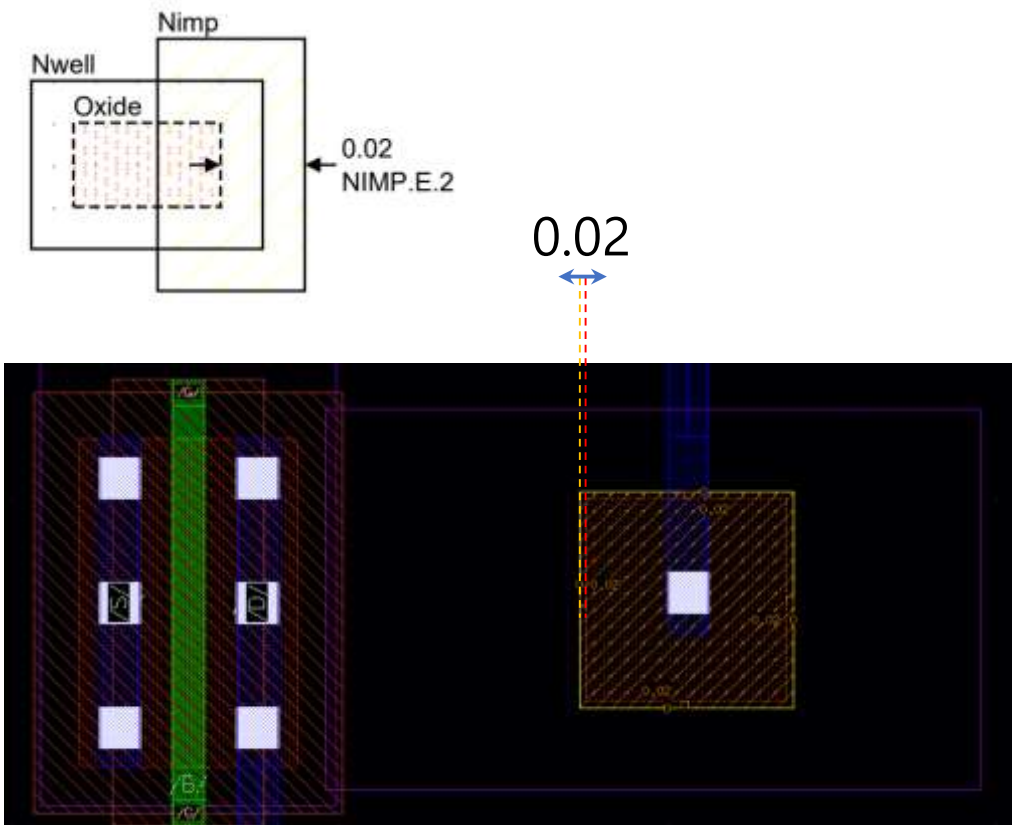
✓ Cadence Reproducing and vertical structure



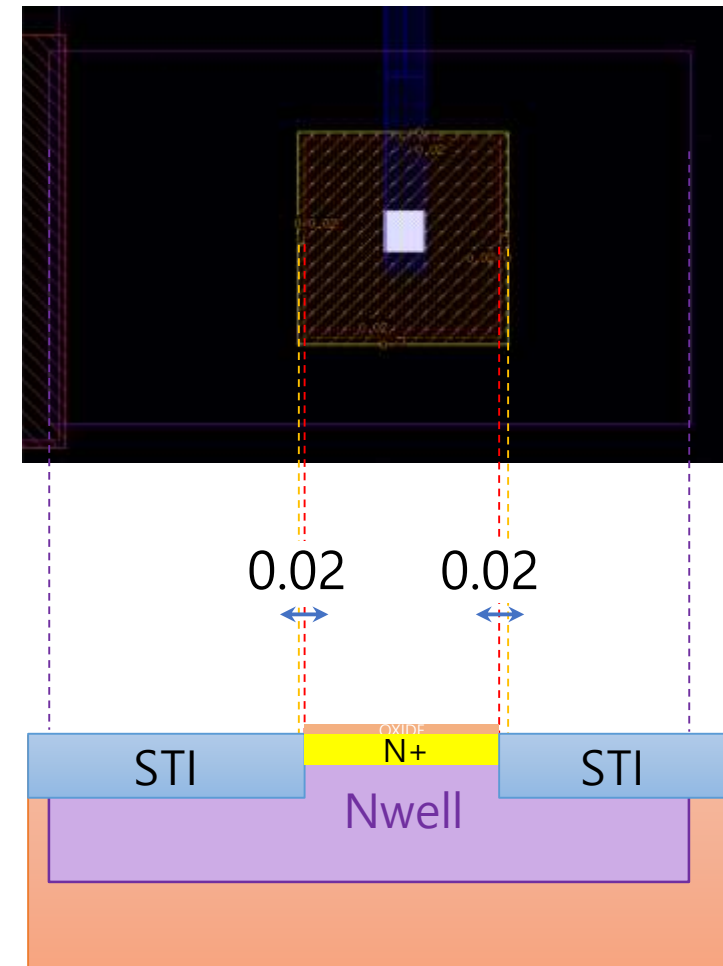
Rule Name	Value	Description
NIMP.E.2	0.02um	Minimum N+ Implant to Active Area (Nwell tie) enclosure

✓ Schematic in Design Rule Manual

Nwell tie의 안정적인 전위 유지를 위한
최소 enclosure



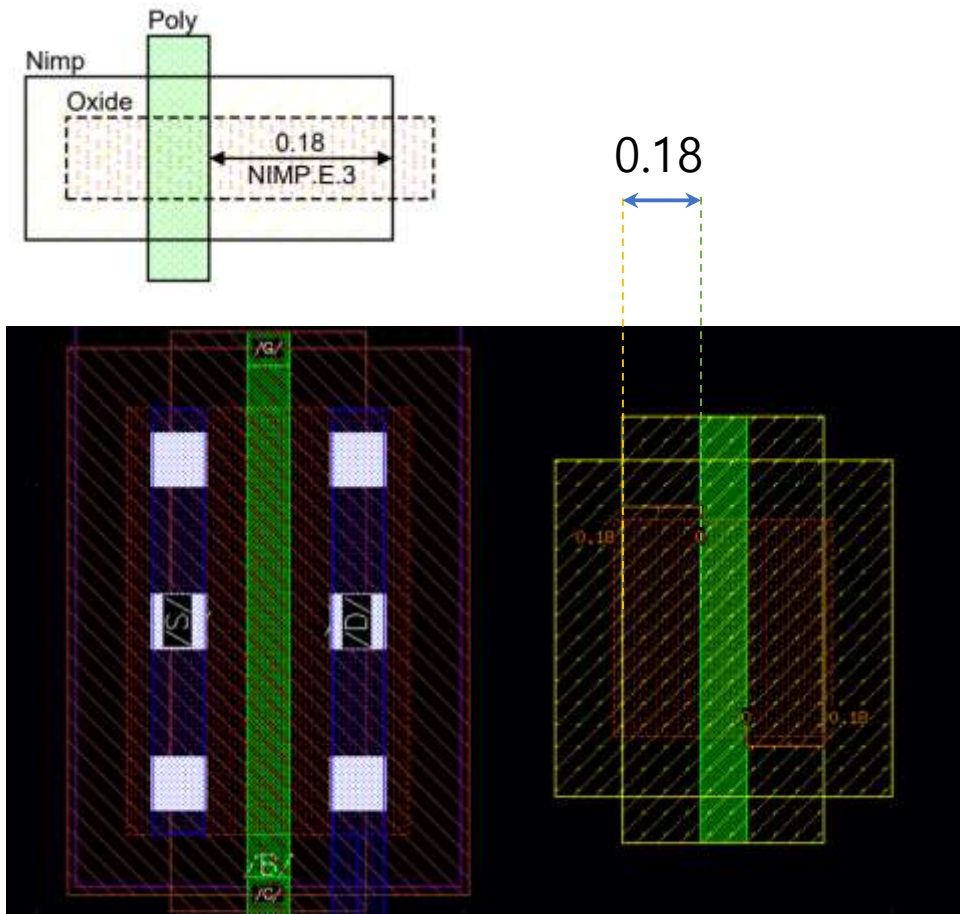
✓ Cadence Reproducing and vertical structure



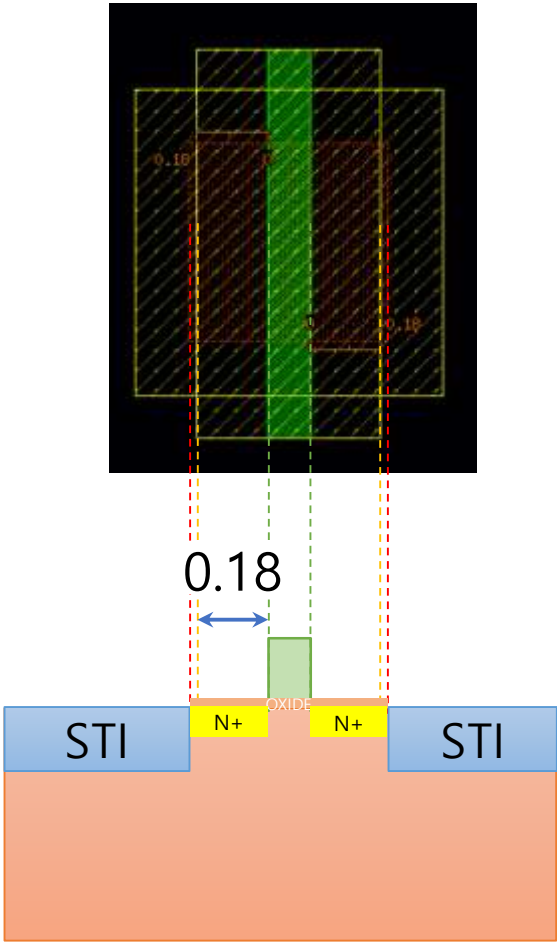
Rule Name	Value	Description
NIMP.E.3	0.18um	Minimum N+ Implant to gate side enclosure

✓ Schematic in Design Rule Manual

Gate 주변 Source/Drain 형성 보장을 위한
최소 enclosure



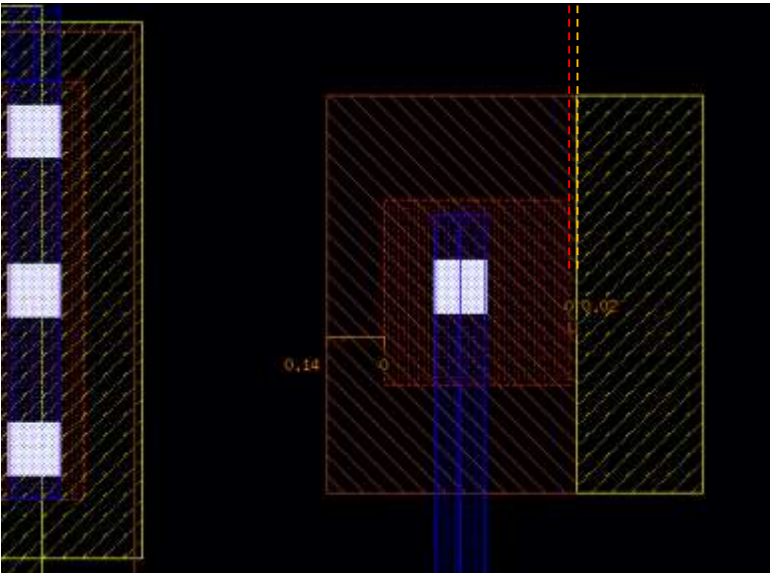
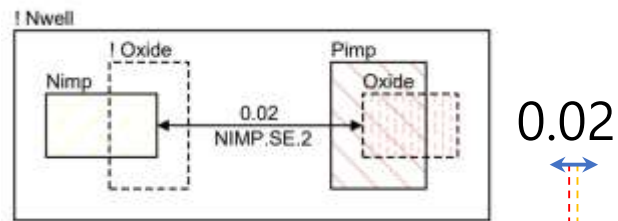
✓ Cadence Reproducing and vertical structure



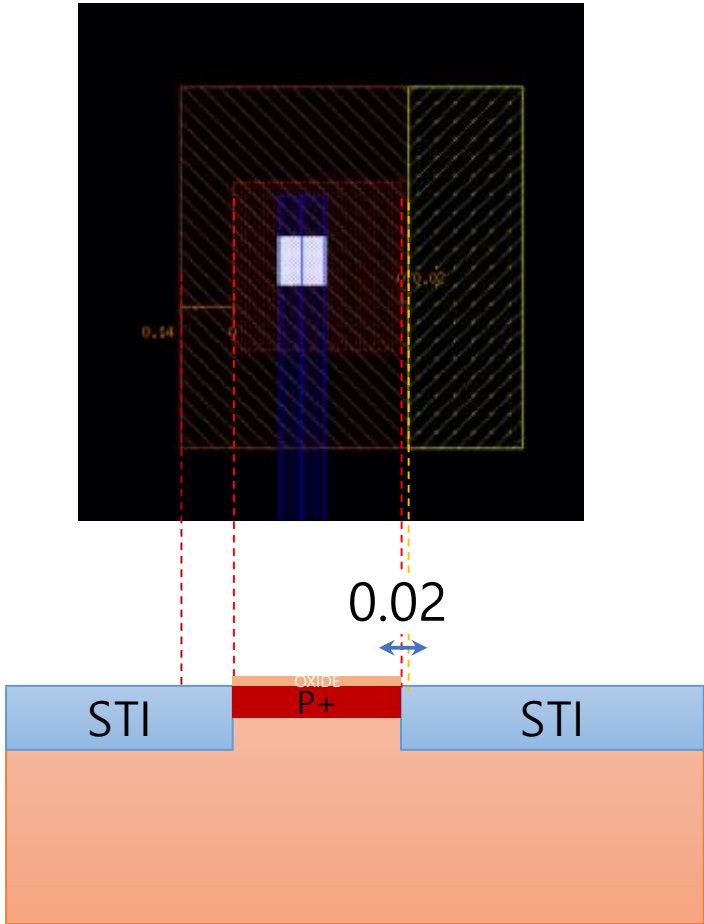
Rule Name	Value	Description
NIMP.SE.2	0.02um	Minimum N+ Implant to P+ Active Area (substrate tie) spacing

✓ Schematic in Design Rule Manual

Substrate Tie의 안정적인 기판 전위 유지를 위한
최소 space



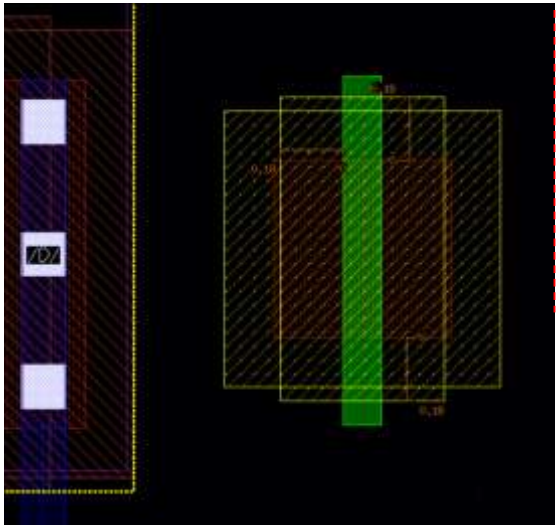
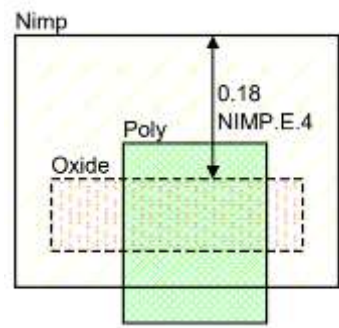
✓ Cadence Reproducing and vertical structure



Rule Name	Value	Description
NIMP.E.4	0.18um	Minimum N+ to gate (endcap) enclosure

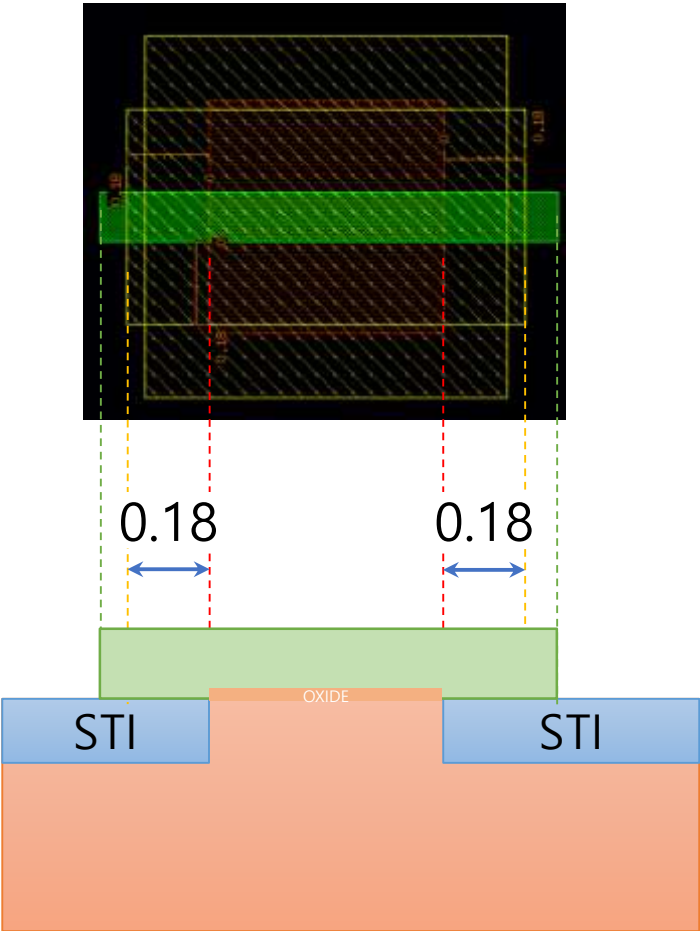
✓ Schematic in Design Rule Manual

Gate Endcap에서 안정적인 Source/Drain 형성을 위한
최소 Enclosure



↕ 0.18

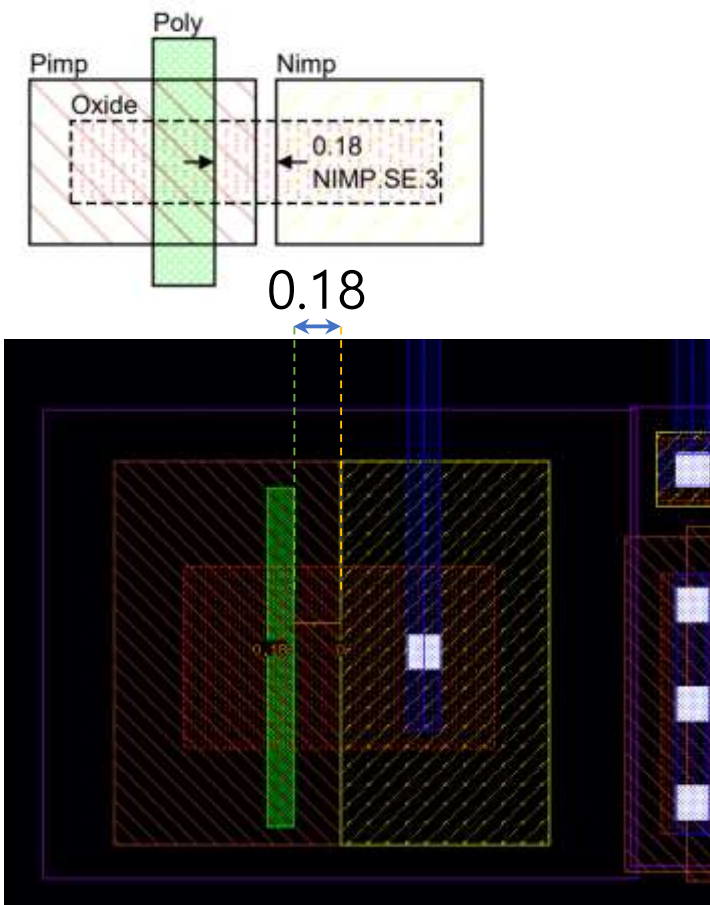
✓ Cadence Reproducing and vertical structure



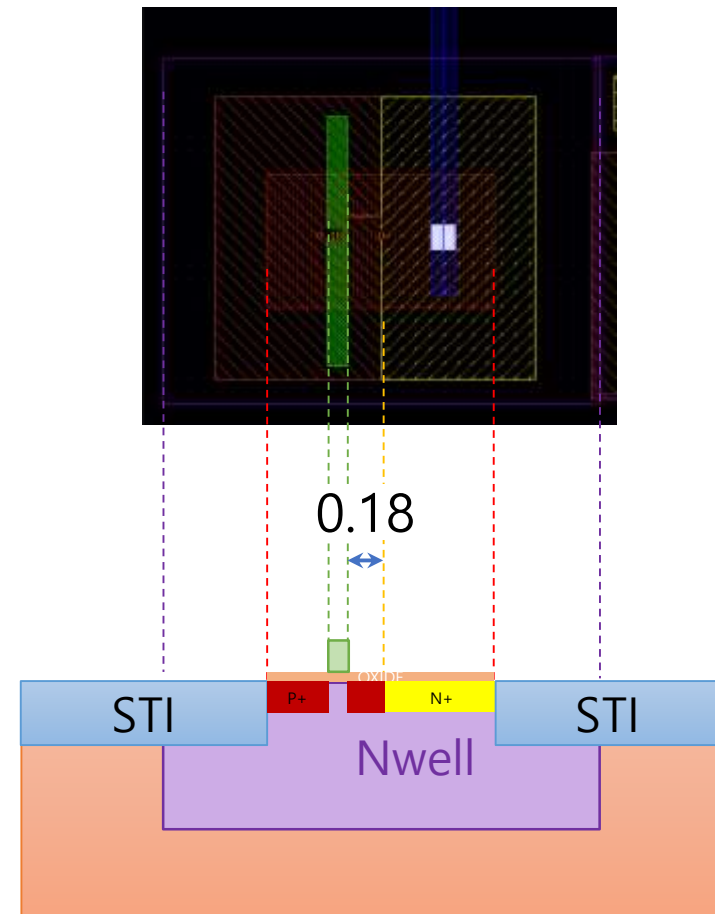
Rule Name	Value	Description
NIMP.SE.3	0.18um	Minimum N+ Implant to P+ gate side (butted Implant) spacing

✓ Schematic in Design Rule Manual

N+/P+ 도핑 간 간섭 방지를 위한 최소 space



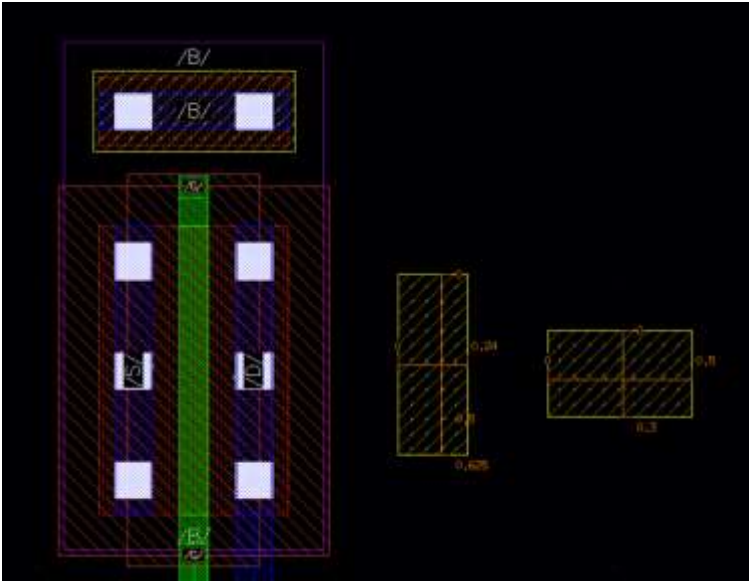
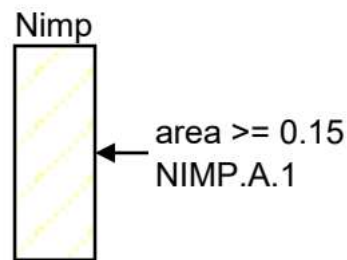
✓ Cadence Reproducing and vertical structure



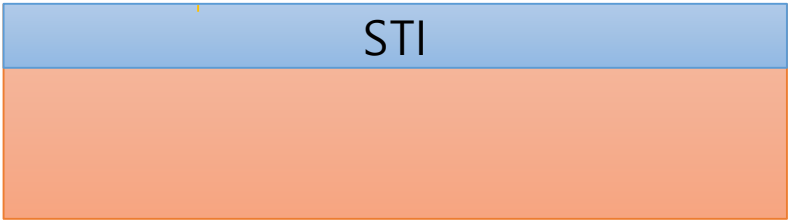
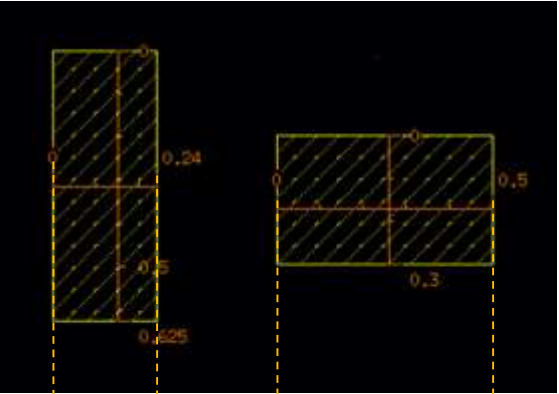
Rule Name	Value	Description
NIMP.A.1	0.15um	Minimum area for N+ Implant

✓ Schematic in Design Rule Manual

공정 재현성 확보를 위한 최소 Implant area



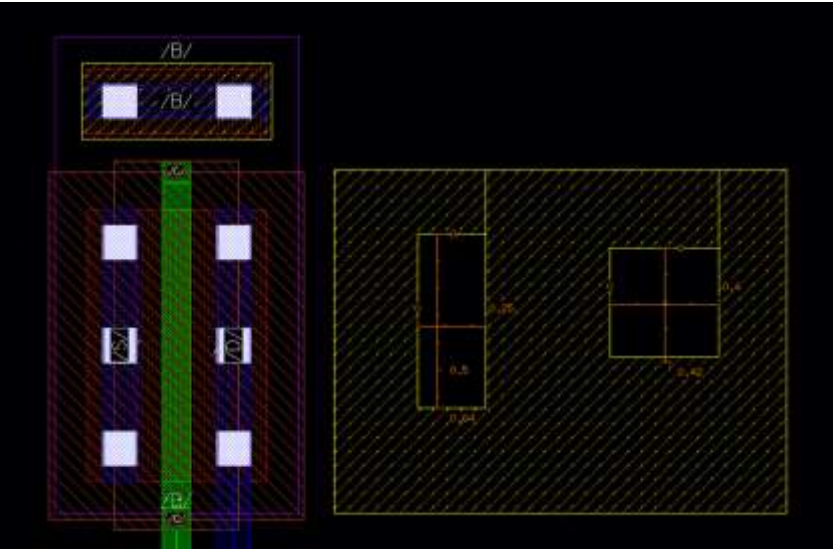
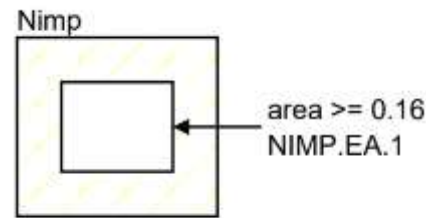
✓ Cadence Reproducing and vertical structure



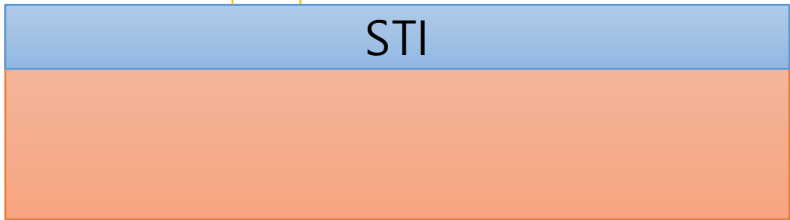
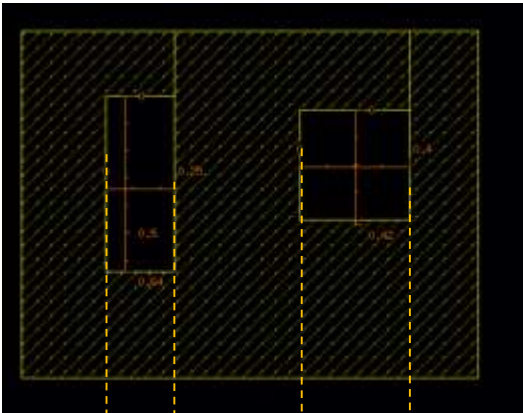
Rule Name	Value	Description
NIMP.EA.1	0.16um	Minimum N+ Implant ring enclosed area ("donut" hole surrounded by N+ Implant)

✓ Schematic in Design Rule Manual

Donut Implant의 제조 가능성을 위한
최소 Enclosed Area



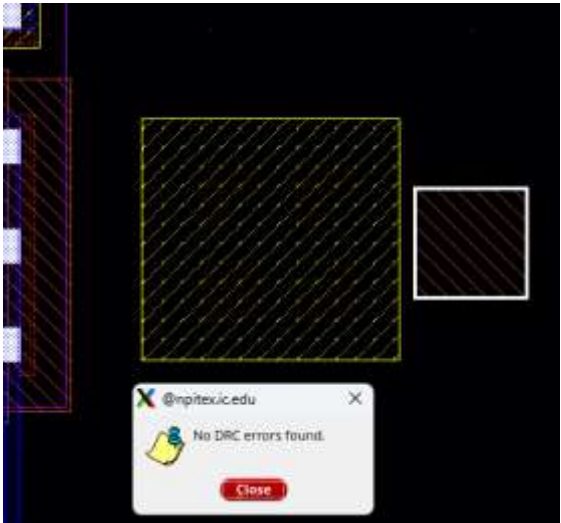
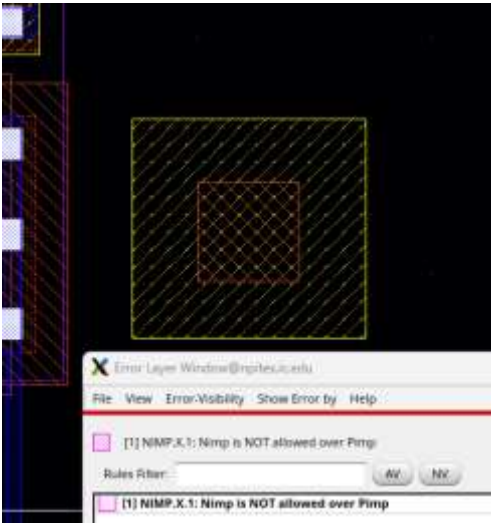
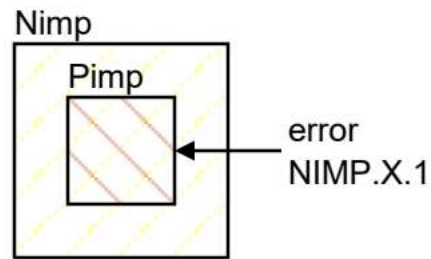
✓ Cadence Reproducing and vertical structure



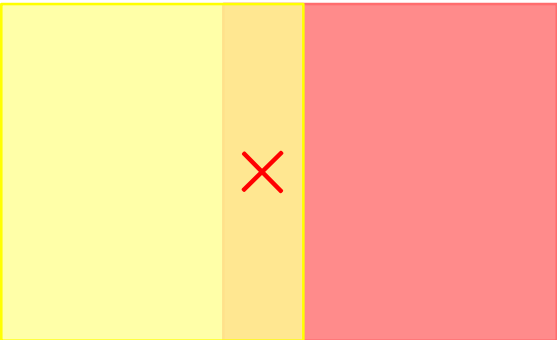
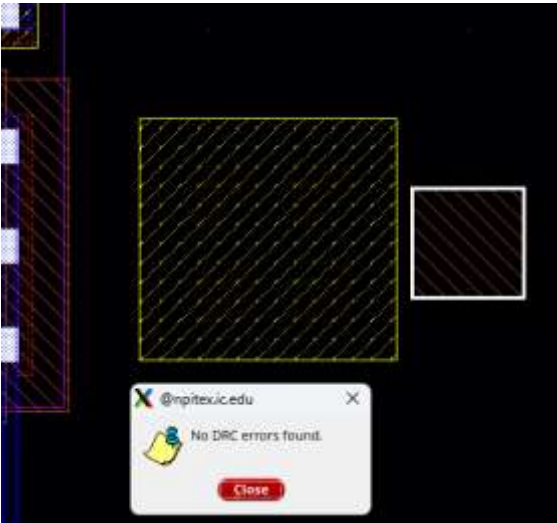
Rule Name	Value	Description
NIMP.X.1	---	N+ Implant is NOT allowed over P+ Implant

✓ Schematic in Design Rule Manual

정상적인 PN Junction 형성을 위한
N+/P+ Implant 중첩 금지



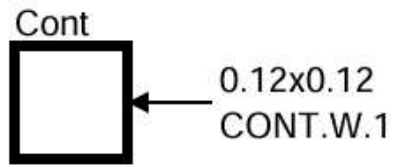
✓ Cadence Reproducing and vertical structure



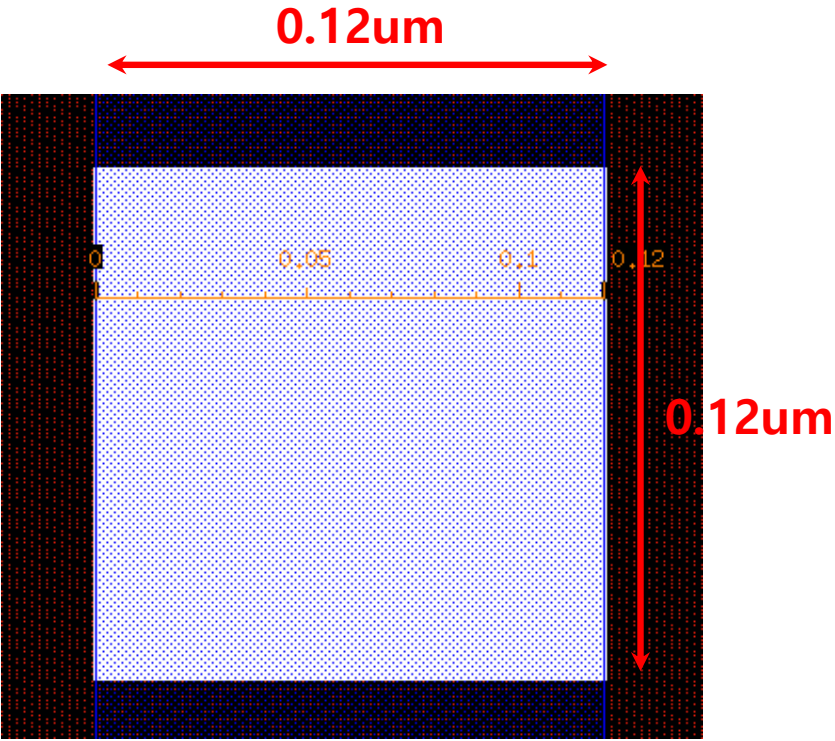
Contact

Rule Name	Value	Description
CONT.W.1	0.12um	Minimun Contact to Contact width/length

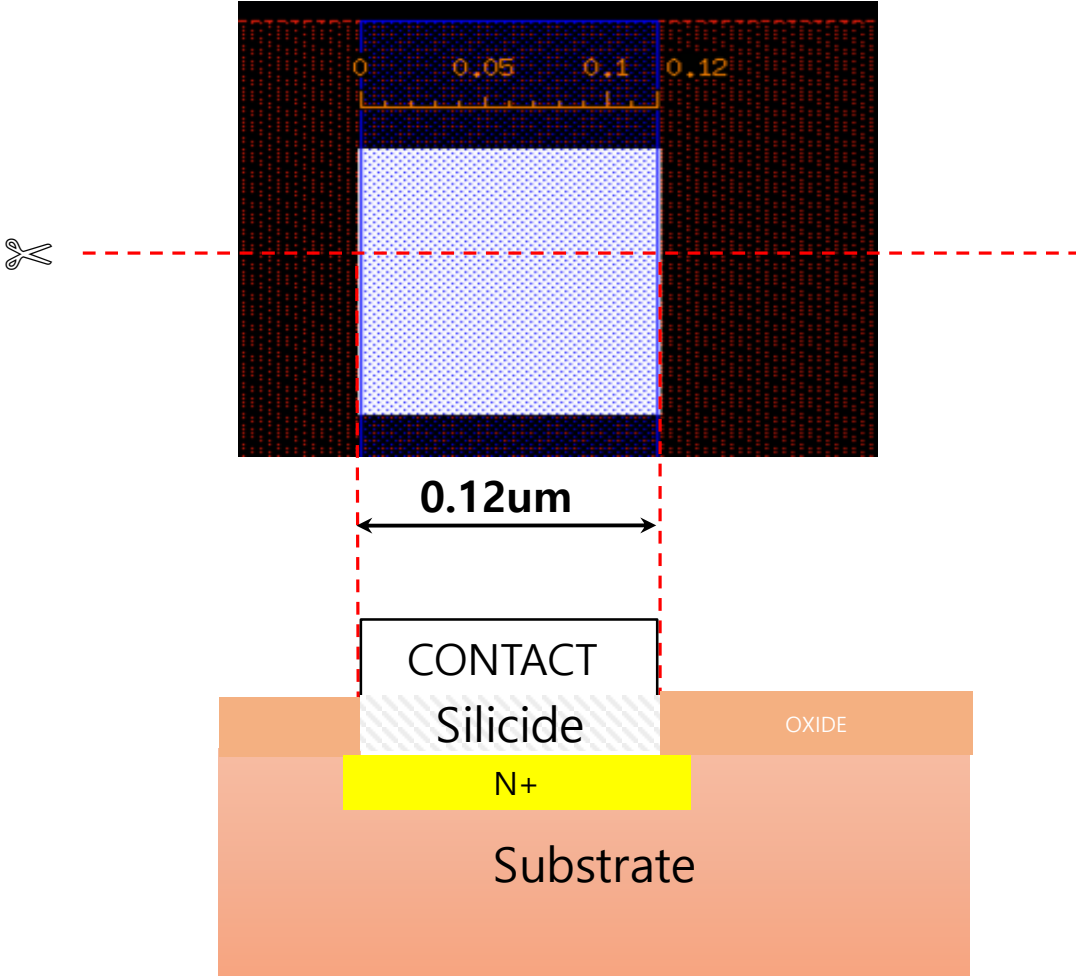
✓ Schematic in Design Rule Manual



공정 안정성 및 Contact 저항 감소를 위한
최소 Contact 크기

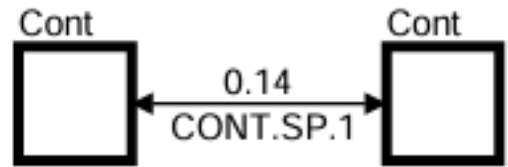


✓ Cadence Reproducing and vertical structure

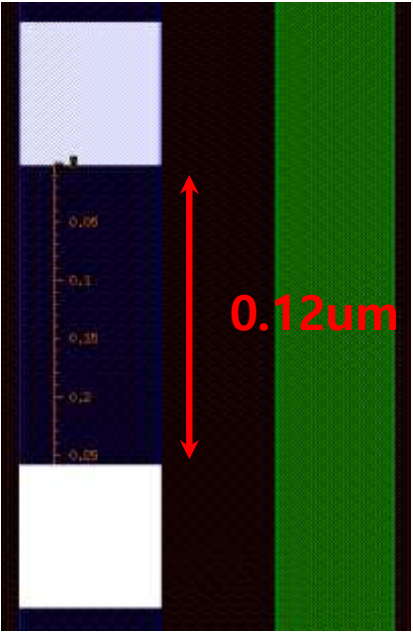


Rule Name	Value	Description
CONT.SP.1	0.14um	Minimun Contact to Contact spacing

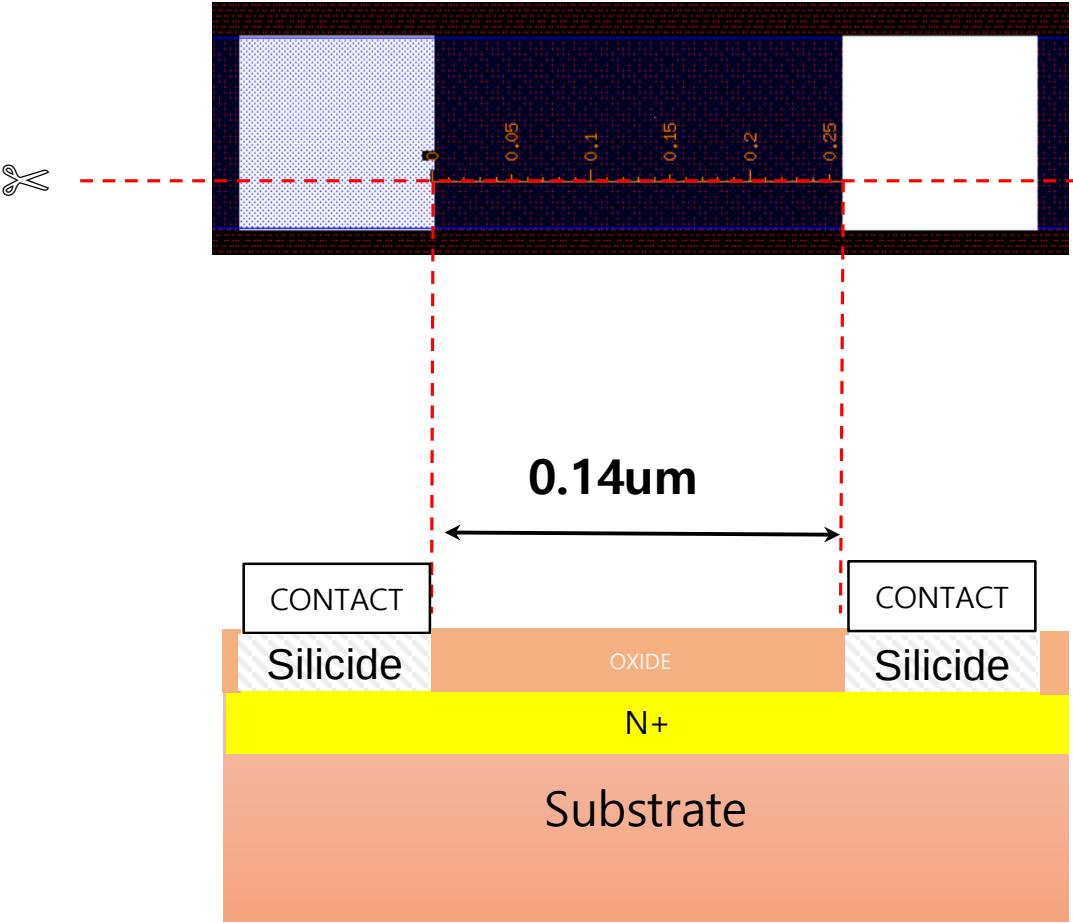
✓ Schematic in Design Rule Manual



공정 안정성 및 Short 방지를 위한
최소 Contact 간격

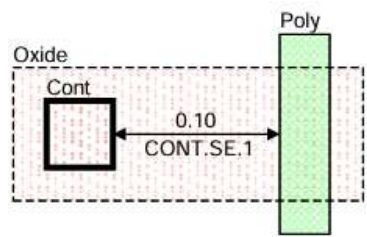


✓ Cadence Reproducing and vertical structure

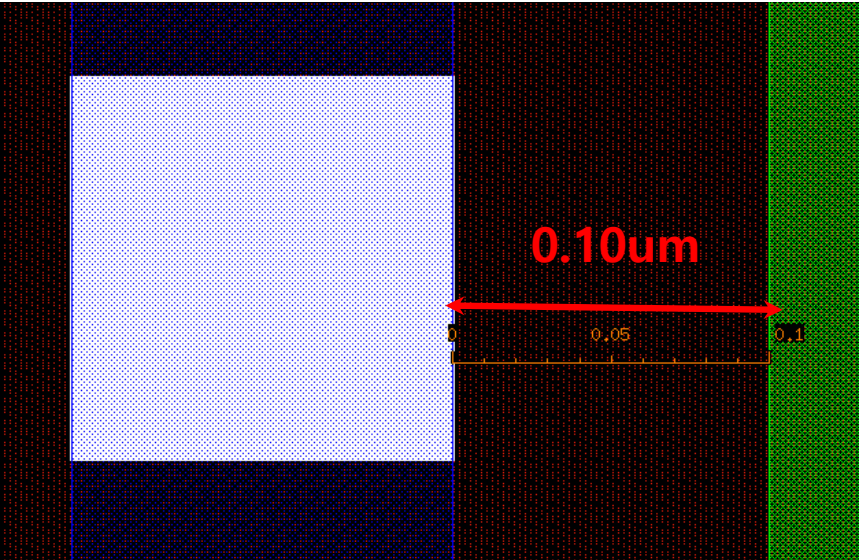


Rule Name	Value	Description
CONT.SE.1	0.10um	Minimum Contact on Active Area to gate spacing

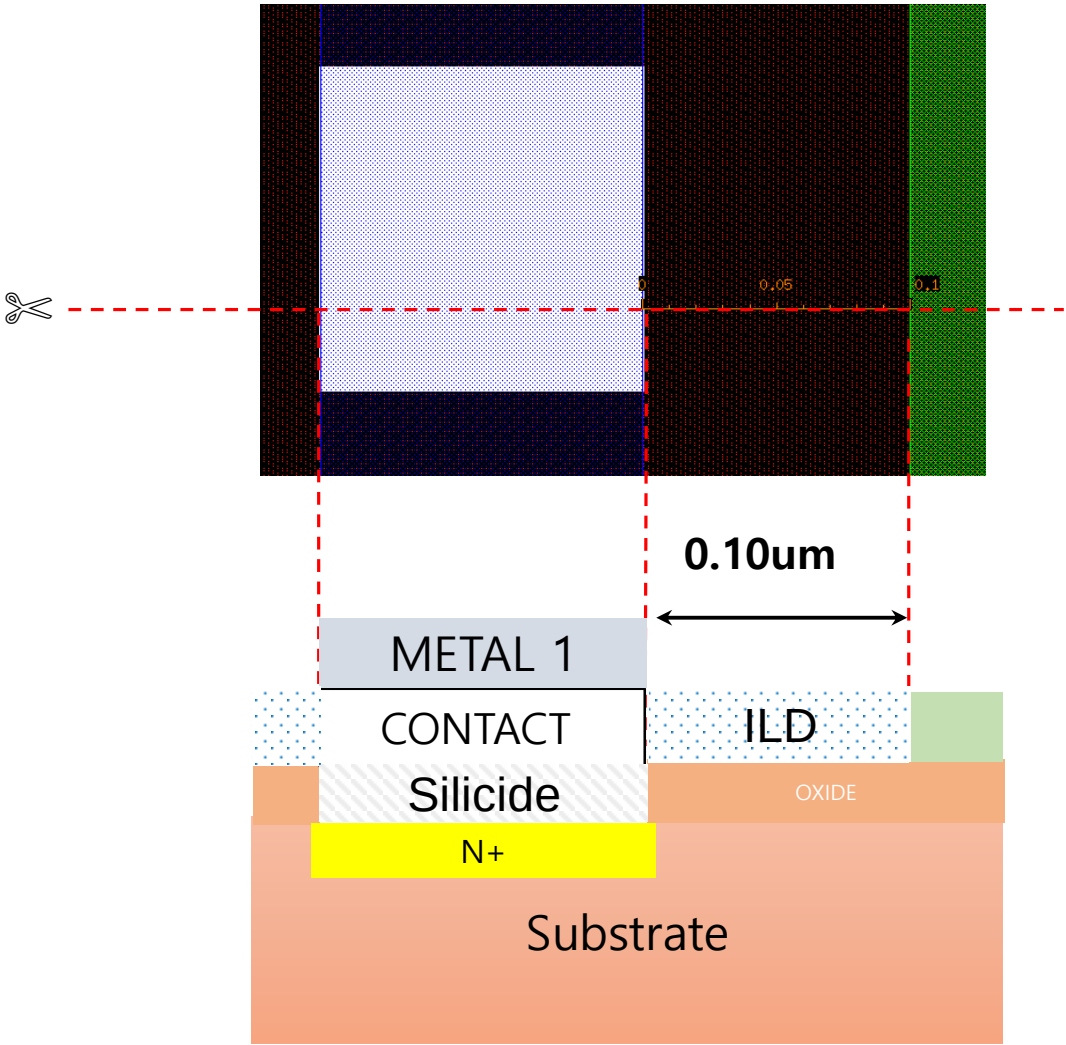
✓ Schematic in Design Rule Manual



Gate 손상·Short 방지를 위한
최소 Contact-Gate 간격

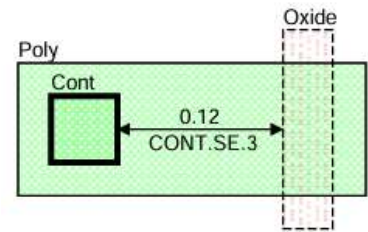


✓ Cadence Reproducing and vertical structure

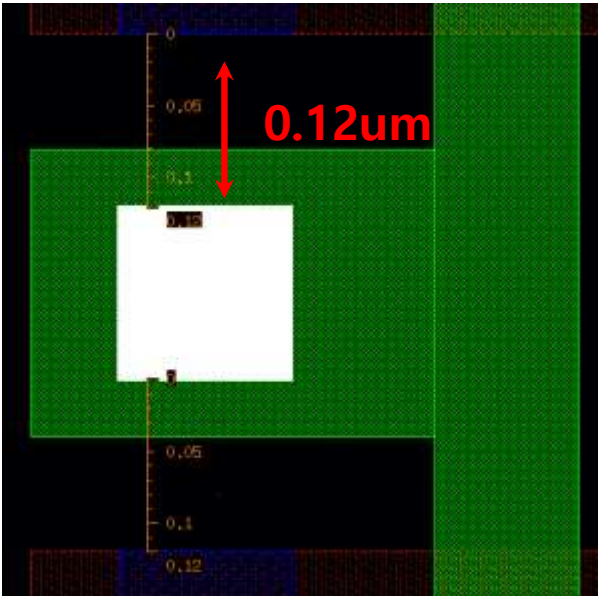


Rule Name	Value	Description
CONT.SE.3	0.12um	Minimum gate Contact to Active Area spacing

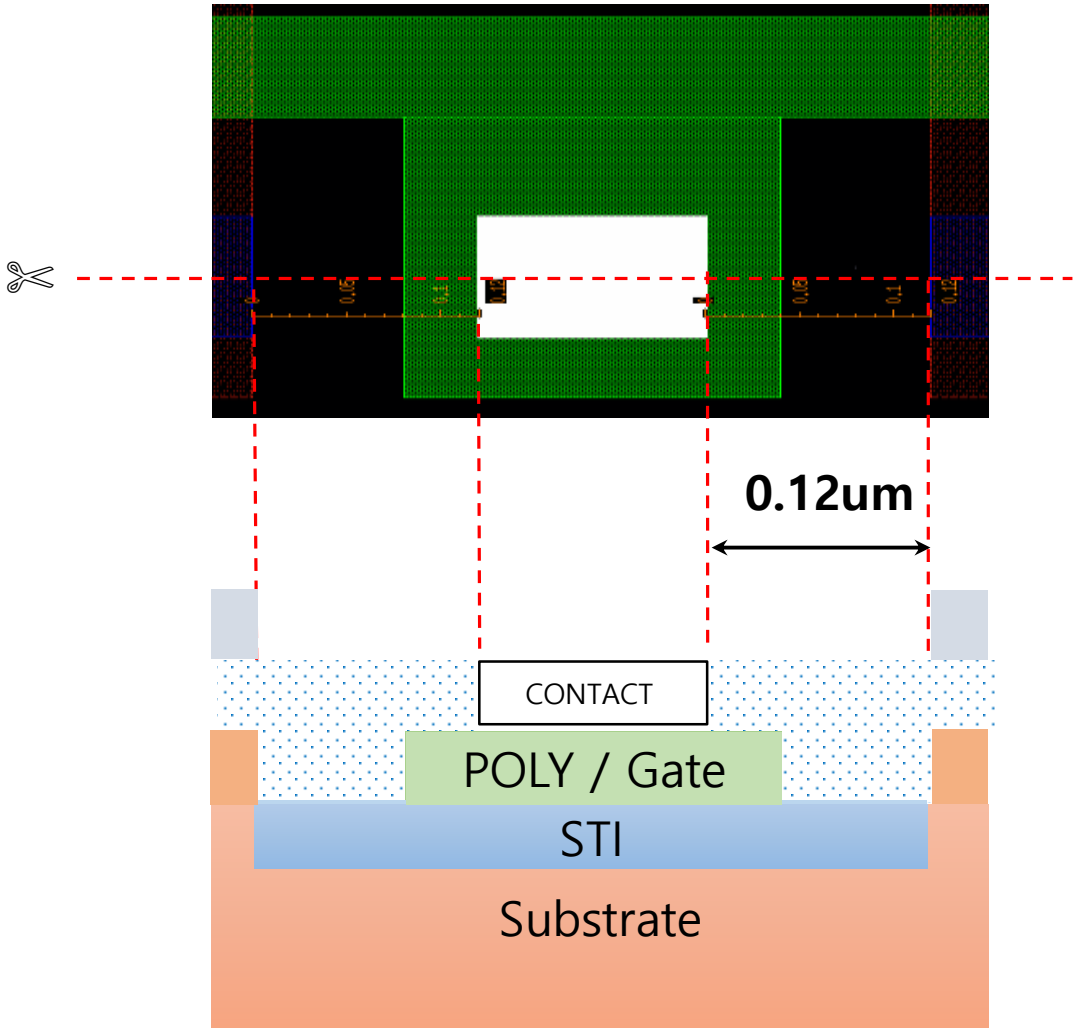
✓ Schematic in Design Rule Manual



Gate-Active Short 방지를 위한 최소
Gate Contact 간격

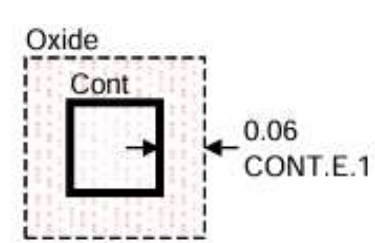


✓ Cadence Reproducing and vertical structure

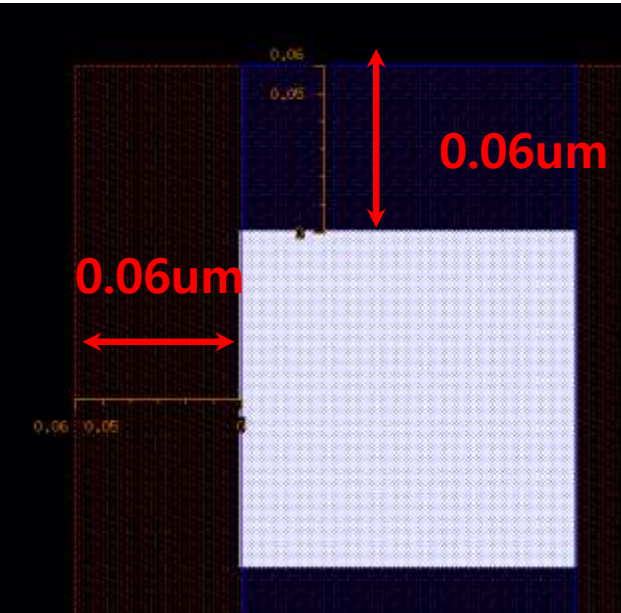


Rule Name	Value	Description
CONT.E.1	0.06um	Minimum Active Area to Contact enclosure

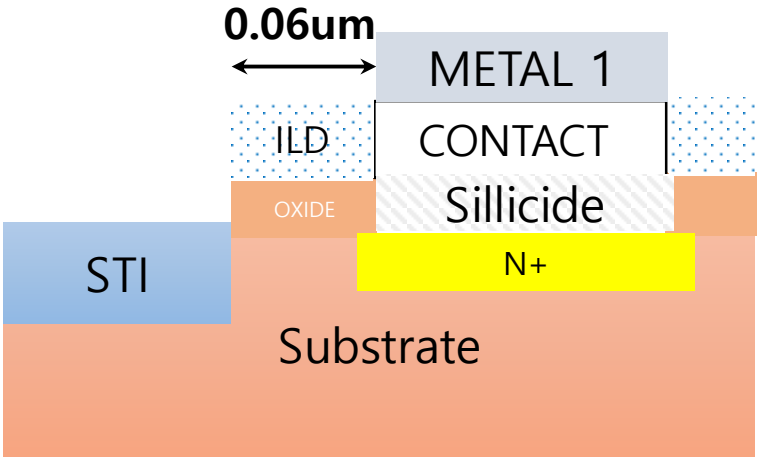
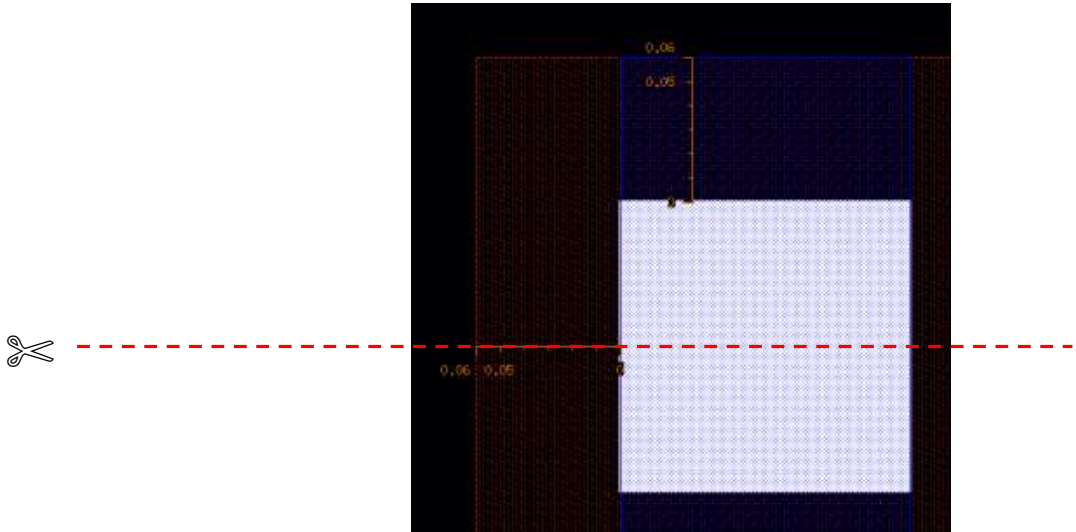
✓ Schematic in Design Rule Manual



안정적인 Contact 형성 및 Contact 저항 감소를 위한
Active Area Enclosure 규정

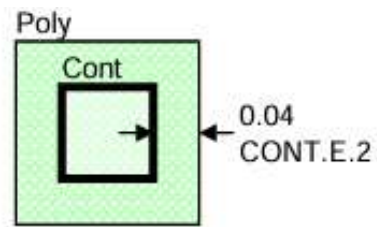


✓ Cadence Reproducing and vertical structure

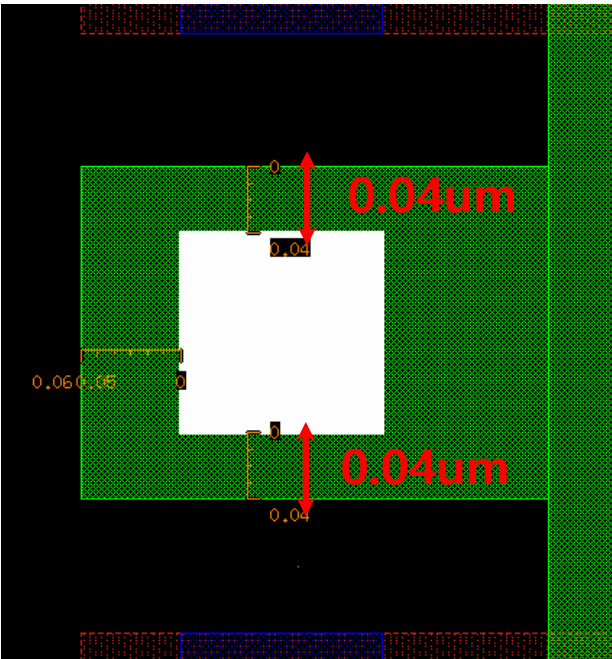


Rule Name	Value	Description
CONT.E.2	0.04um	Minimum Poly to Contact enclosure

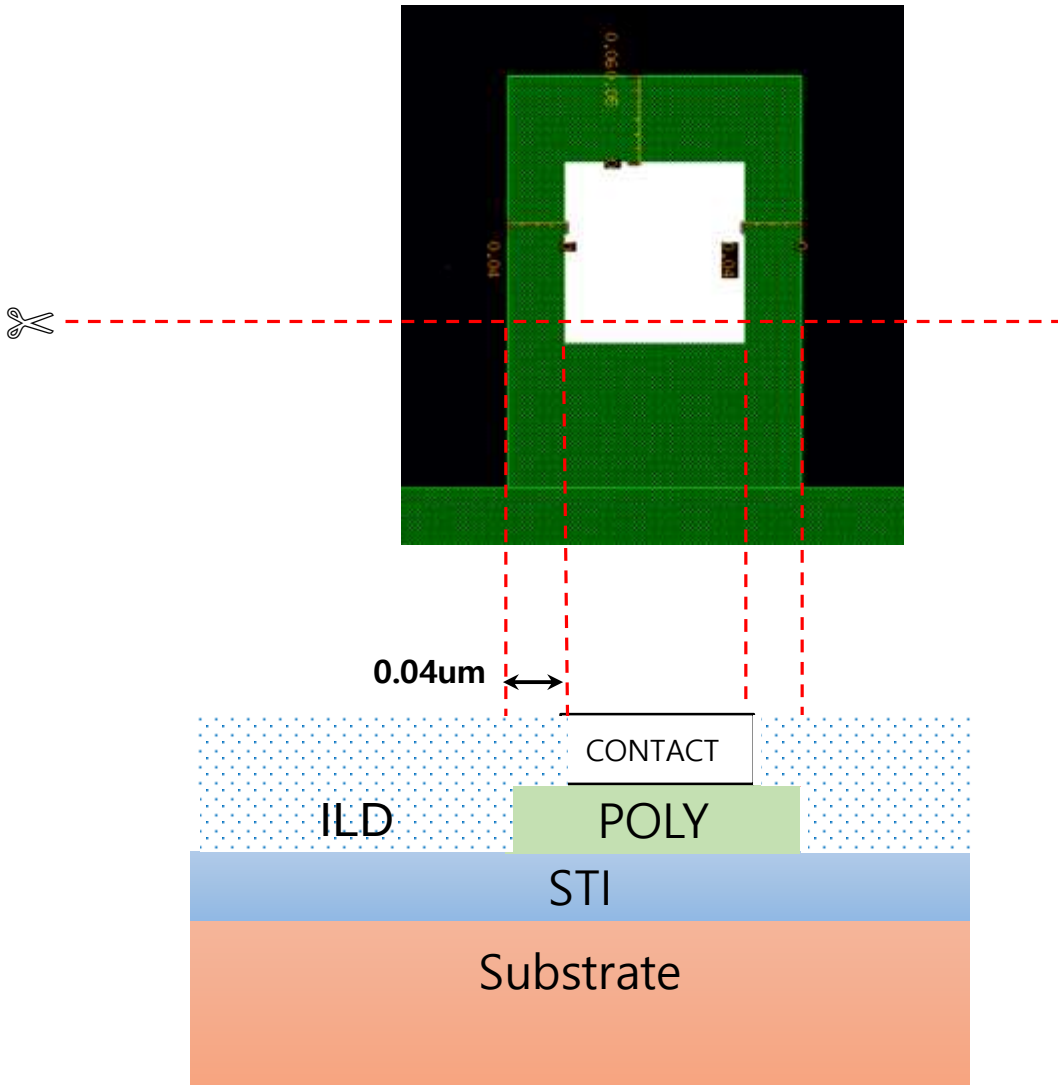
✓ Schematic in Design Rule Manual



안정적인 Gate Contact 형성 및 Contact 저항 감소를 위한 Poly Enclosure 규정

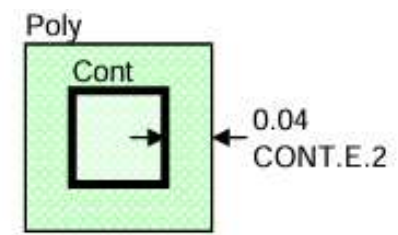


✓ Cadence Reproducing and vertical structure

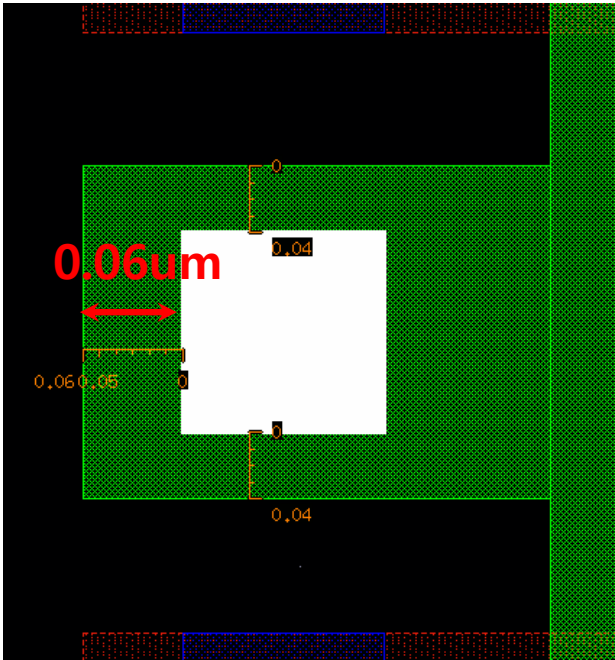


Rule Name	Value	Description
CONT.E.3	0.06um	Minimum Poly to Contact enclosure on at least two opposite sides (end of line)

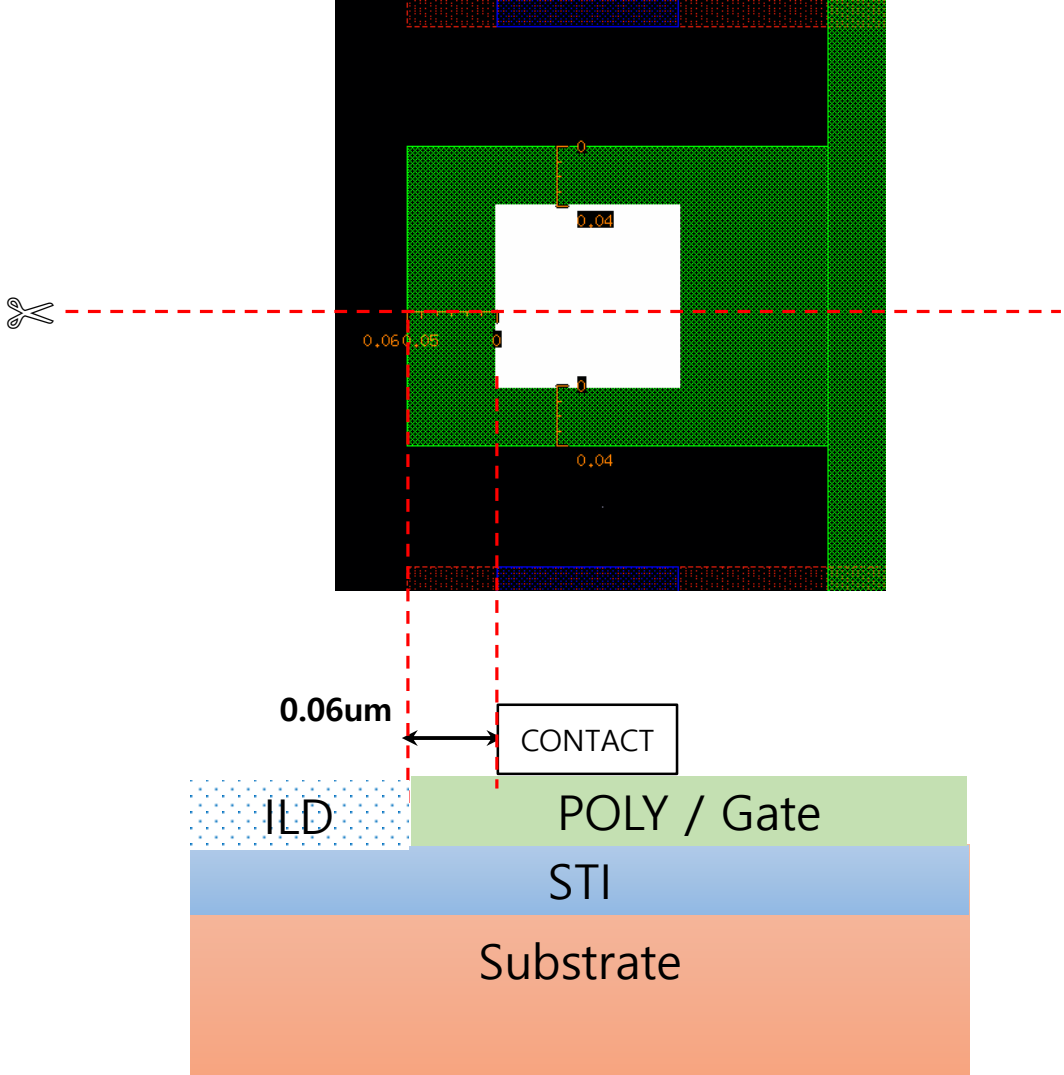
✓ Schematic in Design Rule Manual



안정적인 Contact 형성 및 Open 방지를 위한
Poly Enclosure 규정

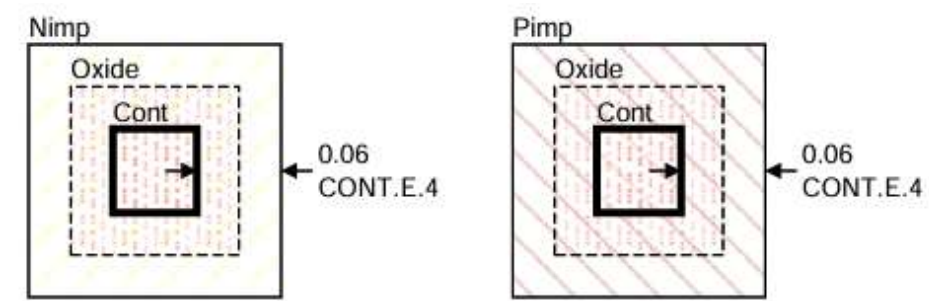


✓ Cadence Reproducing and vertical structure

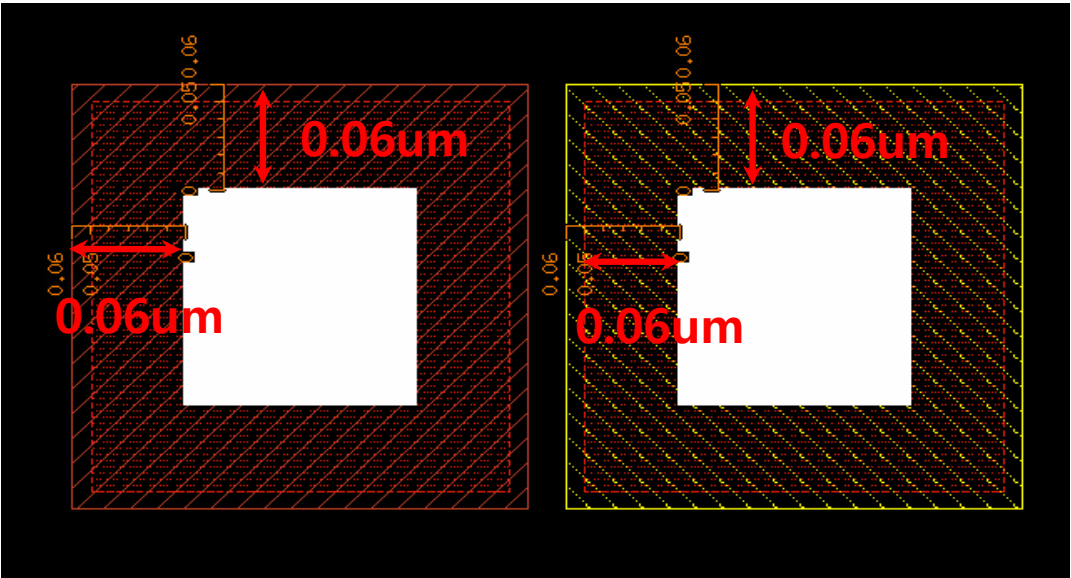


Rule Name	Value	Description
CONT.E.4	0.06um	Minimum N+/P+ Implant on Active Area to Contact enclosure

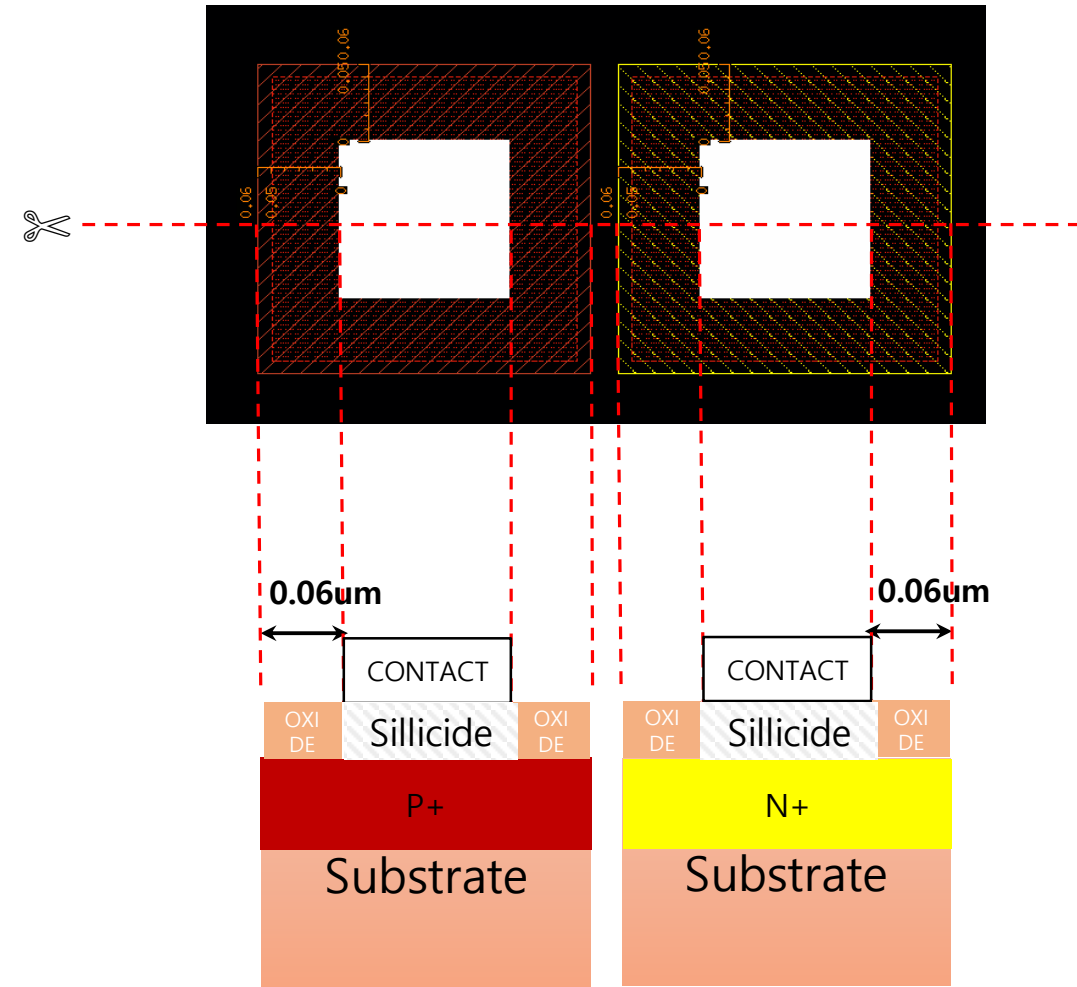
✓ Schematic in Design Rule Manual



안정적인 Ohmic Contact 형성 및 Contact 저항 감소를 위한 Implant Enclosure 규정



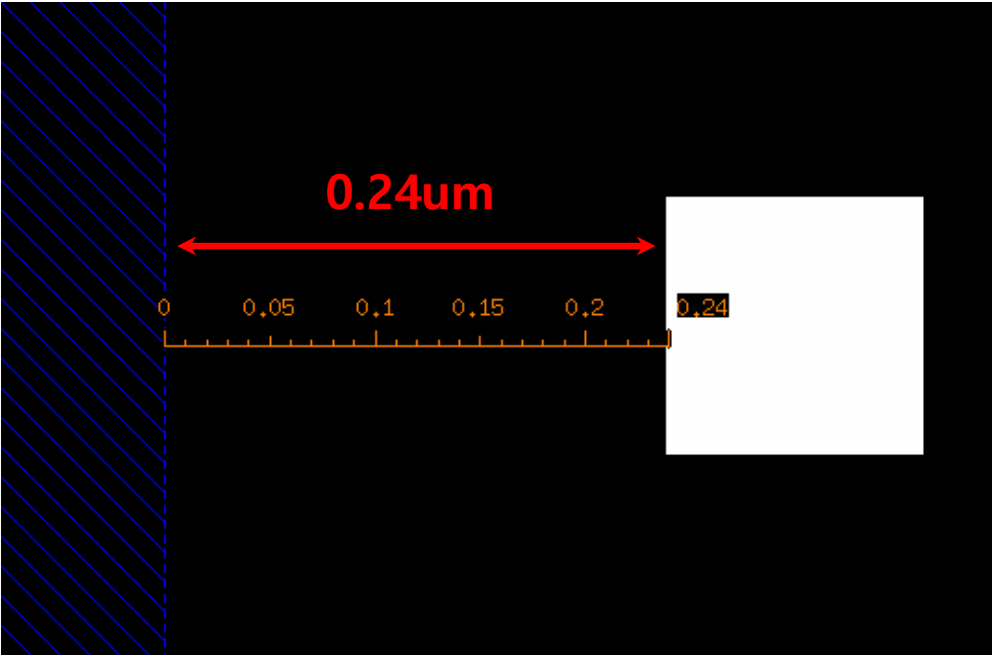
✓ Cadence Reproducing and vertical structure



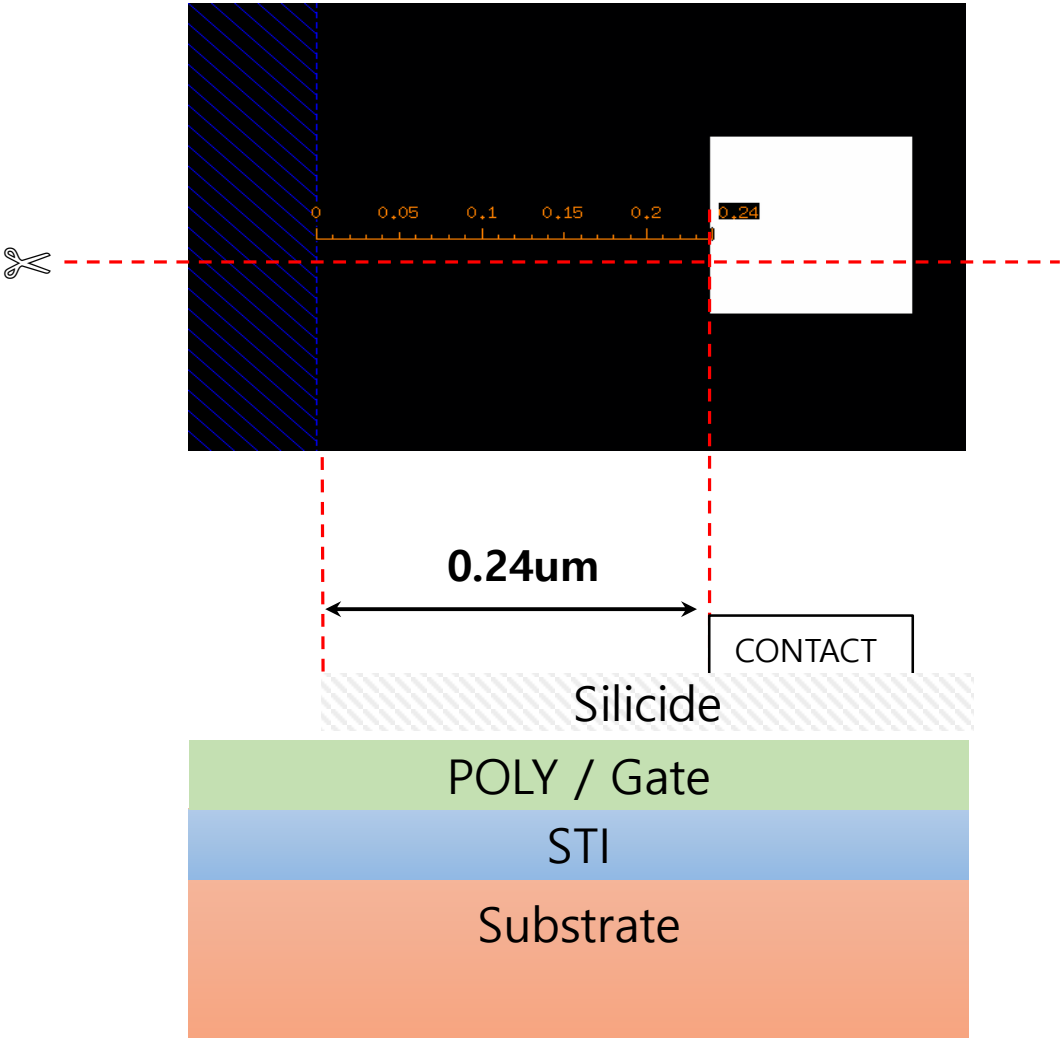
Rule Name	Value	Description
CONT.SE.5	0.24um	Minimum Poly Contact to non-salacided Poly resistor or Active Contact to non-salacided Active resistor spacing

✓ Schematic in Design Rule Manual

저항 특성 유지 및 Salicide 침범 방지를 위한 최소 Contact-Resistor 간격 규정



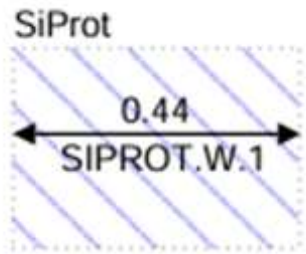
✓ Cadence Reproducing and vertical structure



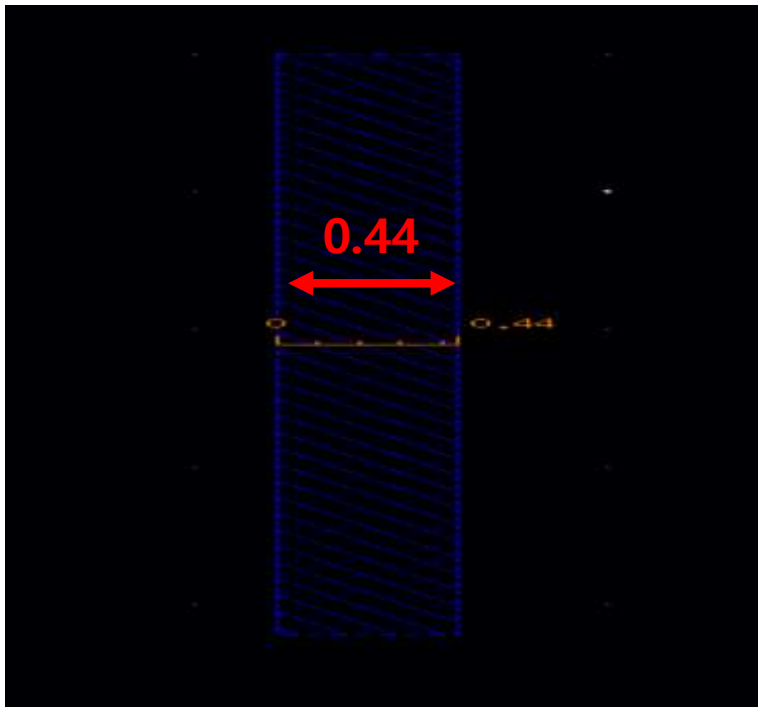
Salicide

Rule Name	Value	Description
SIPROT.W.1	0.44um	Minimum Salicide Block width

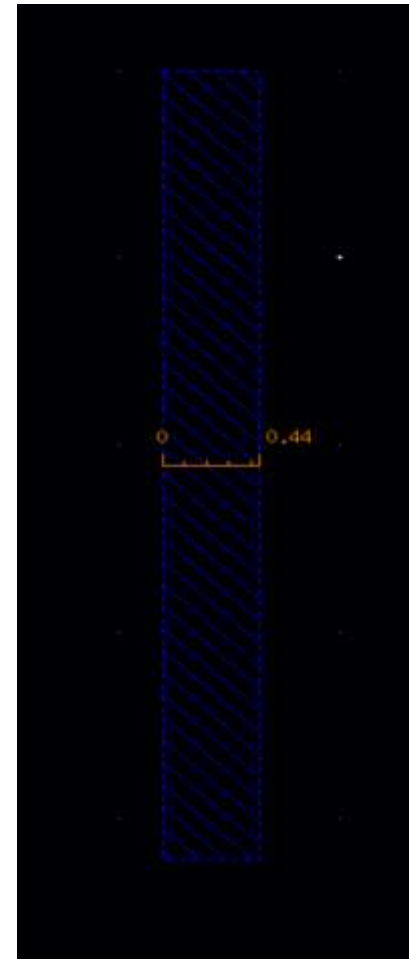
✓ Schematic in Design Rule Manual



얇으면 공정 오차 ↑

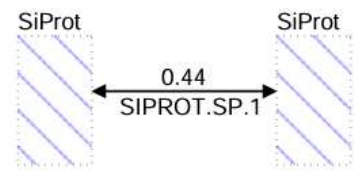


✓ Cadence Reproducing and vertical structure

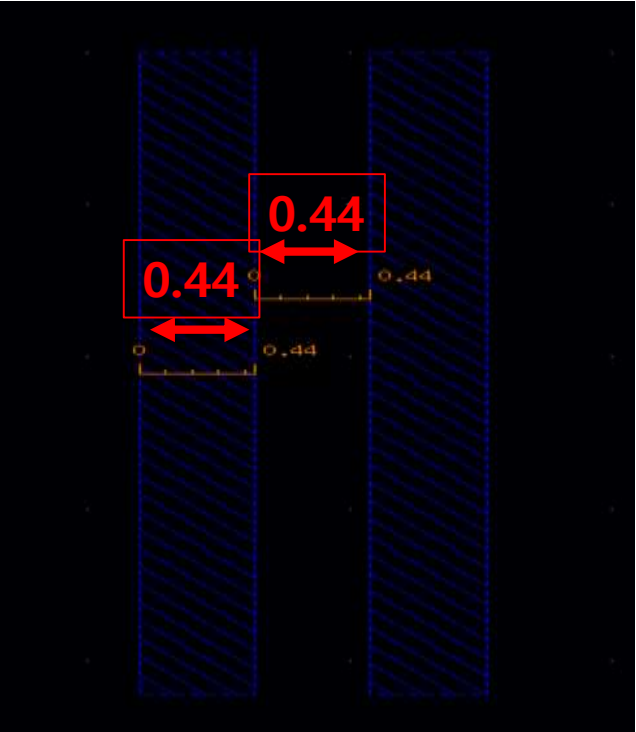


Rule Name	Value	Description
SIPROT.SP.1	0.44um	Minimum Salicide Block space

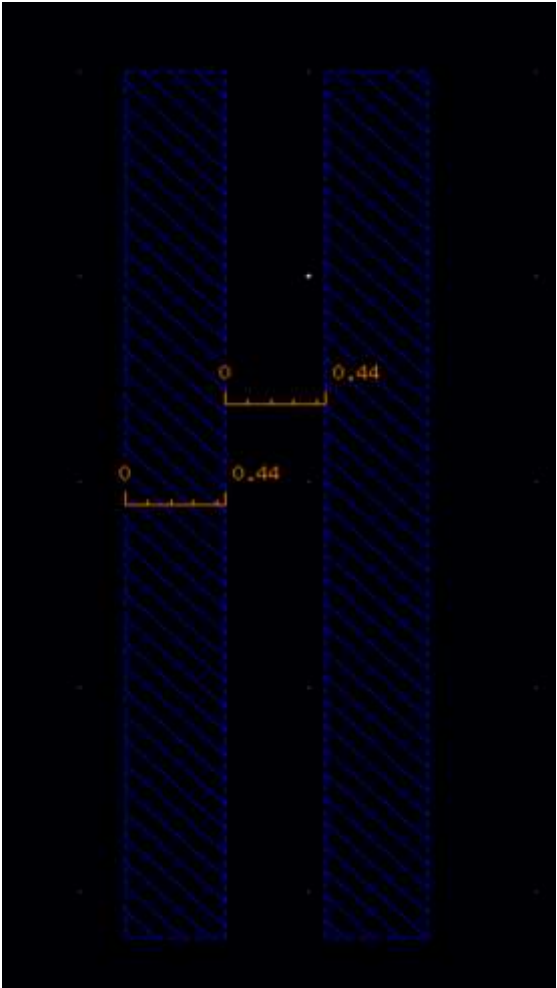
✓ Schematic in Design Rule Manual



가까우면 공정 오차 ↑

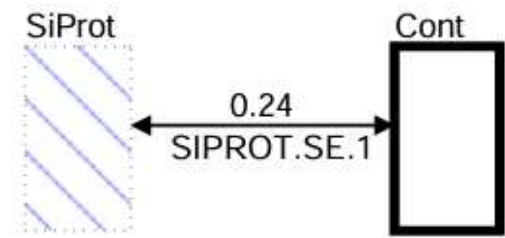


✓ Cadence Reproducing and vertical structure

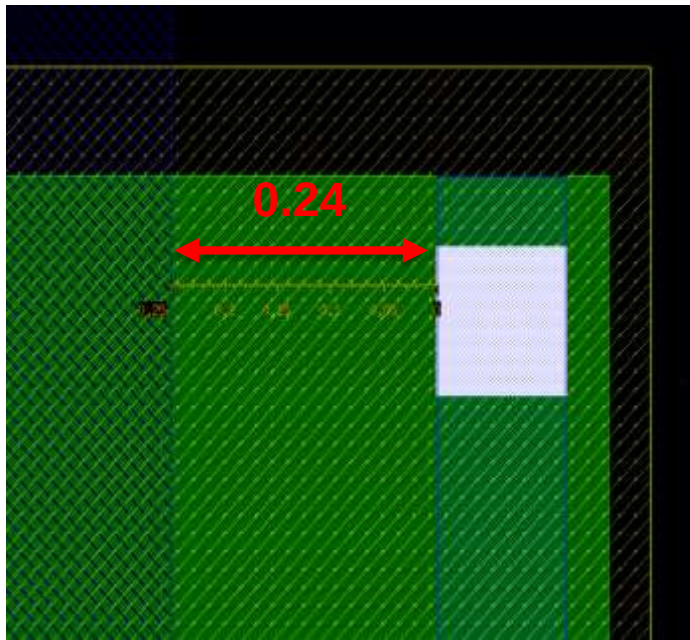


Rule Name	Value	Description
SIPROT.SE.1	0.24um	Minimum Salicide Block to Contact spacing

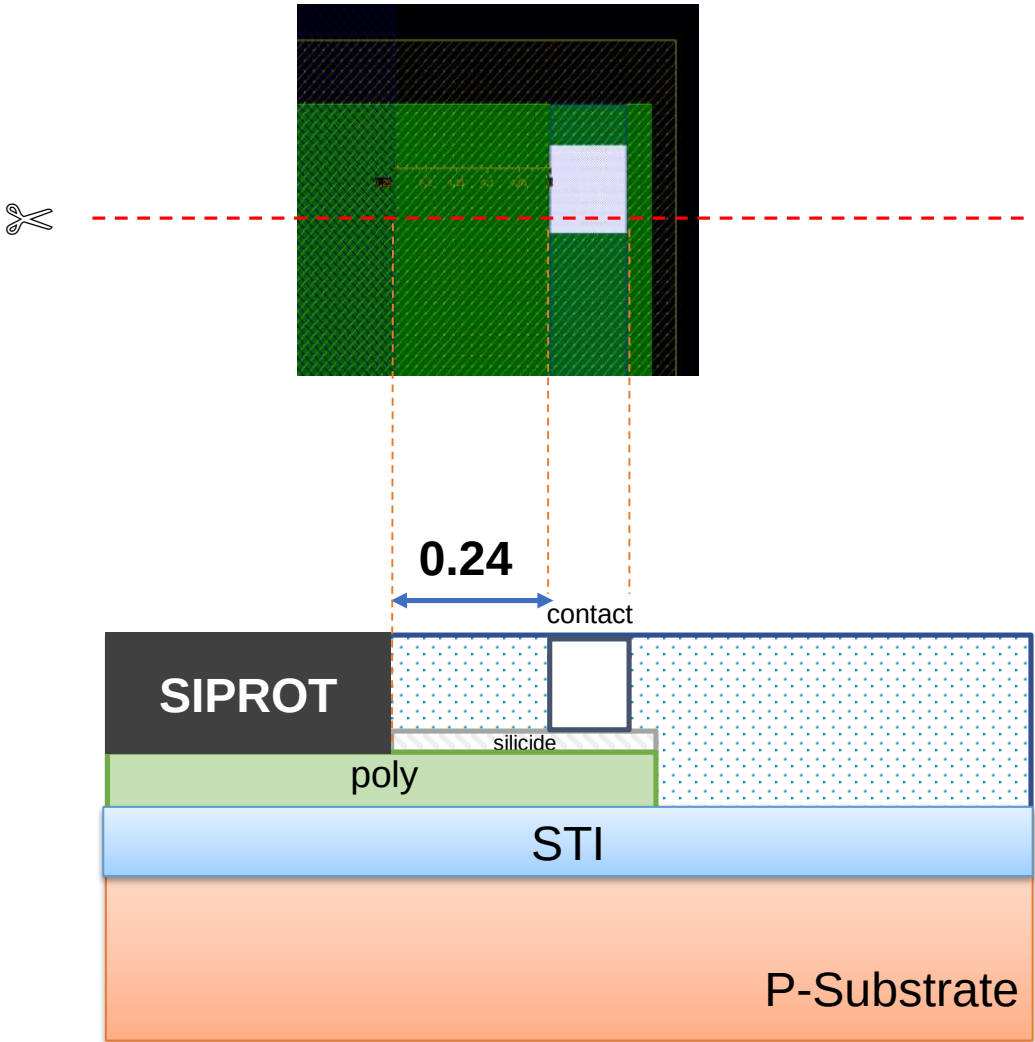
✓ Schematic in Design Rule Manual



• Contact 근처에는 저항,신뢰성 문제가 큼

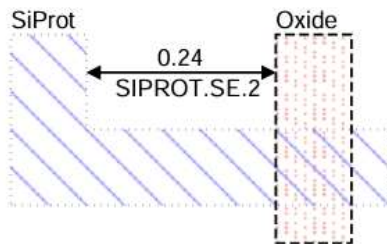


✓ Cadence Reproducing and vertical structure

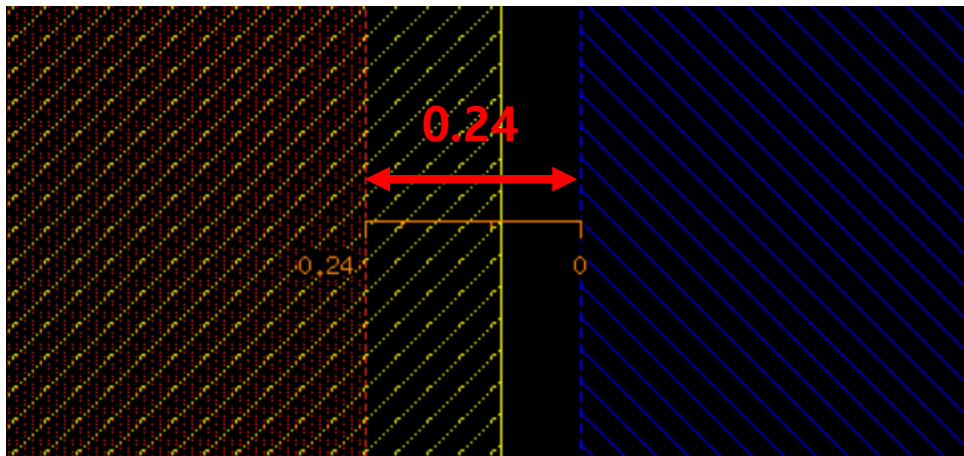


Rule Name	Value	Description
SIPROT.SE.2	0.24um	Minimum Salicide Block to Unrelated Active Area spacing

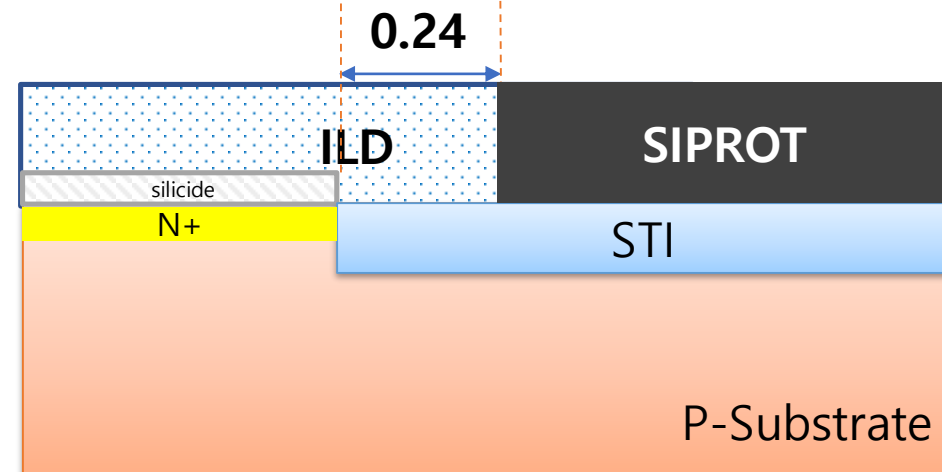
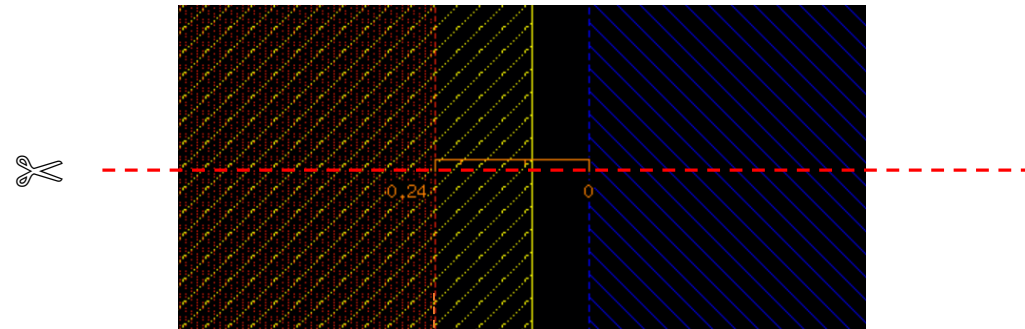
✓ Schematic in Design Rule Manual



Active 영역의 edge에서 도핑과 실리사이드 형성이 불안정해지는 것을 방지

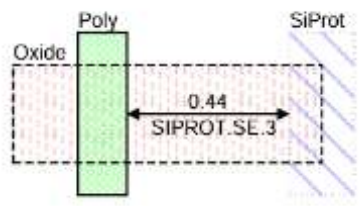


✓ Cadence Reproducing and vertical structure

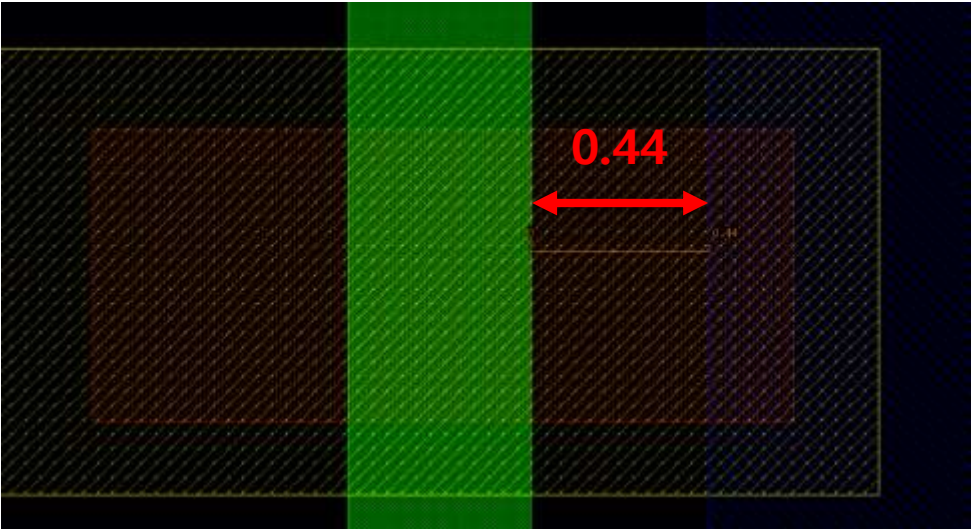


Rule Name	Value	Description
SIPROT.SE.3	0.44um	Minimum Salicide Block to gate spacing

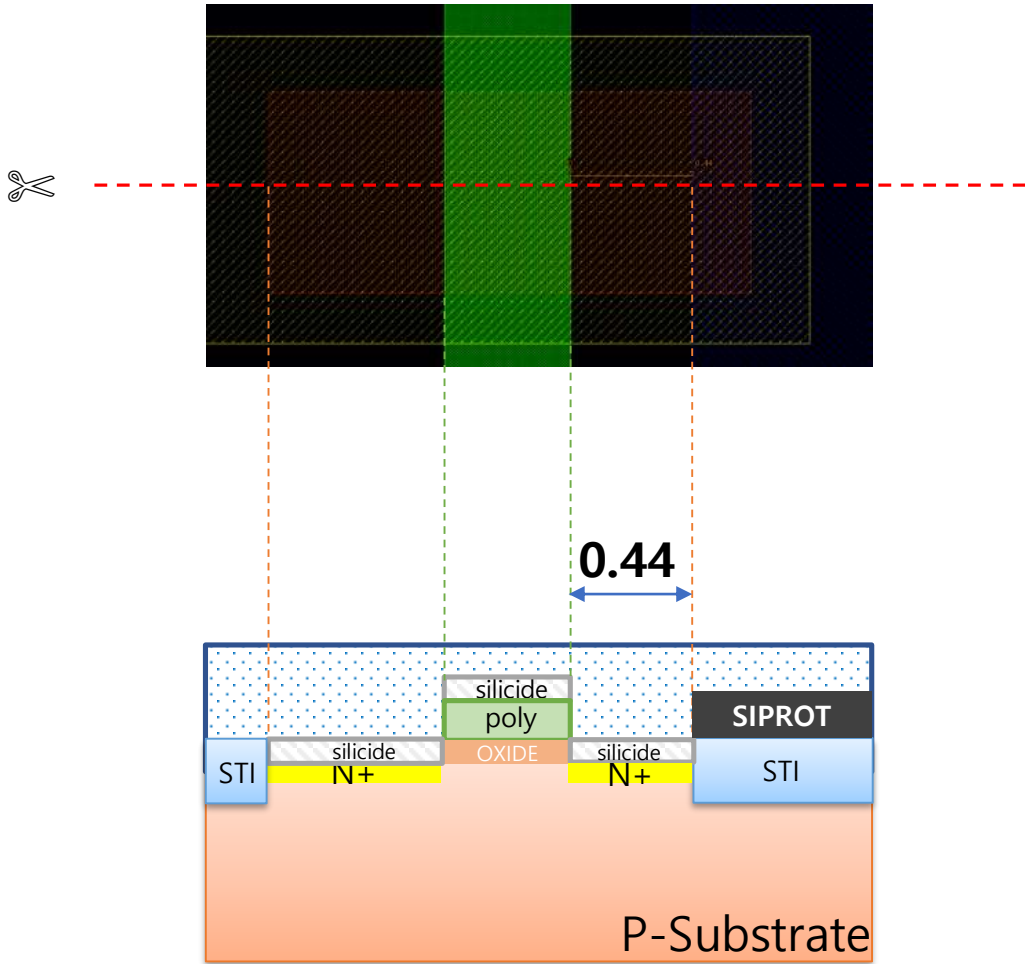
✓ Schematic in Design Rule Manual



Gate 위 silicide가 불균일해지는 것을 방지

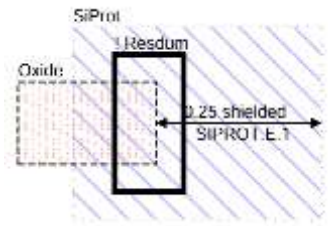


✓ Cadence Reproducing and vertical structure

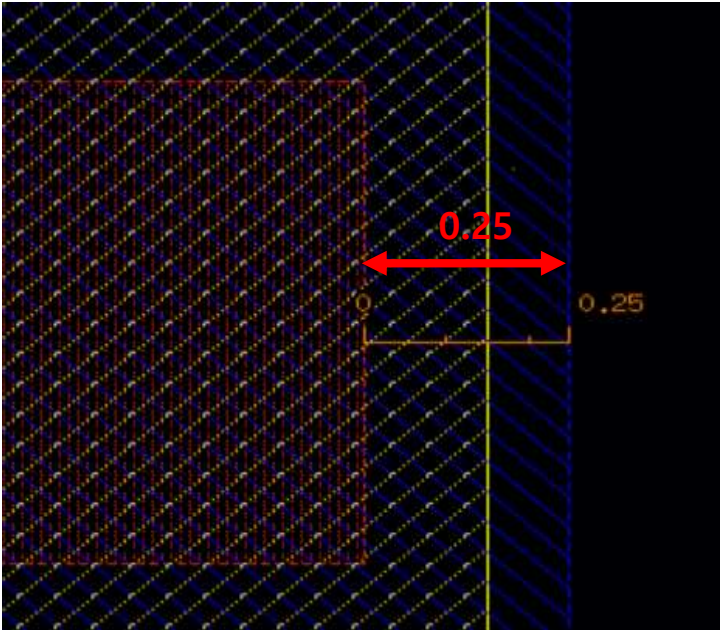


Rule Name	Value	Description
SIPROT.E.1	0.25um	Minimum Salicide Block to Active Area enclosure

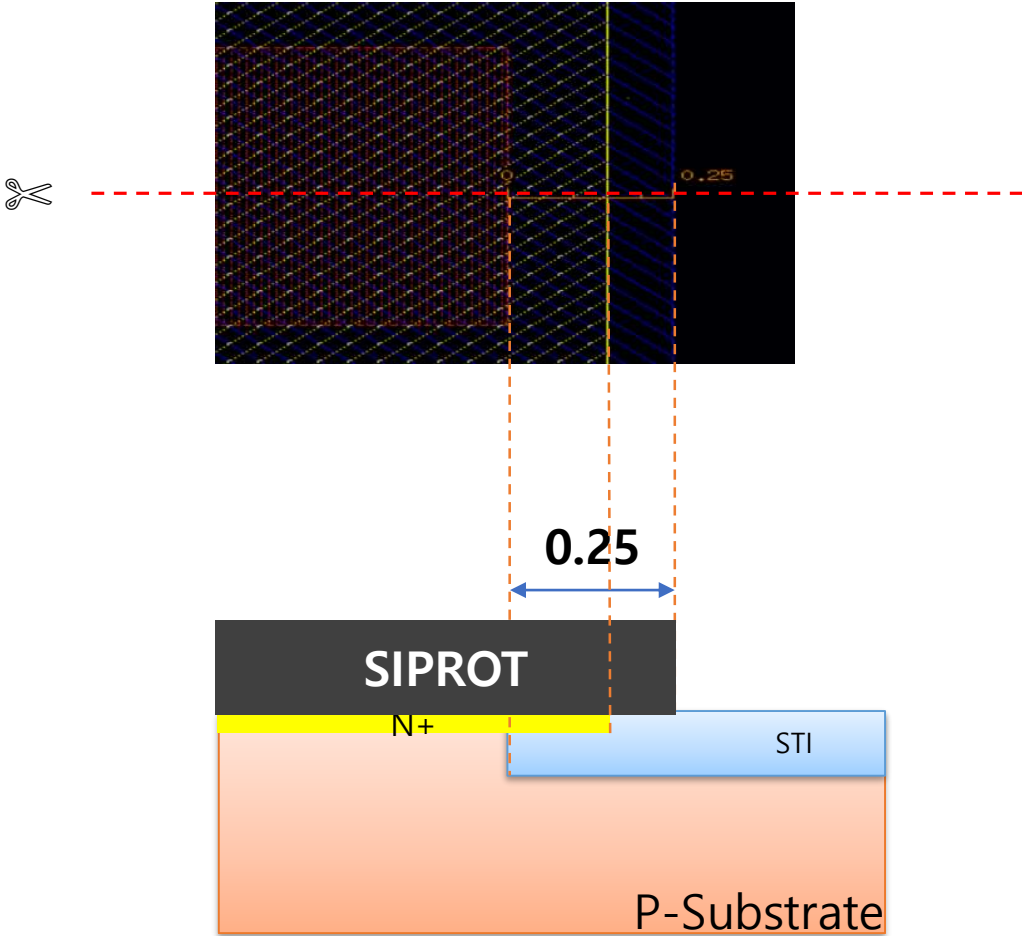
✓ Schematic in Design Rule Manual



Silicide를 확실하게 막기 위해서

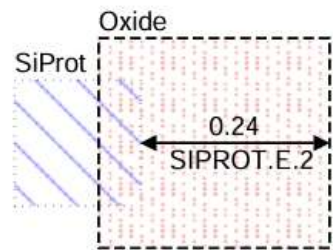


✓ Cadence Reproducing and vertical structure

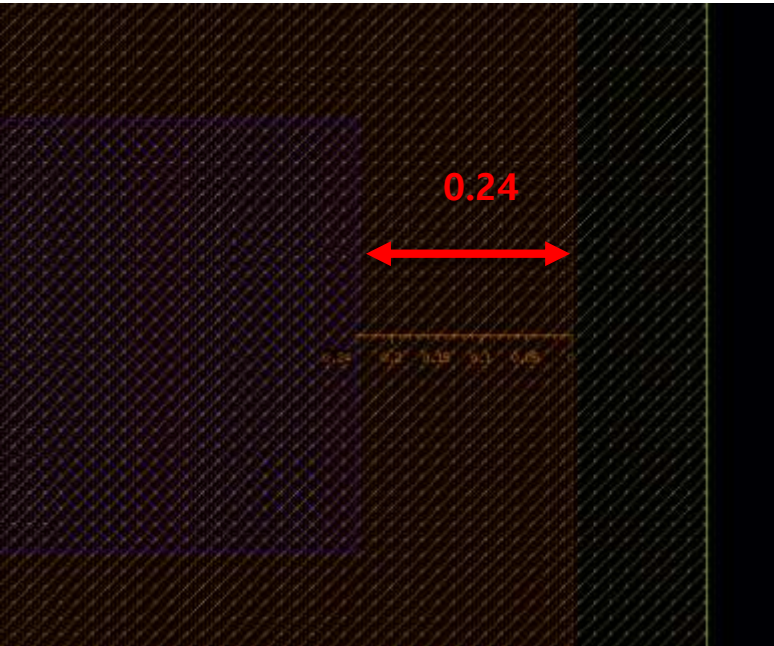


Rule Name	Value	Description
SIPROT.E.2	0.24um	Minimum Active Area to Salicide Block enclosure

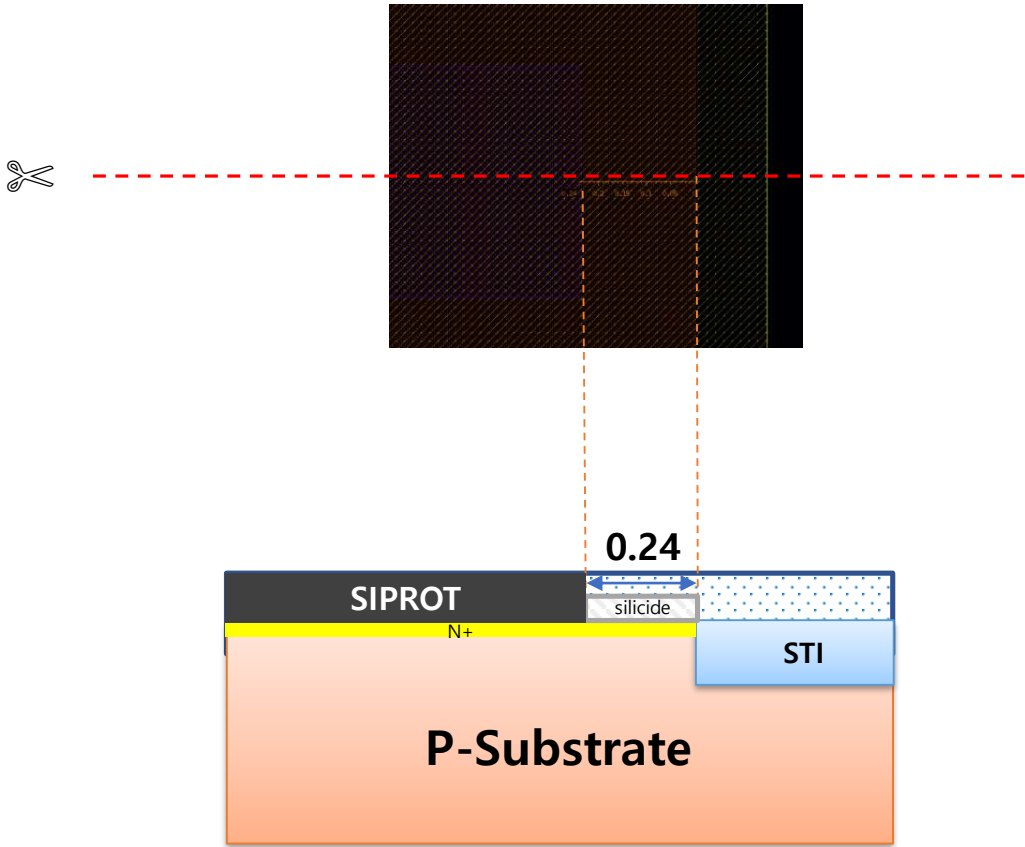
✓ Schematic in Design Rule Manual



Active edge를 확실히 덮기 위해서

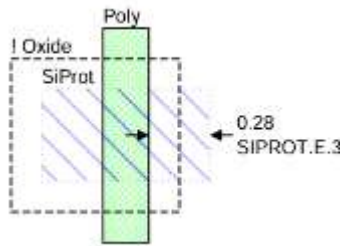


✓ Cadence Reproducing and vertical structure

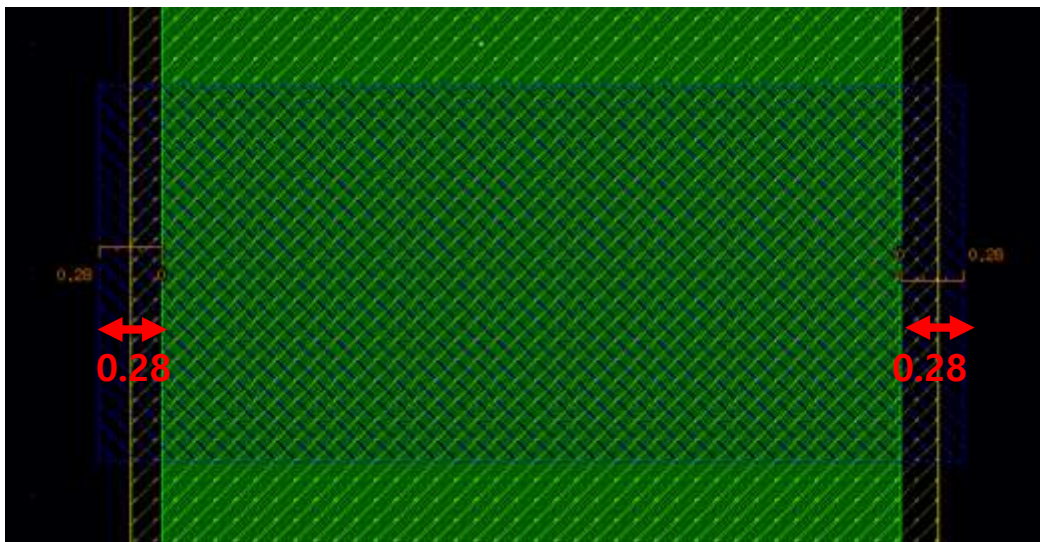


Rule Name	Value	Description
SIPROT.E.3	0.28um	Minimum Salicide Block to Poly (on field) enclosure

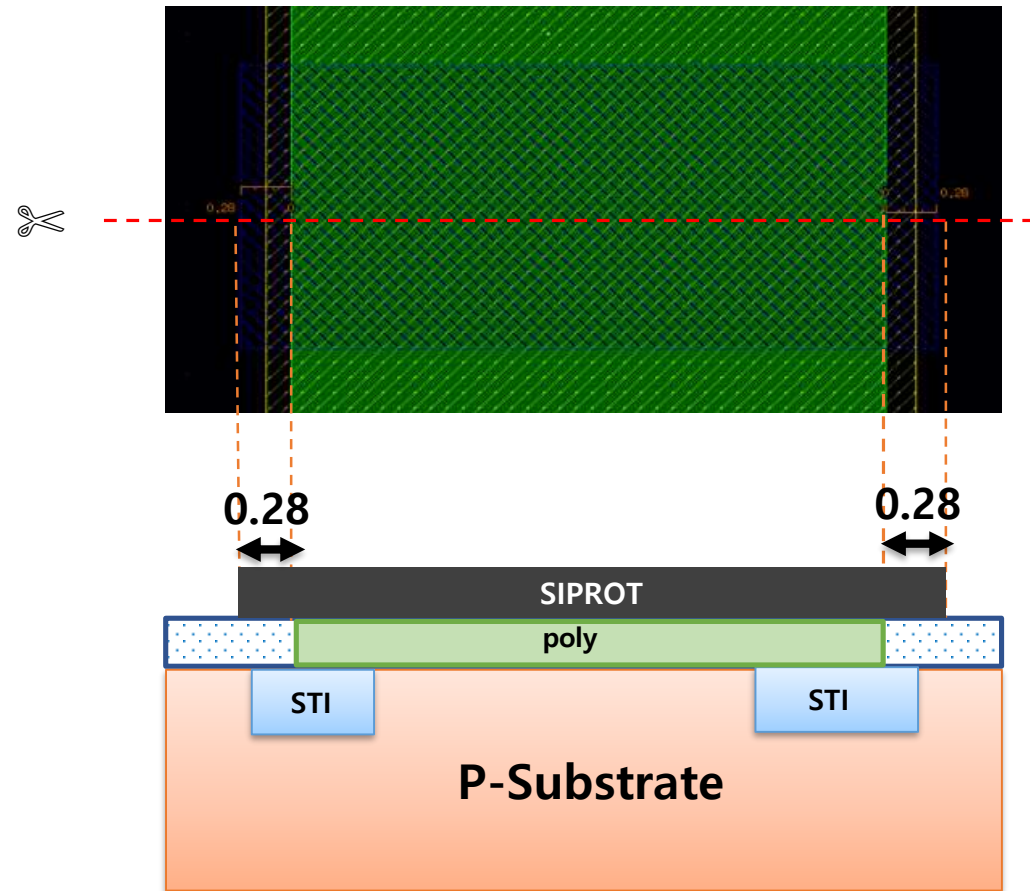
✓ Schematic in Design Rule Manual



STI 위 Poly는 silicide 대상이 아니므로 충분히 덮어 오동작 방지



✓ Cadence Reproducing and vertical structure

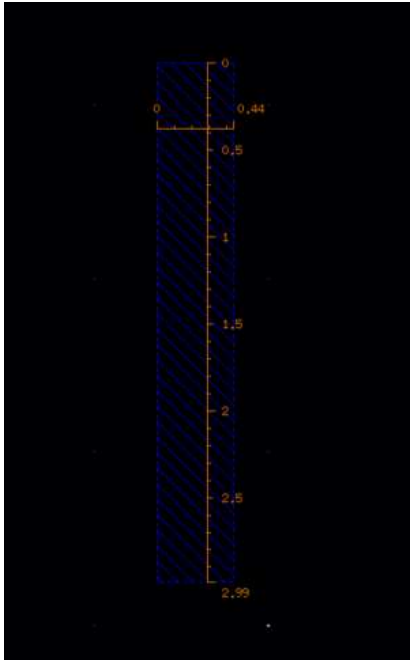
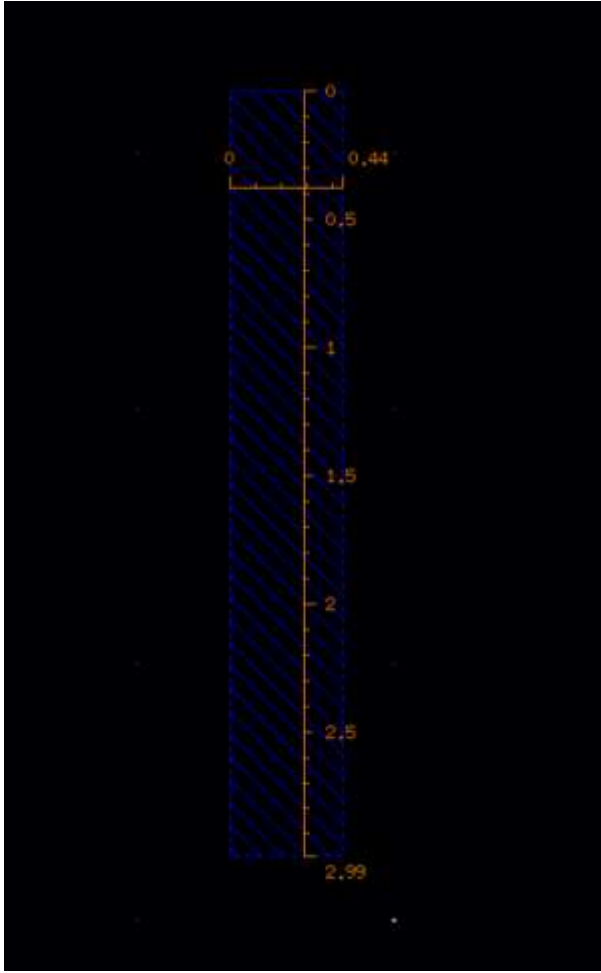
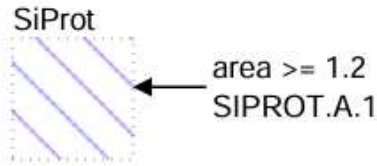


Rule Name	Value	Description
SIPROT.A.1	1.2um	Minimum Salicide Block area

✓ Schematic in Design Rule Manual

✓ Cadence Reproducing and vertical structure

마스크/공정 오차를 위해

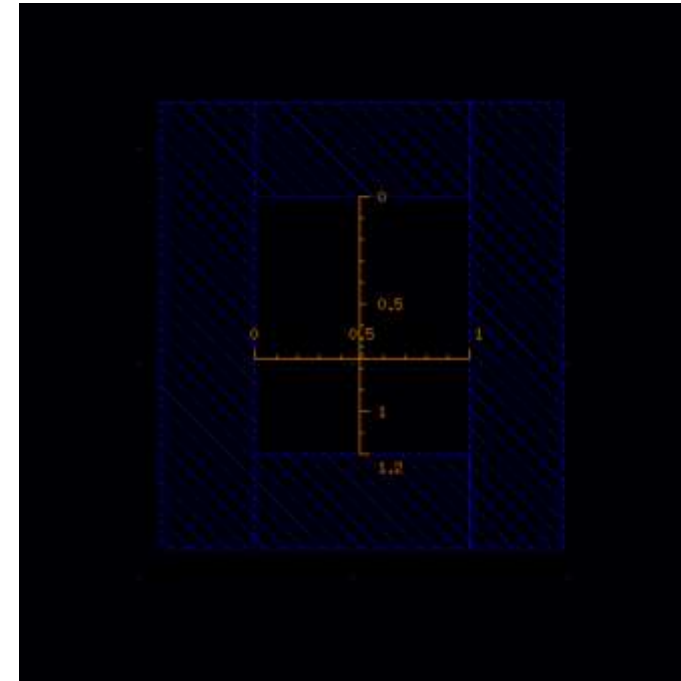
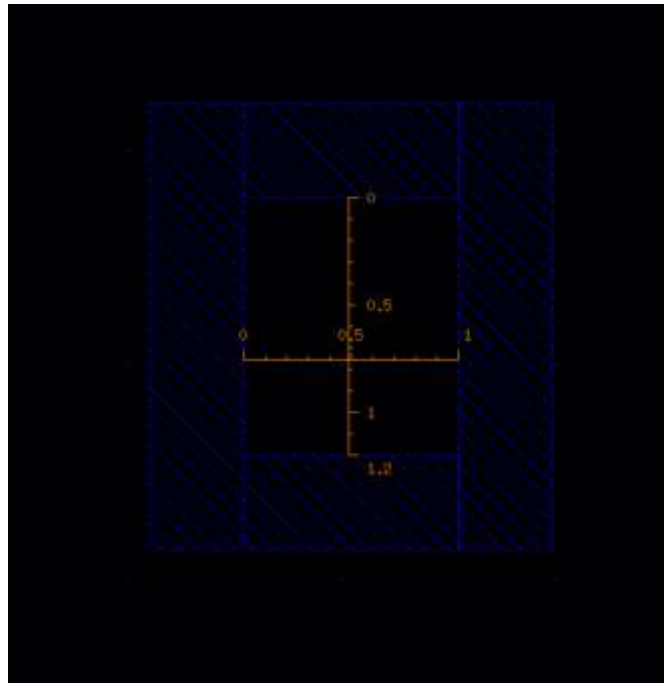
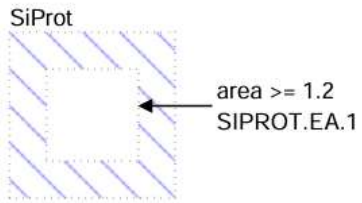


Rule Name	Value	Description
SIPROT.EA.1	1.2um	Minimum Salicide Block enclosed area ("donut" hole surrounded by Salicide Block)

✓ Schematic in Design Rule Manual

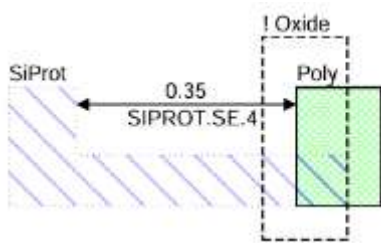
공정 신뢰성을 확보하기 위해

✓ Cadence Reproducing and vertical structure

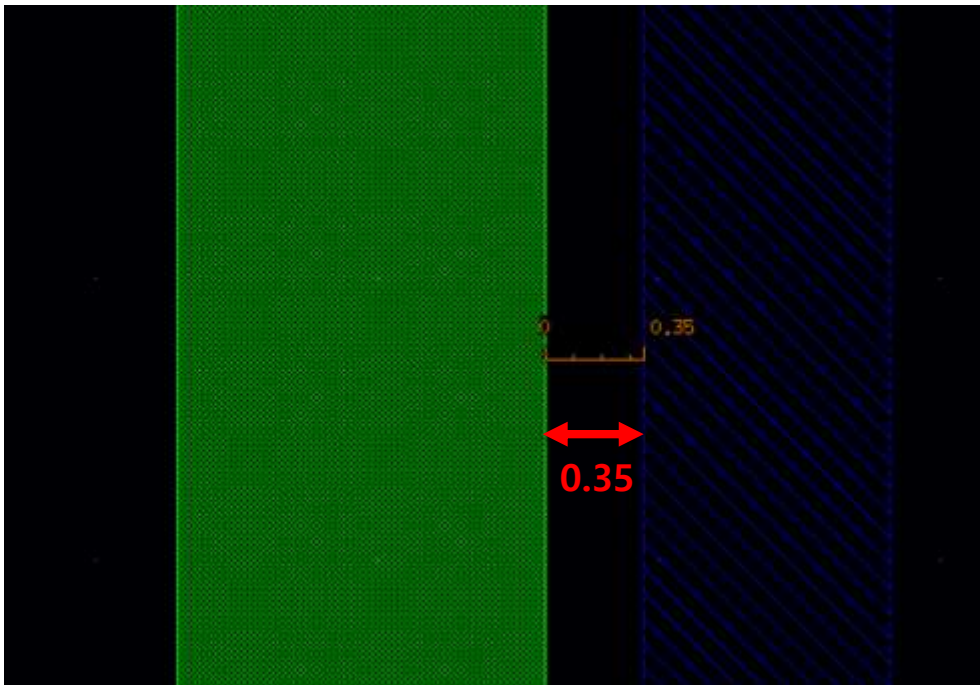


Rule Name	Value	Description
SIPROT.SE.4	0.35um	Minimum Salicide Block to Poly (on field) spacing

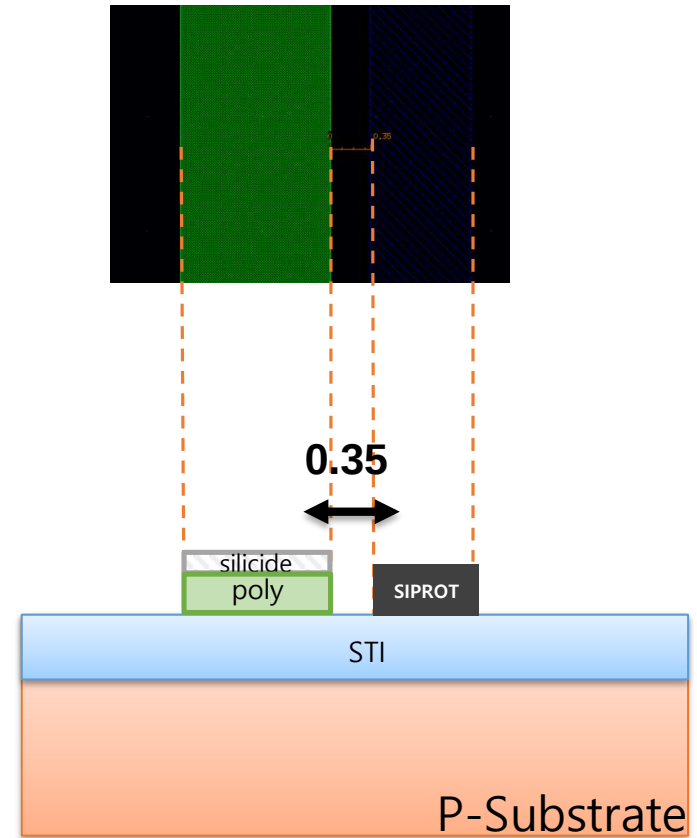
✓ Schematic in Design Rule Manual



STI 위 Poly 근처에 silicide가 생겨 일어날 저항 문제 방지



✓ Cadence Reproducing and vertical structure



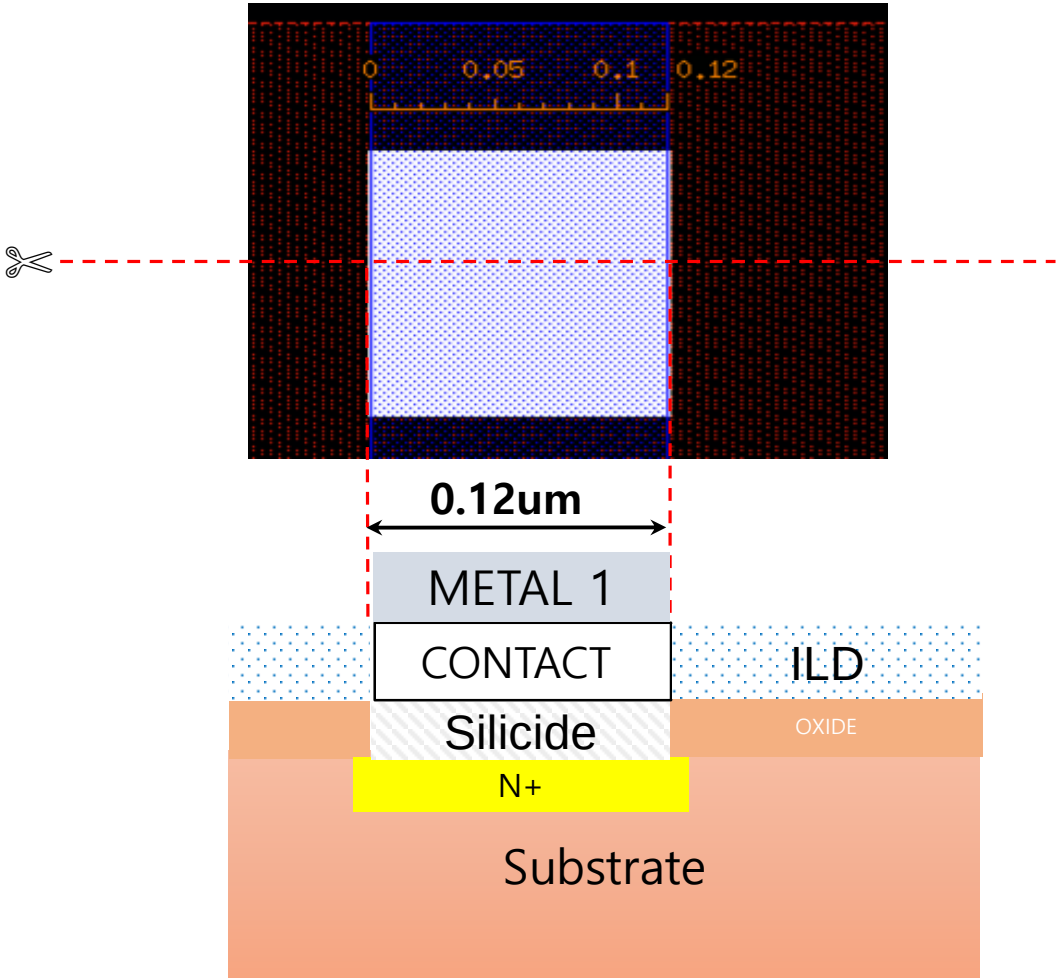
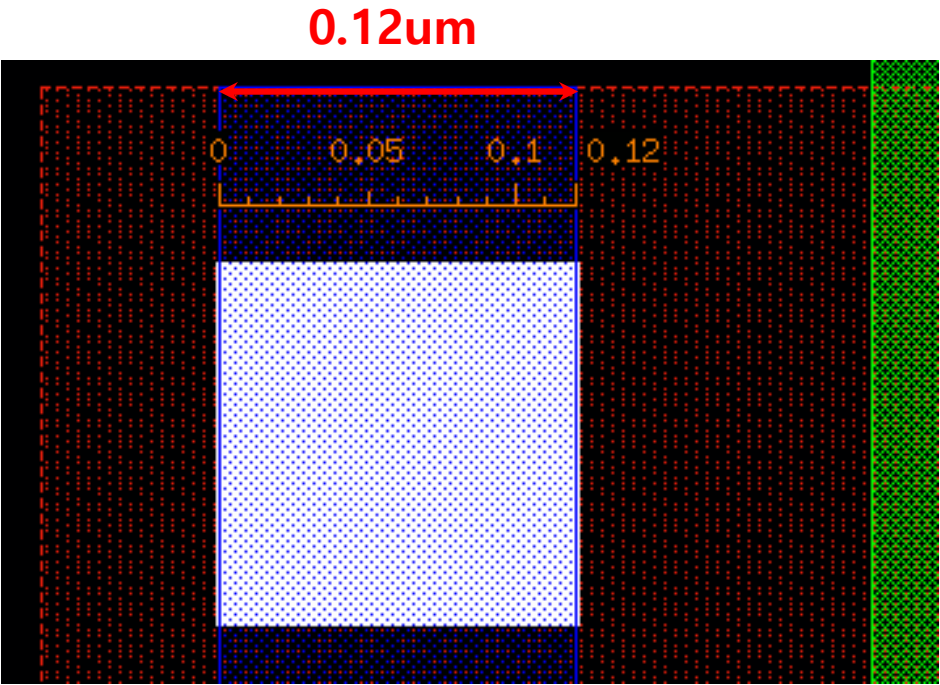
Metal

Rule Name	Value	Description
METAL1.W.1	0.12um	Minimum Metal 1 width

✓ Schematic in Design Rule Manual

✓ Cadence Reproducing and vertical structure

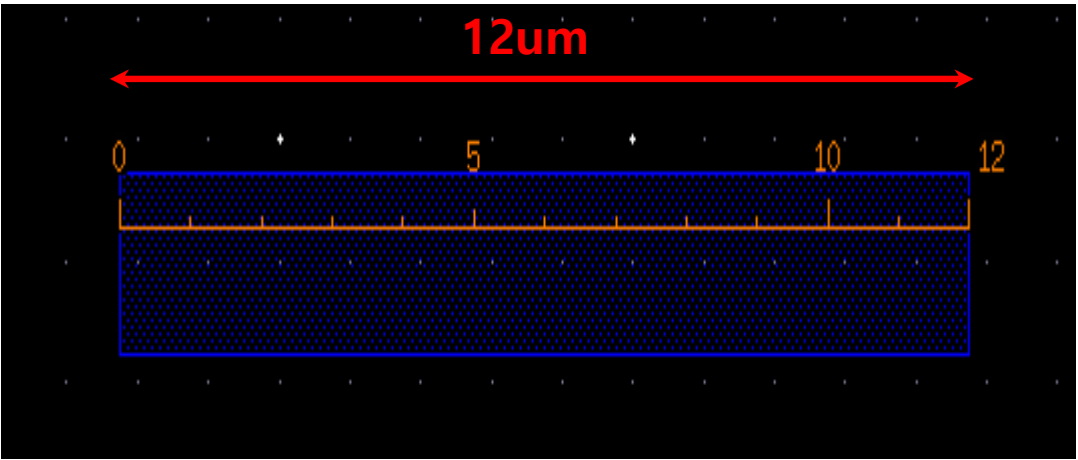
공정 안정성 및 배선 신뢰성 확보를 위한 최소 Metal1 폭 규정



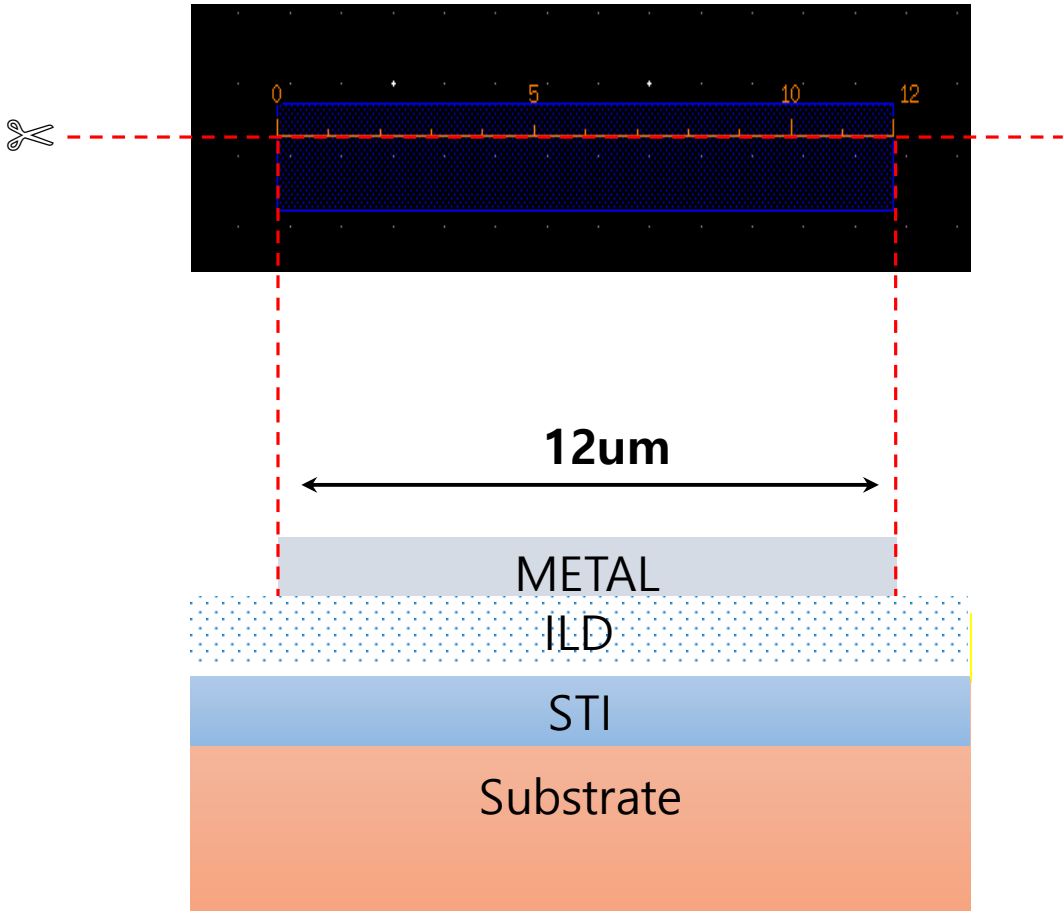
Rule Name	Value	Description
METAL1.W.2	12um	Maximum Metal 1 width

✓ Schematic in Design Rule Manual

공정 안정성 및 배선 신뢰성 확보를 위한
최대 Metal1 폭 규정

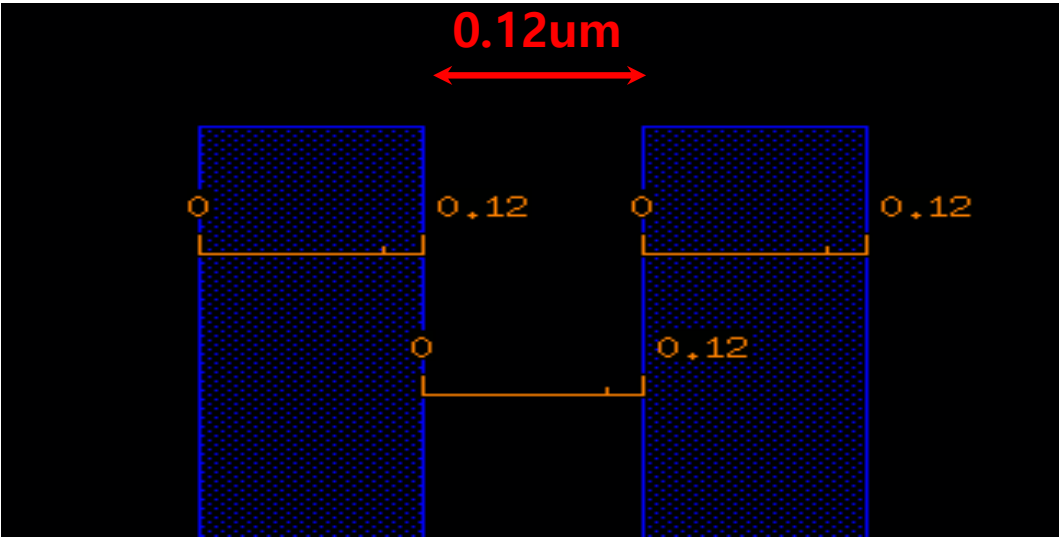


✓ Cadence Reproducing and vertical structure

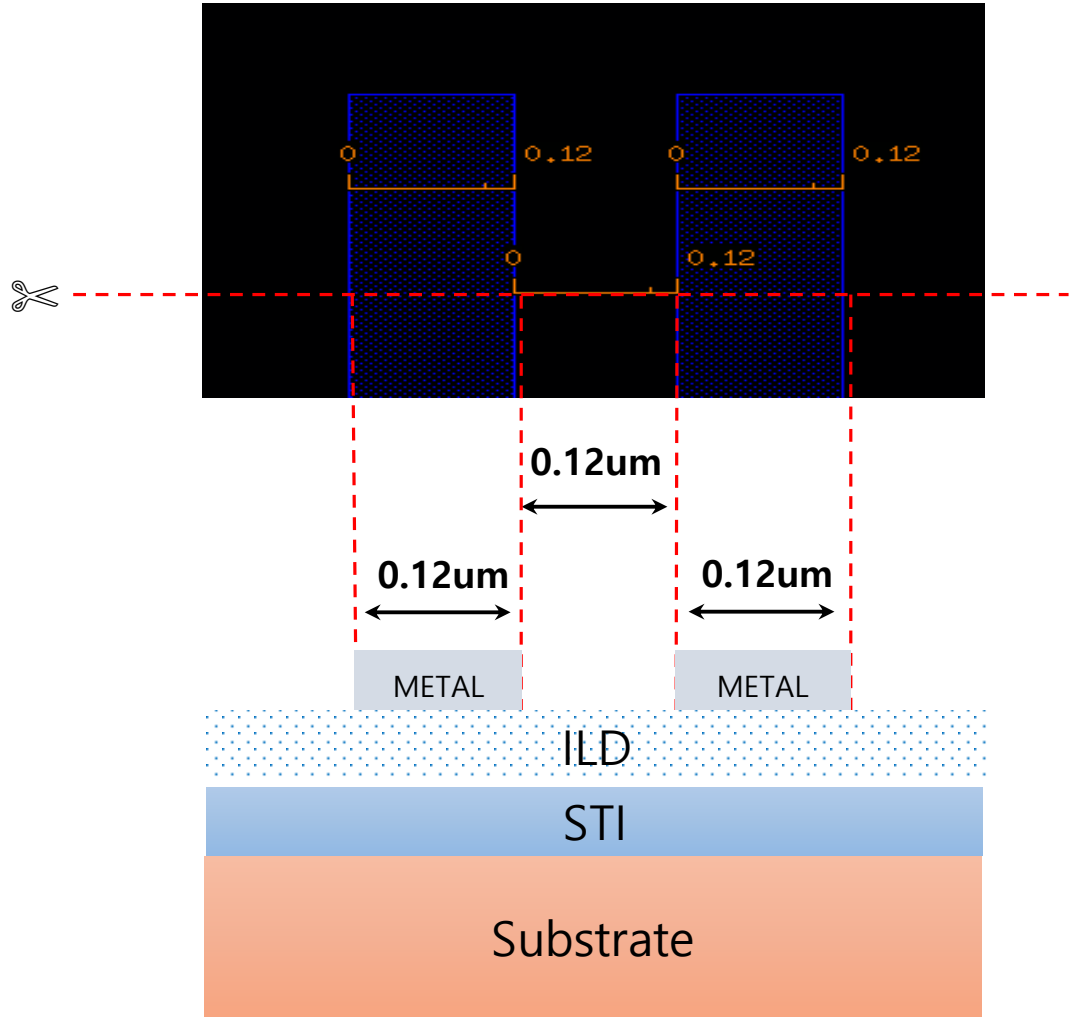


Rule Name	Value	Description
METAL1.SP.1.1	0.12um	Minimum Metal 1 to Metal 1 spacing

- ✓ Schematic in Design Rule Manual
- 공정 안정성 및 배선 간 Short 방지를 위한 최소 Metal1 간격 규정



- ✓ Cadence Reproducing and vertical structure



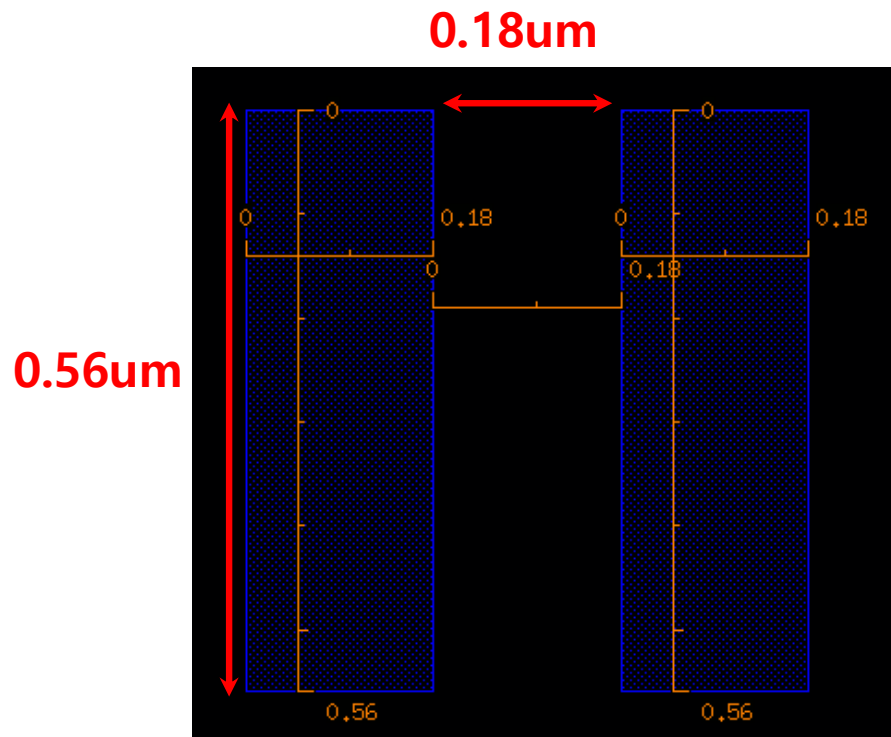
Rule Name	Value	Description
METAL1.SP.1.2	0.18um	Minimum Metal 1 to Metal 1 spacing

✓ Schematic in Design Rule Manual

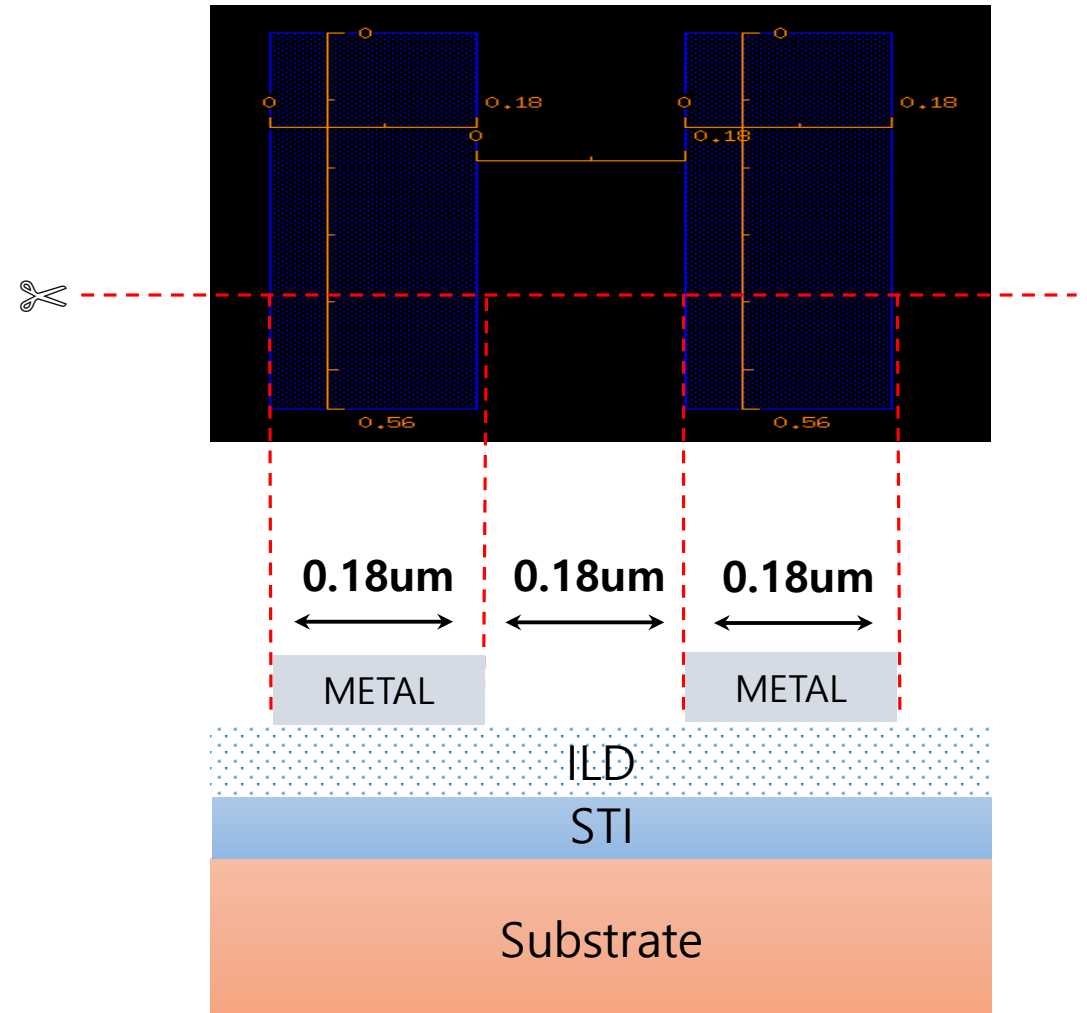
❖ If one metal width > 0.18 and parallel length > 0.56

➡ 0.18um

공정 안정성 및 Crosstalk 감소를 위한 최소 Metal1 간격 규정



✓ Cadence Reproducing and vertical structure



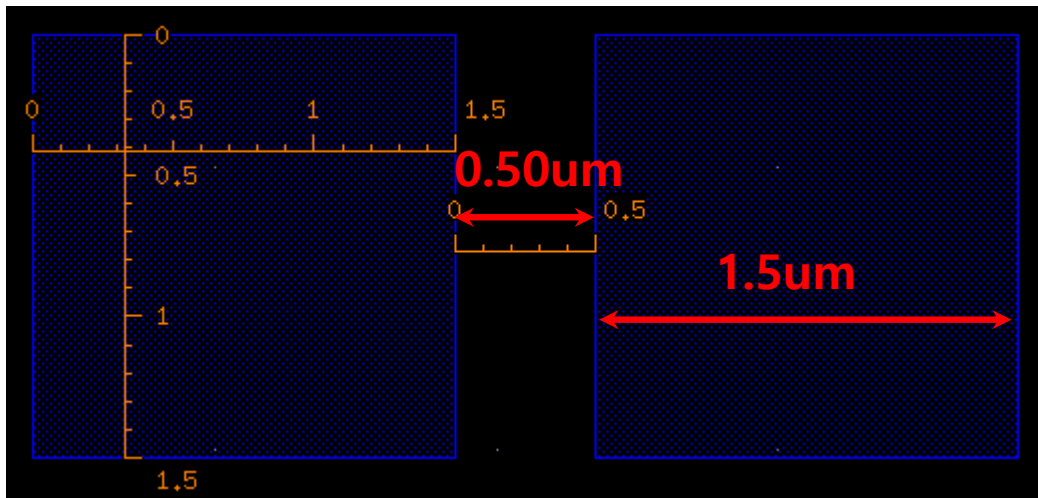
Rule Name	Value	Description
METAL1.SP.1.3	0.50um	Minimum Metal 1 to Metal 1 spacing

✓ Schematic in Design Rule Manual

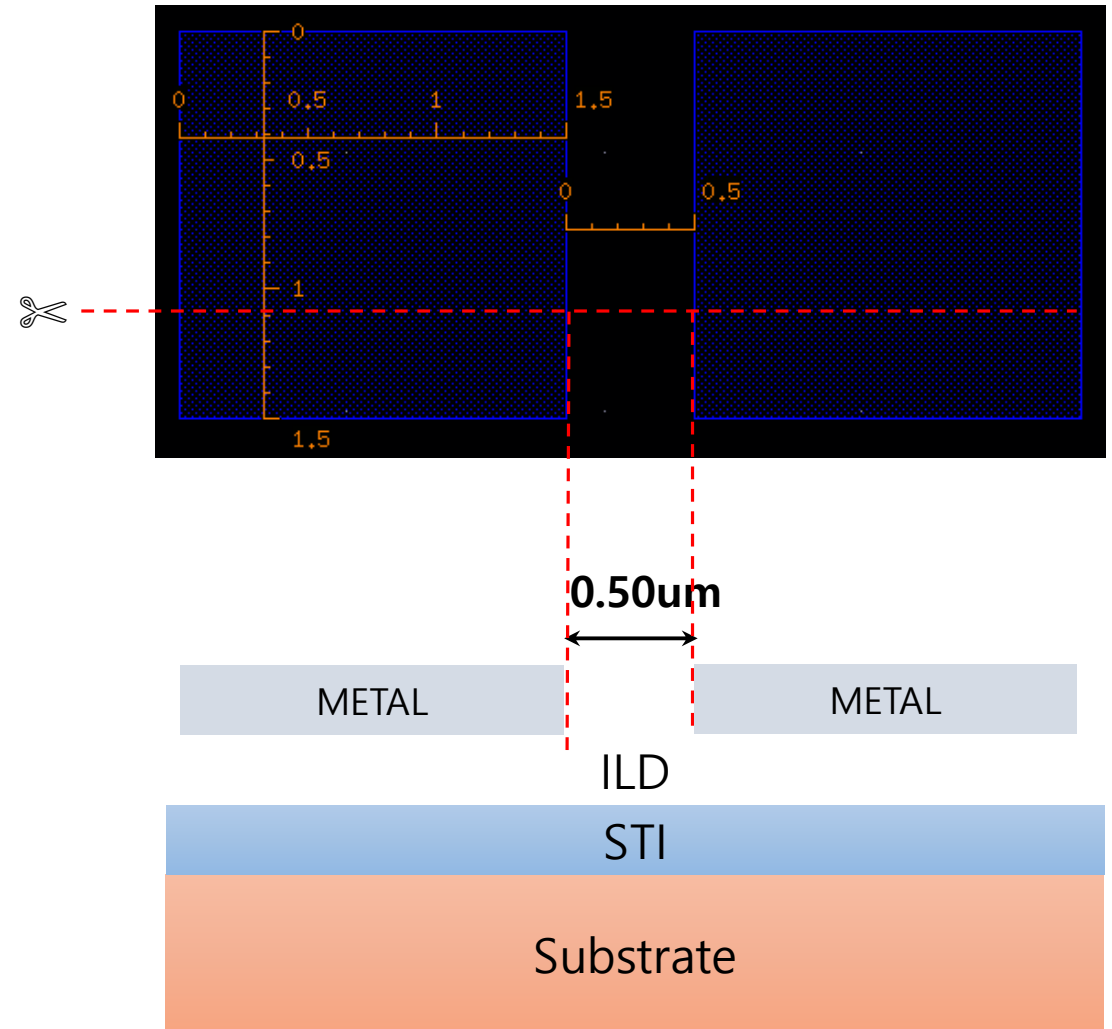
❖ If one metal width > 1.5 and parallel length > 1.5

➡ 0.50um

1. 폭 증가 → Bridge 위험 증가
2. 평행 길이 증가 → Short 위험 증가
3. 평행 길이 증가 → Crosstalk 증가



✓ Cadence Reproducing and vertical structure

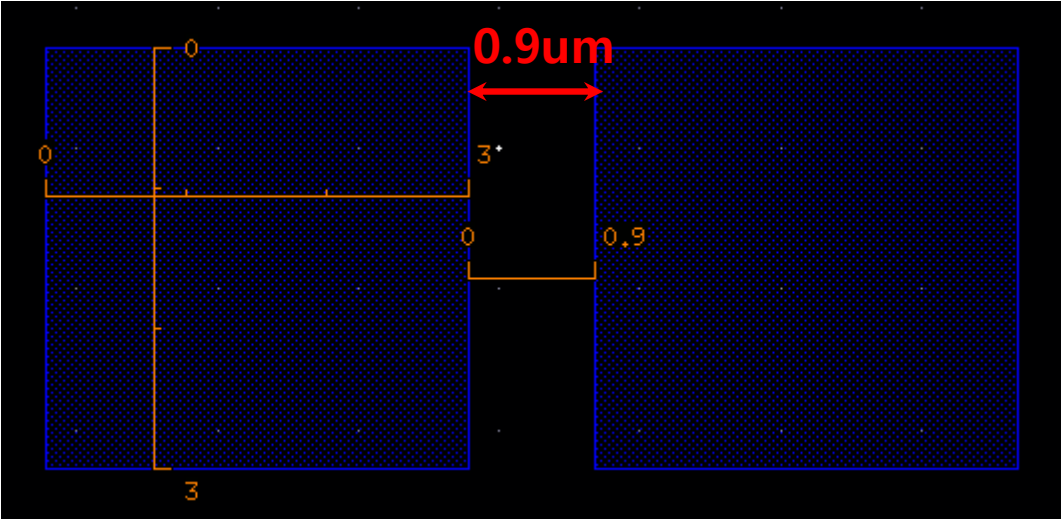


Rule Name	Value	Description
METAL1.SP.1.4	0.90um	Minimum Metal 1 to Metal 1 spacing

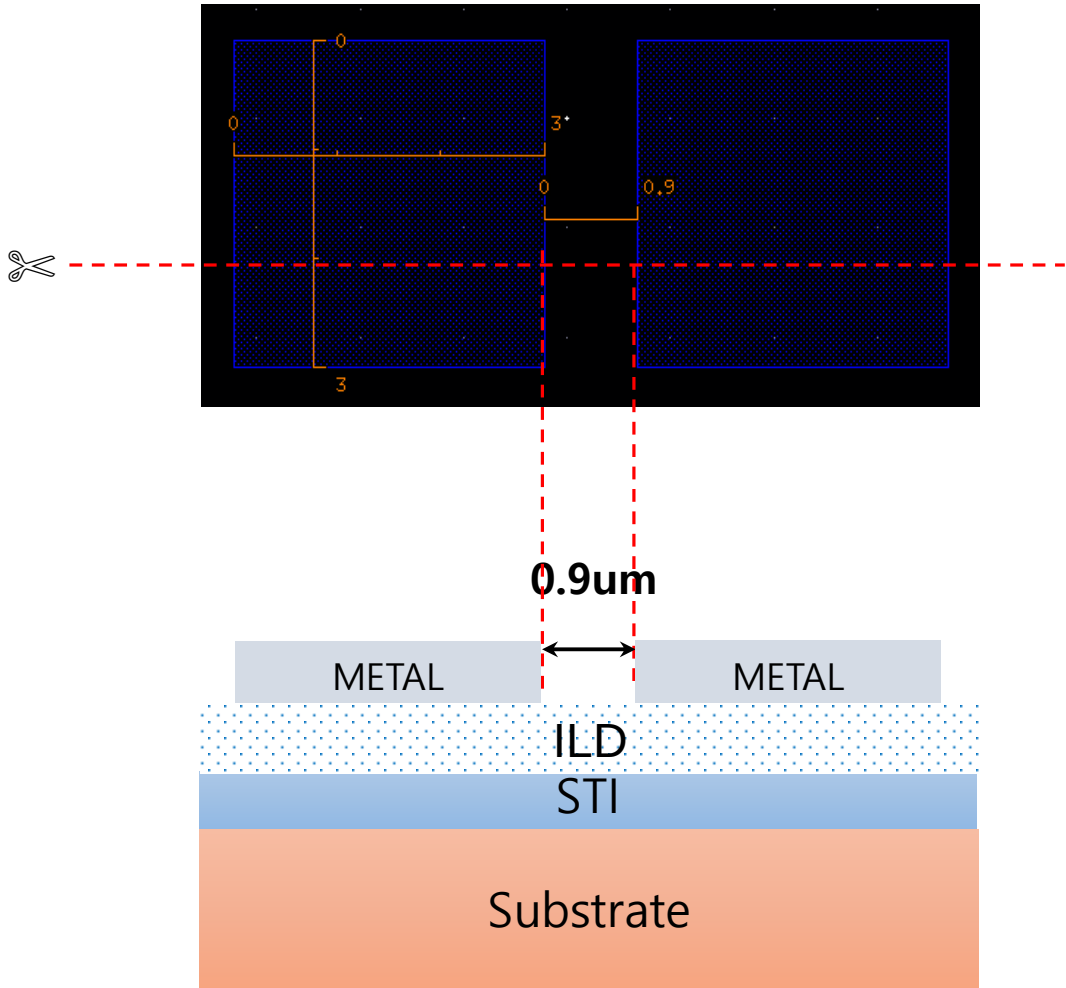
✓ Schematic in Design Rule Manual

❖ If one metal width > 3.0 and parallel length > 3.0

➡ 0.90um



✓ Cadence Reproducing and vertical structure

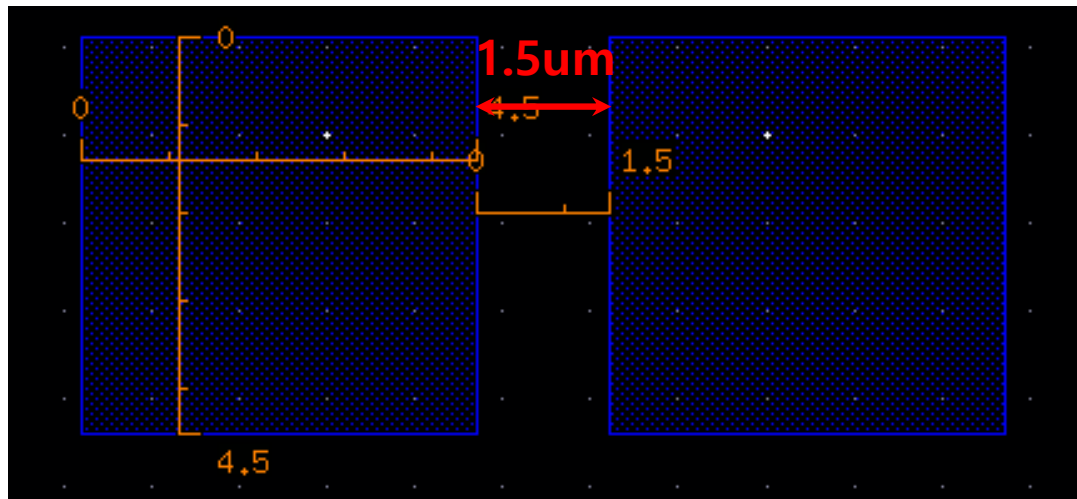


Rule Name	Value	Description
METAL1.SP.1.5	1.50um	Minimum Metal 1 to Metal 1 spacing

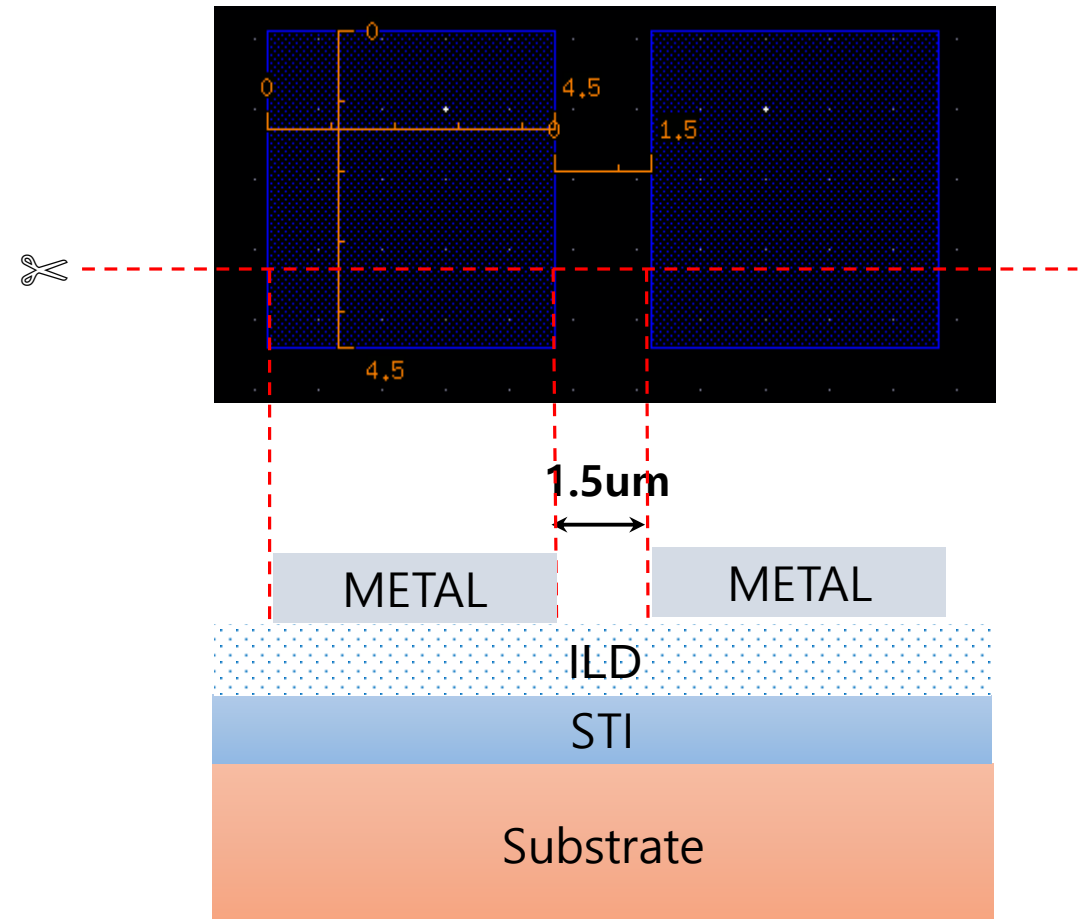
✓ Schematic in Design Rule Manual

❖ If one metal width > 4.5 and parallel length > 4.5

➡ 1.50um



✓ Cadence Reproducing and vertical structure

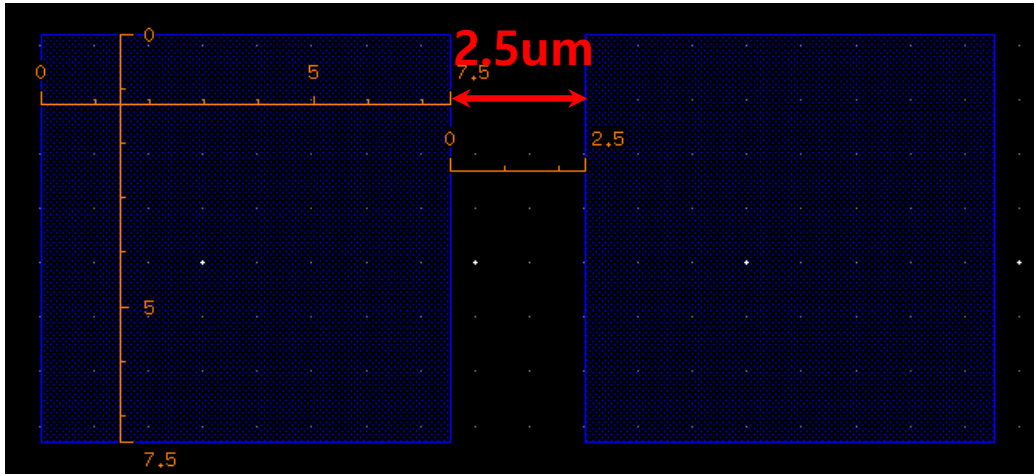


Rule Name	Value	Description
METAL1.SP.1.6	2.50um	Minimum Metal 1 to Metal 1 spacing

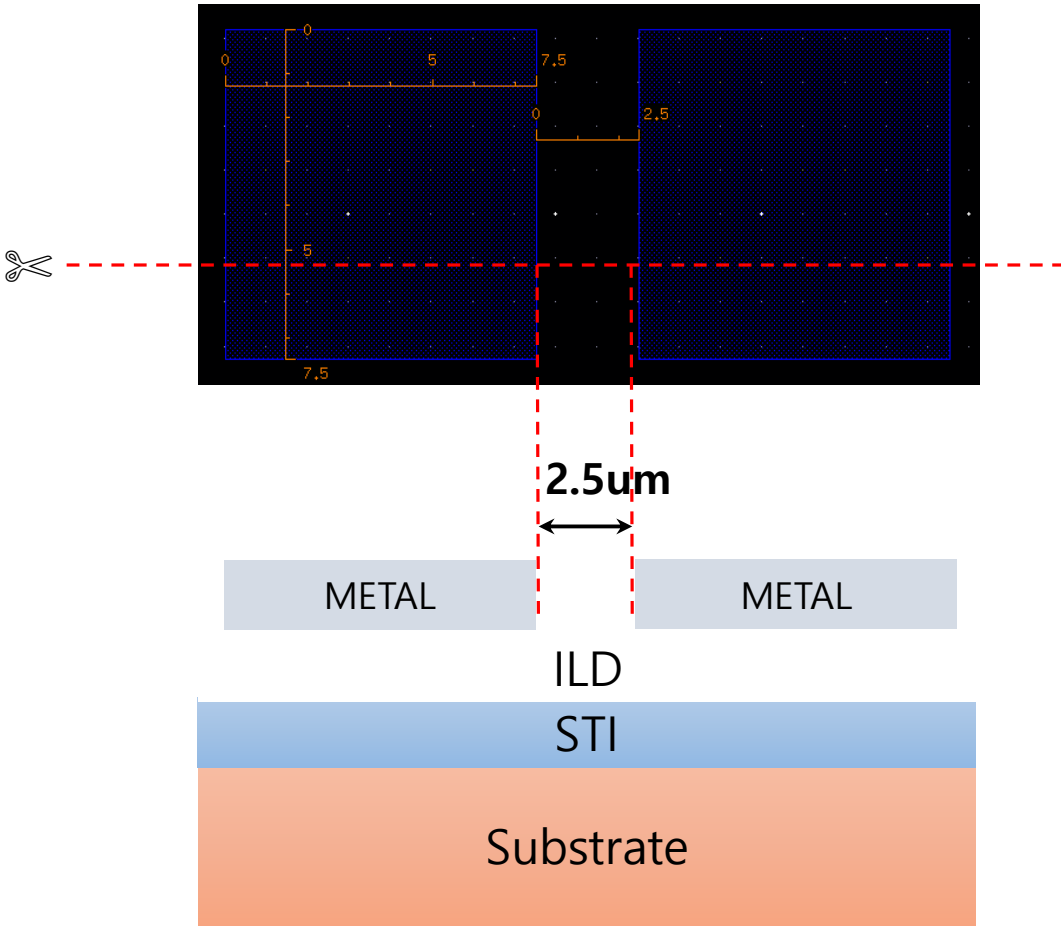
✓ Schematic in Design Rule Manual

❖ If one metal width > 7.5 and parallel length > 7.5

➡ 2.50um



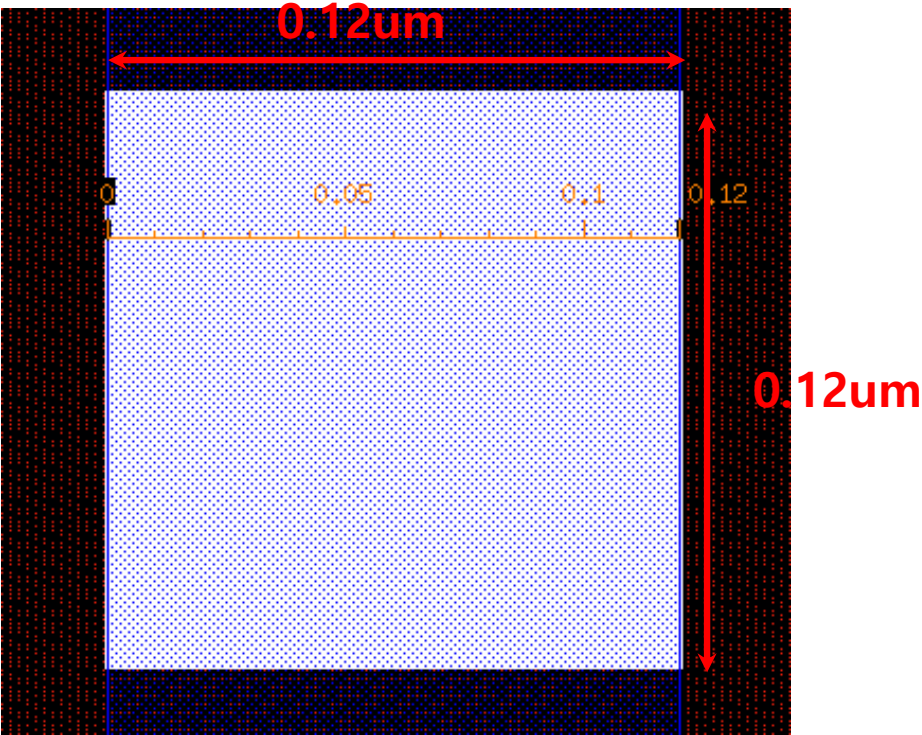
✓ Cadence Reproducing and vertical structure



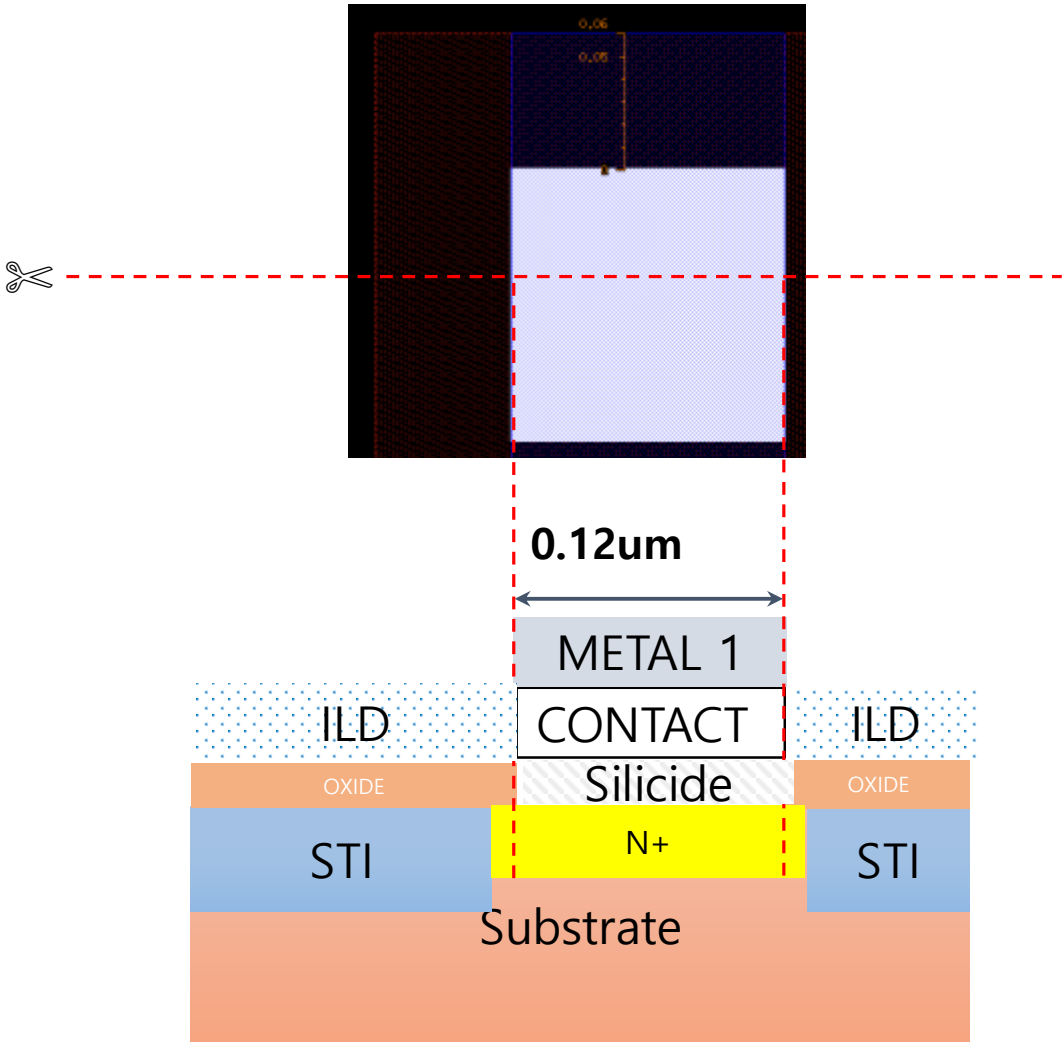
Rule Name	Value	Description
METAL1.E.1	0.00um	Minimun Metal 1 to Contact enclosure

✓ Schematic in Design Rule Manual

안정적인 Contact 형성 및 Open 방지를 위한
Metal1 Enclosure 규정



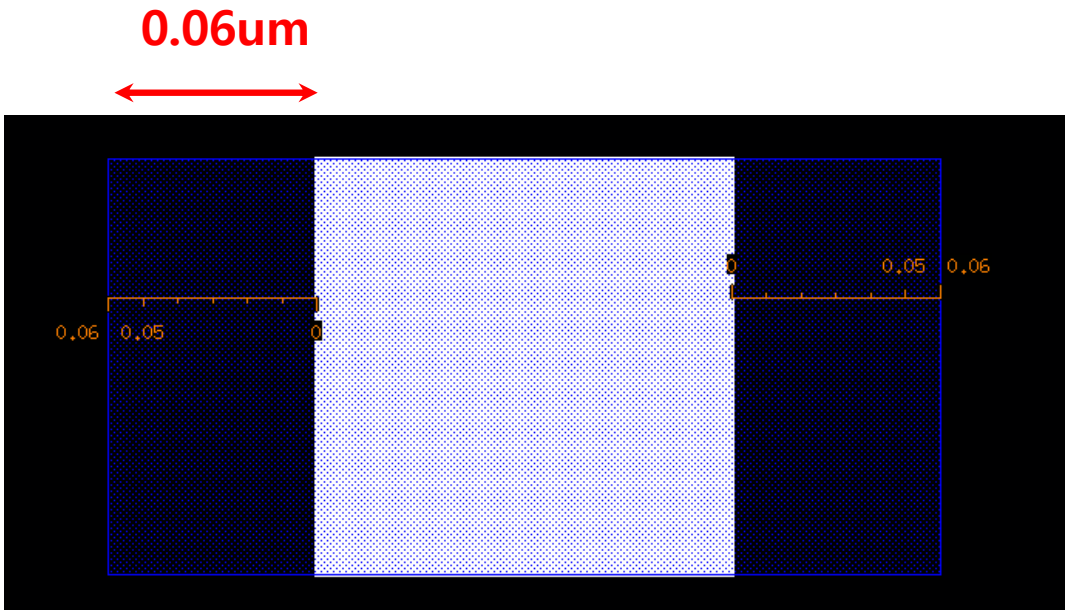
✓ Cadence Reproducing and vertical structure



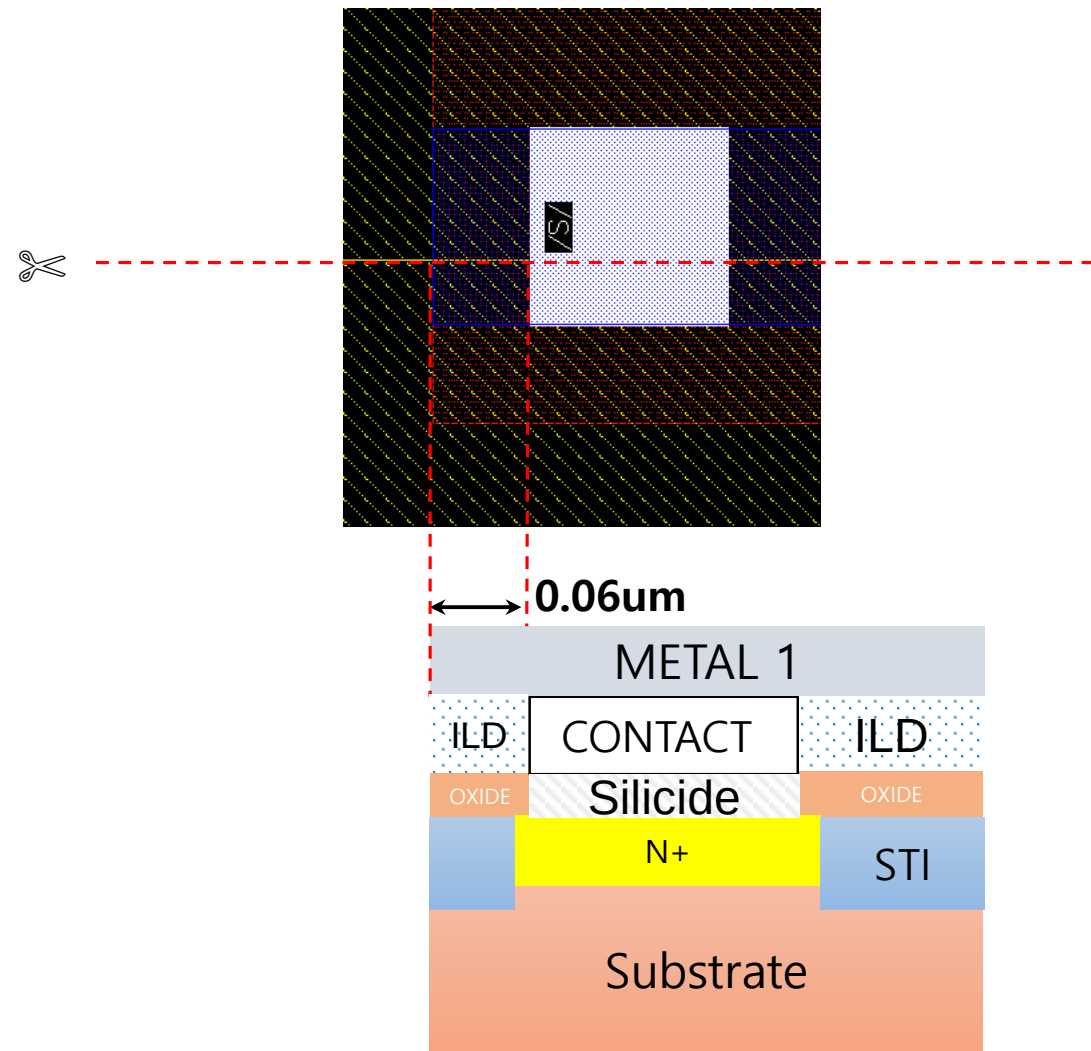
Rule Name	Value	Description
METAL.E.2	0.06um	Minimun Metal 1 to Contact enclosure on two opposite sides of the Contact

✓ Schematic in Design Rule Manual

- 정렬 오차 보정
- 안정적인 전류 전달 확보
- Open 및 contact 저항 증가 방지



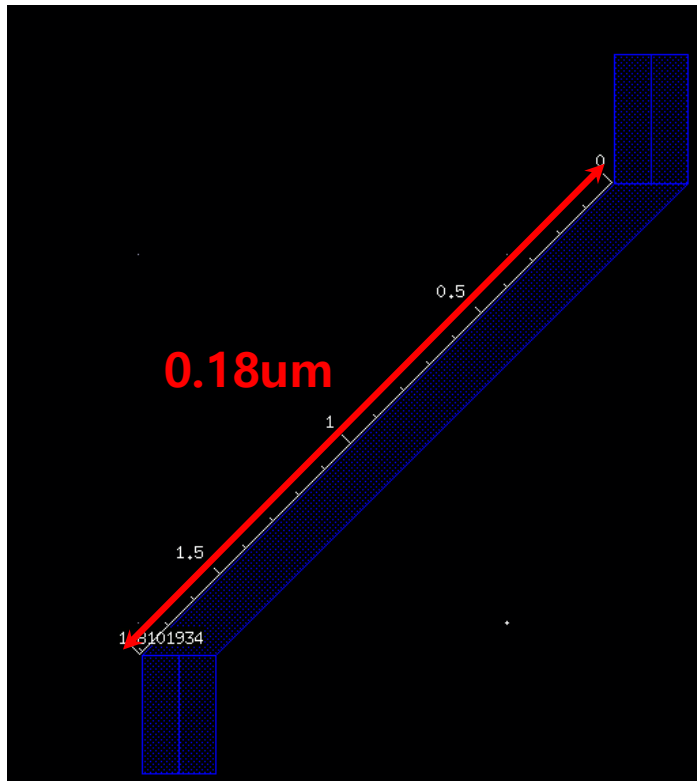
✓ Cadence Reproducing and vertical structure



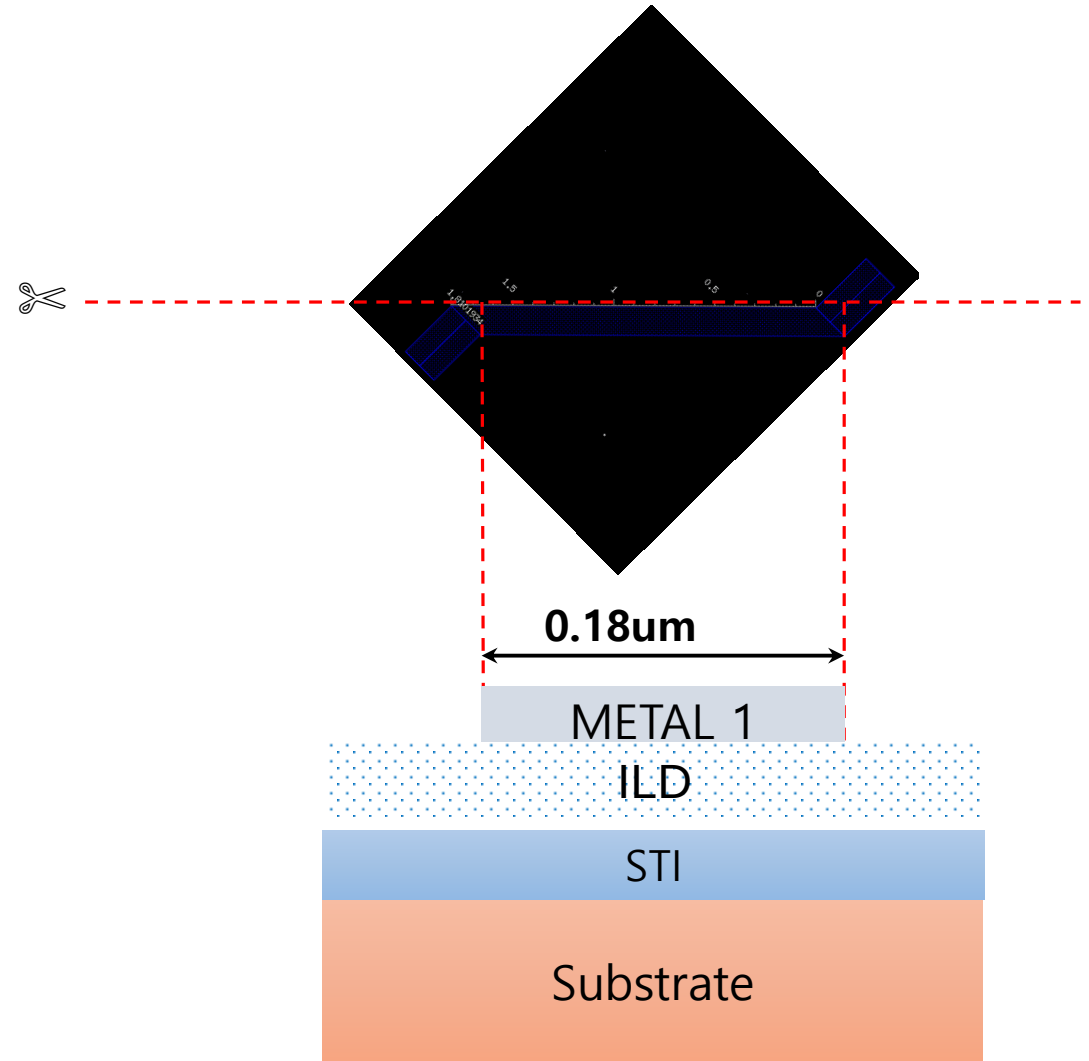
Rule Name	Value	Description
METAL1.L.1	0.18um	Minimun bent Metal1(45 degree angle) length

✓ Schematic in Design Rule Manual

1. 45° 패턴 형성 안정성 확보
2. 식각 공정 안정성 확보
3. Open 및 배선 불량 방지



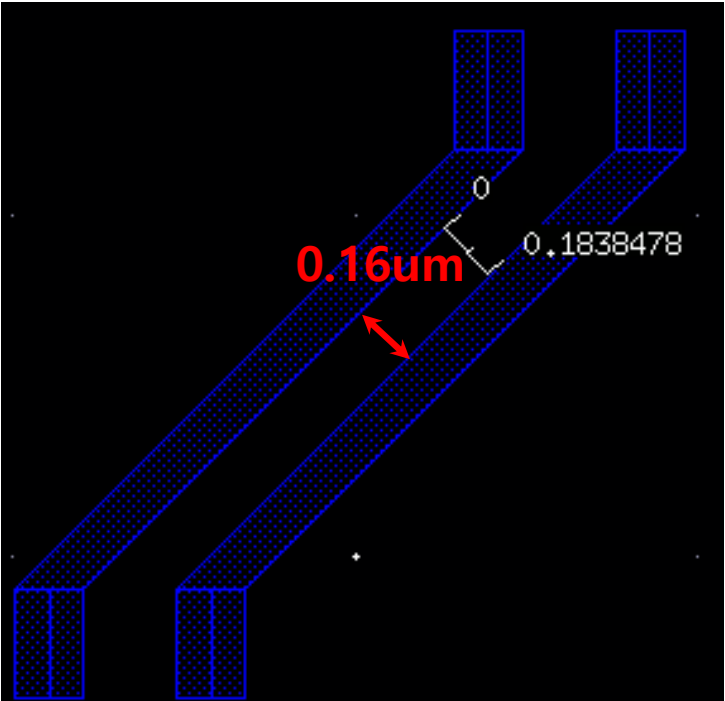
✓ Cadence Reproducing and vertical structure



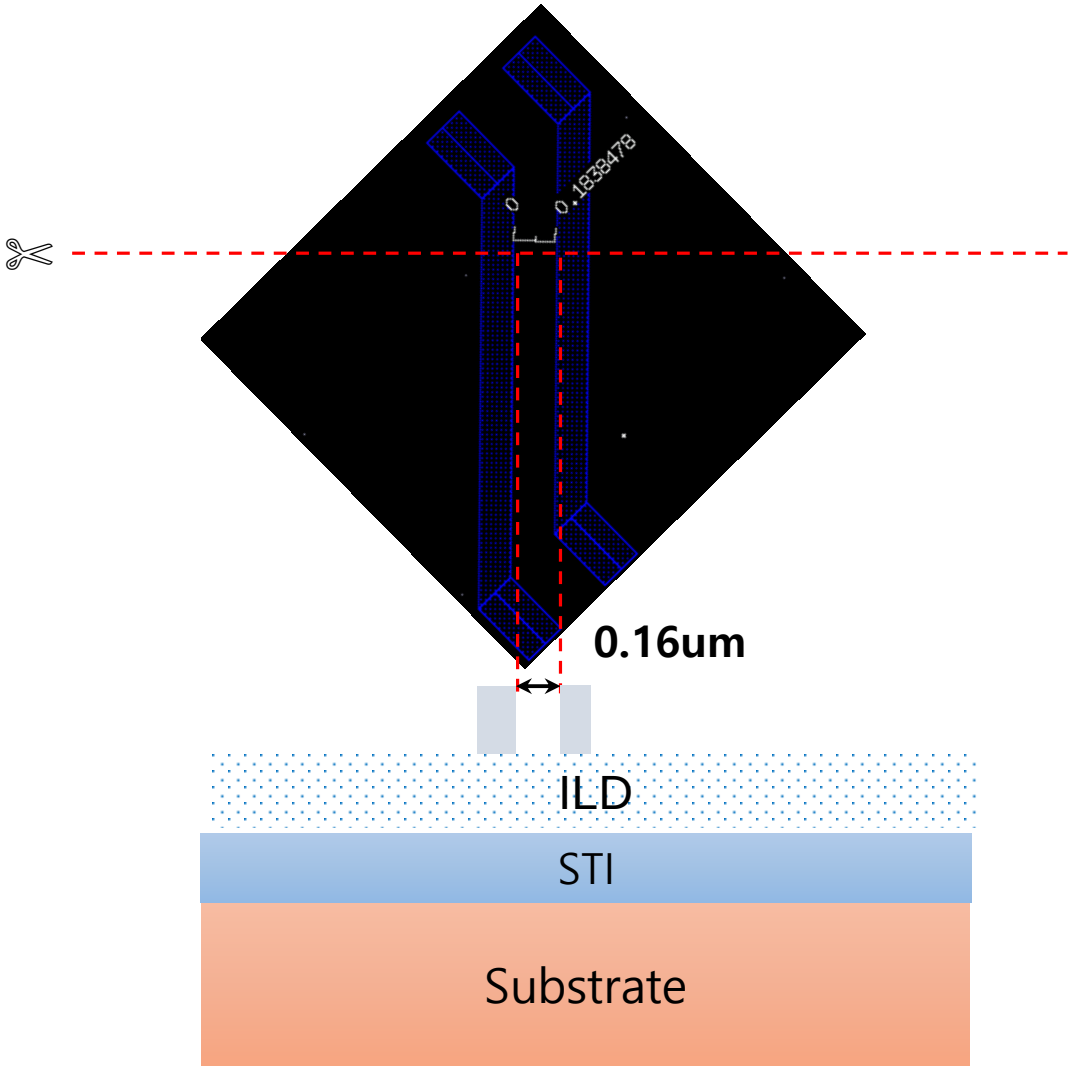
Rule Name	Value	Description
METAL1.SP.2	0.16um	Minimun bent Metal1(45 degree angle) space

✓ Schematic in Design Rule Manual

공정 안정성 및 Short 방지를 위한 최소 45°
Metal1 간격 규정



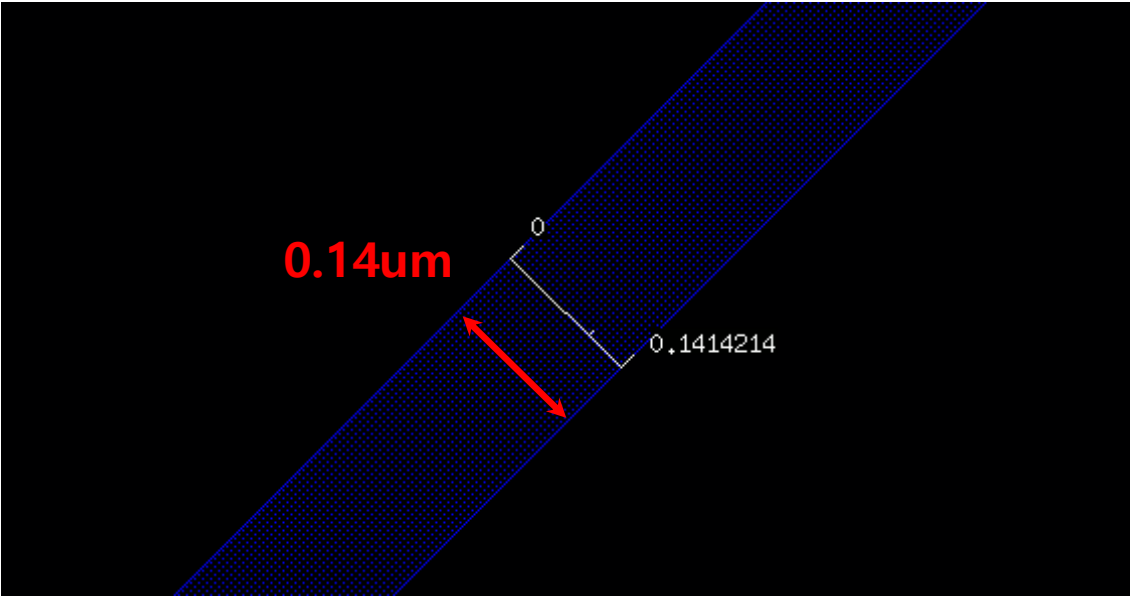
✓ Cadence Reproducing and vertical structure



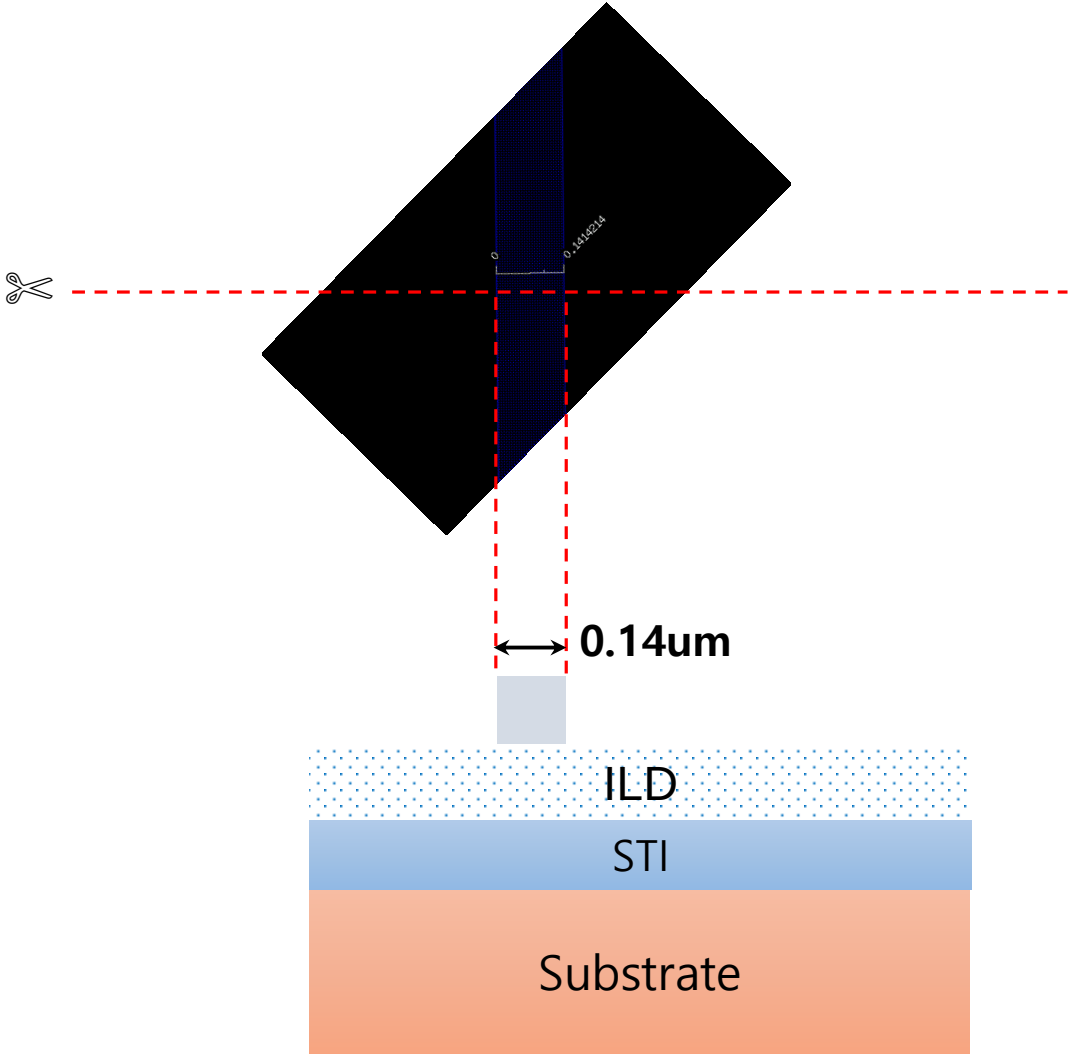
Rule Name	Value	Description
METAL1.W.3	0.14um	Minimun bent Metal1(45 degree angle) width

✓ Schematic in Design Rule Manual

공정 안정성 및 배선 신뢰성 확보를 위한 최소
45° Metal1 폭 규정



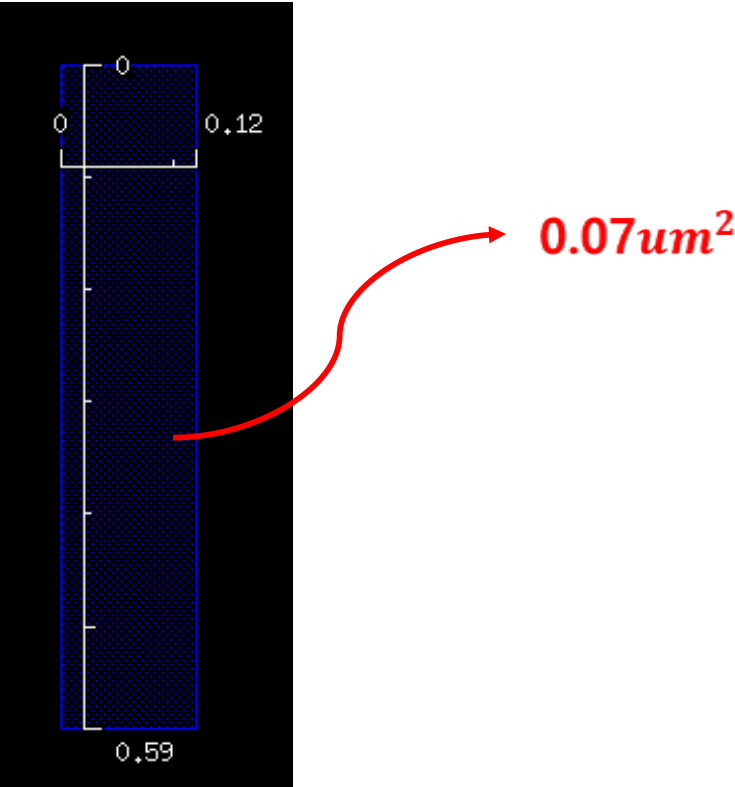
✓ Cadence Reproducing and vertical structure



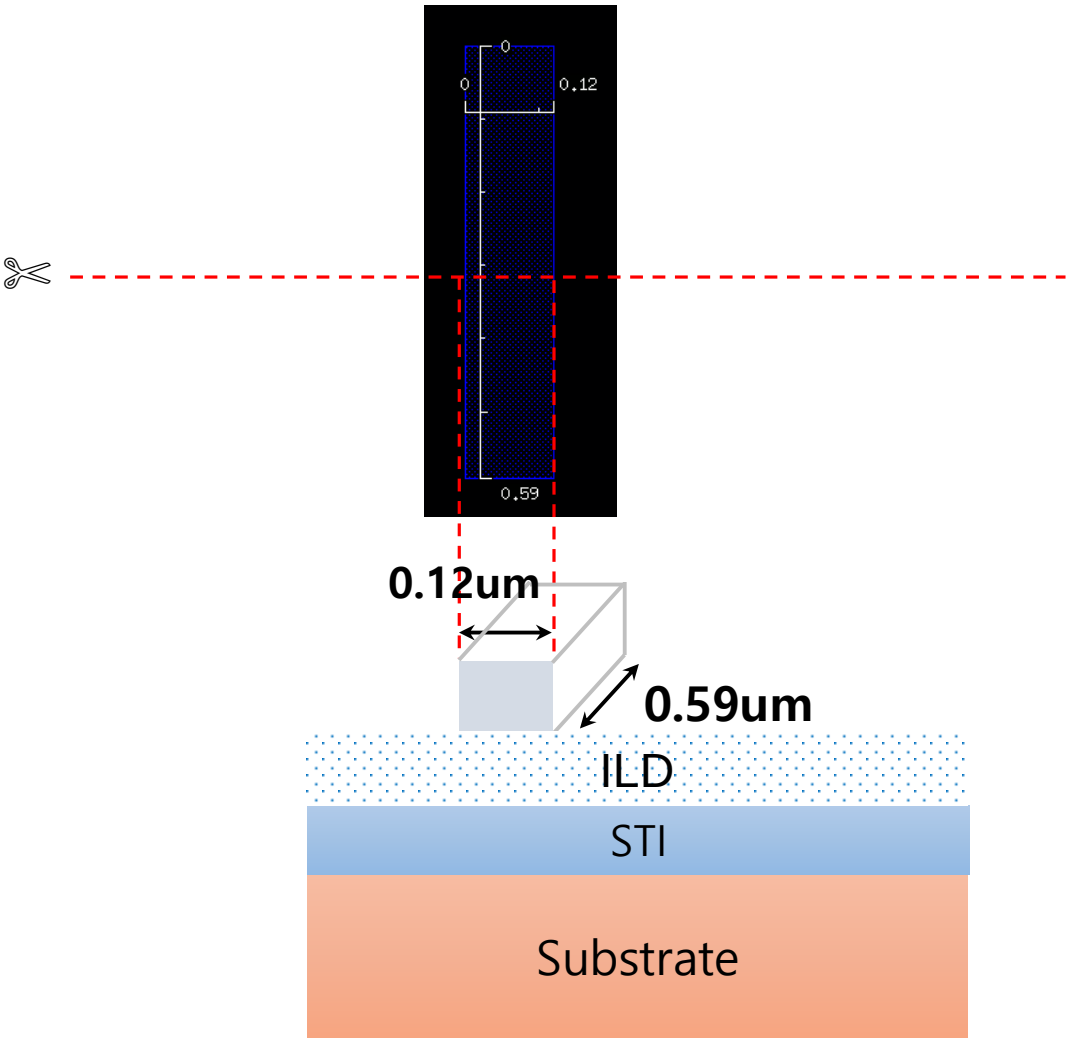
Rule Name	Value	Description
METAL1.A.1	---	Minimum bent Metal1 area

✓ Schematic in Design Rule Manual

공정 안정성 및 배선 신뢰성 확보를 위한 최소 Metal1 면적 규정



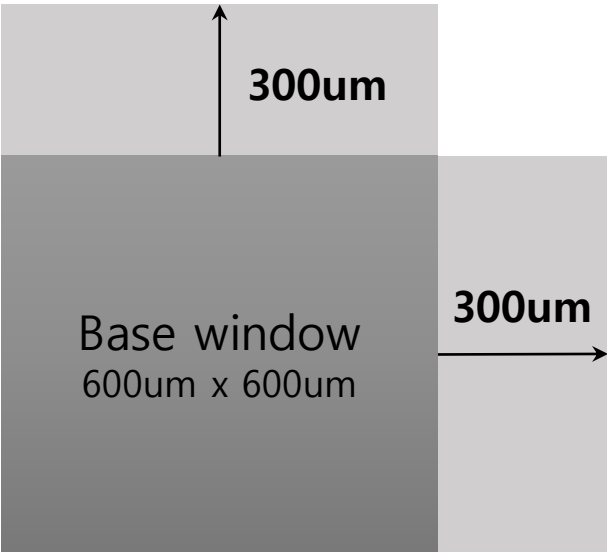
✓ Cadence Reproducing and vertical structure



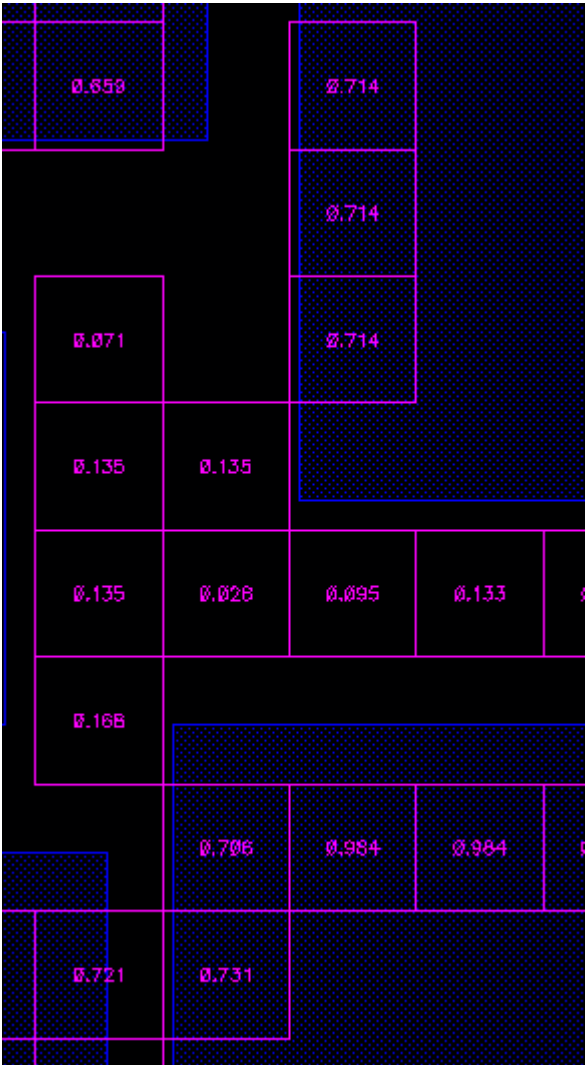
Rule Name	Value	Description
METAL1.D.2	<60%	Metal 1 Density range over any 600um x 600um area (checked by stepping in 300um increments)

✓ Density Window Stepping(300um increments)

- 1. 대면적 CMP 안정성 확보
- 2. 공정 균일성 확보
- 3. 배선 신뢰성 확보



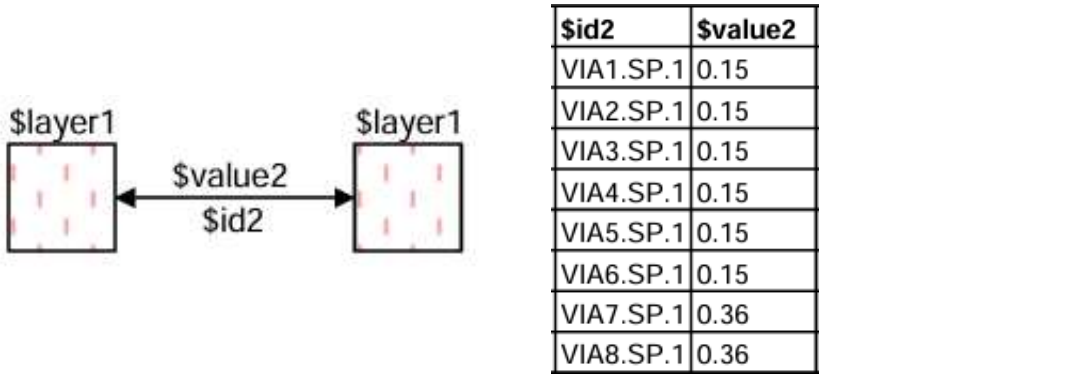
✓ Cadence Reproducing and vertical structure



Via

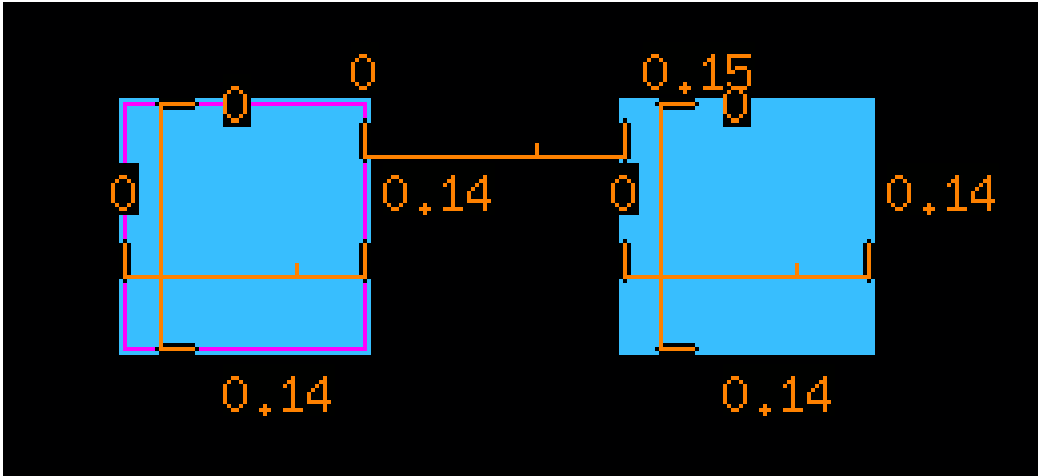
Rule Name	Value	Description
VIAk.SP.1	0.15um	Minimum Via k to Via k spacing

✓ Schematic in Design Rule Manual

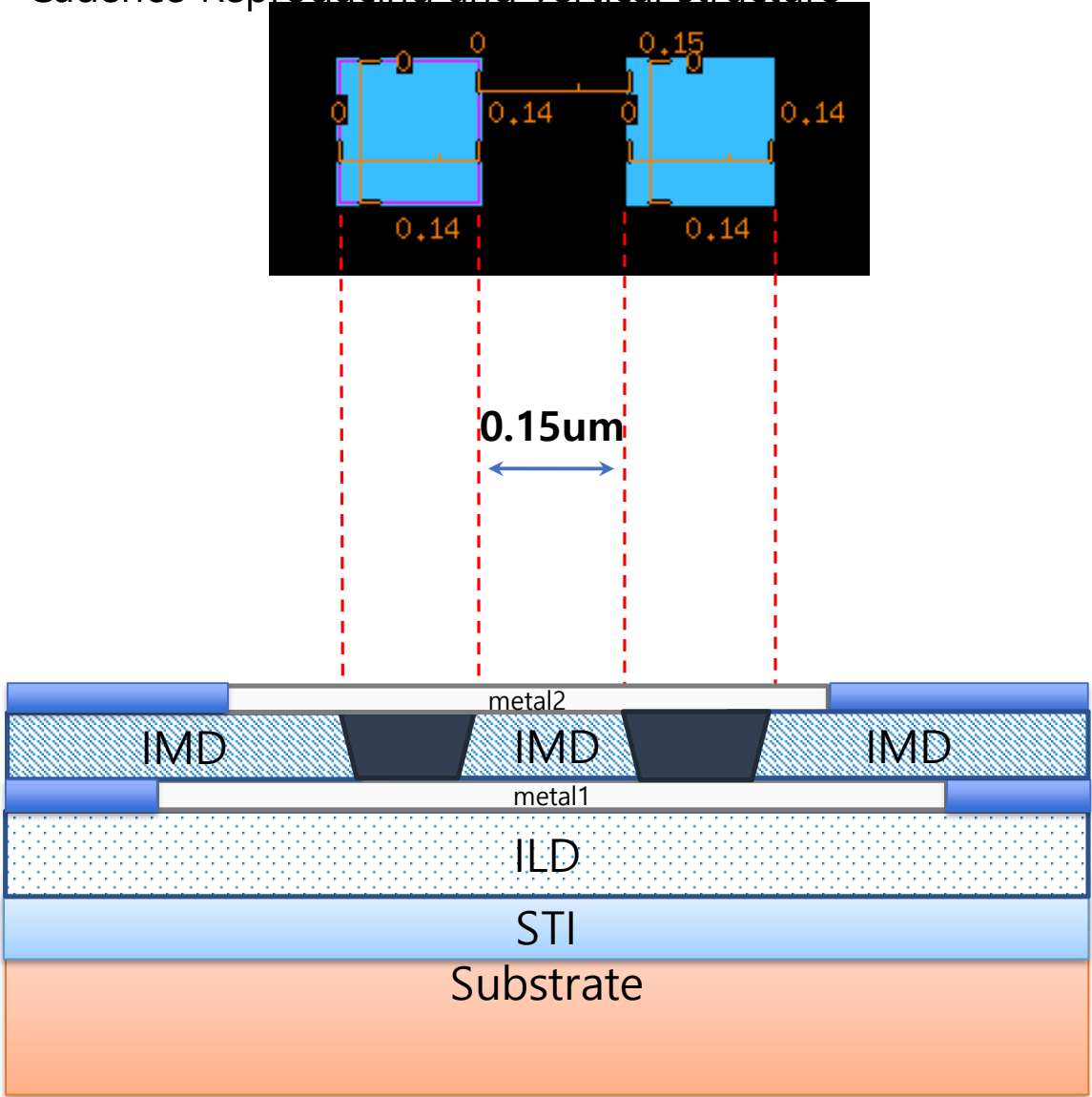


Via7, Via8 spacing = 0.36um

“인접한 Via 사이의 최소 간격을 확보하여 브리징을 방지하고
공정 수율 및 전기적 신뢰성을 확보”

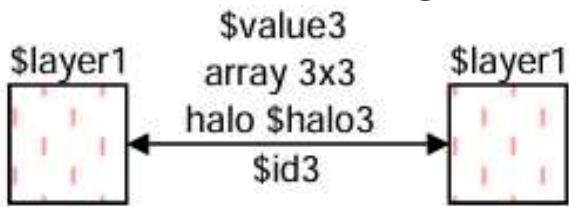


✓ Cadence Reproducing and vertical structure

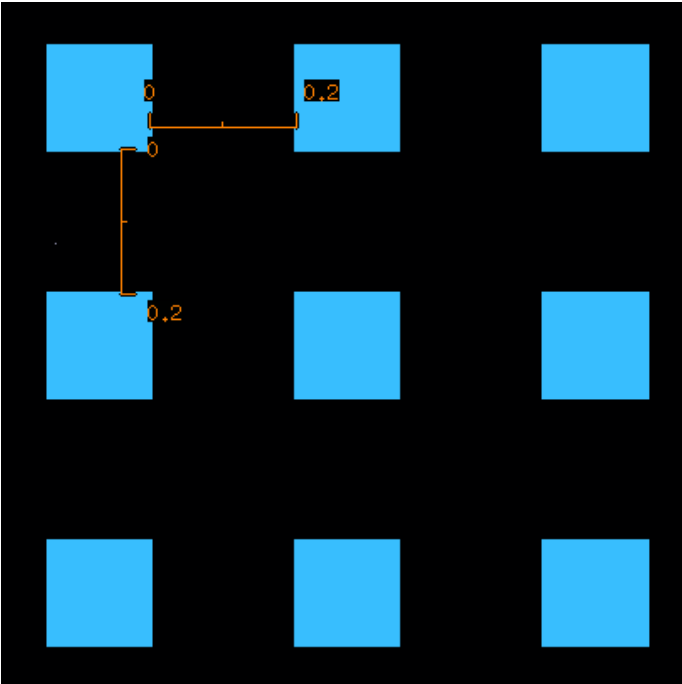


Rule Name	Value	Description
VIAk.SP.2	0.20um	Minimum Via k to Via k spacing when the Via ks are in a 3x3 or larger array (minimum dimension on one side of array is 3). Via ks spaced less than 0.21um should be considered for array spacing check.

✓ Schematic in Design Rule Manual

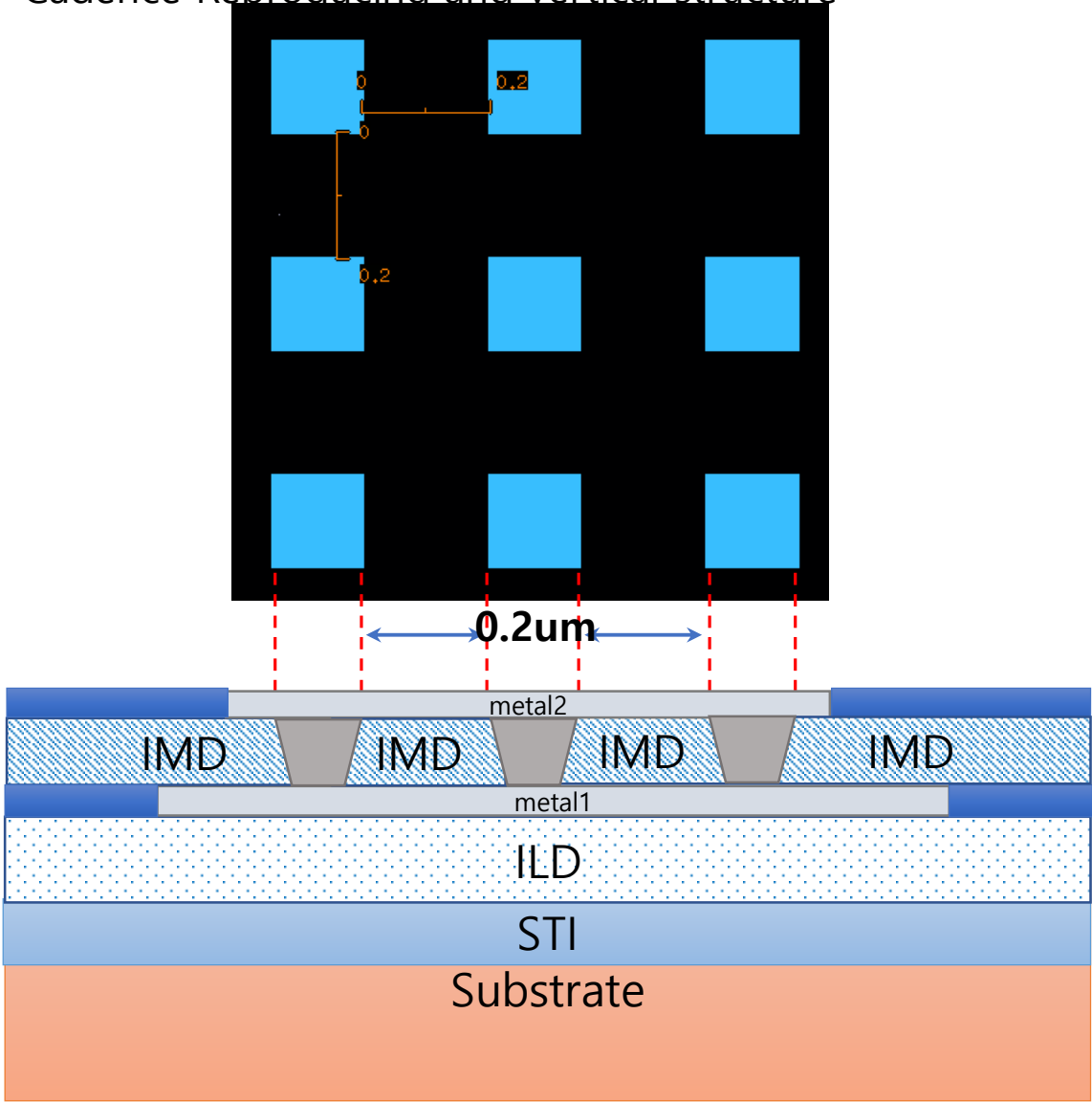


\$id3	\$value3	\$halo3
VIA1.SP.2	0.20	0.10
VIA2.SP.2	0.20	0.10
VIA3.SP.2	0.20	0.10
VIA4.SP.2	0.20	0.10
VIA5.SP.2	0.20	0.10
VIA6.SP.2	0.20	0.10
ignore	ignore	ignore
ignore	ignore	ignore



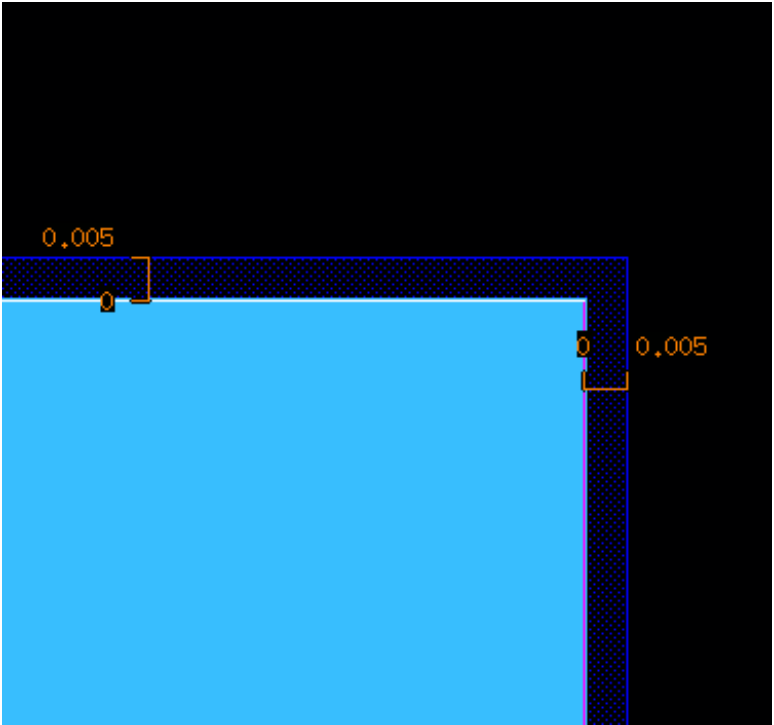
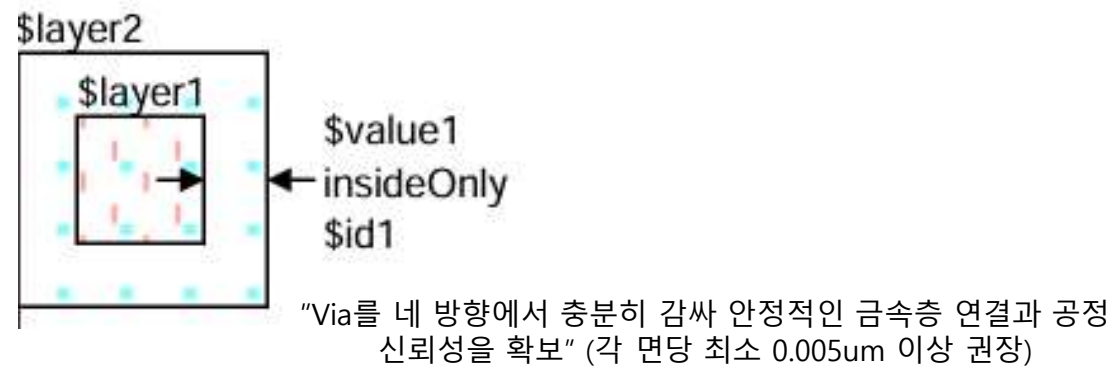
“3×3 이상의 Via Array에서는 Via가 밀집되면서 식각 및 금속 매립 공정의 균일성이 저하되어 Void, 브리징 등의 결함이 발생할 수 있으므로, 공정 신뢰성과 수율을 확보하기 위해 더 큰 최소 간격을 적용”

✓ Cadence Reproducing and vertical structure

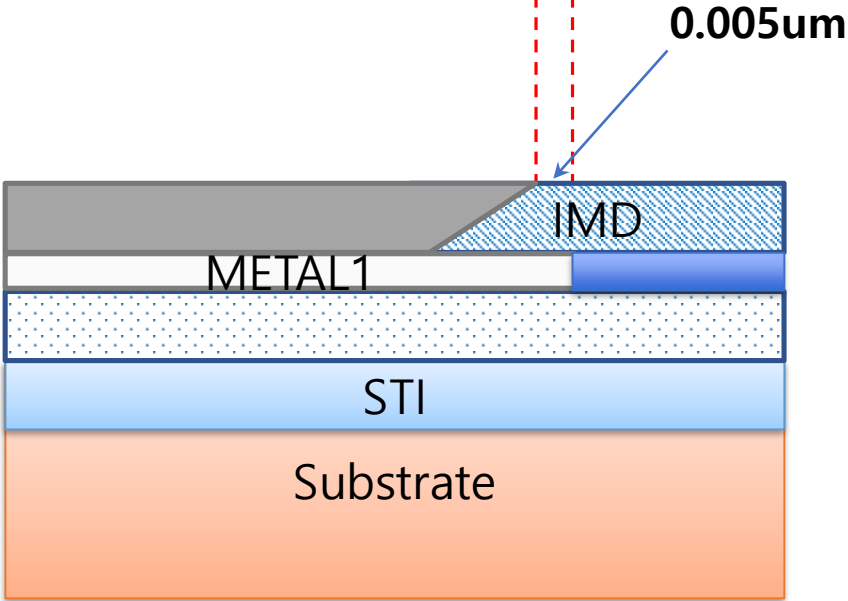
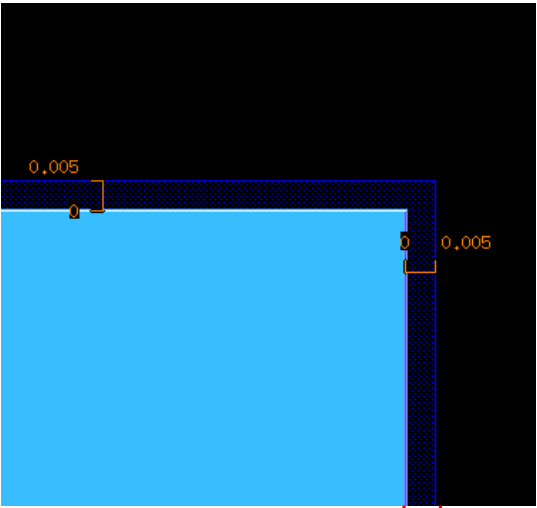


Rule Name	Value	Description
VIAk.E.1	0.005um	Minimum Metal k to Via k enclosure

✓ Schematic in Design Rule Manual



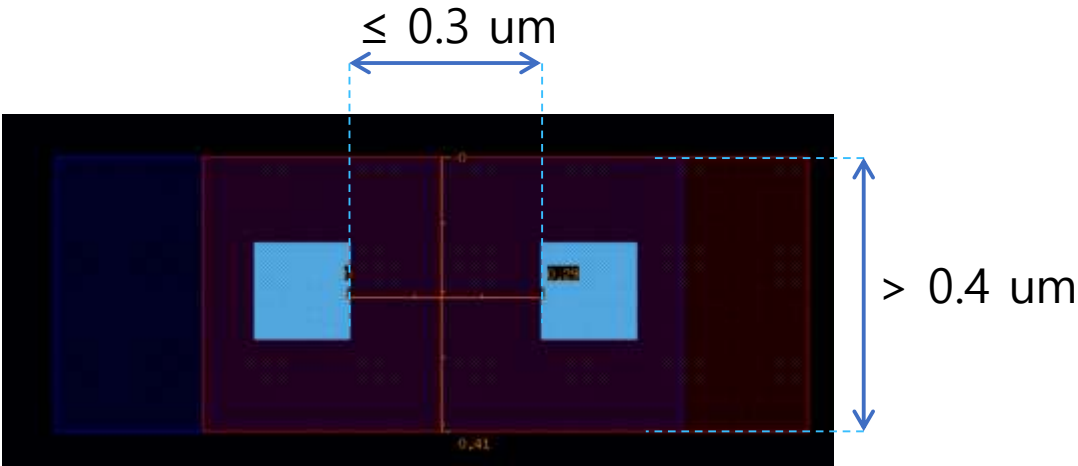
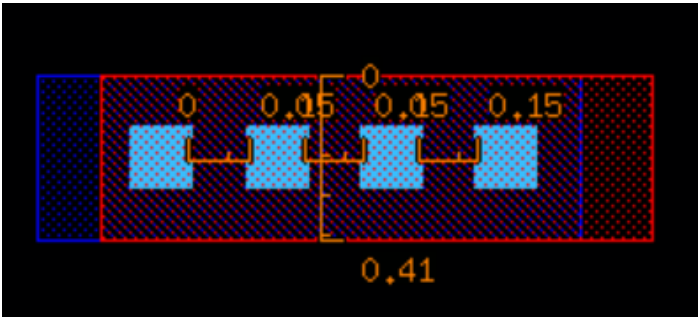
✓ Cadence Reproducing and vertical structure



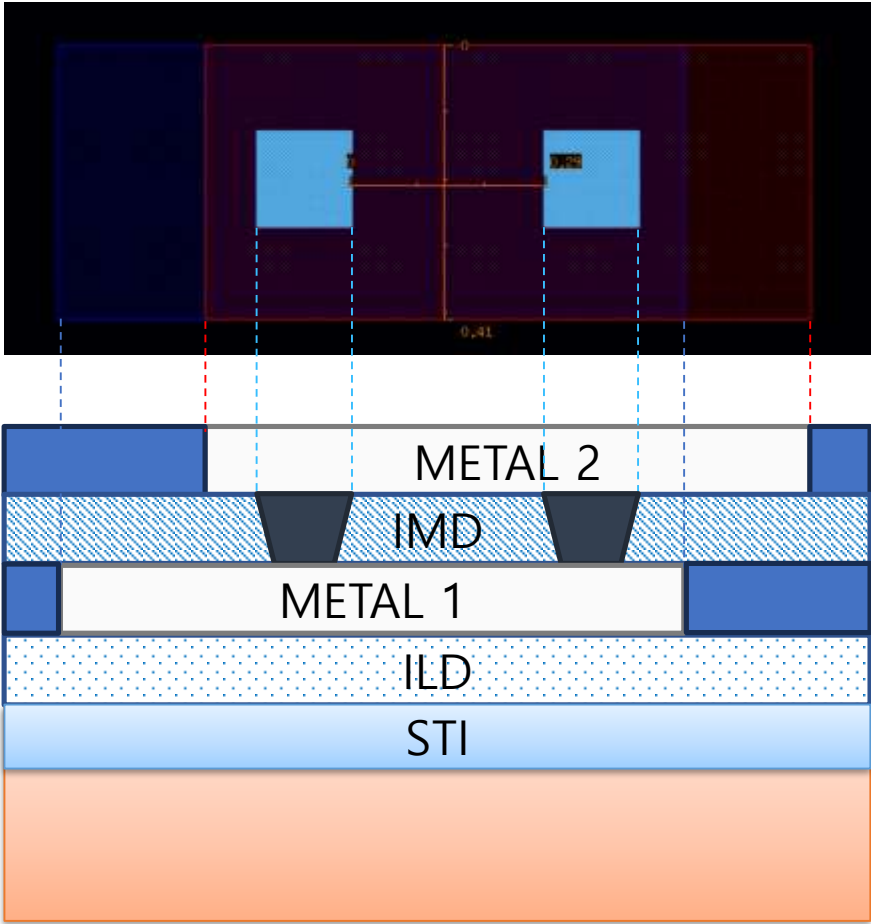
Rule Name	Value	Description
VIAk.X.1	---	Minimum of two Via k with spacing $\leq 0.30\mu\text{m}$ or four Via k with spacing $\leq 0.60\mu\text{m}$ are required when connecting Metal k and Metal k+1 when one of the Metals has a width $> 0.40\mu\text{m}$ at the connection point

✓ Schematic in Design Rule Manual

넓은 Metal 연결부의 전류 분산과 신뢰성 확보를 위해
최소 개수의 Via를 배치



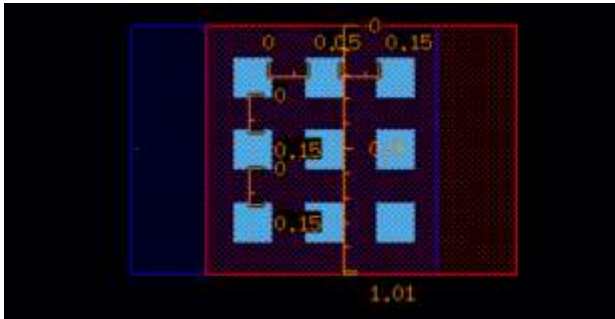
✓ Cadence Reproducing and vertical structure



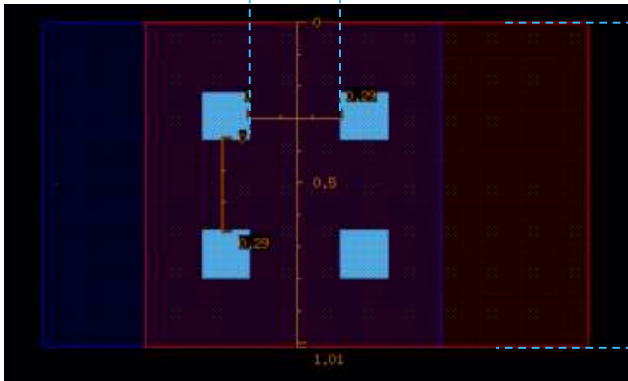
Rule Name	Value	Description
VIAk.X.2	---	Minimum of four Via k with spacing $\leq 0.30\mu\text{m}$ or nine Via k with spacing $\leq 0.60\mu\text{m}$ are required when connecting Metal k and Metal k+1 when one of the Metals has a width $> 1.0\mu\text{m}$ at the connection point

✓ Schematic in Design Rule Manual

더 넓은 Metal에서는 더 큰 전류를 고려하여
더 많은 Via 개수를 요구

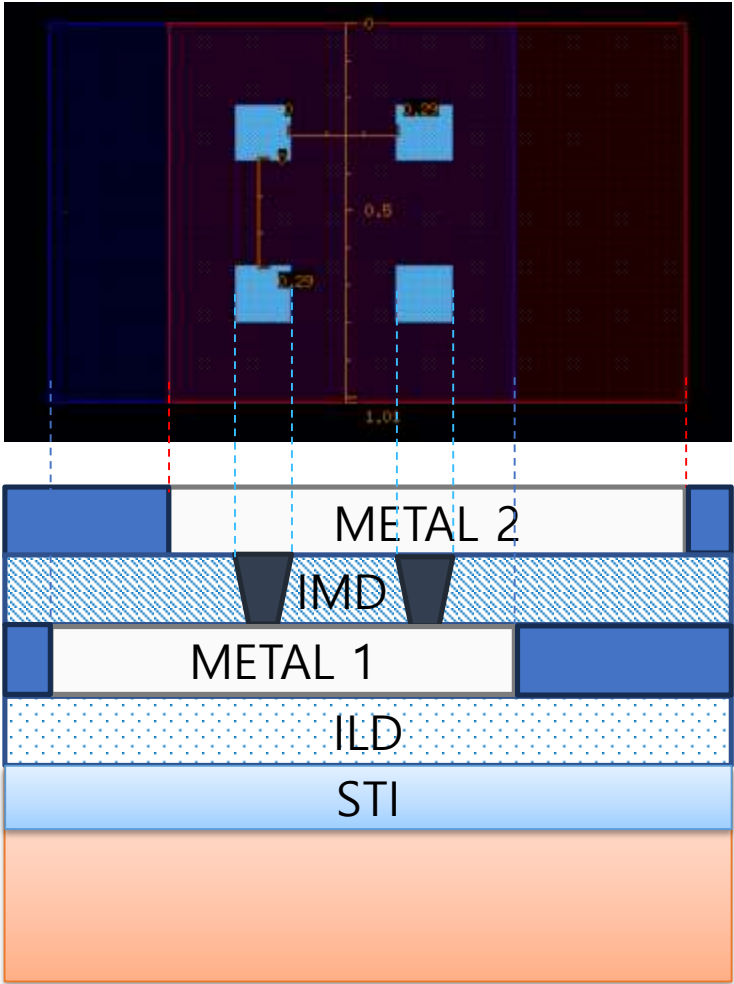


$\leq 0.3 \mu\text{m}$



$> 0.4 \mu\text{m}$

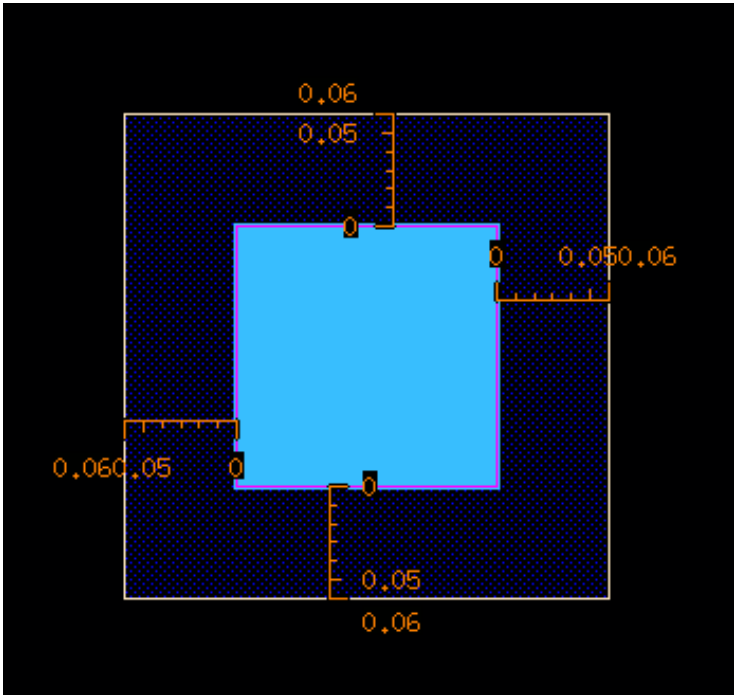
✓ Cadence Reproducing and vertical structure



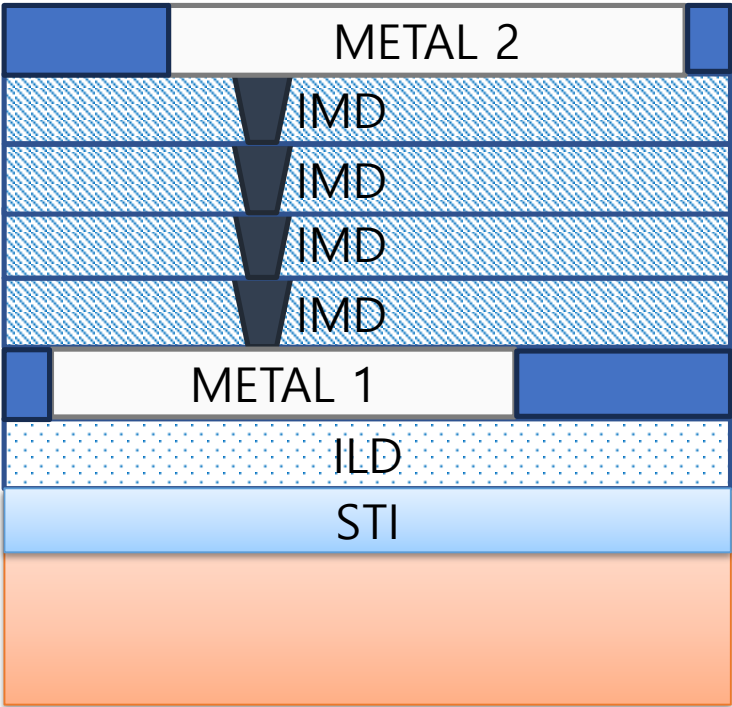
Rule Name	Value	Description
VIAk.X.3	---	Vias 1 through 6 may be consecutively stacked up to four high when only one Via is connecting two Metal layers for any level of the stack

✓ Schematic in Design Rule Manual

단일 Via Stack은 신뢰성 확보를 위해
최대 4층까지만 허용



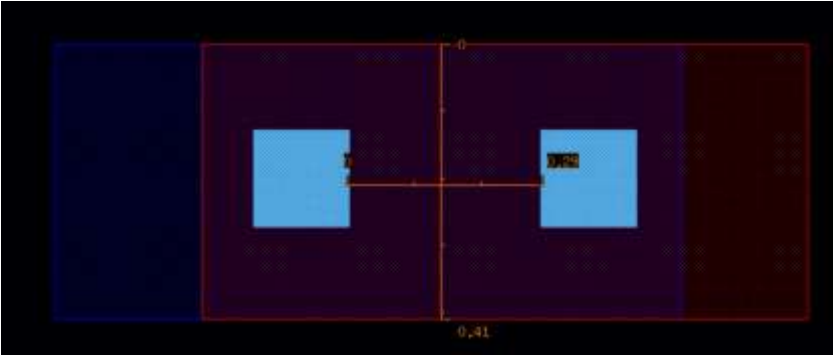
✓ Cadence Reproducing and vertical structure



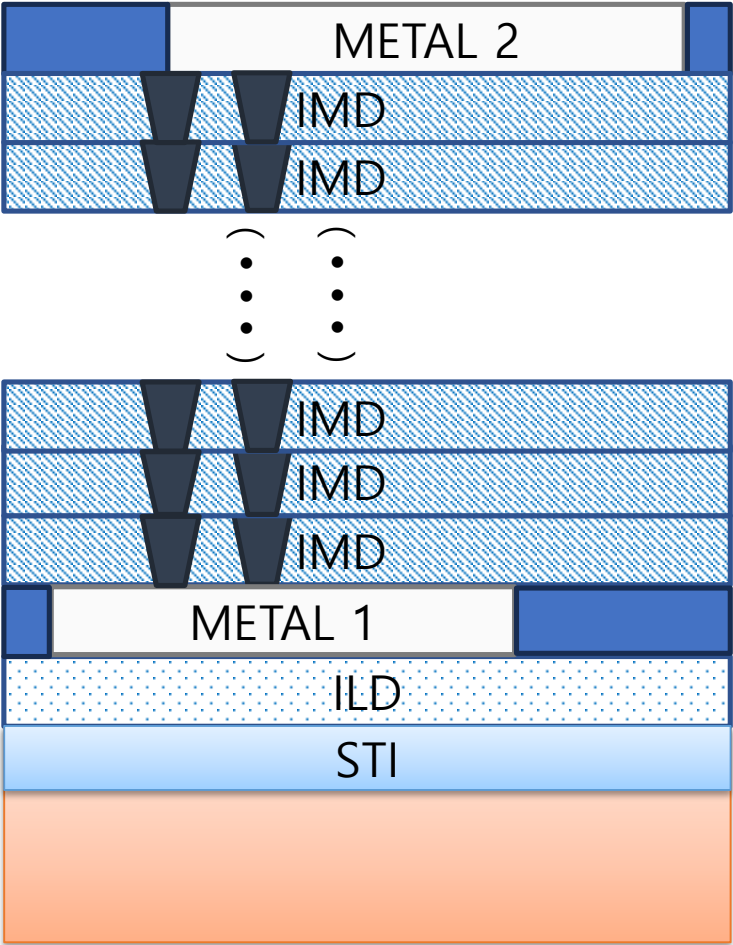
Rule Name	Value	Description
VIAk.X.4	---	Vias 1 through 6 may be consecutively stacked up more than four high when at least two Vias are connecting two Metal layers for all levels of the stack

✓ Schematic in Design Rule Manual

각 층에 2개 이상의 Via를 사용하면
전류 분산을 통해 4층 이상의 Via Stack을 허용



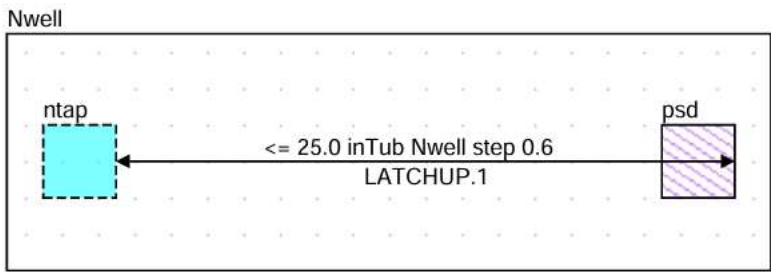
✓ Cadence Reproducing and vertical structure



Latch up

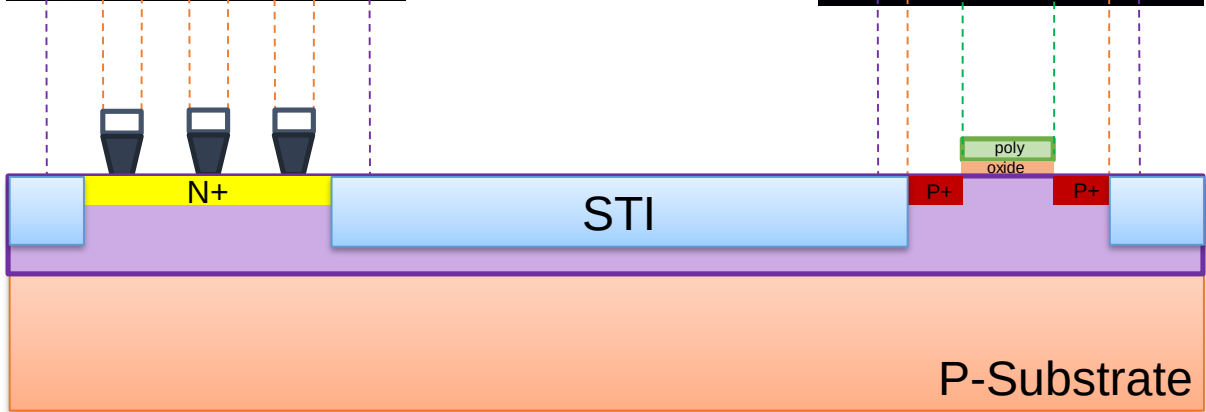
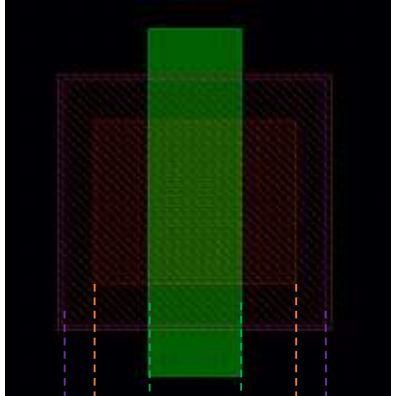
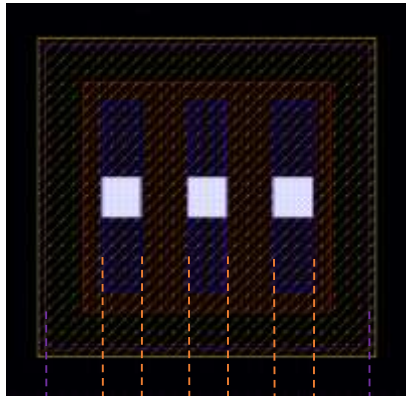
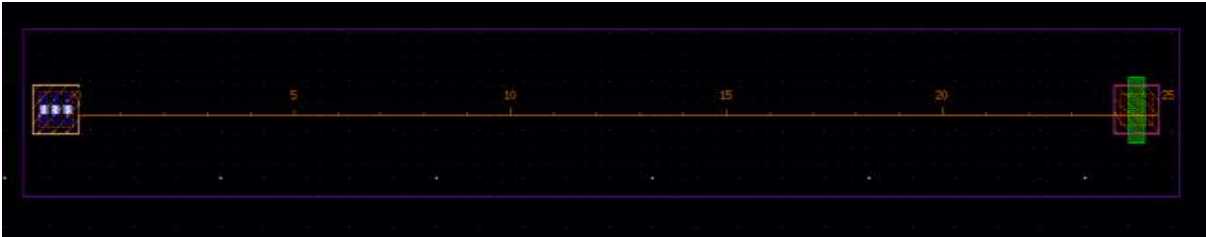
Rule Name	Value	Description
LATCHUP.1	25um	The maximum distance from any point in a P+ source/drain Active Area to the nearest Nwell pick-up in the same Nwell.

✓ Schematic in Design Rule Manual



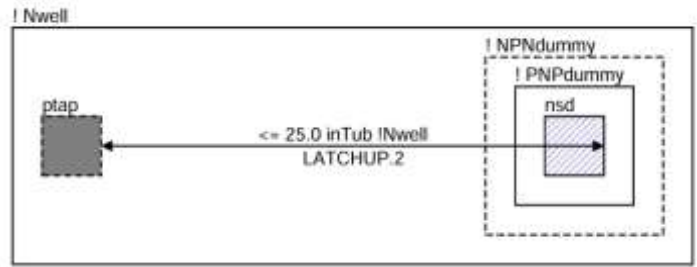
기생 BJT가 켜져서 latch-up이 발생하는 걸 막기 위해

✓ Cadence Reproducing and vertical structure



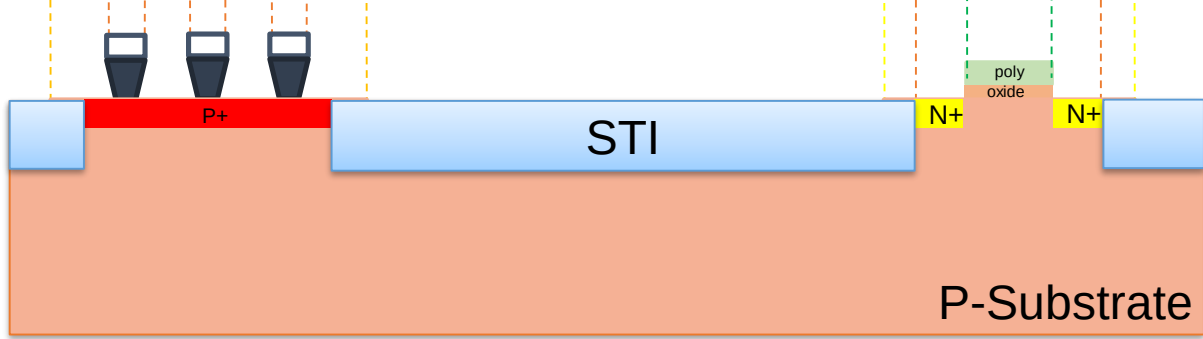
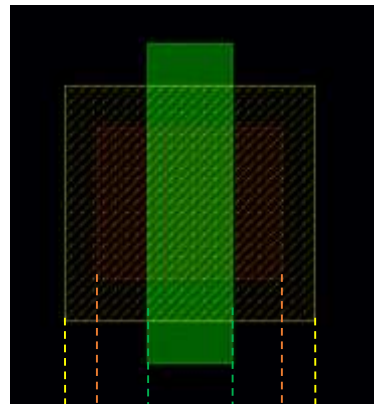
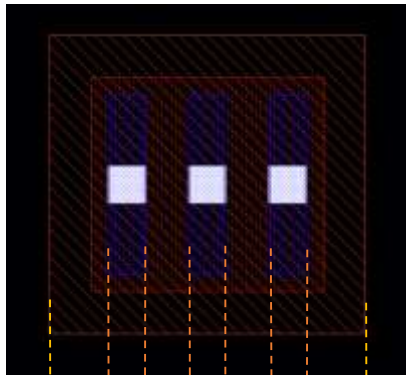
Rule Name	Value	Description
LATCHUP.2	25um	The maximum distance from any point in an N+ source/drain Active Area to the nearest Psub pick-up in the same Psub

✓ Schematic in Design Rule Manual



기생 BJT가 켜져서 latch-up이 발생하는 걸 막기 위해

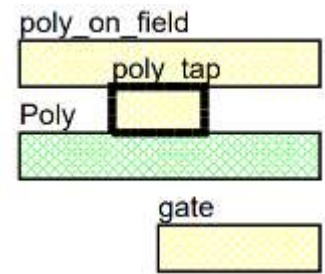
✓ Cadence Reproducing and vertical structure



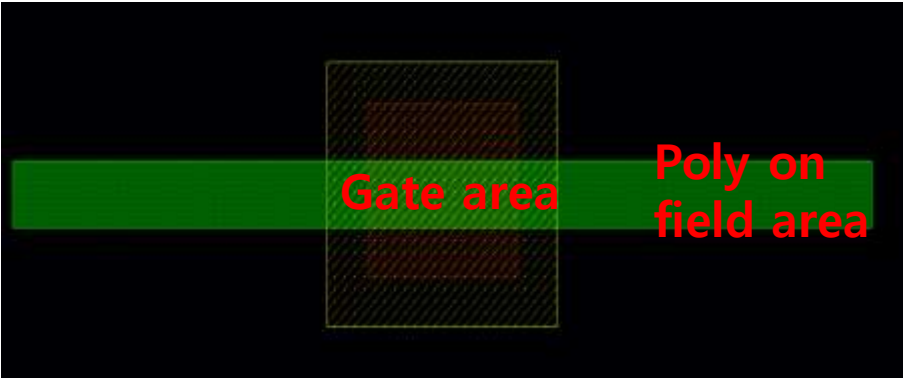
Antenna

Rule Name	Value	Description
ANT.1	275um	Maximum ratio of Poly area to the gate area the Poly is connected to

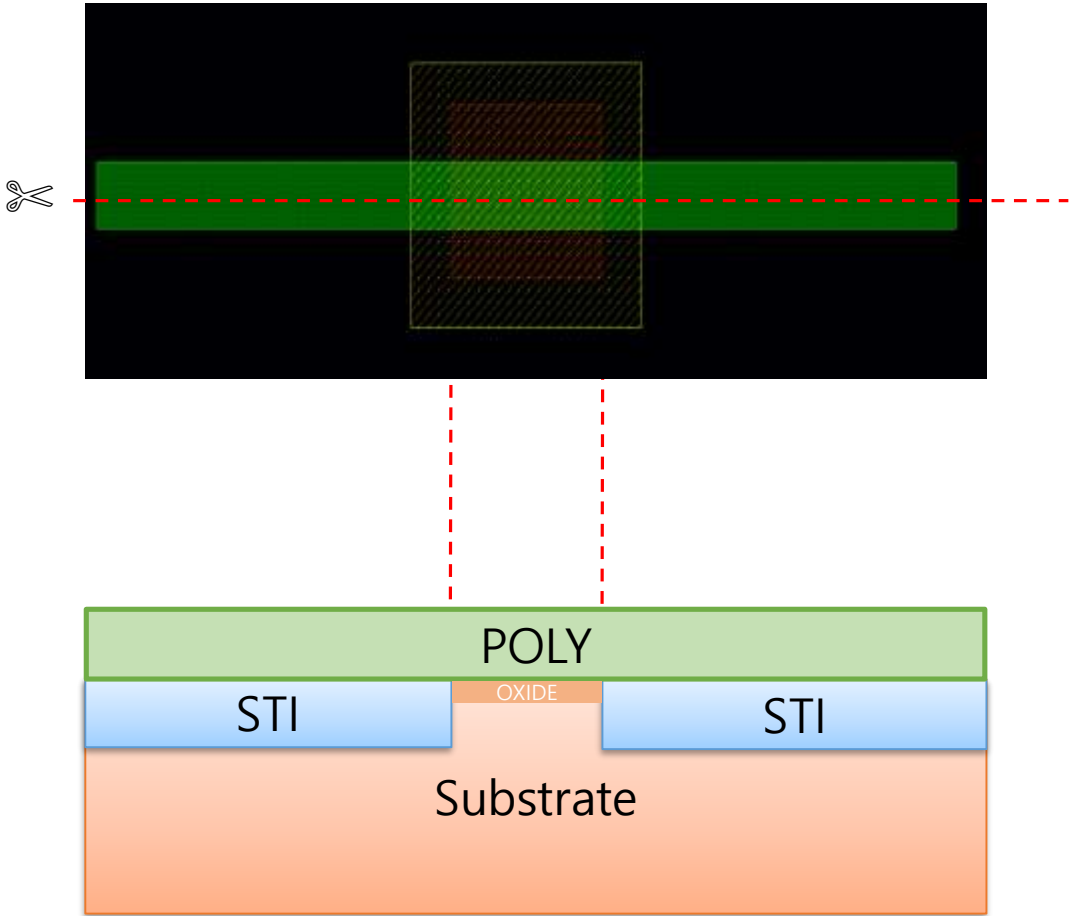
✓ Schematic in Design Rule Manual



공정 중 poly에 축적된 전하로 인해 gate oxide가 손상되는 것을 방지하기 위함이다.

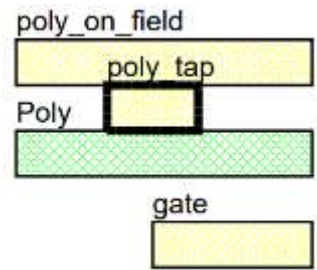


✓ Cadence Reproducing and vertical structure

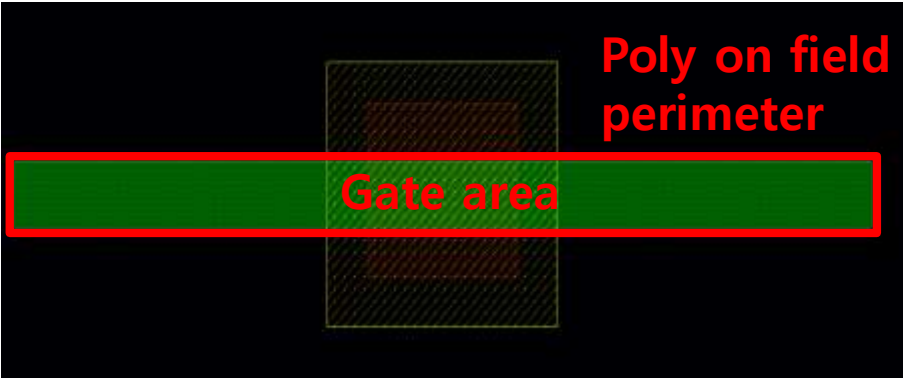


Rule Name	Value	Description
ANT.2	550um	Maximum ratio of Poly sidewall area to the gate area the Poly is connected to

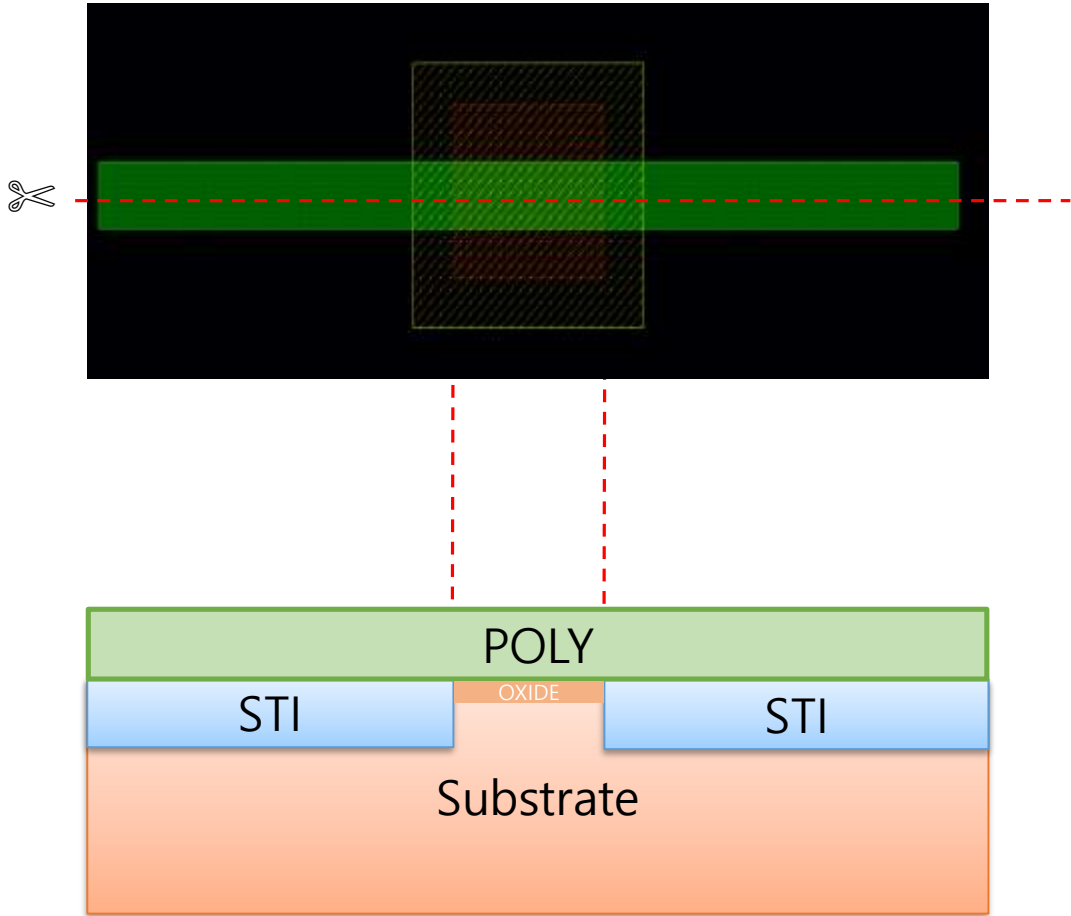
✓ Schematic in Design Rule Manual



Poly 측면에 축적되는 전하로 인해 gate oxide가 손상되지 않게 하기 위함이다.

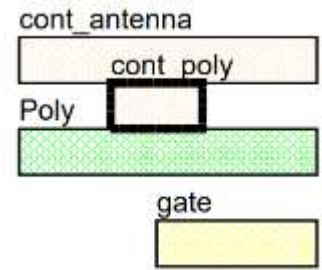


✓ Cadence Reproducing and vertical structure

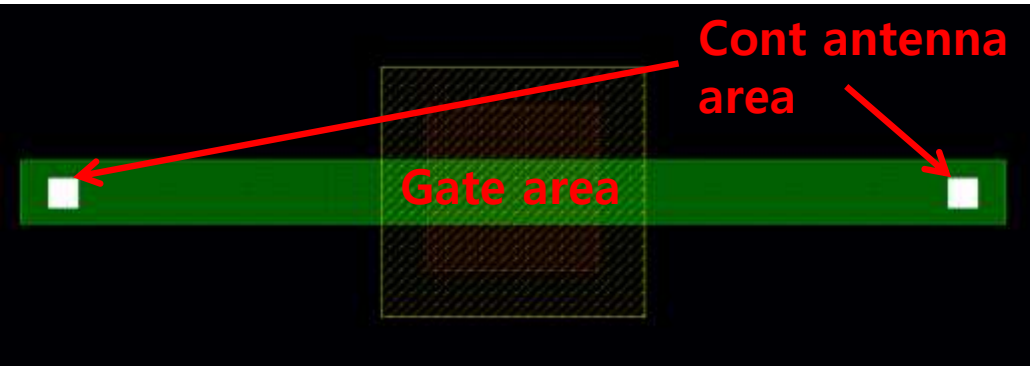


Rule Name	Value	Description
ANT.3	15	Maximum ratio of Poly Contact area to the gate area the Contact is connected with

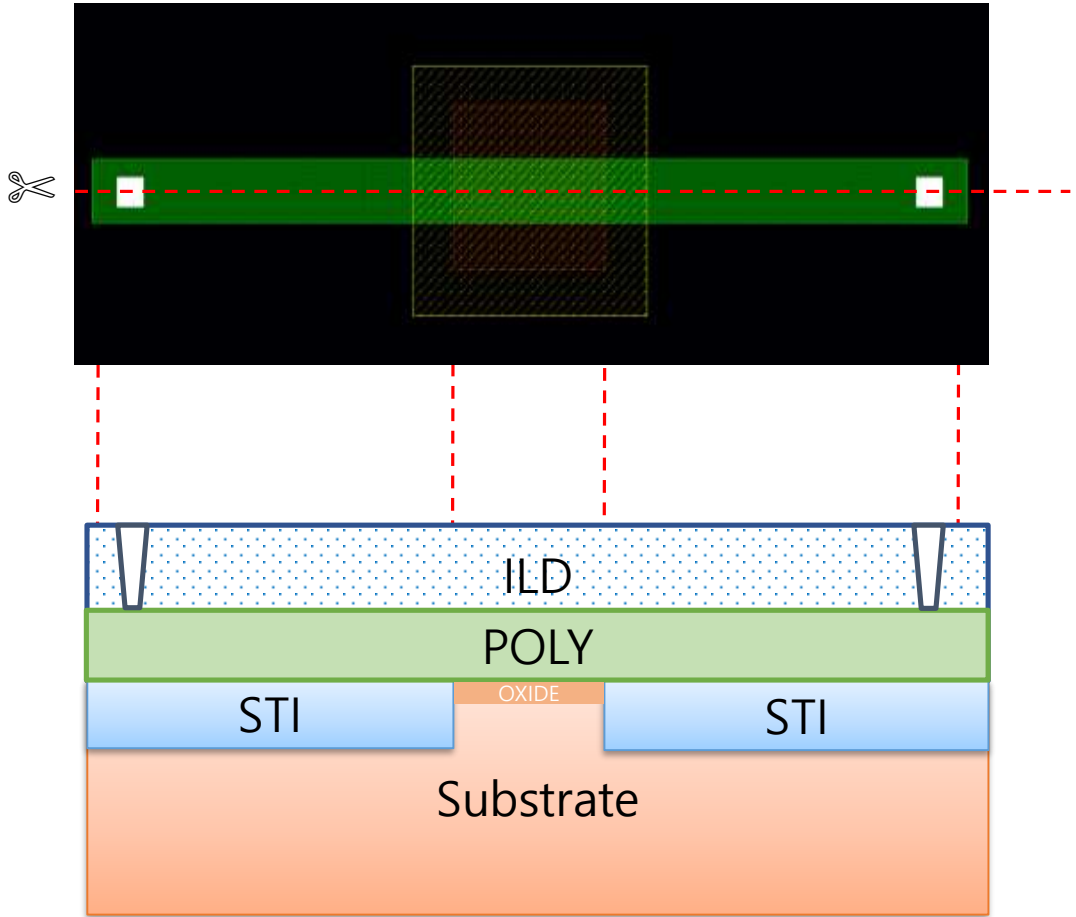
✓ Schematic in Design Rule Manual



Contact을 통해 전달되는 과도한 전하로부터 gate oxide를 보호하기 위함이다.

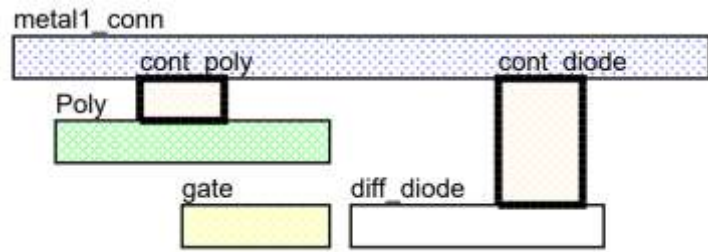


✓ Cadence Reproducing and vertical structure

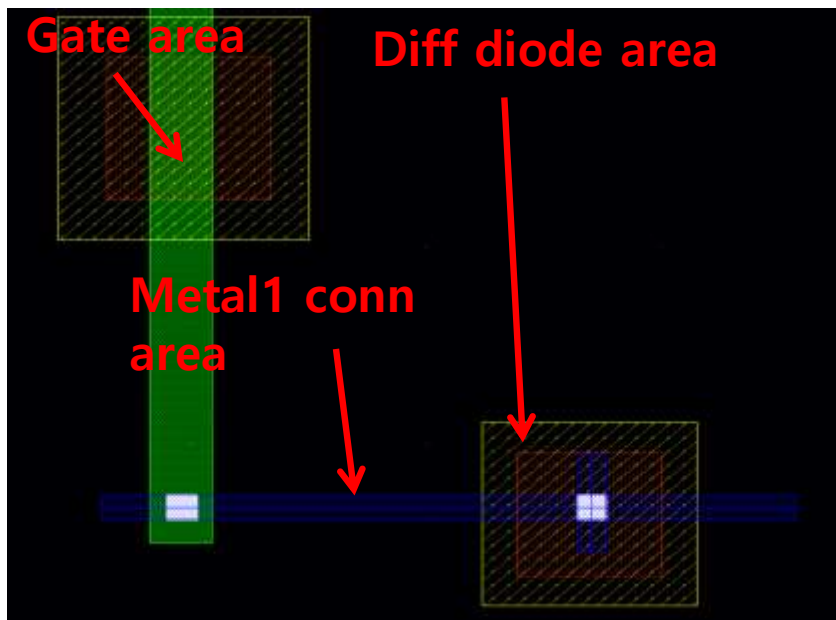


Rule Name	Value	Description
ANT.4	475	Maximum ratio of single level Metal x (x = 1, 2, 3, 4, 5, 6, 7, 8, 9) area to the (gate area + 2*Diff area)

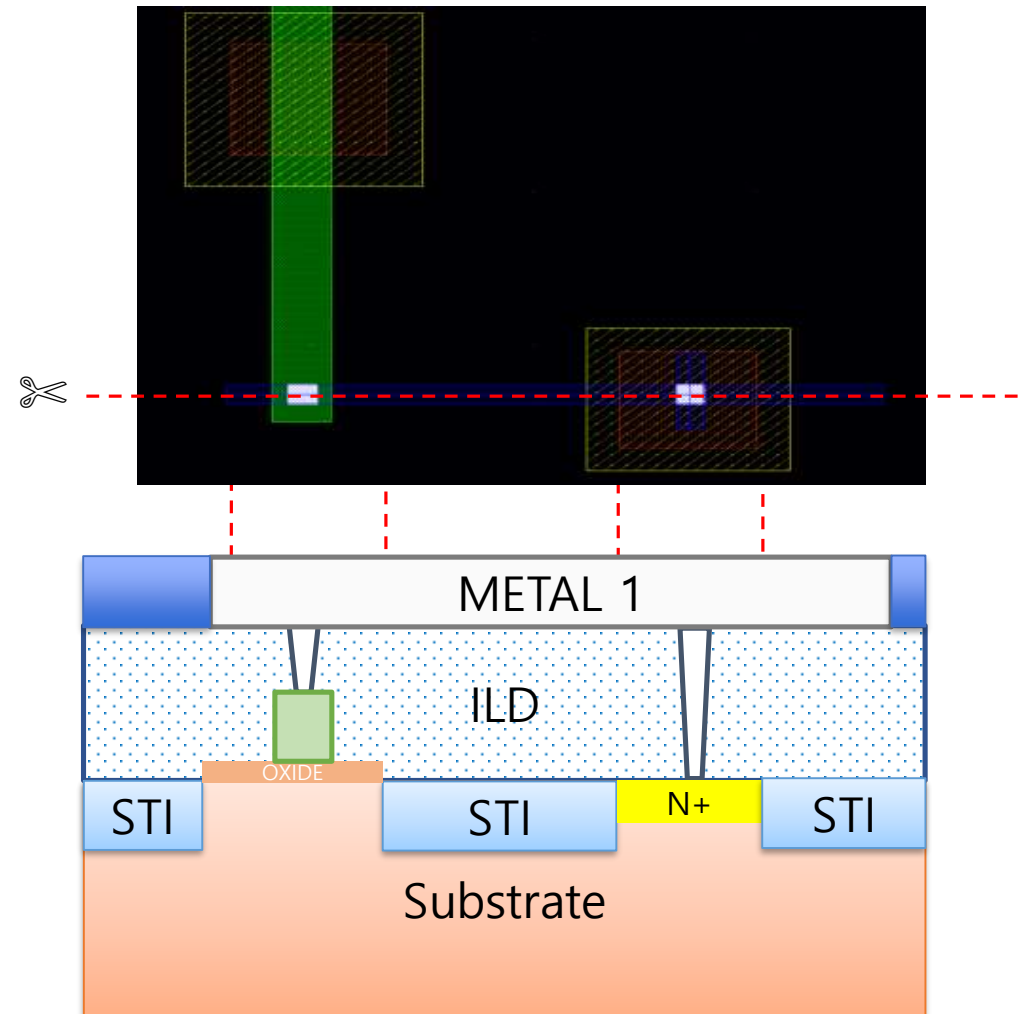
✓ Schematic in Design Rule Manual



공정 중 metal에 축적되는 전하로 부터 gate를 보호하기 위함이다.

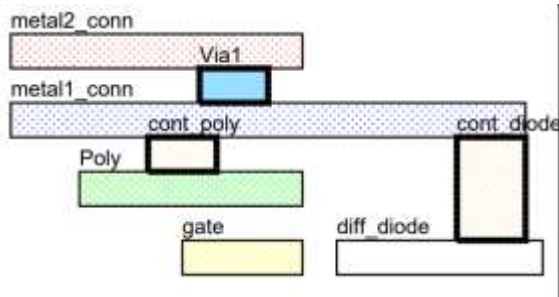


✓ Cadence Reproducing and vertical structure

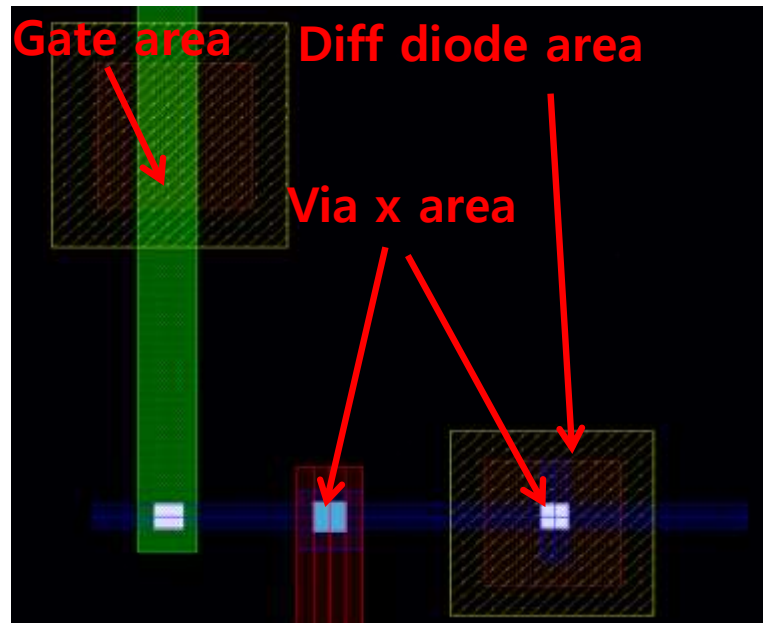


Rule Name	Value	Description
ANT.5	25	Maximum ratio of single level Via x (x = 1, 2, 3, 4, 5, 6, 7, 8) area to the (gate area + 2*Diff area)

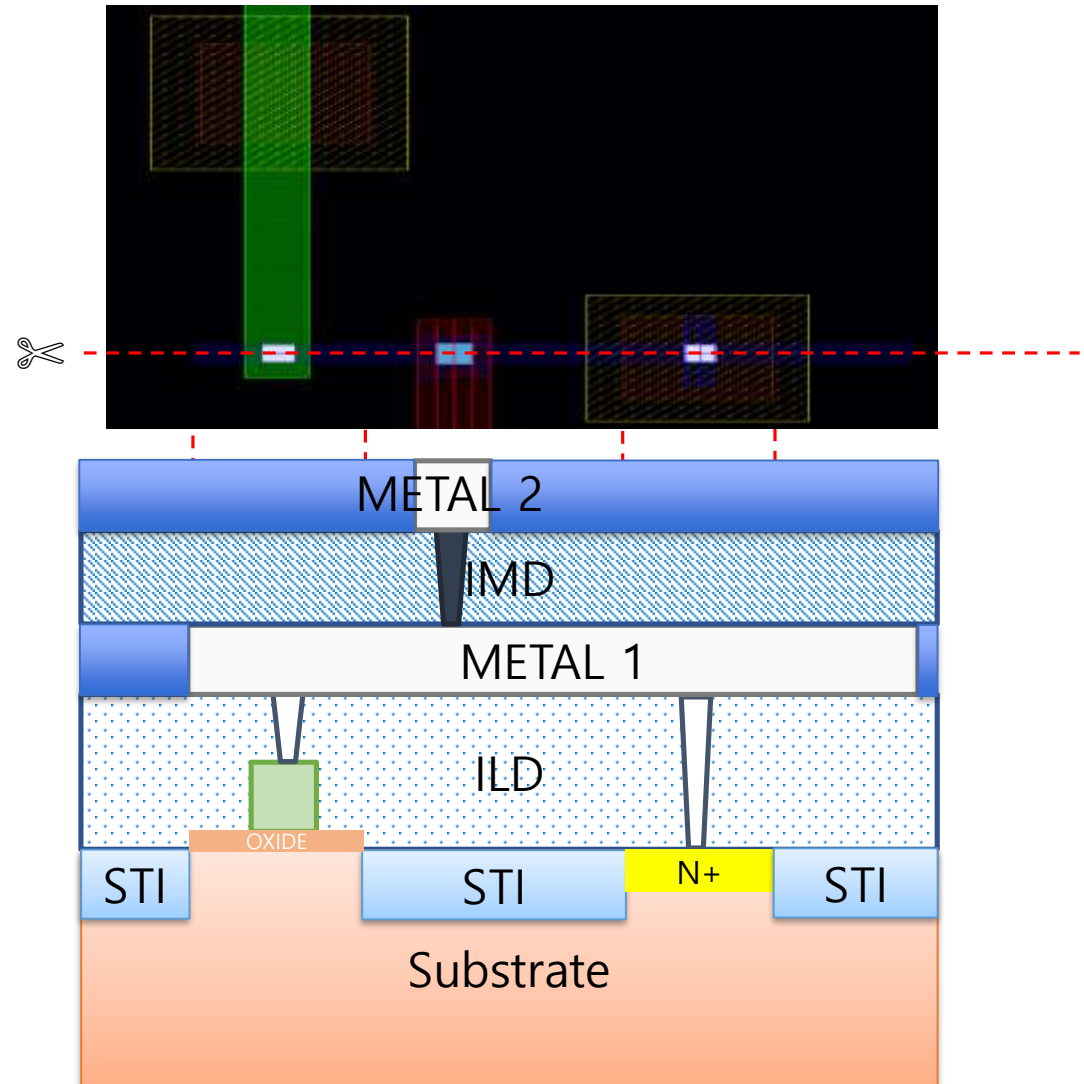
✓ Schematic in Design Rule Manual



공정 중 via를 통해 축적되는 전하로 부터 gate를 보호하기 위함이다.

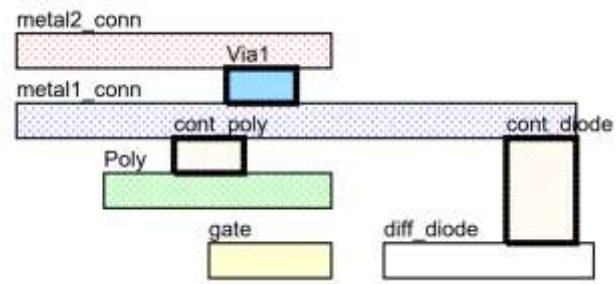


✓ Cadence Reproducing and vertical structure

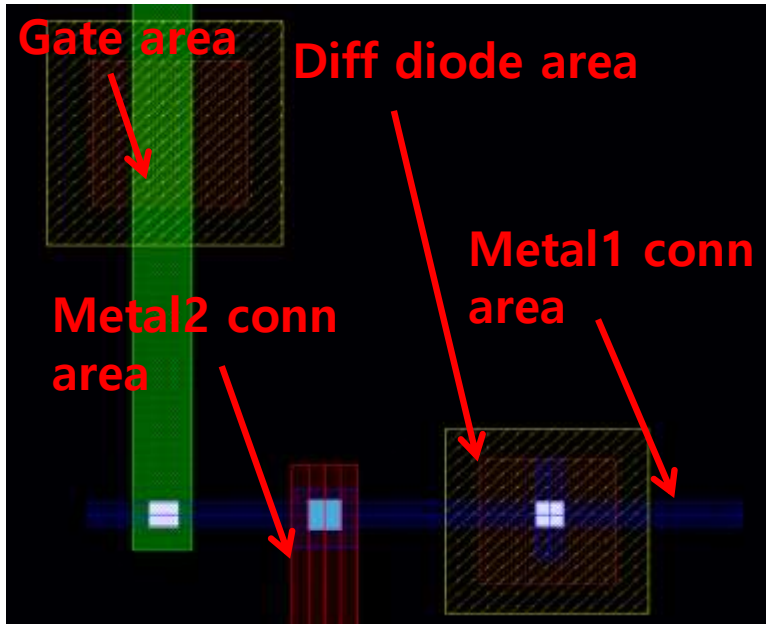


Rule Name	Value	Description
ANT.6	1200	Maximum ratio of cumulative multi level Metal areas to the (gate area + 2*Diff area)

✓ Schematic in Design Rule Manual



공정 중 여러 metal 층에 축적된 전하가 gate oxide를 손상시키는 것을 방지하기 위함이다.



✓ Cadence Reproducing and vertical structure

